

May 7, 2020

Mr. Jonathon Travis
Ryan, LLC
2800 Post Oak Boulevard, Suite 4200
Houston, TX 77056

**RE: Kline Federal #5300 11-18 2B
Lot 1 Sec. 18, T.153N., R.100W.
McKenzie County, North Dakota
Baker Field
Well File No. 29334
STRIPPER WELL DETERMINATION**

Dear Mr. Travis:

Oasis Petroleum North America LLC (Oasis) filed with the North Dakota Industrial Commission – Oil and Gas Division (Commission) on November 19, 2019 an application for a Stripper Well Determination for the above captioned well.

Information contained in the application indicates that the above mentioned well is a stripper well pursuant to statute and rule, and Oasis has elected to designate said well as a stripper well. The well produced from a well depth greater than 10000 feet and was completed after June 30, 2013. During the qualifying period, February 1, 2018 through January 31, 2019, the well produced at a maximum efficient rate or was not capable of exceeding the production threshold. The average daily production from the well was 32.8 barrels of oil per day during this period.

It is therefore determined that the above captioned well qualifies as a “Stripper Well” pursuant to Section 57-51.1-01 of the North Dakota Century Code. This determination is applicable only to the Bakken Pool in and under said well.

The Commission shall have continuing jurisdiction, and shall have the authority to review the matter, and to amend or rescind the determination if such action is supported by additional or newly discovered information. If you have any questions, do not hesitate to contact me.

Sincerely,


David J. McCusker
Petroleum Engineer

Cc: ND Tax Department

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

Well File No.
29334

FEB 12 2016

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

ND Oil & Gas Division

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed December 11, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☒ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Well is now on pump

Well Name and Number Kline Federal 5300 11-18 2B					
Footages	Qtr-Qtr	Section	Township	Range	
960 F N L 318 F W L	Lot 1	18	153 N	100 W	
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 12/11/2015 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 9924'

Pump: ESP @ 9924'

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Specialist	Date February 8, 2016		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date 3-3-2016	
By 	
Title TAYLOR ROTH	
Engineering Technician	



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE – Form 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)

Well File No.
29334
NDIC CTR No.
0

228651

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number KLINE FEDERAL 5300 11-18 2B	Qtr-Qtr LOT1	Section 18	Township 153	Range 100	County Mckenzie
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Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
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Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
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Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective September 20, 2015
--	------------------------------------	---------------------	--------------------------------------

Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
--	-----------------	-------------	-------------------

Field Address	City	State	Zip Code
---------------	------	-------	----------

Transporter Hiland Crude, LLC	Telephone Number (580) 616-2058	% Transported 75%	Date Effective September 20, 2015
----------------------------------	------------------------------------	----------------------	--------------------------------------

Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
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The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other Transporters Transporting From This Lease Power Crude Transport	% Transported 25%	Date Effective September 20, 2015
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Other Transporters Transporting From This Lease	% Transported	Date Effective September 20, 2015
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Comments

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Date October 22, 2015
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Signature 	Printed Name Brianna Salinas	Title Marketing Assistant
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Above Signature Witnessed By: Signature 	Printed Name Laura Whitten	Title Marketing Analyst II
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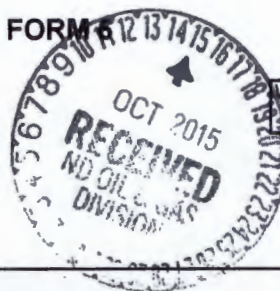


FOR STATE USE ONLY	
Date Approved OCT 27 2015	
By 	
Title Oil & Gas Production Analyst	



WELL COMPLETION OR RECOMPLETION REPORT - FORM 1

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No.
29334

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Kline Federal 5300 11-18 2B		Spacing Unit Description Sec. 17/18 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Field Baker	
City Houston		Pool Bakken	
State TX		Zip Code 77002	
Permit Type ☐ Wildcat ☒ Development ☐ Extension			

LOCATION OF WELL

At Surface 960 F N L	318 F W L	Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W	County McKenzie
Spud Date 2.25-15 March 29, 2015	Date TD Reached April 16, 2015	Drilling Contractor and Rig Number Xtreme 21		KB Elevation (Ft) 2078	Graded Elevation (Ft) 2053	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD; CBL from int. TD to surface						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	Type	String Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	13 3/8	0	2135	17 1/2	54.5			1226	0
Vertical Hole	Intermediate	7	0	11024	8 3/4	32			825	2150
Lateral1	Liner	4 1/2	10193	20445	6	13.5			500	10193

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft)		Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20450	Perforations	11024	20445	10225		07/16/2015			
	17597			17597						
ST 1	20450		17420	20445						

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1- 11024' to 20445'						Name of Zone (If Different from Pool Name)	
Date Well Completed (SEE INSTRUCTIONS) September 20, 2015		Producing Method Flowing		Pumping-Size & Type of Pump		Well Status (Producing or Shut-In) Producing	
Date of Test 09/20/2015	Hours Tested 24	Choke Size 48 /64	Production for Test	Oil (Bbls) 661	Gas (MCF) 498	Water (Bbls) 4579	Oil Gravity-API (Corr.) 42.0 °
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI) 750		Calculated 24-Hour Rate	Oil (Bbls) 661	Gas (MCF) 498	Water (Bbls) 4579
							Gas-Oil Ratio 753

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
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Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 07/16/2015	Stimulated Formation Bakken	Top (Ft) 11024	Bottom (Ft) 20445	Stimulation Stages 36	Volume 209640	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 4168220	Maximum Treatment Pressure (PSI) 9141		Maximum Treatment Rate (BBLS/Min) 73.0	
Details 100 Mesh White: 295320 40/70 white: 1555360 30/50 White: 1740010 30/50 Resin Coated: 504920 20/40 Resin Coated: 69610						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.			Email Address jswenson@oasispetroleum.com	Date 10/14/2015
Signature 	Printed Name Jennifer Swenson	Title Regulatory Specialist		

Industrial Commission of North Dakota
Oil and Gas Division

Well or Facility No
29334

Verbal Approval To Purchase and Transport Oil

Tight Hole **Yes**

OPERATOR

Operator	Representative	Rep Phone
OASIS PETROLEUM NORTH AMERICA LL	Todd Hanson	(701) 577-1632

WELL INFORMATION

WELL INFORMATION

Well Name						Inspector
KLINE FEDERAL 5300 11-18 2B						Richard Dunn
Well Location	QQ	Sec	Twp	N	Rng	County
	LOT1	18	153	N	100	W MCKENZIE
Footages	960	Feet From the N Line				Field BAKER
	318	Feet From the W Line				Pool BAKKEN
Date of First Production Through Permanent Wellhead						This Is Not The First Sales

PURCHASER / TRANSPORTER

Purchaser	Transporter
Kinder Morgan	Kinder Morgan

TANK BATTERY

Single Well Tank Battery Number :

SALES INFORMATION This Is Not The First Sales

ESTIMATED BARRELS TO BE SOLD		ACTUAL BARRELS SOLD	DATE
1500	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	

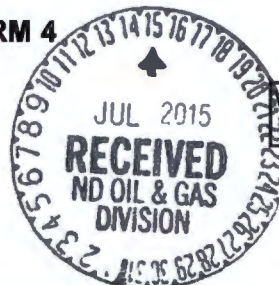
DETAILS

Must E-Mail or Call Inspector at 701-770-3554/rsdunn@nd.gov on first date of sales and report amount sold, date sold, and first date of production through the permanent wellhead. Must also forward Forms 6 & 8 to State prior to reaching 15000 Bbl estimate or no later than required time frame for submitting those forms.

Start Date **8/25/2015**
Date Approved **8/25/2015**
Approved By **Richard Dunn**

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

29334

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date July 14, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	<u>Waiver from tubing/packer requirement</u>

Well Name and Number Kline Federal 5300 11-18 2B						
Footages		Qtr-Qtr	Section	Township	Range	
960 F N L 318 F W L		LOT1	18	153 N	100 W	
Field Baker		Pool Bakken		County McKenzie		

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by monitoring equipment;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date July 14, 2015	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

☐ Received	☒ Approved
Date July 31, 2015	
By 	
Title PETROLEUM ENGINEER	



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (08-2008)



Well File No.
29334

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date August 9, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporary Abandon	☐ Reclamation
☒ Other	Change well status to CONFIDENTIAL

Well Name and Number Kline Federal 5300 11-18 2B					
Footages	Qtr-Qtr	Section	Township	Range	
960 F N L 318 F W L LOT 1		18	153 N	100 W	
Field Baker	Pool BAKKEN	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective Immediately, we request **CONFIDENTIAL STATUS** for the above referenced well.

This well has not been completed

OFF CONFIDENTIAL 2/10/16.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date August 10, 2015	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date	8/12/15
By	
Title	Engineering Technician



Oasis Petroleum North America, LLC.

Kline Federal 5300 11-18 2B

960' FNL & 318' FWL

Lot 1 Section 18, T153N, R100W

Baker Field / Middle Bakken

McKenzie County, North Dakota

BOTTOM HOLE LOCATION:

111.86' south & 9,932.10' east of surface location or approx.

1,071.86' FNL & 296.01' FEL, NE NE Section 17, T153N, R100W

Prepared for:

Curtis Johnson
Oasis Petroleum
North America, LLC.
1001 Fannin, Suite 1500
Houston, TX 77002

Prepared by:

Dillon Johnson, Matt Hegland,
Ryan O'Donnell
PO Box 80507; Billings, MT 59108
(406) 259-4124
geology@sunburstconsulting.com
www.sunburstconsulting.com

WELL EVALUATION

Oasis Petroleum North America, LLC *Kline Federal 5300 11-18 2B*



Figure 1. Xtreme 21 at the Oasis Petroleum North America, LLC *Kline Federal 5300 11-18 2B* well during June of 2015 in McKenzie County, North Dakota.

INTRODUCTION

The Oasis Petroleum, North America LLC, *Kline Federal 5300 11-18 2B* well is located approximately 4 miles south of Williston in McKenzie County, North Dakota [Lot 1 Section 18, T153N, R100W]. The *Kline Federal 5300 11-18 2B* is a horizontal, Middle Bakken Member well drilled by Xtreme rig #21 (**Figure 1**). The lateral section consists of one west to east trending, 9,932' long wellbore penetrating underneath Lake Sakakawea. The surface, vertical and curve sections were previously drilled, and this evaluation will consist of primarily operations and observations made during the lateral section.

ENGINEERING

The *Kline Federal 5300 11-18 2B* surface, vertical and curve sections were drilled earlier in 2015 by separate drilling rig. Xtreme 21 returned to location in May of 2015 to drill the laterals of the *Kline Federal 5300 11-18 2B*, 3T, 2B, and re-entered the *Kline Federal 5300 11-18 2B* on May 30, 2015.

Drilling began with a 6" Halliburton PDC bit, Ryan Directional Services MWD tools, and 1.5° adjustable Baker Hughes mud motor. This assembly drilled to 15,484' MD where a trip was made due to low penetration rates. Another complete assembly drilled to 17,597' MD, where at 23:30 CDT on June 2, 2015 the wellbore made contact with the upper Bakken shale at 17,540' MD, 10,766' TVD (-8,698'). A sidetrack was initiated at 17,420' MD, using the same assembly, and successfully kicked off. After the sidetrack, drilling continued to 20,149' MD where another trip was made due to slowing penetration rates. A new motor was picked up and the well later reached its total depth at 20,450' MD, at 08:45 CDT on June 6, 2015, generating an exposure of 9,568' linear feet of 6" hole through the Middle Bakken Member. The bottom hole location lies 111.86' south & 9,932.10' east of surface location or approximately 1,071.86' FNL & 296.01' FEL, NE NE Section 17, T153N, R100W.

GEOLOGY

Lithology

Sunburst Geology, Inc. was not present for the vertical and curve sections, therefore no lithology is discussed prior to the Middle Bakken Member.

Middle Bakken Member

The Middle Bakken Member was penetrated at 10,882' MD, 10,727' TVD (-8,659'). The middle member in this locality is generally composed of a light gray-brown to light gray, very fine to fine grained, moderately to well cemented, calcite cemented silty sandstone. It was apparent that samples observed higher in the member were often light brown, off white to cream colored, while the lower portion of the member was predominantly darker, gray-brown. Throughout the lateral, trace amounts of both disseminated and nodular pyrite were observed. Samples also displayed *trace, poor to fair intergranular porosity* and *trace spotty and occasionally even, light brown oil staining* (**Figure 2**).



Figure 2. Middle Bakken silty sandstone, observed in the upper portion of the target interval.

Geosteering

The potential pay zone for the *Kline Federal 5300 11-18 2B* was identified by evaluating gamma data collected while drilling the *Oasis Petroleum, Kline Federal 5300 41-18 11 T2*, and the steering practices from the *Oasis Petroleum, Kline Federal 5300 11-18 5B*. The target was determined to be 12' thick, and located 12' below the Upper Bakken Shale and 20' above the Lower Bakken Shale. The Middle Bakken Member displayed gamma counts ranging from 70 to 140 (API). The target was identified by having a higher gamma marker at the very top of target, moderate markers throughout the target zone and a cool marker at the base (**Figure 3**). Despite a single, upper shale strike, geologists were successful in exposing 9,568' of potentially productive Middle Bakken Member the well bore. The well was within the targeted zone for 67% of the lateral.

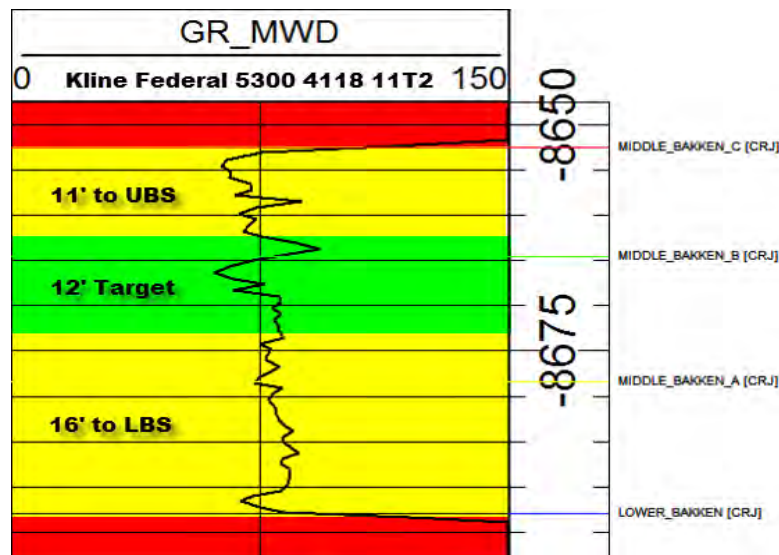


Figure 3. Gamma characteristics observed throughout the Middle Bakken Member while drilling the *Oasis Petroleum, Kline Federal 5300 41-18 11 T2*.

Shale Strike Summary

At measured depths 16,750' through 16,820', erratic high gamma counts were logged. This high count anomaly was interpreted as being the warm gamma associated with being near the top of target. Given the well bore was descending at inclinations ranging from 88.9° to 88.2° (16,750' through 16,820'), it was anticipated that this higher gamma would repeat as the well bore leveled off or slowly rose in zone. As the wells descent was controlled and the inclination raised above 90° (~17,050' MD), the dip profile displayed the well bore being ~3' from the top of target (13.5' from base of UBS). From this point forward the well bore slowly rose in section and geologists did not see any great concern with the well bore position, given that we had yet to see any warm gamma signatures associated with the top of target. In addition, a majority of the wells within this two-section corridor, drilling east, had a synclinal feature after the halfway point of the lateral. Between the surveys at 17,395' and 17,489' MD, the inclination increased from 89.9° to 91.7°. While sliding down to correct the ascending inclination gamma counts rose. It was at this time that decision to stop and circulate a bottoms-up was made. Samples from the bottoms up confirmed the Upper Bakken Shale had been contacted.

Gas and Oil Shows

Hydrocarbon shows at the beginning of the lateral were relatively low. During the first ~1,500' of lateral operations the well bore remained primarily near the base if not below the target zone. While drilling this interval background gasses rarely exceeded 1,000 units, but connection, survey and slide gasses regularly exceeded 1,000 units. Oil show throughout this period of drilling was moderate, with most cuttings reacting slowly with a *pale yellow diffuse cut fluorescence*. As the well bore reached the top of the target interval hydrocarbon shows increased. These brief periods of higher gas shows occurred three times while drilling along the upper portion of the target zone (12,250' - 12,510' MD, 13,200' - 13,950' MD, and 16,300' - 16,430' MD) (**Figure 4**). Although these optimistic shows were intermittent, samples displayed a more promising oil staining that that observed in lower samples. Samples along the upper portion displayed a light to *medium brown even oil staining*. Fluorescent cuts were also slightly more promising than those observed lower in zone were. Cuts were described as: *moderate, pale yellow, streaming cut fluorescence*. While drilling from 11,510' to 11,656' MD there were issues with the gas detection, this caused low gas readings. When the system was repaired, gas shows immediately climbed. For the remainder of the well, despite of the well being high or low in zone, background gasses were low (400-1,200 units), while connection gasses occasionally exceeded 1,300 units. Unfortunately, due to heavy contamination from lubricants there were no cuts performed after 14,950' MD. Although hydrocarbon shows remained consistent despite of position within the target zone, the most promising oil staining remained in the upper most portion of the target interval.

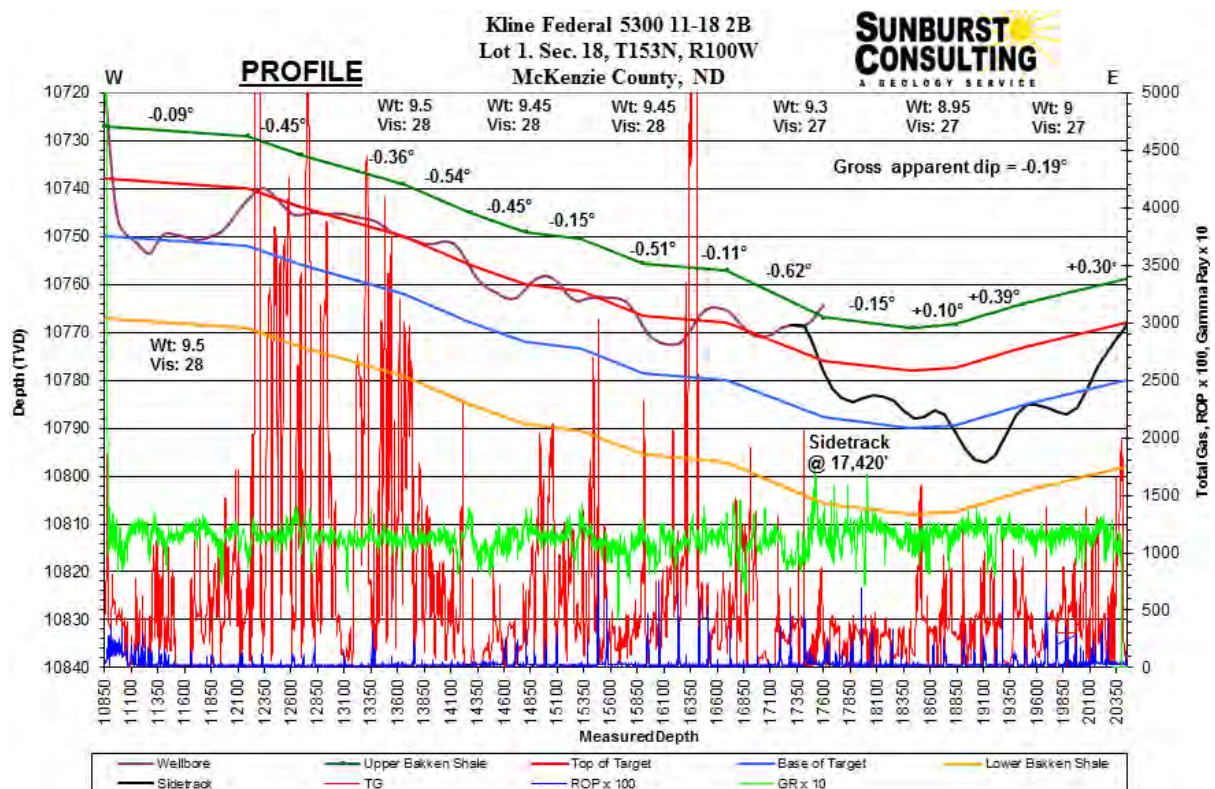


Figure 4. Cross-sectional interpretation of the *Kline Federal 5300 11-18 2B* borehole with total gas units and estimated dip based on lithology, MWD data, drill rate, and regional structural data.

SUMMARY

The Oasis Petroleum North America, LLC, *Kline Federal 5300 11-18 2B* lateral section was drilled by Xtreme 21, from re-entry to total depth, in 7 days. The well reached a total depth of 20,450' MD on June 6, 2015. Geologic data, hydrocarbon gas measurements, and sample examination indicate an encouraging Middle Bakken Member well. Multiple sections of higher gas were recorded through the lateral section. Positive connection and survey shows, exceeding 1,000 units, were recorded throughout the lateral. Samples from the target zone consisted of light brown to off white, silty sandstone and yielded a *moderate, pale yellow, streaming cut fluorescence* and poor to fair intergranular porosity. The well was within the targeted zone for 67% of the lateral and exposed 9,568' of potentially productive Middle Bakken Member to the well bore. The well currently awaits completion operations.

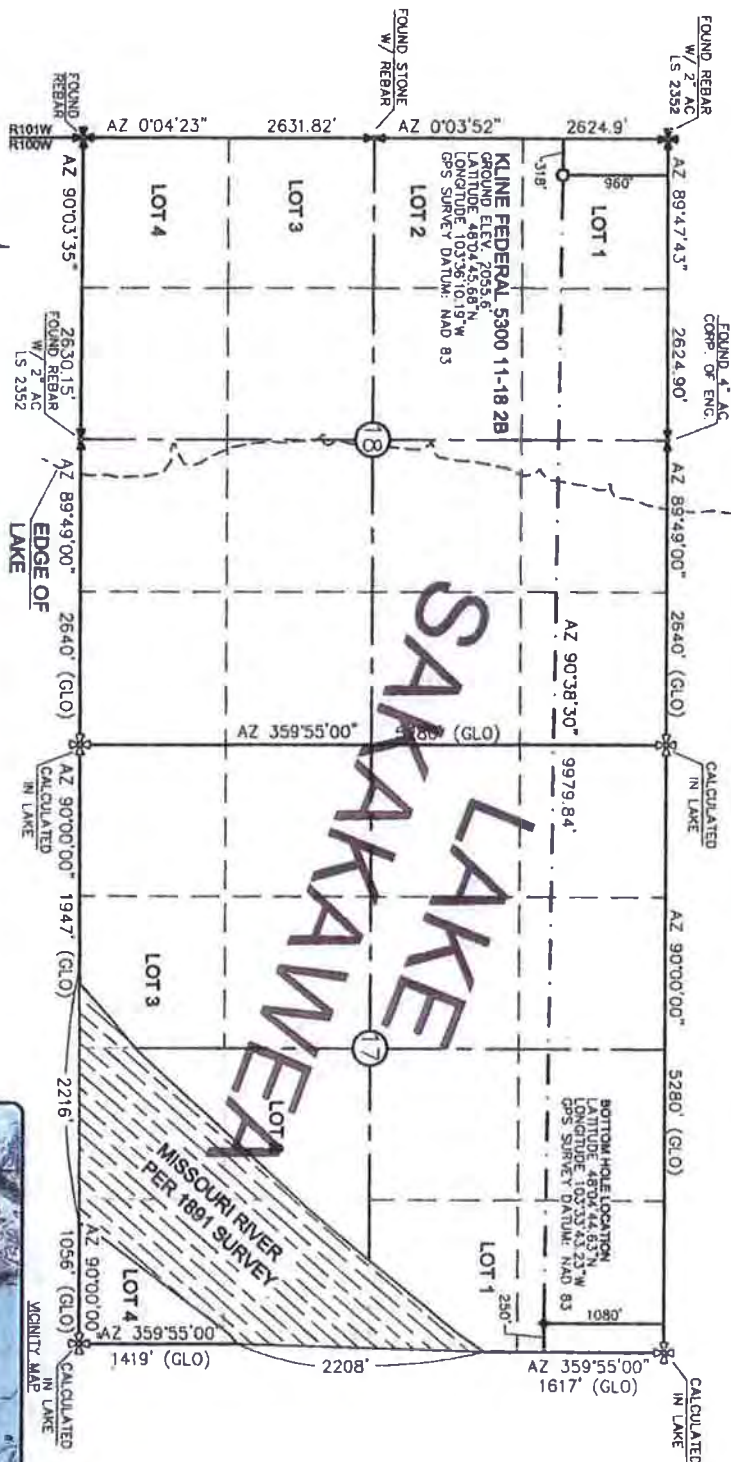
Respectfully submitted,
Ryan O'Donnell
Well Site Geologist
Sunburst Consulting, Inc.
June 6, 2015

WELL DATA SUMMARY

<u>OPERATOR:</u>	Oasis Petroleum North America, LLC.
<u>ADDRESS:</u>	1001 Fannin, Suite 1500 Houston, TX 77002
<u>WELL NAME:</u>	Kline Federal 5300 11-18 2B
<u>API #:</u>	33-053-06243-00-00
<u>WELL FILE #:</u>	29334
<u>SURFACE LOCATION:</u>	960' FNL & 318' FWL Lot 1 Section 18, T153N, R100W
<u>FIELD/ OBJECTIVE:</u>	Baker Field / Middle Bakken
<u>COUNTY. STATE</u>	McKenzie County, North Dakota
<u>BASIN:</u>	Williston
<u>WELL TYPE:</u>	Middle Bakken Horizontal
<u>ELEVATION:</u>	GL: 2,052' KB: 2,068'
<u>RE-ENTRY DATE:</u>	May 30, 2015
<u>BOTTOM HOLE LOCATION:</u>	111.86' south & 9,932.10' east of surface location or approx. 1,071.86 ' FNL & 296.01' FEL, NE NE Section 17, T153N, R100W
<u>CLOSURE COORDINATES:</u>	Closure Azimuth: 90.65° Closure Distance: 9,932.73'
<u>TOTAL DEPTH / DATE:</u>	20,450' on June 6, 2015 67% within target interval
<u>TOTAL DRILLING DAYS:</u>	8 Days
<u>CONTRACTOR:</u>	Xtreme #21

<u>PUMPS:</u>	Continental-Emsco F1600 (12" stroke length; 5 ½" liners) Output: 0.0838 bbls/stk at 95% efficiency
<u>TOOLPUSHERS:</u>	Josh Barkell, Allen Franklin
<u>FIELD SUPERVISORS:</u>	Dan Sandaker, Marty Amsbaugh, Miles Gordon
<u>CHEMICAL COMPANY:</u>	Mi Swaco
<u>MUD ENGINEER:</u>	Justin McNicholas
<u>MUD TYPE:</u>	Salt water in lateral
<u>MUD LOSSES:</u>	Not tracked
<u>PROSPECT GEOLOGIST:</u>	Curtis Johnson
<u>WELLSITE GEOLOGISTS:</u>	Dillon Johnson, Matt Hegland, Ryan O'Donnell
<u>GEOSTEERING SYSTEM:</u>	Sunburst Digital Wellsite Geological System
<u>ROCK SAMPLING:</u>	50' from 11,070' - 8,614'
<u>SAMPLE EXAMINATION:</u>	Binocular microscope & fluoroscope
<u>SAMPLE CUTS:</u>	Trichloroethylene
<u>GAS DETECTION:</u>	MSI (Mudlogging Systems, Inc.) TGC - total gas with chromatograph Serial Number(s): ML-488
<u>DIRECTIONAL DRILLERS:</u>	RPM Consulting, Inc. Marty Amsbaugh, Bruce Jorgenson, Derrick Ramsdell
<u>MWD:</u>	Gyro/data in vertical in curve Ryan Directional Service in lateral David Foley, David Unger
<u>CASING:</u>	Surface: 9 5/8" 36# J-55 set to 2,172' Intermediate: 7" 29# & 32# HCP-110 set to 11,024'
<u>SAFETY/ H₂S MONITORING:</u>	Oilind Safety

WELL LOCATION PLAT
OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 TULANE FEDERAL, S300 11-18 2B
 900 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
 SECTION 18, T163N, R100W, 8th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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STAKED ON 6/18/14
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'
 THIS SURVEY AND PLAT IS BEING PROVIDED AT THE
 REQUEST OF ERIC BAYES OF OASIS PETROLEUM, I
 CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
 WORK PERFORMED BY ME OR UNDER MY SUPERVISION
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[Signature]
 DART, D. KASEMAN LS-3980



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Interstate Engineering, Inc.
 P.O. Box 440
 420 East Main Street
 Minn, Minnesota 55270
 Ph (408) 433-0817
 Fax (408) 433-0818
 www.interstateengineering.com

OASIS PETROLEUM NORTH AMERICA, LLC
 WELL LOCATION PLAT
 SECTION 18, T163N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

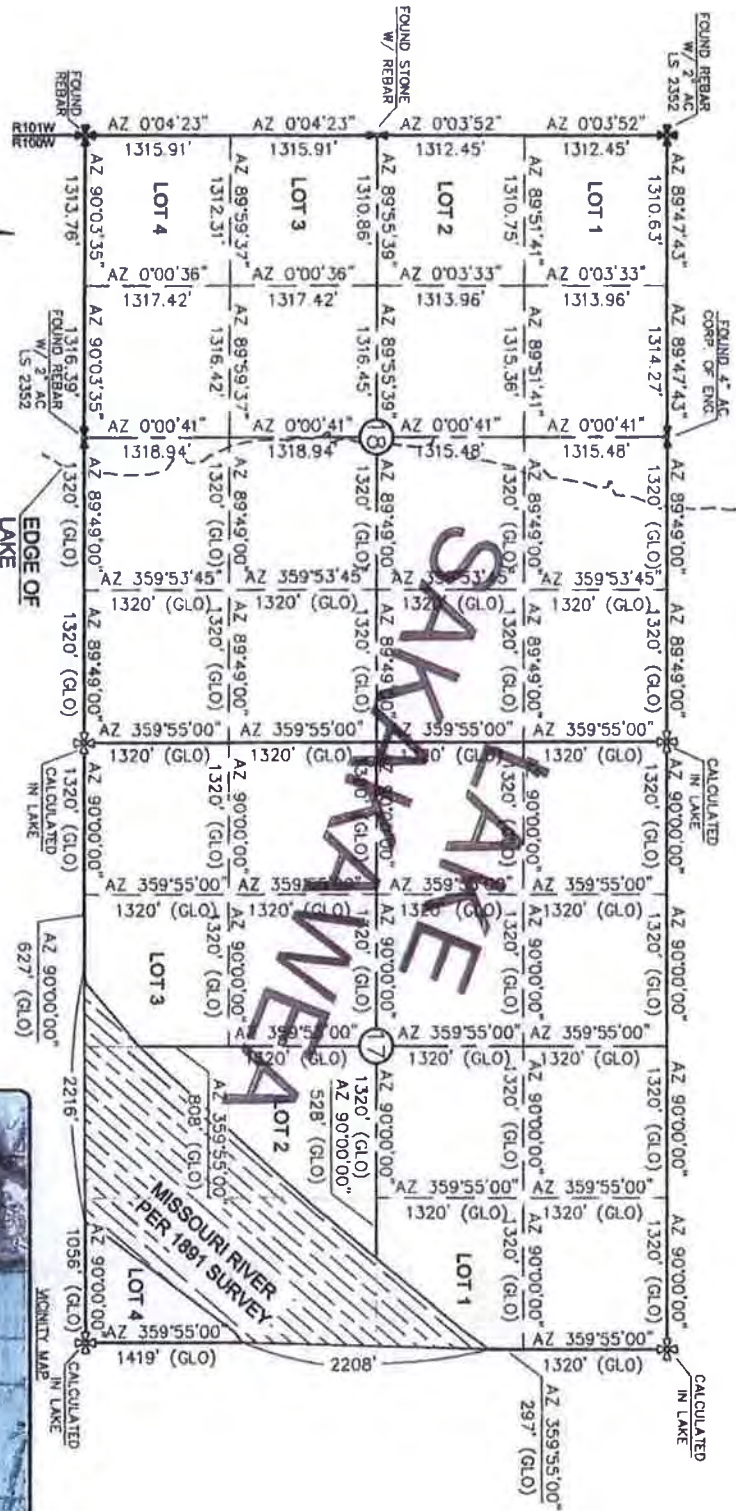
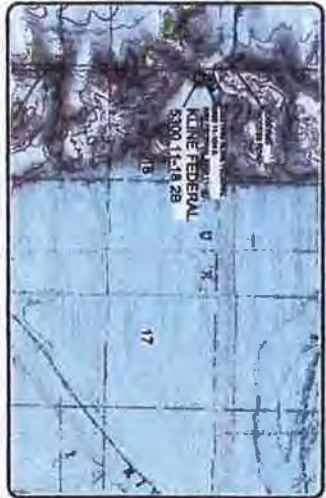
Drawn By: J.M.H. Project No.: 114-09-127
 Checked By: J.M.H. Date: 08/08/2014

Revision	By	Date	Description
REV 1	J.M.H.	08/08/2014	ISSUED WELLS
REV 2	J.M.H.	08/08/2014	REVISED LAYOUT
REV 3	J.M.H.	08/08/2014	ADDED CONTROL POINT TO PLAT
REV 4	J.M.H.	08/08/2014	REVISED WELL NAME & ID
REV 5	J.M.H.	08/08/2014	REVISED WELL NAME & ID

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 - MONUMENT - NOT RECOVERED



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 OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
 AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
 WAS 1891. THE CORNERS FOUND ARE AS
 INDICATED AND ALL OTHERS ARE COMPUTED FROM
 THOSE CORNERS FOUND AND BASED ON G.L.O.
 DATA. THE MAPPING ANGLE FOR THIS AREA IS
 APPROXIMATELY 003°.

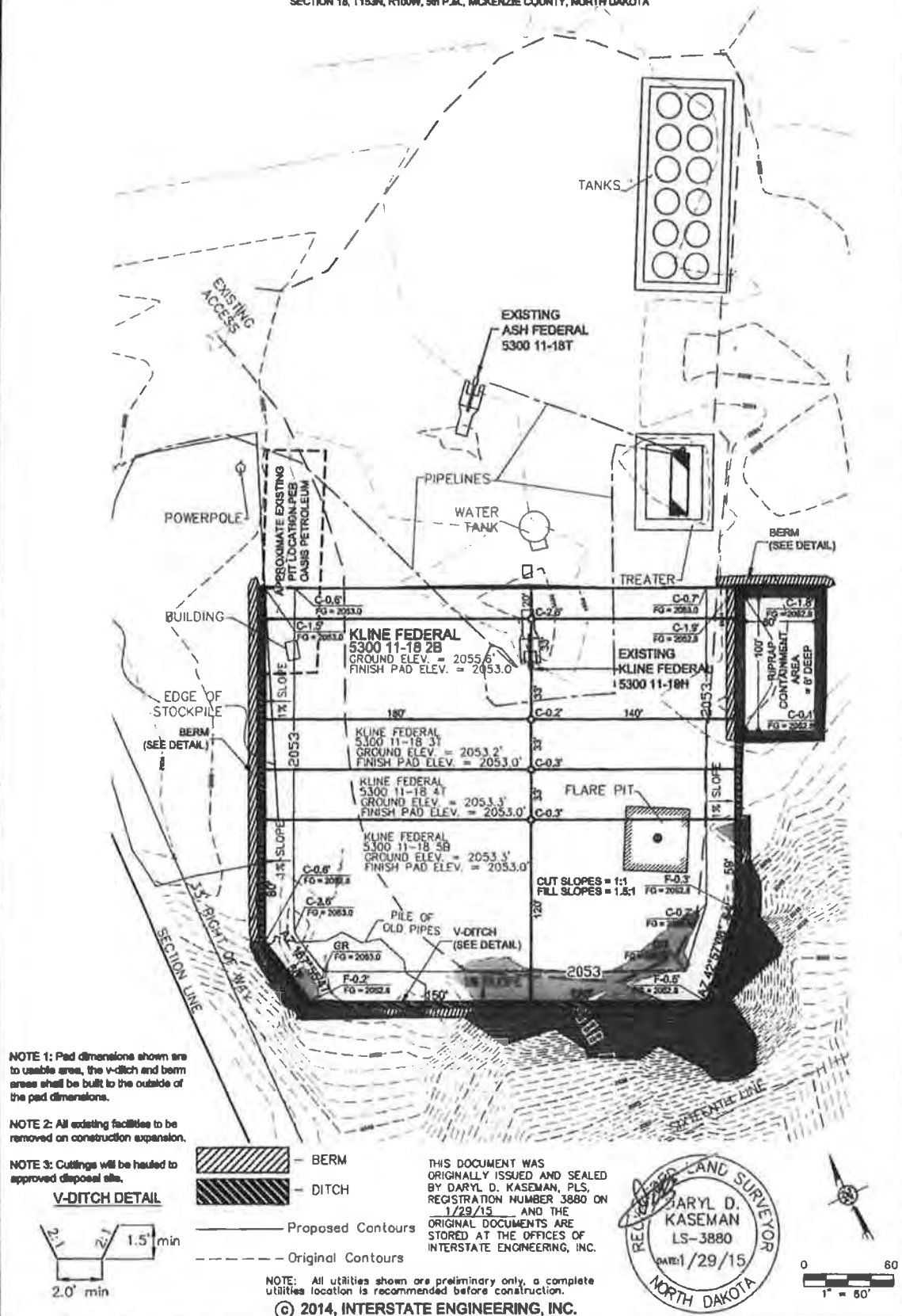


SECTION BREAKDOWN
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 KLINE FEDERAL 5300 11-18 2B
 800 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
 SECTIONS 17 & 18, T185N, R100W, 50-P.M., MCKENZIE COUNTY, NORTH DAKOTA

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Rev	By	Date	Description
REV 1	CAJAN	02/11/14	MISSOURI RIVER
REV 2	CAJAN	02/11/14	MISSOURI RIVER
REV 3	CAJAN	02/11/14	MISSOURI RIVER
REV 4	CAJAN	02/11/14	MISSOURI RIVER
REV 5	CAJAN	02/11/14	MISSOURI RIVER

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 6300 11-18 28"
860 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



 — BERM
 — DITCH

— Proposed Contours
- - - Original Contours

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NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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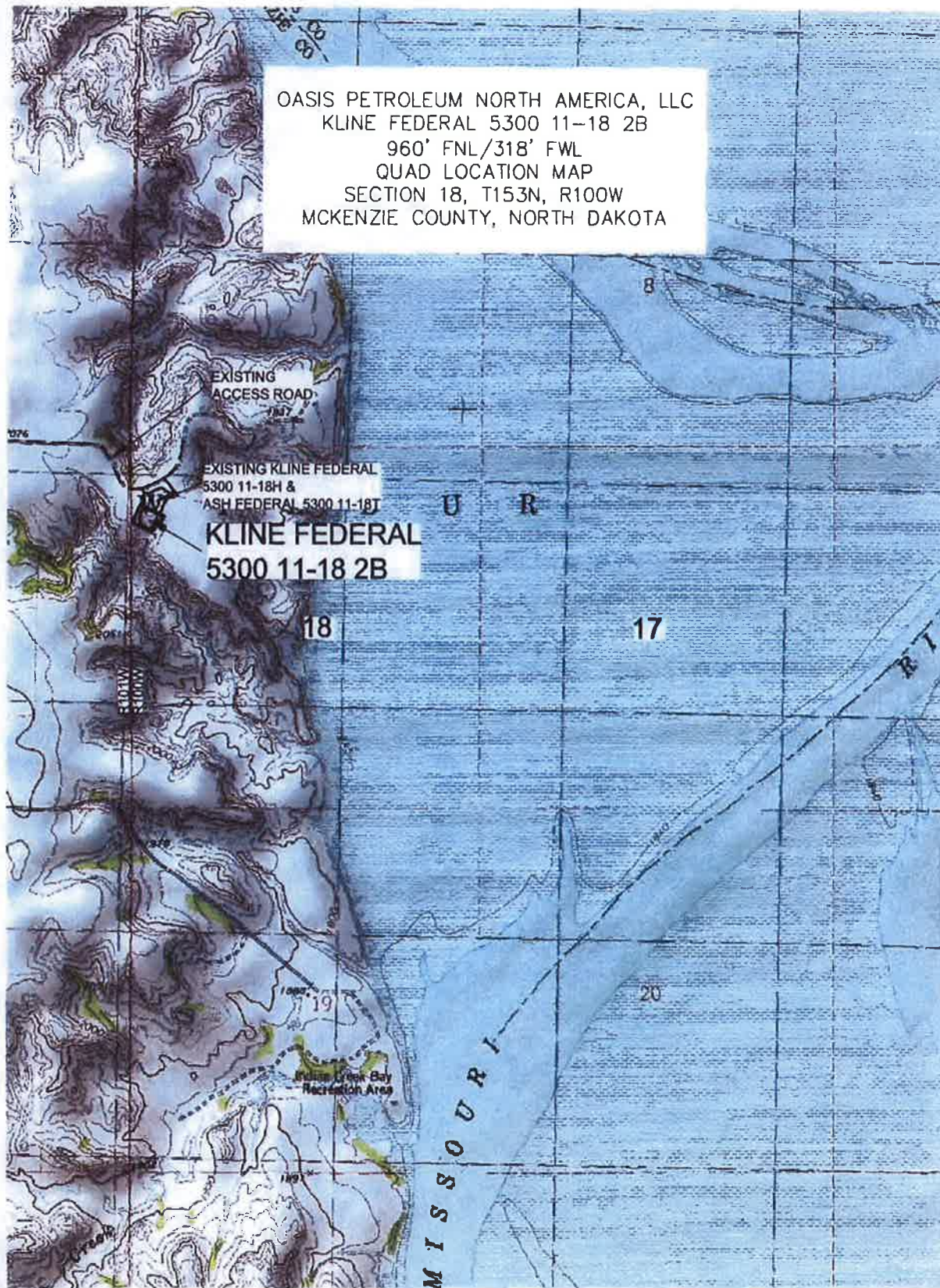
Innovative Engineering, Inc.
P.O. Box 648
426 East Main Street
Morris, Minnesota 56270
Ph (408) 433-8817
Fax (408) 433-8818
www.innovativeeng.com

QASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Contract No. DA-36-039-MD-0001 Project No. 2-4-52-17

Session No.	Date	By	Description
SESS 1	6/14/74	BW	ARMED ROBBERY
SESS 2	6/19/74	AB	ARMED LARCENY
SESS 3	6/25/74	AB	ARMED CHARGING RT TO PAD
SESS 4	7/2/74	AB	CHARGING WOLA ARMED & DFI

OASIS PETROLEUM NORTH AMERICA, LLC
 KLINE FEDERAL 5300 11-18 2B
 960' FNL/318' FWL
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



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 425 East Main Street
 Sidney, Montana 59270
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 www.interstateeng.com

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

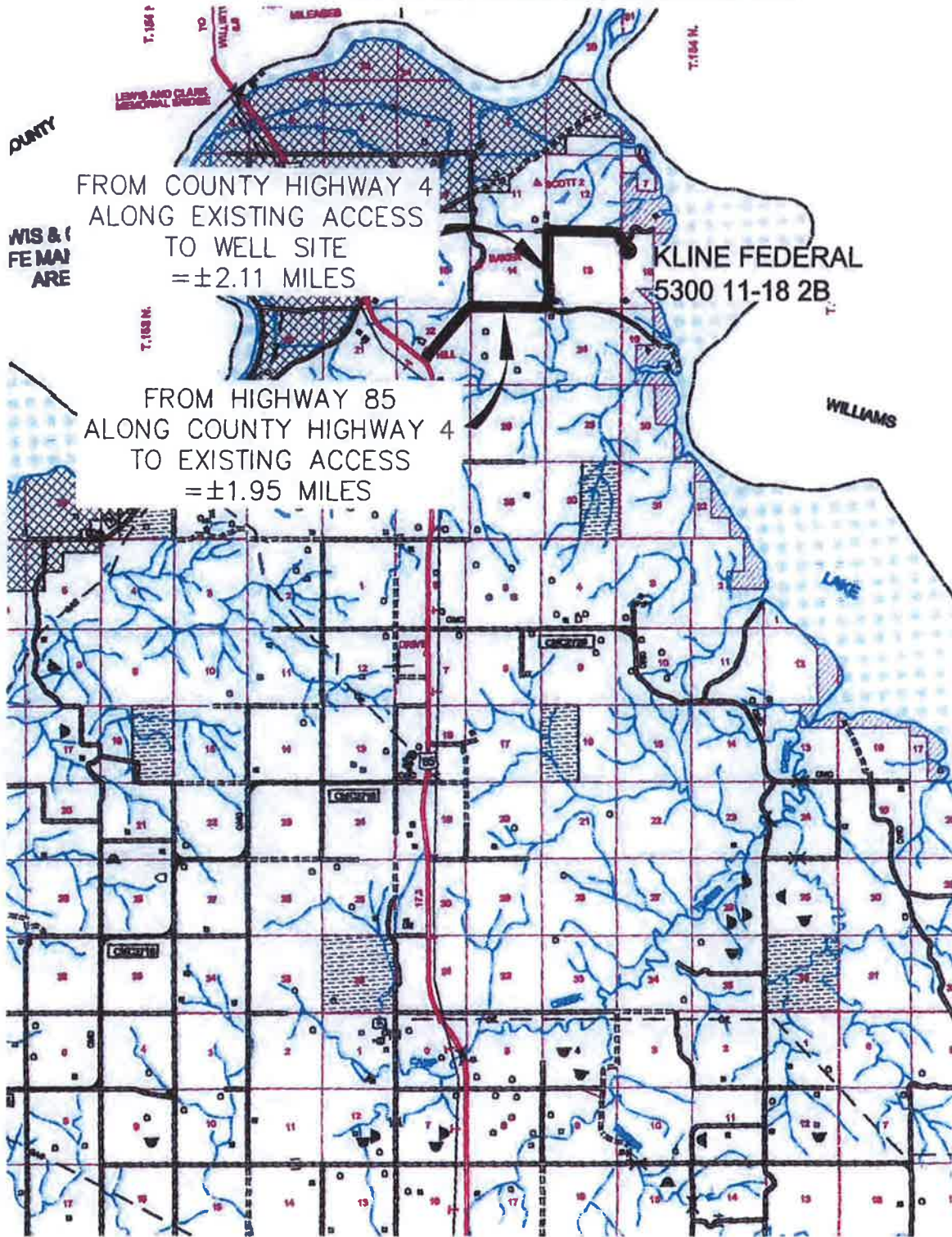
Drawn By: B.H.H. Project No.: 314-08-177
 Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	4/16/14	B.H.H.	MOVED WELL
REV 2	4/29/14	J.A.S.	REVISED LAYOUT
REV 3	5/28/14	J.A.S.	ADDED EXISTING PVT TO PAB LAYOUT
REV 4	7/29/14	J.A.S.	CHANGED WELL NAME & SH
REV 5	1/22/15	B.H.H.	CHANGED WELL NAME & SH

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 2B"

960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

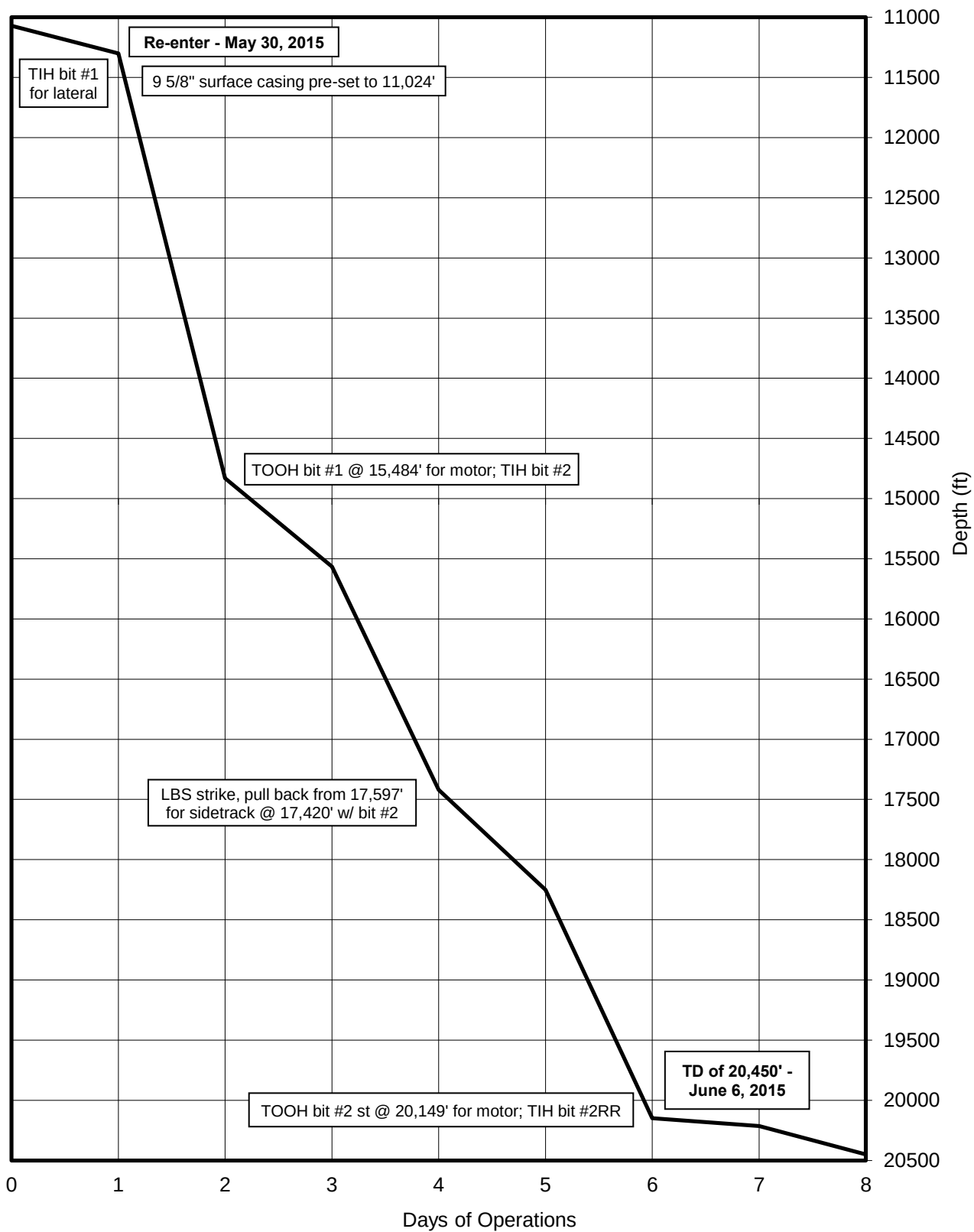
Drawn By: B.H.M. Project No: 514-08-127
Checked By: D.O.K. Date: APRIL 2014

Revision	Drawn By	By	Description
REV 1	5/15/14	BH	ISSUED WELLS
REV 2	5/24/14	JAB	REVISED LAYOUT
REV 3	5/24/14	JAB	ADDED EXISTING RPT TO PAD LAYOUT
REV 4	12/26/14	JAB	CHANGED WELL NAME & ID
REV 5	1/27/15	BH	CHANGED WELL NAMES & ID

TIME VS. DEPTH

Oasis Petroleum North America, LLC.

Kline Federal 5300 11-18 2B



MORNING REPORT SUMMARY

Day	Date 2015	Depth (0600 Hrs)	24 Hr Footage	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity Summary	Formation
0	5/30	11,070'	-	-	-	-	-	-	-	-	-	Prepare to skid rig.	Middle Bakken
1	5/31	11,300'	230	10	44	15	328	3300	0	89	318	Skid rig. Test BOP. Pick up BHA. TIH. Slip and cut. Drill out cement. FIT test. Drill F/11,070'-11,265'.	Middle Bakken
2	6/1	14,830'	3530	14	45	28	334	4150	0	90	324	Drill F/11,300'-12,882'. Lubricate rig. Drill F/12,882'-14,830'.	Middle Bakken
3	6/2	15,565'	735	12	44	67	334	4130	0	90	324	Drill F/14,830'-15,484'. Pump dry pill. TOOH. Pick up new BHA. TIH. Drill F/15,484'-15,565'.	Middle Bakken
4	6/3	17,420'	1855	11	40	26	331	3900	0	90	321	Drill F/15,565'-16,824'. Lubricate rig. Drill F/16,824'-17,597'. Pull two stands and trough at 160 left. Time drilling.	Middle Bakken
5	6/4	18,252'	832	12	40	32	331	3950	0	90	321	Time drilling. Lubricate rig. Drill F/17,438'-18,252'.	Middle Bakken
6	6/5	20,149'	1897	10	39	30	331	4050	89	0	321	Drill F/18,252'-19,507'. Lubricate rig. Drill F/19,507'-20,149'. Circulate. TOOH.	Middle Bakken
7	6/6	20,214'	65	15	40	40	143	2900	81	0	292	Ream. Verify sidetrack. Pump dry pill. TOOH. Cut & slip. Pick up BHA. TIH. Circulate. Ream sidetrack. Lubricate rig. Drill F/20,149'-20,214'.	Middle Bakken
8	6/7	20,450'	236	15	40	50	143	2900	81	0	292	Drill F/20,214'-20,450'. Circulate bottoms up. TOOH.	Middle Bakken

DAILY MUD SUMMARY

[illegible]

BOTTOM HOLE ASSEMBLY RECORD

Bit Data																Motor Data					Reason For Removal
Bit #	Size (in.)	Type	Make	Model	Depth In	Depth Out	Footage	Hours	Σ hrs	Vert. Dev.	Make	Model	Bend	Rev/Gal							
1	6	PDC	Halliburton	MM64	11,070'	15,484'	4,414'	31	31	Lateral	Baker	-	1.50°	1.03	Low ROP						
2	6	PDC	Halliburton	MM64	15,484'	17,597'	2,113'	17	48	Lateral	Baker	-	1.50°	1.03	-						
2 RR	6	PDC	Halliburton	MM64	17,420'	20,149'	2,729'	32.5	80.5	Sidetrack	Baker	-	1.50°	1.03	Low ROP						
2 RR	6	PDC	Halliburton	MM64	20,149'	20,450'	301'	3.2	83.7	Sidetrack	Baker	XLLS	1.50°	0.49	TD lateral						



Oasis Petroleum North America LLC
Kline Federal 5300 11-18 2B

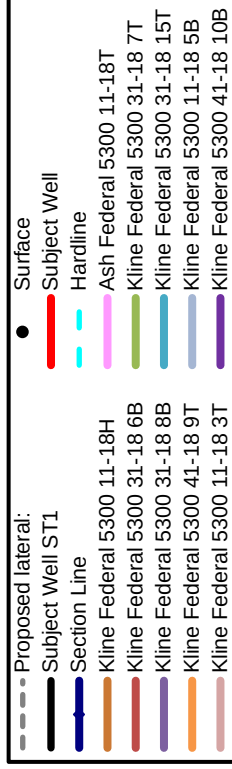
Surface Location

960' FNL & 318' FWL

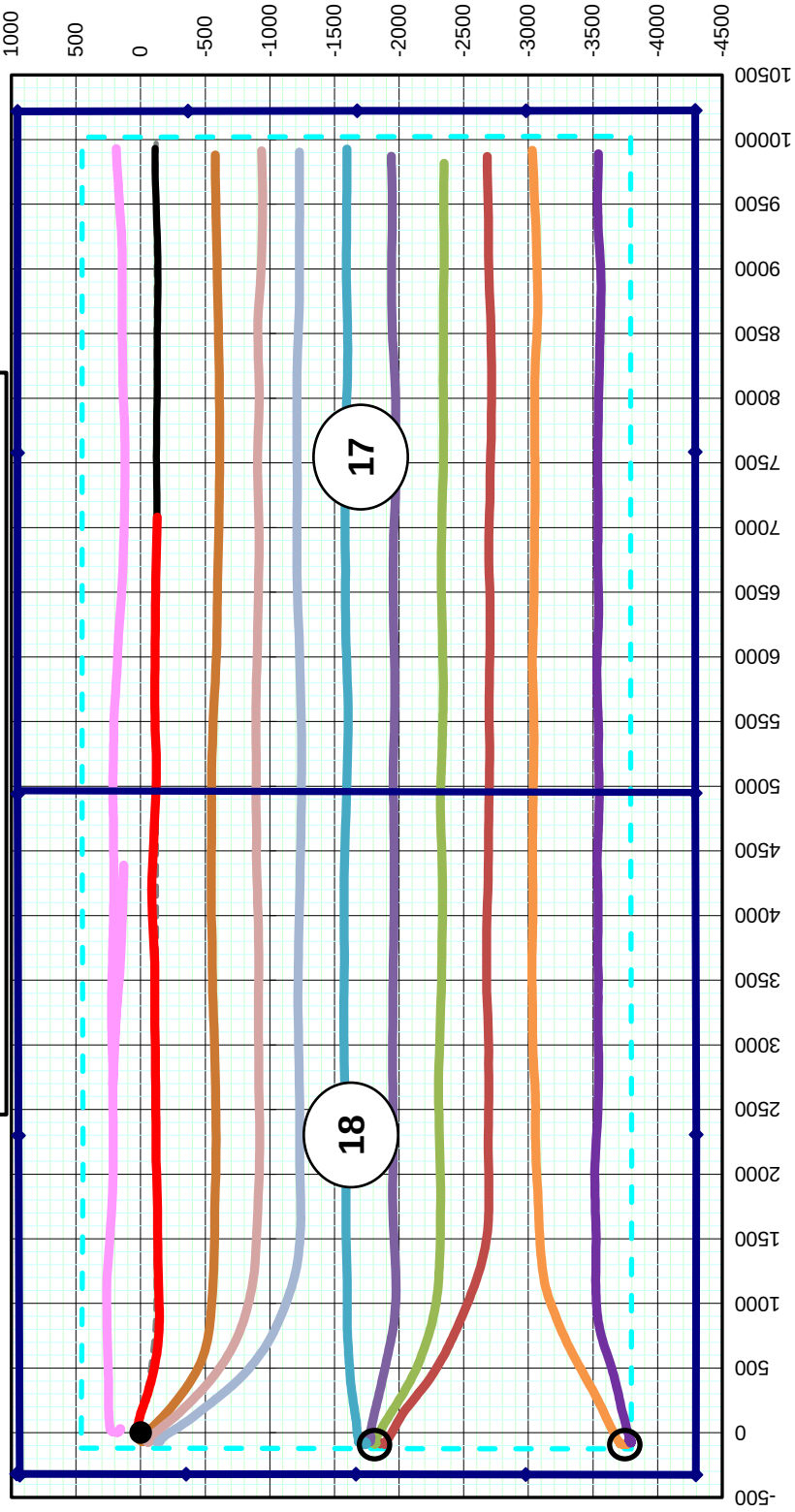
Lot 1, Sec. 18, T153N, R100W

PLAN VIEW

Note: 1,280 acre laydown spacing unit
with 500' N/S & 200' E/W setbacks



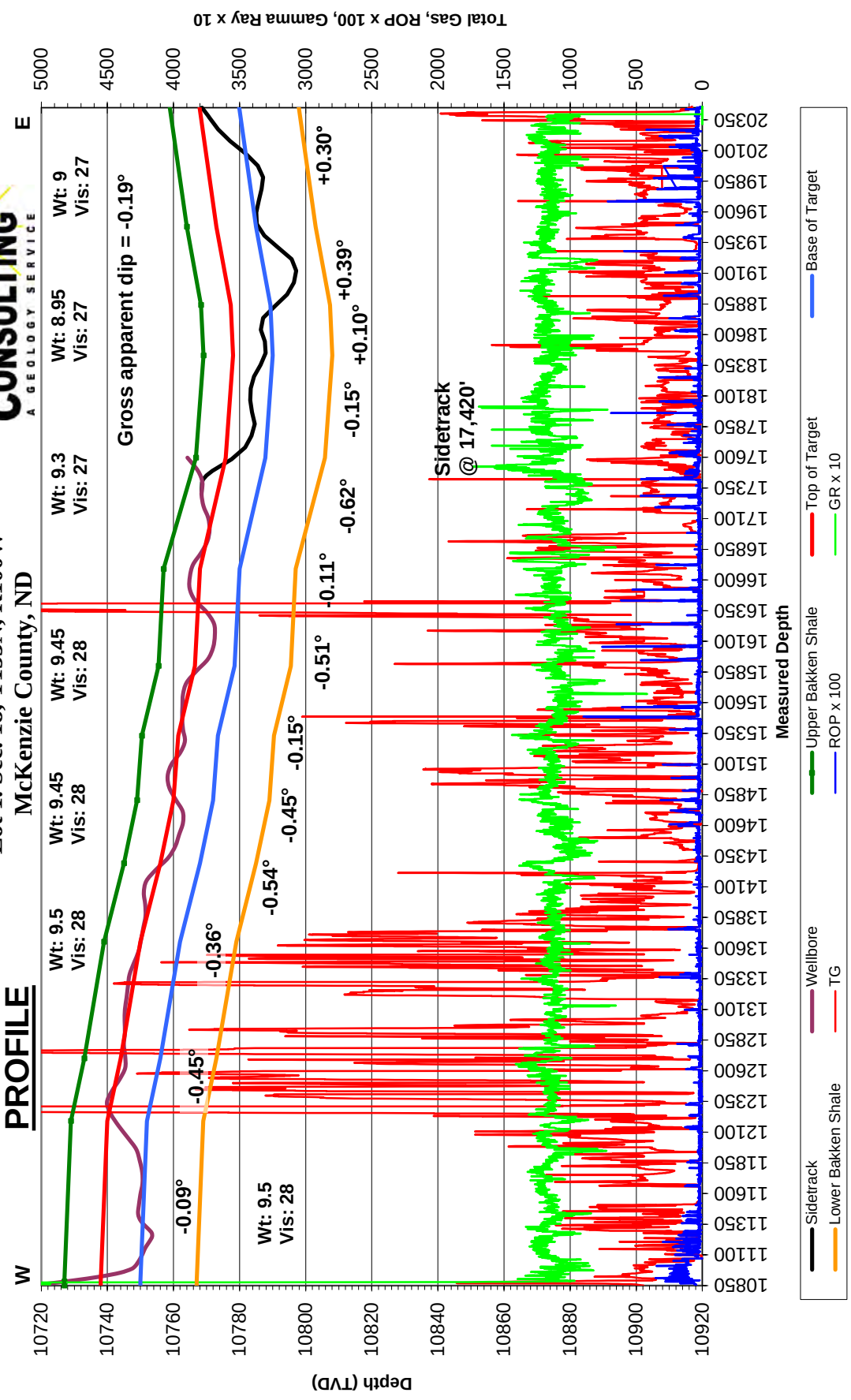
Bottom Hole Location
111.86' S & 9,932.10' E
of surface location or approx.
1,071.86' FNL & 296.01' FEL
NE NE Sec. 17, T153N, R100W





Oasis Petroleum North America LLC
Kline Federal 5300 11-18 2B
Lot 1. Sec. 18, T153N, R100W
McKenzie County, ND

PROFILE



FORMATION MARKERS & DIP ESTIMATES

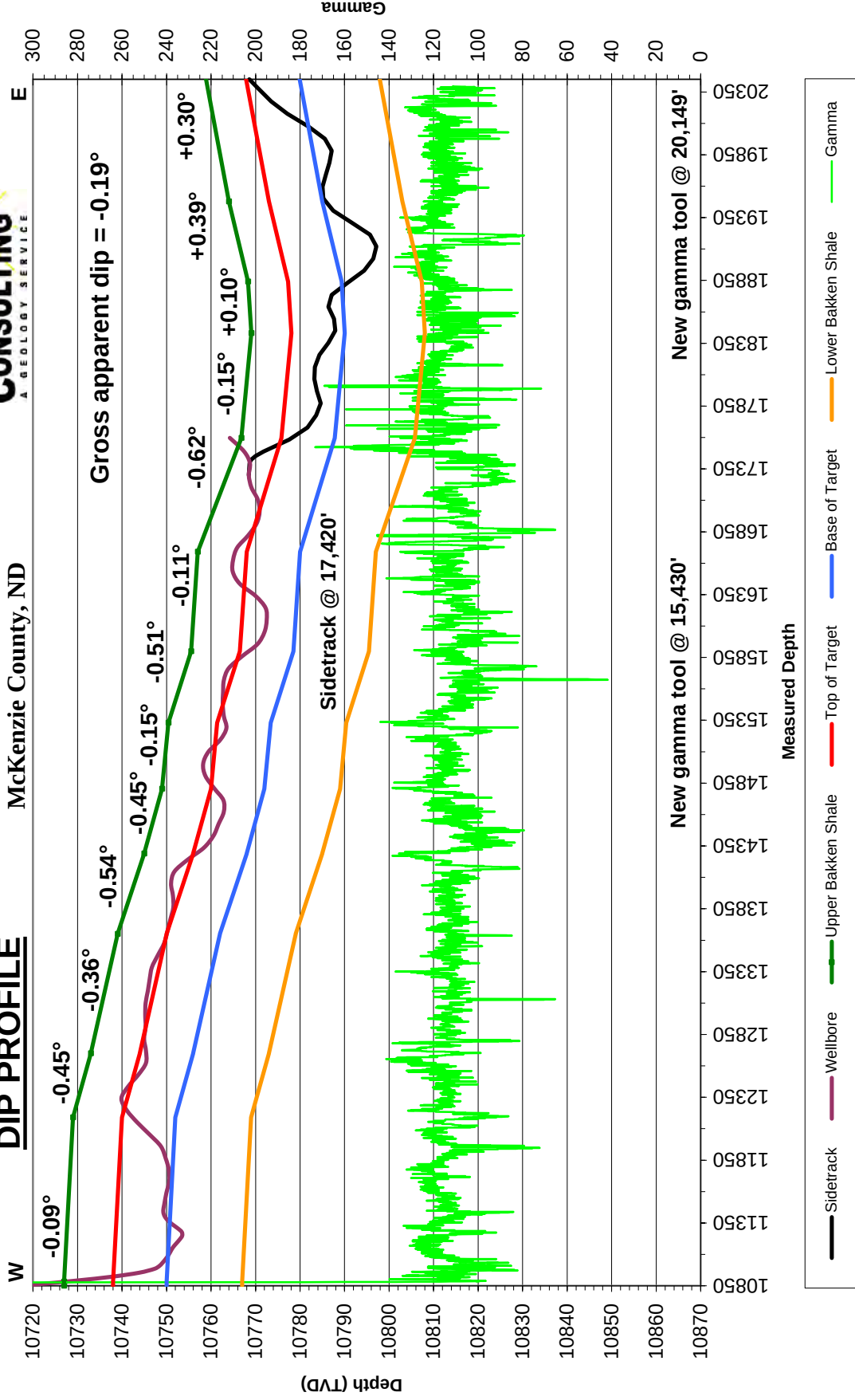
Oasis Petroleum North America LLC - Kline Federal 5300 11-18 2B

Dip Change Points	MD	TVD	TVD diff.	MD diff.	Dip	Dipping up/down	Type of Marker
Marker							
Middle Bakken Top	10,882'	10,727.00					Gamma
Projected Middle Bakken Top	12,190'	10,729.00	2.00	1308.00	-0.09	Down	Gamma
Projected Middle Bakken Top	12,700'	10,733.00	4.00	510.00	-0.45	Down	Gamma/Deflect
Projected Middle Bakken Top	13,650'	10,739.00	6.00	950.00	-0.36	Down	Gamma/Deflect
Projected Middle Bakken Top	14,290'	10,745.00	6.00	640.00	-0.54	Down	Gamma
Projected Middle Bakken Top	14,805'	10,749.00	4.00	515.00	-0.45	Down	Gamma
Projected Middle Bakken Top	15,330'	10,750.40	1.40	525.00	-0.15	Down	Gamma
Projected Middle Bakken Top	15,900'	10,755.50	5.10	570.00	-0.51	Down	Gamma
Projected Middle Bakken Top	16,690'	10,757.00	1.50	790.00	-0.11	Down	Gamma
Projected Middle Bakken Top	17,597'	10,766.81	9.81	907.00	-0.62	Down	Gamma
Projected Middle Bakken Top	18,430'	10,769.04	2.23	833.00	-0.15	Down	Gamma
Projected Middle Bakken Top	18,840'	10,768.33	-0.72	410.00	0.10	Up	Gamma
Projected Middle Bakken Top	19,478'	10,764.00	-4.33	638.00	0.39	Up	Gamma
Projected Middle Bakken Top	20,450'	10,758.90	-5.10	972.00	0.30	Up	Gamma
Gross Dip							
Initial Middle Bakken Top	10,882'	10,727.00					
Projected Final M. Bakken Top	20,450'	10,758.90	31.90	9568.00	-0.19	Down	Projection

Oasis Petroleum North America LLC
 Kline Federal 5300 11-18 2B
 Lot 1. Sec. 18, T153N, R100W
 McKenzie County, ND



DIP PROFILE



<

SUNBURST CONSULTING, INC.

>

Operator: Oasis Petroleum North America LLC
 Well: Kline Federal 5300 11-18 2B
 Surface Coordinates: 960' FNL & 318' FWL
 Surface Location: Lot 1. Sec. 18, T153N, R100W
 County State: McKenzie County, ND

Kick-off: 5/30/2015
 Finish: 6/3/2015

Directional Supervision:
 RPM Directional Inc.

Minimum Curvature Method (SPE-3362)

Proposed dir 90

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
Tie	2081.00	0.90	135.20	2080.88	5.76	-5.35	-5.35	0.38
1	2176.00	1.60	146.20	2175.86	4.13	-4.09	-4.09	0.78
2	2270.00	1.70	152.90	2269.82	1.80	-2.72	-2.72	0.23
3	2365.00	1.00	160.10	2364.79	-0.24	-1.80	-1.80	0.76
4	2459.00	0.70	163.90	2458.78	-1.56	-1.36	-1.36	0.32
5	2553.00	0.50	263.40	2552.78	-2.16	-1.61	-1.61	0.98
6	2647.00	0.80	284.50	2646.77	-2.04	-2.65	-2.65	0.40
7	2742.00	1.50	298.80	2741.75	-1.28	-4.38	-4.38	0.79
8	2836.00	1.90	300.00	2835.71	0.10	-6.81	-6.81	0.43
9	2930.00	2.30	305.70	2929.65	1.97	-9.69	-9.69	0.48
10	3024.00	1.10	321.10	3023.60	3.78	-11.79	-11.79	1.35
11	3119.00	1.40	328.70	3118.58	5.48	-12.96	-12.96	0.36
12	3213.00	1.20	34.40	3212.56	7.27	-13.00	-13.00	1.51
13	3307.00	1.20	47.50	3306.54	8.75	-11.72	-11.72	0.29
14	3401.00	0.90	36.20	3400.53	10.01	-10.56	-10.56	0.39
15	3495.00	1.00	36.90	3494.51	11.26	-9.63	-9.63	0.11
16	3590.00	0.70	43.50	3589.50	12.35	-8.74	-8.74	0.33
17	3684.00	0.80	33.50	3683.49	13.31	-7.98	-7.98	0.17
18	3778.00	0.60	41.60	3777.49	14.22	-7.29	-7.29	0.24
19	3872.00	0.50	37.40	3871.48	14.92	-6.71	-6.71	0.11
20	3966.00	0.40	14.70	3965.48	15.56	-6.38	-6.38	0.22
21	4060.00	0.30	25.70	4059.48	16.10	-6.19	-6.19	0.13
22	4154.00	0.40	346.40	4153.48	16.64	-6.16	-6.16	0.27
23	4248.00	0.50	76.80	4247.47	17.05	-5.84	-5.84	0.68
24	4343.00	0.50	84.90	4342.47	17.19	-5.02	-5.02	0.07
25	4437.00	0.40	36.40	4436.47	17.49	-4.42	-4.42	0.41
26	4531.00	0.30	48.60	4530.47	17.91	-4.04	-4.04	0.13
27	4625.00	0.40	217.00	4624.47	17.81	-4.05	-4.05	0.74
28	4720.00	0.40	225.60	4719.46	17.32	-4.49	-4.49	0.06
29	4813.00	0.40	330.20	4812.46	17.37	-4.88	-4.88	0.68
30	4907.00	0.50	128.90	4906.46	17.40	-4.73	-4.73	0.94
31	5001.00	0.60	143.20	5000.46	16.75	-4.11	-4.11	0.18
32	5095.00	0.40	207.70	5094.45	16.06	-3.97	-3.97	0.60
33	5190.00	0.30	193.10	5189.45	15.53	-4.18	-4.18	0.14
34	5284.00	0.60	181.00	5283.45	14.79	-4.25	-4.25	0.33
35	5378.00	0.80	158.90	5377.44	13.69	-4.02	-4.02	0.35
36	5472.00	0.80	177.40	5471.43	12.42	-3.75	-3.75	0.27
37	5567.00	1.10	185.80	5566.42	10.85	-3.81	-3.81	0.35

<

SUNBURST CONSULTING, INC.

>

Operator: Oasis Petroleum North America LLC
 Well: Kline Federal 5300 11-18 2B
 Surface Coordinates: 960' FNL & 318' FWL
 Surface Location: Lot 1. Sec. 18, T153N, R100W
 County State: McKenzie County, ND

Kick-off: 5/30/2015
 Finish: 6/3/2015

Directional Supervision:
 RPM Directional Inc.

Minimum Curvature Method (SPE-3362)

Proposed dir 90

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
38	5661.00	1.10	202.30	5660.40	9.12	-4.25	-4.25	0.34
39	5755.00	0.60	98.20	5754.40	8.22	-4.10	-4.10	1.46
40	5850.00	0.60	89.90	5849.39	8.15	-3.11	-3.11	0.09
41	5944.00	0.70	54.60	5943.39	8.48	-2.15	-2.15	0.43
42	6038.00	0.70	36.50	6037.38	9.27	-1.34	-1.34	0.23
43	6132.00	0.80	38.60	6131.37	10.25	-0.59	-0.59	0.11
44	6226.00	1.20	26.10	6225.36	11.64	0.25	0.25	0.48
45	6320.00	1.40	22.00	6319.33	13.59	1.11	1.11	0.23
46	6414.00	1.20	62.70	6413.31	15.11	2.42	2.42	0.98
47	6509.00	1.30	47.90	6508.29	16.29	4.10	4.10	0.35
48	6603.00	1.40	37.00	6602.26	17.92	5.58	5.58	0.29
49	6697.00	1.10	52.80	6696.24	19.38	6.99	6.99	0.48
50	6791.00	1.10	71.10	6790.22	20.22	8.57	8.57	0.37
51	6885.00	0.90	72.50	6884.21	20.73	10.12	10.12	0.21
52	6979.00	1.10	38.50	6978.19	21.66	11.39	11.39	0.65
53	7073.00	1.20	73.50	7072.18	22.65	12.90	12.90	0.74
54	7167.00	0.80	101.80	7166.16	22.79	14.48	14.48	0.66
55	7262.00	1.00	127.00	7261.15	22.16	15.79	15.79	0.46
56	7356.00	0.90	118.50	7355.14	21.31	17.10	17.10	0.18
57	7450.00	1.00	121.00	7449.13	20.54	18.45	18.45	0.12
58	7544.00	0.70	113.20	7543.12	19.89	19.68	19.68	0.34
59	7639.00	0.90	108.40	7638.11	19.43	20.92	20.92	0.22
60	7733.00	1.00	107.40	7732.09	18.95	22.40	22.40	0.11
61	7827.00	1.10	95.80	7826.08	18.61	24.08	24.08	0.25
62	7921.00	0.80	82.00	7920.06	18.61	25.63	25.63	0.40
63	8015.00	0.80	89.90	8014.06	18.70	26.94	26.94	0.12
64	8109.00	0.70	82.20	8108.05	18.78	28.16	28.16	0.15
65	8203.00	0.60	108.20	8202.04	18.71	29.20	29.20	0.33
66	8298.00	0.50	118.60	8297.04	18.35	30.04	30.04	0.15
67	8392.00	0.50	108.30	8391.03	18.03	30.79	30.79	0.10
68	8486.00	0.50	89.80	8485.03	17.90	31.58	31.58	0.17
69	8580.00	0.60	102.80	8579.03	17.79	32.48	32.48	0.17
70	8674.00	0.70	110.80	8673.02	17.48	33.49	33.49	0.14
71	8769.00	0.60	115.20	8768.01	17.06	34.48	34.48	0.12
72	8863.00	0.60	133.30	8862.01	16.52	35.29	35.29	0.20
73	8957.00	0.40	114.20	8956.00	16.04	35.95	35.95	0.27
74	9051.00	0.40	118.30	9050.00	15.75	36.53	36.53	0.03
75	9145.00	0.70	97.20	9144.00	15.53	37.39	37.39	0.38

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SUNBURST CONSULTING, INC.

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Operator: Oasis Petroleum North America LLC
 Well: Kline Federal 5300 11-18 2B
 Surface Coordinates: 960' FNL & 318' FWL
 Surface Location: Lot 1. Sec. 18, T153N, R100W
 County State: McKenzie County, ND

Kick-off: 5/30/2015
 Finish: 6/3/2015

Directional Supervision:
 RPM Directional Inc.

Minimum Curvature Method (SPE-3362)

Proposed dir 90

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
76	9240.00	0.20	343.00	9239.00	15.61	37.92	37.92	0.85
77	9334.00	0.20	335.80	9333.00	15.92	37.80	37.80	0.03
78	9428.00	0.40	38.40	9426.99	16.32	37.94	37.94	0.38
79	9523.00	0.50	27.20	9521.99	16.95	38.34	38.34	0.14
80	9617.00	0.60	10.90	9615.99	17.80	38.62	38.62	0.20
81	9711.00	0.50	2.40	9709.98	18.69	38.73	38.73	0.14
82	9805.00	0.50	354.20	9803.98	19.51	38.70	38.70	0.08
83	9899.00	0.50	330.90	9897.98	20.28	38.46	38.46	0.21
84	9993.00	0.60	353.10	9991.97	21.13	38.20	38.20	0.25
85	10087.00	0.70	0.00	10085.97	22.19	38.14	38.14	0.13
86	10171.00	1.00	11.90	10169.96	23.42	38.30	38.30	0.41
87	10201.00	1.00	4.30	10199.95	23.94	38.37	38.37	0.44
88	10232.00	1.50	58.70	10230.94	24.42	38.74	38.74	3.96
89	10263.00	4.40	88.80	10261.90	24.65	40.27	40.27	10.30
90	10295.00	7.40	97.40	10293.73	24.41	43.54	43.54	9.75
91	10326.00	11.10	97.50	10324.32	23.77	48.48	48.48	11.94
92	10357.00	14.50	98.30	10354.55	22.82	55.28	55.28	10.98
93	10389.00	17.50	100.70	10385.30	21.34	63.98	63.98	9.60
94	10420.00	20.90	103.40	10414.58	19.20	73.94	73.94	11.33
95	10452.00	24.20	104.50	10444.12	16.23	85.85	85.85	10.40
96	10483.00	27.20	104.60	10472.06	12.85	98.86	98.86	9.68
97	10514.00	30.20	104.80	10499.24	9.07	113.25	113.25	9.68
98	10546.00	33.30	105.00	10526.45	4.74	129.52	129.52	9.69
99	10577.00	36.40	105.90	10551.89	0.02	146.59	146.59	10.14
100	10609.00	39.70	107.80	10577.09	-5.71	165.46	165.46	10.94
101	10640.00	43.20	109.50	10600.32	-12.28	184.90	184.90	11.86
102	10671.00	46.60	110.40	10622.27	-19.75	205.46	205.46	11.16
103	10703.00	50.10	110.40	10643.54	-28.08	227.87	227.87	10.94
104	10734.00	53.80	109.50	10662.64	-36.40	250.81	250.81	12.15
105	10766.00	57.80	108.00	10680.62	-44.90	275.87	275.87	13.09
106	10797.00	62.90	106.70	10695.96	-52.92	301.58	301.58	16.85
107	10828.00	67.30	106.40	10709.00	-60.93	328.53	328.53	14.22
108	10860.00	71.50	105.70	10720.26	-69.21	357.31	357.31	13.28
109	10891.00	73.50	105.60	10729.58	-77.18	385.78	385.78	6.46
110	10922.00	76.40	104.50	10737.63	-84.95	414.68	414.68	9.96
111	10954.00	81.30	103.40	10743.82	-92.52	445.14	445.14	15.68
112	10985.00	85.90	102.10	10747.27	-99.31	475.18	475.18	15.41
113	11007.00	88.40	100.60	10748.37	-103.64	496.72	496.72	13.25

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SUNBURST CONSULTING, INC.

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Operator: Oasis Petroleum North America LLC
 Well: Kline Federal 5300 11-18 2B
 Surface Coordinates: 960' FNL & 318' FWL
 Surface Location: Lot 1. Sec. 18, T153N, R100W
 County State: McKenzie County, ND

Kick-off: 5/30/2015
 Finish: 6/3/2015

Directional Supervision:
 RPM Directional Inc.

Minimum Curvature Method (SPE-3362)

Proposed dir 90

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
114	11032.00	88.40	101.10	10749.06	-108.34	521.27	521.27	2.00
115	11063.00	88.90	100.50	10749.79	-114.15	551.71	551.71	2.52
116	11094.00	88.80	100.70	10750.42	-119.85	582.17	582.17	0.72
117	11124.00	89.30	98.80	10750.91	-124.93	611.73	611.73	6.55
118	11155.00	88.80	97.30	10751.43	-129.27	642.42	642.42	5.10
119	11186.00	88.50	96.10	10752.16	-132.89	673.20	673.20	3.99
120	11217.00	88.70	95.40	10752.92	-135.99	704.04	704.04	2.35
121	11248.00	89.10	93.90	10753.51	-138.50	734.93	734.93	5.01
122	11279.00	91.30	93.60	10753.40	-140.53	765.86	765.86	7.16
123	11310.00	92.30	92.90	10752.43	-142.29	796.79	796.79	3.94
124	11341.00	92.10	92.80	10751.24	-143.83	827.73	827.73	0.72
125	11372.00	91.60	91.70	10750.24	-145.04	858.69	858.69	3.90
126	11403.00	90.80	90.80	10749.59	-145.72	889.68	889.68	3.88
127	11434.00	90.30	89.10	10749.29	-145.69	920.68	920.68	5.72
128	11465.00	89.60	88.10	10749.32	-144.93	951.67	951.67	3.94
129	11495.00	89.80	88.20	10749.48	-143.97	981.65	981.65	0.75
130	11588.00	89.60	87.80	10749.96	-140.72	1074.59	1074.59	0.48
131	11681.00	89.60	88.50	10750.61	-137.72	1167.54	1167.54	0.75
132	11773.00	90.50	88.30	10750.53	-135.15	1259.50	1259.50	1.00
133	11865.00	90.40	89.30	10749.81	-133.22	1351.48	1351.48	1.09
134	11960.00	90.90	89.50	10748.73	-132.23	1446.47	1446.47	0.57
135	12055.00	92.20	89.70	10746.16	-131.57	1541.43	1541.43	1.38
136	12150.00	91.10	89.80	10743.43	-131.15	1636.39	1636.39	1.16
137	12245.00	91.70	89.10	10741.11	-130.24	1731.35	1731.35	0.97
138	12340.00	89.80	89.20	10739.86	-128.83	1826.33	1826.33	2.00
139	12434.00	88.40	87.80	10741.34	-126.37	1920.28	1920.28	2.11
140	12529.00	88.80	87.60	10743.66	-122.56	2015.18	2015.18	0.47
141	12623.00	89.10	88.80	10745.38	-119.61	2109.11	2109.11	1.32
142	12719.00	90.90	90.30	10745.38	-118.85	2205.10	2205.10	2.44
143	12814.00	89.60	89.90	10744.97	-119.02	2300.10	2300.10	1.43
144	12910.00	90.00	89.50	10745.30	-118.52	2396.10	2396.10	0.59
145	13005.00	90.10	89.50	10745.22	-117.69	2491.09	2491.09	0.11
146	13100.00	89.80	89.30	10745.30	-116.69	2586.09	2586.09	0.38
147	13195.00	89.60	89.60	10745.80	-115.78	2681.08	2681.08	0.38
148	13290.00	89.90	89.60	10746.21	-115.12	2776.08	2776.08	0.32
149	13384.00	89.50	89.40	10746.71	-114.30	2870.07	2870.07	0.48
150	13480.00	88.90	89.00	10748.05	-112.96	2966.05	2966.05	0.75
151	13574.00	89.40	89.10	10749.44	-111.40	3060.03	3060.03	0.54

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SUNBURST CONSULTING, INC.

>

Operator: Oasis Petroleum North America LLC
 Well: Kline Federal 5300 11-18 2B
 Surface Coordinates: 960' FNL & 318' FWL
 Surface Location: Lot 1. Sec. 18, T153N, R100W
 County State: McKenzie County, ND

Kick-off: 5/30/2015
 Finish: 6/3/2015

Directional Supervision:
 RPM Directional Inc.

Minimum Curvature Method (SPE-3362)

Proposed dir 90

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
152	13671.00	89.70	88.70	10750.20	-109.54	3157.01	3157.01	0.52
153	13765.00	89.50	90.30	10750.86	-108.72	3251.00	3251.00	1.72
154	13860.00	89.70	90.20	10751.52	-109.13	3346.00	3346.00	0.24
155	13956.00	90.40	90.40	10751.44	-109.63	3442.00	3442.00	0.76
156	14051.00	90.10	89.90	10751.02	-109.88	3536.99	3536.99	0.61
157	14146.00	89.00	88.00	10751.77	-108.14	3631.97	3631.97	2.31
158	14240.00	87.40	86.80	10754.72	-103.88	3725.82	3725.82	2.13
159	14335.00	88.20	87.30	10758.37	-98.99	3820.62	3820.62	0.99
160	14431.00	89.20	87.60	10760.55	-94.72	3916.50	3916.50	1.09
161	14525.00	89.40	86.90	10761.70	-90.22	4010.39	4010.39	0.77
162	14620.00	89.20	88.60	10762.86	-86.49	4105.30	4105.30	1.80
163	14714.00	90.90	90.40	10762.78	-85.67	4199.29	4199.29	2.63
164	14809.00	91.60	92.20	10760.70	-87.82	4294.24	4294.24	2.03
165	14907.00	90.70	92.60	10758.74	-91.92	4392.13	4392.13	1.00
166	15003.00	89.90	92.90	10758.23	-96.53	4488.02	4488.02	0.89
167	15099.00	88.70	92.60	10759.41	-101.13	4583.90	4583.90	1.29
168	15194.00	88.10	92.20	10762.06	-105.11	4678.78	4678.78	0.76
169	15288.00	90.20	92.80	10763.45	-109.21	4772.68	4772.68	2.32
170	15382.00	90.50	92.90	10762.88	-113.88	4866.56	4866.56	0.34
171	15478.00	89.80	92.40	10762.63	-118.32	4962.45	4962.45	0.90
172	15573.00	90.10	91.00	10762.71	-121.14	5057.41	5057.41	1.51
173	15668.00	89.70	89.80	10762.88	-121.80	5152.40	5152.40	1.33
174	15763.00	89.00	88.40	10763.95	-120.31	5247.38	5247.38	1.65
175	15859.00	87.40	87.00	10766.97	-116.46	5343.25	5343.25	2.21
176	15955.00	88.50	88.30	10770.40	-112.53	5439.11	5439.11	1.77
177	16051.00	89.60	89.40	10772.00	-110.60	5535.07	5535.07	1.62
178	16148.00	89.80	89.20	10772.50	-109.42	5632.06	5632.06	0.29
179	16244.00	90.50	90.10	10772.25	-108.83	5728.06	5728.06	1.19
180	16340.00	92.20	91.30	10769.99	-110.00	5824.02	5824.02	2.17
181	16436.00	92.00	90.80	10766.47	-111.76	5919.94	5919.94	0.56
182	16532.00	89.90	90.00	10764.88	-112.43	6015.92	6015.92	2.34
183	16628.00	90.00	90.30	10764.96	-112.68	6111.92	6111.92	0.33
184	16724.00	88.90	90.20	10765.89	-113.10	6207.91	6207.91	1.15
185	16820.00	88.20	90.30	10768.31	-113.52	6303.88	6303.88	0.74
186	16916.00	89.50	90.80	10770.24	-114.44	6399.85	6399.85	1.45
187	17012.00	89.70	90.90	10770.91	-115.87	6495.84	6495.84	0.23
188	17106.00	90.80	91.30	10770.50	-117.67	6589.82	6589.82	1.25
189	17202.00	90.90	91.10	10769.08	-119.68	6685.79	6685.79	0.23

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SUNBURST CONSULTING, INC.

>

Operator: Oasis Petroleum North America LLC
 Well: Kline Federal 5300 11-18 2B
 Surface Coordinates: 960' FNL & 318' FWL
 Surface Location: Lot 1. Sec. 18, T153N, R100W
 County State: McKenzie County, ND

Kick-off: 5/30/2015
 Finish: 6/3/2015

Directional Supervision:
 RPM Directional Inc.

Minimum Curvature Method (SPE-3362)

Proposed dir 90

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
190	17299.00	89.80	91.30	10768.49	-121.71	6782.76	6782.76	1.15
191	17395.00	89.90	91.30	10768.74	-123.89	6878.74	6878.74	0.10
192	17489.00	91.70	91.70	10767.42	-126.35	6972.69	6972.69	1.96
193	17597.00	91.70	91.70	10764.22	-129.55	7080.60	7080.60	0.00

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SUNBURST CONSULTING, INC.

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Operator: Oasis Petroleum North America LLC
 Sidetrack 1
 Well: Kline Federal 5300 11-18 2B
 Surface Coordinates: 960' FNL & 318' FWL
 Surface Location: Lot 1. Sec. 18, T153N, R100W
 County State: McKenzie County, ND

Kick-off: 6/3/2015
 Finish: 6/6/2015
 Directional Supervision:
 RPM Directional Inc.

Minimum Curvature Method (SPE-3362)

Proposed dir 90

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
Tie	17299.00	89.80	91.30	10768.49	-121.71	6782.76	6782.76	1.15
1	17394.00	90.00	91.40	10768.65	-123.95	6877.74	6877.74	0.24
2	17420.00	88.90	91.00	10768.90	-124.49	6903.73	6903.73	4.50
3	17453.00	87.30	90.90	10769.99	-125.04	6936.71	6936.71	4.86
4	17489.00	87.00	90.70	10771.78	-125.54	6972.66	6972.66	1.00
5	17584.00	86.00	90.30	10777.58	-126.37	7067.48	7067.48	1.13
6	17680.00	89.10	89.90	10781.69	-126.54	7163.38	7163.38	3.26
7	17776.00	88.60	89.50	10783.61	-126.04	7259.35	7259.35	0.67
8	17872.00	90.20	89.10	10784.62	-124.86	7355.34	7355.34	1.72
9	17969.00	90.80	89.10	10783.77	-123.34	7452.32	7452.32	0.62
10	18065.00	89.90	89.90	10783.19	-122.50	7548.32	7548.32	1.25
11	18161.00	89.90	90.20	10783.35	-122.59	7644.32	7644.32	0.31
12	18258.00	88.90	90.60	10784.37	-123.26	7741.31	7741.31	1.11
13	18354.00	88.70	90.70	10786.38	-124.35	7837.28	7837.28	0.23
14	18450.00	89.50	91.20	10787.89	-125.94	7933.25	7933.25	0.98
15	18544.00	90.80	91.10	10787.64	-127.83	8027.23	8027.23	1.39
16	18640.00	90.70	90.40	10786.39	-129.09	8123.21	8123.21	0.74
17	18735.00	88.40	89.70	10787.13	-129.17	8218.20	8218.20	2.53
18	18833.00	87.40	89.50	10790.72	-128.48	8316.13	8316.13	1.04
19	18928.00	88.20	90.10	10794.37	-128.15	8411.06	8411.06	1.05
20	19023.00	89.30	90.70	10796.44	-128.82	8506.04	8506.04	1.32
21	19120.00	89.90	90.70	10797.12	-130.00	8603.03	8603.03	0.62
22	19215.00	91.80	91.20	10795.71	-131.58	8698.00	8698.00	2.07
23	19311.00	93.20	90.90	10791.52	-133.33	8793.89	8793.89	1.49
24	19406.00	91.70	90.20	10787.46	-134.24	8888.79	8888.79	1.74
25	19502.00	91.00	89.70	10785.20	-134.16	8984.77	8984.77	0.90
26	19596.00	89.20	88.60	10785.04	-132.77	9078.75	9078.75	2.24
27	19692.00	89.90	88.50	10785.79	-130.34	9174.72	9174.72	0.74
28	19787.00	89.10	88.00	10786.62	-127.44	9269.67	9269.67	0.99
29	19882.00	90.30	88.40	10787.12	-124.45	9364.62	9364.62	1.33
30	19977.00	91.60	88.40	10785.54	-121.80	9459.56	9459.56	1.37
31	20073.00	92.70	88.00	10781.94	-118.79	9555.45	9555.45	1.22
32	20177.00	92.60	87.70	10777.13	-114.89	9659.26	9659.26	0.30
33	20273.00	91.70	89.80	10773.53	-112.80	9755.17	9755.17	2.38
34	20369.00	91.50	89.60	10770.85	-112.30	9851.13	9851.13	0.29
35	20382.00	91.60	89.70	10770.50	-112.22	9864.12	9864.12	1.09
36	20450.00	91.60	89.70	10768.60	-111.86	9932.10	9932.10	0.00

FORMATION TOPS & STRUCTURAL RELATIONSHIPS

Operator: Well Name: Location: Elevation:	Subject Well:							
	Oasis Petroleum North America, LLC. Kline Federal 5300 11-18 2B 960' FNL & 318' FWL Lot 1 Section 18, T153N, R100W 2,053' Sub: 25' KB: 2,078'							
Formation/ Marker	Prog. Top	Prog. Datum (MSL)	Driller's Depth Top (MD)	Driller's Depth Top (TVD)	Datum (MSL)	Interval Thickness	Thickness to Target	Dip To Prog.
Kibbey Lime	8,367'	-6,289'	8,355'	8,354'	-6,276'	153'	2,386'	13'
First Charles Salts	8,517'	-6,439'	8,508'	8,507'	-6,429'	616'	2,233'	10'
Upper Berentson	9,141'	-7,063'	9,124'	9,123'	-7,045'	76'	1,617'	18'
Base Last Salt	9,216'	-7,138'	9,200'	9,199'	-7,121'	223'	1,541'	17'
Mission Canyon	9,421'	-7,343'	8,423'	9,422'	-7,344'	559'	1,318'	-1'
Lodgepole	9,999'	-7,921'	9,982'	9,981'	-7,903'	721'	759'	18'
False Bakken	10,696'	-8,618'	10,811'	10,702'	-8,624'	8'	38'	-6'
Upper Bakken Shale	10,708'	-8,630'	10,830'	10,710'	-8,632'	17'	30'	-2'
Middle Bakken	10,722'	-8,644'	10,882'	10,727'	-8,649'	13'	13'	-5'

LITHOLOGY

Rig crews caught lagged samples in 50' intervals under the supervision of a Sunburst geologist. A detailed list of sampling intervals is included in the well data summary page. Samples were examined wet and dry conditions under a binocular microscope and checked for hydrocarbon cut fluorescence with Entron. Sample descriptions began at re-entry point within the Middle Bakken Member. The drilling fluid was salt water solution in the lateral.

Middle Bakken

10,882' MD; 10,727' TVD (-8,649')

11,070-11,100 SILTY SANDSTONE: light to medium gray-brown, off white, occasional tan, very fine grained, firm, sub-angular, moderately to well sorted, calcite cemented, moderately cemented, occasional disseminated pyrite, occasional nodular pyrite, trace very fine to fine intergranular porosity, trace spotty brown oil stain, slow pale yellow diffuse cut fluorescence

11,100-11,150 SILTY SANDSTONE: tan, light to medium gray-brown, off white, very fine grained, firm, sub-angular, moderately to well sorted, calcite cemented, moderately cemented, occasional disseminated pyrite, occasional nodular pyrite, trace very fine to fine intergranular porosity, trace spotty brown oil stain, slow pale yellow streaming cut fluorescence

11,150-11,200 SILTY SANDSTONE: light gray, medium to dark gray, off white, light gray-brown, very fine grained, firm, rarely hard, sub-angular, moderately to well sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

11,200-11,250 SILTY SANDSTONE: light brown, light gray, medium to dark gray, off white, light gray-brown, very fine grained, firm, rarely hard, sub-angular, occasionally sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

11,250-11,300 SILTY SANDSTONE: light to medium gray, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

11,300-11,350 SILTY SANDSTONE: light to medium gray, light brown, light brown-gray, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow to moderately pale yellow diffuse cut fluorescence

11,350-11,400 SILTY SANDSTONE: off white, light brown, light brown-gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

11,400-11,450 SILTY SANDSTONE: off white, light to medium gray, very fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

11,450-11,500 SILTY SANDSTONE: light brown-gray, off white, light gray, occasional dark gray, very fine grained, firm, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, moderately pale yellow diffuse cut fluorescence

11,500-11,550 SILTY SANDSTONE: light to medium gray, light brown, light brown-gray, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow to moderately pale yellow diffuse cut fluorescence

11,550-11,600 SILTY SANDSTONE: dark gray, light to medium gray, light brown, light brown-gray, off white, very fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, moderately pale yellow diffuse cut fluorescence

11,600-11,650 SILTY SANDSTONE: light brown, light brown-gray, light gray, off white, trace dark gray, very fine grained, firm, common hard, sub-angular, occasionally sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow to moderately pale yellow diffuse cut fluorescence

11,650-11,700 SILTY SANDSTONE: light brown, light brown-gray, light gray, off white, very fine grained, firm, common hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

11,700-11,750 SILTY SANDSTONE: light gray, medium to dark gray, off white, light gray-brown, very fine grained, firm, rarely hard, sub-angular, moderately to well sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

11,750-11,800 SILTY SANDSTONE: dark gray, light gray, medium gray, off white, light brown-gray, very fine grained, firm, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

11,800-11,850 SILTY SANDSTONE: off white, light to medium gray, light brown, very fine grained, firm, moderately sorted, calcite cemented, moderately to well cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, moderately pale yellow diffuse trace streaming cut fluorescence

11,850-11,900 SILTY SANDSTONE: off white, light brown, light brown-gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

11,900-11,950 SILTY SANDSTONE: off white, light brown, light brown-gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

11,950-12,000 SILTY SANDSTONE: light to medium gray, light brown, light brown-gray, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow to moderately pale yellow diffuse cut fluorescence

12,000-12,050 SILTY SANDSTONE: light brown, light brown-gray, off white, very fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow to moderately pale yellow diffuse cut fluorescence

12,050-12,100 SILTY SANDSTONE: light brown, light brown-gray, off white, very fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow to moderately pale yellow diffuse cut fluorescence

12,100-12,150 SILTY SANDSTONE: dark gray, light brown, light brown-gray, off white, very fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, moderately pale yellow diffuse cut fluorescence

12,150-12,200 SILTY SANDSTONE: light brown, light to medium gray, off white, cream, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, moderately pale yellow diffuse cut fluorescence

12,200-12,250 SILTY SANDSTONE: off white, light brown, light brown-gray, very fine grained, firm, sub-angular, rarely sub-rounded, moderately to well sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, moderately pale yellow streaming cut fluorescence

12,250-12,300 SILTY SANDSTONE: light brown, light brown-gray, light gray, off white, trace dark gray, very fine grained, firm, common hard, sub-angular, occasionally sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow to moderately pale yellow diffuse cut fluorescence

12,300-12,350 SILTY SANDSTONE: light brown, light gray, off white, very fine grained, firm, common hard, sub-angular, occasionally sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow to moderately pale yellow diffuse cut fluorescence

12,350-12,400 SILTY SANDSTONE: light brown, light brown-gray, off white, very fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow to moderately pale yellow diffuse cut fluorescence

12,400-12,450 SILTY SANDSTONE: off white, light to medium gray, light brown, very fine grained, firm, moderately sorted, calcite cemented, moderately to well cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, moderately pale yellow diffuse trace streaming cut fluorescence

12,450-12,500 SILTY SANDSTONE: off white, light brown, light brown-gray, occasional dark gray, trace tan, very fine to fine grained, firm, moderately sorted, calcite cemented, moderately to well cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, moderately pale yellow diffuse trace streaming cut fluorescence

12,500-12,550 SILTY SANDSTONE: off white, light brown, light brown-gray, very fine to fine grained, firm, moderately sorted, calcite cemented, moderately to well cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, moderately pale yellow diffuse trace streaming cut fluorescence

12,550-12,600 SILTY SANDSTONE: off white, light brown, light brown-gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

12,600-12,650 SILTY SANDSTONE: off white, light brown, light brown-gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

12,650-12,700 SILTY SANDSTONE: medium brown, light gray, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, occasional nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

12,700-12,750 SILTY SANDSTONE: medium brown, medium to dark gray, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

12,750-12,800 SILTY SANDSTONE: medium brown, medium to dark gray, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, rare spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

12,800-12,850 SILTY SANDSTONE: light to medium gray, medium brown, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, rare spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

12,850-12,900 SILTY SANDSTONE: medium gray-brown, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

12,900-12,950 SILTY SANDSTONE: tan, off white, medium gray-brown, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

12,950-13,000 SILTY SANDSTONE: tan, off white, medium gray-brown, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, occasional nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

13,000-13,050 SILTY SANDSTONE: tan, off white, medium gray-brown, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

13,050-13,100 SILTY SANDSTONE: medium gray-brown, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

13,100-13,150 SILTY SANDSTONE: medium gray, medium to dark brown, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

13,150-13,200 SILTY SANDSTONE: medium gray-brown, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

13,200-13,250 SILTY SANDSTONE: medium gray-brown, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

13,250-13,300 SILTY SANDSTONE: medium gray-brown, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

13,300-13,350 SILTY SANDSTONE: medium to dark gray, tan-medium brown, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

13,350-13,400 SILTY SANDSTONE: light to medium brown, gray, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, occasional nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

13,400-13,450 SILTY SANDSTONE: light to medium brown, gray, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, occasional nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,450-13,500 SILTY SANDSTONE: medium brown-gray, off white, tan, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,500-13,550 SILTY SANDSTONE: light to medium brown, gray, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,550-13,600 SILTY SANDSTONE: off white, medium gray, light to medium brown, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,600-13,650 SILTY SANDSTONE: medium gray, light to medium brown, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,650-13,700 SILTY SANDSTONE: medium gray-brown, off white, light gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,700-13,750 SILTY SANDSTONE: medium gray-brown, off white, light gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,750-13,800 SILTY SANDSTONE: tan, light to medium gray, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,800-13,850 SILTY SANDSTONE: medium brown, light to medium gray, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,850-13,900 SILTY SANDSTONE: medium to dark gray-brown, off white, very fine grained, firm, trace hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,900-13,950 SILTY SANDSTONE: off white, medium brown, light to medium gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

13,950-14,000 SILTY SANDSTONE: gray, light to medium brown, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, rare spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

14,000-14,050 SILTY SANDSTONE: medium brown, light to medium gray, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow streaming cut fluorescence

14,050-14,100 SILTY SANDSTONE: tan, off white, medium brown, light to medium gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,100-14,150 SILTY SANDSTONE: medium gray-brown, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,150-14,200 SILTY SANDSTONE: medium gray-brown, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,200-14,250 SILTY SANDSTONE: off white, gray, light to medium brown, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,250-14,300 SILTY SANDSTONE: tan, off white, gray, light to medium brown, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,300-14,350 SILTY SANDSTONE: medium brown, dark gray, tan, off white, very fine grained, firm, trace hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,350-14,400 SILTY SANDSTONE: light to medium gray, tan, very fine grained, firm, trace hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,400-14,450 SILTY SANDSTONE: tan, off white, light to medium gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,450-14,500 SILTY SANDSTONE: gray-brown, cream, very fine grained, firm, trace hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,500-14,550 SILTY SANDSTONE: gray-brown, cream, very fine grained, firm, trace hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,550-14,600 SILTY SANDSTONE: off white, tan, medium gray-brown, very fine grained, firm, trace hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, occasional nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, slow pale yellow diffuse cut fluorescence

14,600-14,650 SILTY SANDSTONE: light brown, light brown-gray, light to medium gray, off white, very fine grained, firm, trace hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

14,650-14,700 SILTY SANDSTONE: medium gray-brown, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

14,700-14,750 SILTY SANDSTONE: light brown, off white, light to medium gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

14,750-14,800 SILTY SANDSTONE: medium to dark gray, tan-medium brown, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

14,800-14,850 SILTY SANDSTONE: medium gray, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

14,850-14,900 SILTY SANDSTONE: off white, light brown, light brown-gray, light to medium gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, moderately pale yellow diffuse cut fluorescence

14,900-14,950 SILTY SANDSTONE: light brown, light brown-gray, light gray, very fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, moderately pale yellow diffuse cut fluorescence

14,950-15,000 SILTY SANDSTONE: off white, light brown, gray, very fine grained, firm, rarely hard, sub-angular, common sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,000-15,050 SILTY SANDSTONE: light to medium gray, off white, light brown dark gray, tan, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light to medium brown oil stain, sample moderately contaminated with lube

15,050-15,100 SILTY SANDSTONE: light gray, off white, light brown dark gray, tan, very fine grained, firm, sub-angular, occasionally sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light to medium brown oil stain, sample moderately contaminated with lube

15,100-15,150 SILTY SANDSTONE: light brown, light brown-gray, cream, off white, gray, very fine grained, firm, sub-angular, occasionally sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, sample moderately contaminated with lube

15,150-15,200 SILTY SANDSTONE: cream, light brown, light brown-gray, light gray, very fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, moderately pale yellow diffuse cut fluorescence

15,200-15,250 SILTY SANDSTONE: off white, light gray, light brown, very fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, well cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty to even light brown oil stain, moderately pale yellow diffuse cut fluorescence

15,250-15,300 SILTY SANDSTONE: medium gray-brown, tan, off white, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

15,300-15,350 SILTY SANDSTONE: medium gray-brown, tan, off white, very fine grained, firm, rarely hard, sub-angular, common sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, slow pale yellow diffuse cut fluorescence

15,350-15,400 SILTY SANDSTONE: off white, light brown, gray, very fine grained, firm, rarely hard, sub-angular, common sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,400-15,450 SILTY SANDSTONE: light to medium gray, off white, cream, light brown, occasional dark gray, very fine grained, firm, sub-angular, common sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,450-15,500 SILTY SANDSTONE: light to medium gray, off white, light brown, dark gray, tan, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,500-15,550 SILTY SANDSTONE: light brown, light brown-gray, light to medium gray, off white, very fine grained, firm, sub-angular, occasionally sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,550-15,600 SILTY SANDSTONE: cream, light brown, light brown-gray, light to medium gray, off white, very fine grained, firm, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,600-15,650 SILTY SANDSTONE: dark gray, light to medium gray, very fine grained, firm, sub-angular, trace sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,650-15,700 SILTY SANDSTONE: medium gray-brown, off white, light gray, very fine grained, firm, rarely hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,700-15,750 SILTY SANDSTONE: cream, light brown, medium gray-brown, off white, light gray, very fine grained, firm, occasional hard, sub-angular, common sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,750-15,800 SILTY SANDSTONE: light brown-gray, off white, light to medium gray, dark gray, very fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,800-15,850 SILTY SANDSTONE: cream, off white, light gray, very fine grained, firm, sub-angular, trace sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,850-15,900 SILTY SANDSTONE: light to medium brown, medium gray-brown, light gray, very fine grained, firm, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,900-15,950 SILTY SANDSTONE: gray, medium gray, light brown-gray, occasional off white, very fine to fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

15,950-16,000 SILTY SANDSTONE: gray, medium gray, light brown-gray, occasional off white, very fine to fine grained, firm, rarely hard, sub-angular, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,000-16,050 SILTY SANDSTONE: light gray, off white, light brown dark gray, tan, very fine grained, firm, sub-angular, occasionally sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,050-16,100 SILTY SANDSTONE: cream, off white, light gray, light brown, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty medium brown oil stain, sample moderately contaminated with lube

16,100-16,150 SILTY SANDSTONE: light to medium gray, off white, light brown, dark gray, tan, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,150-16,200 SILTY SANDSTONE: light to medium gray, off white, cream, light brown, occasional dark gray, very fine grained, firm, sub-angular, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,200-16,250 SILTY SANDSTONE: cream, light brown, medium gray-brown, off white, light gray, very fine grained, firm, occasional hard, sub-angular, common sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,250-16,300 SILTY SANDSTONE: light to medium gray, cream, light brown, medium gray-brown, off white, very fine grained, firm, occasional hard, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,300-16,350 SILTY SANDSTONE: light brown, light brown-gray, very fine grained, firm, occasional hard, sub-angular, trace sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,350-16,400 SILTY SANDSTONE: light gray, light brown, light brown-gray, very fine grained, firm, occasional hard, sub-angular, trace sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,400-16,450 SILTY SANDSTONE: light to medium gray, off white, light brown, dark gray, tan, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,450-16,500 SILTY SANDSTONE: cream, light brown, medium gray-brown, off white, light gray, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,500-16,550 SILTY SANDSTONE: cream, light brown, medium gray-brown, off white, light gray, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,550-16,600 SILTY SANDSTONE: light to medium brown, gray, off white, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,600-16,650 SILTY SANDSTONE: tan, medium gray, off white, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,650-16,700 SILTY SANDSTONE: tan, medium gray, off white, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

16,700-16,750 SILTY SANDSTONE: tan, medium gray, off white, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,350-17,400 SILTY SANDSTONE: medium to dark gray, light brown, off white, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,400-17,450 SILTY SANDSTONE: medium to dark gray, light brown, off white, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,450-17,500 SILTY SANDSTONE: medium to dark gray, light brown, off white, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,500-17,550 SILTY SANDSTONE: medium to dark gray, light brown, off white, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, rare SHALE: black, dark gray-brown, firm, blocky, earthy texture, occasional disseminated pyrite, petroliferous, carbonaceous, earthy porosity, sample moderately contaminated with lube

Upper Bakken/Middle Bakken Contact

17,540' MD; 10,766' TVD (-8,698')

17,500-17,597 SHALE: black, dark gray-brown, firm, blocky, earthy texture, occasional disseminated pyrite, petroliferous, carbonaceous, earthy porosity, sample moderately contaminated with lube

SIDETRACK INITIATED AT 17,420' MD

17,450-17,500 SILTY SANDSTONE: light to medium gray, off white, dark gray, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,500-17,550 SILTY SANDSTONE: light to medium gray, blue-gray, off white, dark gray, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,550-17,600 SILTY SANDSTONE: light gray, blue-gray, off white, dark gray, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,600-17,650 SILTY SANDSTONE: medium to dark gray, blue-gray, off white, light brown, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,650-17,700 SILTY SANDSTONE: medium to dark gray, blue-gray, off white, light brown, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,700-17,750 SILTY SANDSTONE: light to medium gray-brown, off white, tan, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,750-17,800 SILTY SANDSTONE: light to medium gray-brown, off white, tan, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, rare nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

17,800-17,850 SILTY SANDSTONE: tan, off white, light blue-gray, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

18,450-18,500 SILTY SANDSTONE: medium gray-brown, light gray, tan, occasional off white, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

19,100-19,150 SILTY SANDSTONE: medium gray-brown, light gray, tan, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

19,750-19,800 SILTY SANDSTONE: light to medium gray, light blue-gray, off white, tan, very fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube

20,400-20,450 SILTY SANDSTONE: medium gray-brown, light gray, tan, occasional off white, very fine to fine grained, firm, sub-angular, rarely sub-rounded, moderately sorted, calcite cemented, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace very fine to fine intergranular porosity, trace spotty light brown oil stain, sample moderately contaminated with lube



Directional Survey Certification

Operator: Oasis Petroleum LLC **Well Name:** Kline Federal 5300 11-18 2B **API:** 33-053-06243
Enseco Job#: S15090-02 **Job Type:** MWD D&I **County, State:** McKenzie County, N. Dakota

Well Surface Hole Location (SHL): Lot 1, Sec. 18, T153N, R100W (960' FNL & 318 FWL)

Latitude: 48° 04' 45.680 N **Longitude:** 103° 36' 10.190 W **Datum:** Nad 83

Final MWD Report Date: Feb. 28, 2015 **MWD Survey Run Date:** Feb. 25, 2015 to Feb. 27, 2015

Tied In to Surveys Provided By: Enseco Directional Drilling D&I MWD **MD:** Surface

MWD Surveyed from 00 ft to 2,081.0 ft MD **Survey Type:** Positive Pulse D&I MWD **Sensor to Bit:** 63 ft

Rig Contractor: Noble **Rig Number:** 2 **RKB Height:** 2,056.0 ft **GL Elevation:** 2,056.0 ft

MWD Surveyor Name: Brett McClain

"The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Enseco Energy Services USA Corp. I am authorized and qualified to review the data, calculations and this report and that the report represents a true and correct Directional Survey of this well based on the original data corrected to True North and obtained at the well site. Wellbore coordinates are calculated using the minimum curvature method."

Jonathan Hovland, Well Planner

Enseco Representative Name, Title

Jonathan Hovland

Signature

March 5th 2015

Date Signed

On this the ___ day of ___, 20___, before me personally appeared First & Last Name, to me known as the person described in and who executed the foregoing instrument and acknowledged the (s)he executed the same as his/her free act and deed.

Seal: _____
Notary Public

Commission Expiry



Enseco Survey Report

05 March, 2015

Oasis Petroleum LLC

**McKenzie County, North Dakota
Lot 1 Sec.18 Twp.153N Rge.100W
Kline Federal 5300 11-18 2B
Job # S15090-02
API#: 33-053-06243**

Survey: Final Surveys Vertical Section



Company:	Oasis Petroleum LLC	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2B
Project:	McKenzie County, North Dakota	Ground Level Elevation:	2,056.00usft
Site:	Lot 1 Sec.18 Twp.153N Rge.100W	Wellhead Elevation:	WELL @ 2056.00usft (Original Well Elev)
Well:	Kline Federal 5300 11-18 2B	North Reference:	True
Wellbore:	Job # S15090-02	Survey Calculation Method:	Minimum Curvature
Design:	Final Surveys Vertical Section	Database:	EDM5000

Project	McKenzie County, North Dakota		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		Using geodetic scale factor

Site	Lot 1 Sec.18 Twp.153N Rge.100W		
Site Position:		Northing:	408,992.30 usft
From:	Lat/Long	Easting:	1,210,243.30 usft
Position Uncertainty:	0.00 usft	Slot Radius:	13-3/16 "
		Latitude:	48° 4' 45.680 N
		Longitude:	103° 36' 10.190 W
		Grid Convergence:	-2.309°

Well	Kline Federal 5300 11-18 2B API#: 33-053-06243		
Well Position	+N/-S	0.00 usft	Northing: 408,992.30 usft
	+E/-W	0.00 usft	Easting: 1,210,243.30 usft
Position Uncertainty	0.00 usft	Wellhead Elevation:	usft
		Latitude:	48° 4' 45.680 N
		Longitude:	103° 36' 10.190 W
		Ground Level:	2,056.00usft

Wellbore	Job # S15090-02				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2015	3/5/2015	8.302	72.945	56,271

Design:	Final Surveys Vertical Section			Survey Error Model:	Standard ISCWSA MWD Tool
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	0.00
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)	
	0.00	0.00	0.00	317.15	

Company:	Oasis Petroleum LLC	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2B
Project:	McKenzie County, North Dakota	Ground Level Elevation:	2,056.00usft
Site:	Lot 1 Sec.18 Twp.153N Rge.100W	Wellhead Elevation:	WELL @ 2056.00usft (Original Well Elev)
Well:	Kline Federal 5300 11-18 2B	North Reference:	True
Wellbore:	Job # S15090-02	Survey Calculation Method:	Minimum Curvature
Design:	Final Surveys Vertical Section	Database:	EDM5000

Survey										
MD (usft)	Inc (°)	Azi (°)	TVD (usft)	SS (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
Tie-in from Surface										
0.00	0.00	0.00	0.00	2,056.00	0.00	0.00	0.00	0.00	0.00	0.00
132.00	0.60	215.50	132.00	1,924.00	-0.56	-0.40	-0.14	0.45	0.45	0.00
224.00	0.50	210.10	223.99	1,832.01	-1.30	-0.88	-0.35	0.12	-0.11	-5.87
314.00	0.70	269.40	313.99	1,742.01	-1.65	-1.63	-0.10	0.69	0.22	65.89
399.00	0.60	281.10	398.98	1,657.02	-1.57	-2.58	0.61	0.19	-0.12	13.76
489.00	0.50	275.00	488.98	1,567.02	-1.44	-3.44	1.28	0.13	-0.11	-6.78
579.00	0.80	278.30	578.97	1,477.03	-1.32	-4.45	2.06	0.34	0.33	3.67
663.00	0.50	268.20	662.97	1,393.03	-1.24	-5.40	2.76	0.38	-0.36	-12.02
751.00	0.60	292.30	750.96	1,305.04	-1.08	-6.21	3.43	0.28	0.11	27.39
840.00	0.40	301.00	839.96	1,216.04	-0.74	-6.91	4.15	0.24	-0.22	9.78
927.00	0.70	307.10	926.96	1,129.04	-0.27	-7.59	4.97	0.35	0.34	7.01
1,015.00	0.70	222.00	1,014.95	1,041.05	-0.34	-8.38	5.45	1.08	0.00	-96.70
1,101.00	0.60	341.60	1,100.95	955.05	-0.31	-8.87	5.81	1.31	-0.12	139.07
1,188.00	0.50	66.70	1,187.95	868.05	0.28	-8.67	6.10	0.86	-0.11	97.82
1,277.00	0.90	38.80	1,276.94	779.06	0.97	-7.87	6.07	0.58	0.45	-31.35
1,365.00	0.90	16.80	1,364.93	691.07	2.17	-7.24	6.52	0.39	0.00	-25.00
1,450.00	0.70	15.30	1,449.92	606.08	3.31	-6.91	7.13	0.24	-0.24	-1.76
1,539.00	0.80	332.70	1,538.91	517.09	4.39	-7.05	8.01	0.62	0.11	-47.87
1,628.00	1.10	343.10	1,627.90	428.10	5.76	-7.58	9.38	0.39	0.34	11.69
1,714.00	0.40	7.80	1,713.89	342.11	6.85	-7.78	10.31	0.88	-0.81	28.72
1,803.00	0.40	120.30	1,802.89	253.11	7.00	-7.47	10.21	0.75	0.00	126.40
1,889.00	0.30	96.80	1,888.89	167.11	6.82	-6.99	9.75	0.20	-0.12	-27.33
1,987.00	0.60	120.10	1,986.89	69.11	6.53	-6.29	9.07	0.35	0.31	23.78
Last MWD Survey										
2,081.00	0.90	135.20	2,080.88	-24.88	5.76	-5.35	7.86	0.38	0.32	16.06

Survey Annotations				
Local Coordinates				
MD (usft)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Comment
0.00	0.00	0.00	0.00	Tie-in from Surface
2,081.00	2,080.88	5.76	-5.35	Last MWD Survey



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Monday, June 15, 2015

State of North Dakota

Subject: **Surveys**

Re: **Oasis**
Kline Federal 5300 11-18 2B
McKenzie, ND

Enclosed, please find the original and one copy of the survey performed on the above-referenced well by Ryan Directional Services, Inc.. Other information required by your office is as follows:

<i>Surveyor Name</i>	<i>Surveyor Title</i>	<i>Borehole Number</i>	<i>Start Depth</i>	<i>End Depth</i>	<i>Start Date</i>	<i>End Date</i>	<i>Type of</i>	<i>TD Straight Line Projection</i>
Nick Brochu	MWD Operator	O.H.	2081'	17489'	05/29/15	06/03/15	MWD	17597'
Nick Brochu	MWD Operator	ST 1	17299'	20382'	06/03/15	06/05/15	MWD	20450'

If any other information is required please contact the undersigned at the letterhead address or phone number.

Douglas Hudson
Well Planner



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Directional Services, Inc.
19510 Oil Center Blvd.
Houston, Texas 77073
Bus: 281.443.1414
Fax: 281.443.1676

Friday, June 05, 2015

State of North Dakota
County of McKenzie

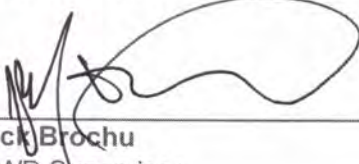
Subject: **Survey Certification Letter**

Survey Company: Ryan Directional Services, Inc.
Job Number: 8983
Survey Job Type: Ryan MWD
Customer: Oasis Petroleum North America
Well Name: Kline Federal 5300 11-18 2B
Rig Name: Xtreme 21

Surface: 48° 4' 45.680 N / 103° 36' 10.190 W
A.P.I. No: 33-053-06243
Location: McKenzie, ND
RKB Height: 2070'
Distance to Bit: 68'

Surveyor Name	Surveyor Title	Borehole Number	Start Depth	End Depth	Start Date	End Date	Type of	TD Straight
								Line Projection
Nick Brochu	MWD Supervisor	OH	11032'	17489'	05/29/15	06/03/15	MWD	17557'
Nick Brochu	MWD Supervisor	ST1	17299'	20382'	06/03/15	06/05/15	MWD	20450'

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.



Nick Brochu
MWD Supervisor
Ryan Directional Services, Inc.

Report #: **1**
Date: **5-Apr-15**



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # **8798**

SURVEY REPORT

Customer: **Oasis Petroleum**
Well Name: **Kline Federal 5300 11-18 2B**
Rig #: **Nabors B27**
API #: **33-053-06243**
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **D.FOLEY / D.OGDEN**
Directional Drillers: **RPM**
Survey Corrected To: **True North**
Vertical Section Direction: **90**
Total Correction: **8.30**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
Tie in to Gyro Surveys									
Tie In	2081	0.90	135.20		2080.88	-5.35	5.76	-5.35	0.38
1	2176	1.60	146.20	77.00	2175.86	-4.09	4.13	-4.09	0.78
2	2270	1.70	152.90	66.00	2269.82	-2.72	1.80	-2.72	0.23
3	2365	1.00	160.10	82.00	2364.79	-1.80	-0.24	-1.80	0.76
4	2459	0.70	163.90	84.00	2458.78	-1.36	-1.56	-1.36	0.32
5	2553	0.50	263.40	87.00	2552.78	-1.61	-2.16	-1.61	0.98
6	2647	0.80	284.50	89.00	2646.77	-2.65	-2.04	-2.65	0.40
7	2742	1.50	298.80	91.00	2741.75	-4.38	-1.28	-4.38	0.79
8	2836	1.90	300.00	93.00	2835.71	-6.81	0.10	-6.81	0.43
9	2930	2.30	305.70	95.00	2929.65	-9.69	1.97	-9.69	0.48
10	3024	1.10	321.10	96.00	3023.60	-11.79	3.78	-11.79	1.35
11	3119	1.40	328.70	100.00	3118.58	-12.96	5.48	-12.96	0.36
12	3213	1.20	34.40	102.00	3212.56	-13.00	7.27	-13.00	1.51
13	3307	1.20	47.50	104.00	3306.54	-11.72	8.75	-11.72	0.29
14	3401	0.90	36.20	105.00	3400.53	-10.56	10.01	-10.56	0.39
15	3495	1.00	36.90	107.00	3494.51	-9.63	11.26	-9.63	0.11
16	3590	0.70	43.50	109.00	3589.50	-8.74	12.35	-8.74	0.33
17	3684	0.80	33.50	111.00	3683.49	-7.98	13.31	-7.98	0.17
18	3778	0.60	41.60	113.00	3777.49	-7.29	14.22	-7.29	0.24
19	3872	0.50	37.40	114.00	3871.48	-6.71	14.92	-6.71	0.11
20	3966	0.40	14.70	114.00	3965.48	-6.38	15.56	-6.38	0.22
21	4060	0.30	25.70	116.00	4059.48	-6.19	16.10	-6.19	0.13
22	4154	0.40	346.40	116.00	4153.48	-6.16	16.64	-6.16	0.27
23	4248	0.50	76.80	118.00	4247.47	-5.84	17.05	-5.84	0.68
24	4343	0.50	84.90	120.00	4342.47	-5.02	17.19	-5.02	0.07
25	4437	0.40	36.40	122.00	4436.47	-4.42	17.49	-4.42	0.41
26	4531	0.30	48.60	122.00	4530.47	-4.04	17.91	-4.04	0.13
27	4625	0.40	217.00	123.00	4624.47	-4.05	17.81	-4.05	0.74
28	4720	0.40	225.60	125.00	4719.46	-4.49	17.32	-4.49	0.06
29	4813	0.40	330.20	125.00	4812.46	-4.88	17.37	-4.88	0.68
30	4907	0.50	128.90	127.00	4906.46	-4.73	17.40	-4.73	0.94
31	5001	0.60	143.20	127.00	5000.46	-4.11	16.75	-4.11	0.18
32	5095	0.40	207.70	129.00	5094.45	-3.97	16.06	-3.97	0.60
33	5190	0.30	193.10	129.00	5189.45	-4.18	15.53	-4.18	0.14
34	5284	0.60	181.00	131.00	5283.45	-4.25	14.79	-4.25	0.33
35	5378	0.80	158.90	131.00	5377.44	-4.02	13.69	-4.02	0.35
36	5472	0.80	177.40	132.00	5471.43	-3.75	12.42	-3.75	0.27
37	5567	1.10	185.80	134.00	5566.42	-3.81	10.85	-3.81	0.35
38	5661	1.10	202.30	134.00	5660.40	-4.25	9.12	-4.25	0.34
39	5755	0.60	98.20	134.00	5754.40	-4.10	8.22	-4.10	1.46
40	5850	0.60	89.90	136.00	5849.39	-3.11	8.15	-3.11	0.09
41	5944	0.70	54.60	138.00	5943.39	-2.15	8.48	-2.15	0.43
42	6038	0.70	36.50	138.00	6037.38	-1.34	9.27	-1.34	0.23
43	6132	0.80	38.60	138.00	6131.37	-0.59	10.25	-0.59	0.11
44	6226	1.20	26.10	138.00	6225.36	0.25	11.64	0.25	0.48
45	6320	1.40	22.00	141.00	6319.33	1.11	13.59	1.11	0.23
46	6414	1.20	62.70	143.00	6413.31	2.42	15.11	2.42	0.98
47	6509	1.30	47.90	145.00	6508.29	4.10	16.29	4.10	0.35
48	6603	1.40	37.00	147.00	6602.26	5.58	17.92	5.58	0.29
49	6697	1.10	52.80	147.00	6696.24	6.99	19.38	6.99	0.48
50	6791	1.10	71.10	152.00	6790.22	8.57	20.22	8.57	0.37
51	6885	0.90	72.50	154.00	6884.21	10.12	20.73	10.12	0.21
52	6979	1.10	38.50	156.00	6978.19	11.39	21.66	11.39	0.65
53	7073	1.20	73.50	158.00	7072.18	12.90	22.65	12.90	0.74
54	7167	0.80	101.80	159.00	7166.16	14.48	22.79	14.48	0.66
55	7262	1.00	127.00	161.00	7261.15	15.79	22.16	15.79	0.46
56	7356	0.90	118.50	161.00	7355.14	17.10	21.31	17.10	0.18
57	7450	1.00	121.00	163.00	7449.13	18.45	20.54	18.45	0.12
58	7544	0.70	113.20	165.00	7543.12	19.68	19.89	19.68	0.34
59	7639	0.90	108.40	165.00	7638.11	20.92	19.43	20.92	0.22
60	7733	1.00	107.40	167.00	7732.09	22.40	18.95	22.40	0.11
61	7827	1.10	95.80	167.00	7826.08	24.08	18.61	24.08	0.25
62	7921	0.80	82.00	167.00	7920.06	25.63	18.61	25.63	0.40
63	8015	0.80	89.90	167.00	8014.06	26.94	18.70	26.94	0.12
64	8109	0.70	82.20	170.00	8108.05	28.16	18.78	28.16	0.15
65	8203	0.60	108.20	172.00	8202.04	29.20	18.71	29.20	0.33

Report #: **1**
Date: **5-Apr-15**



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # **8798**

SURVEY REPORT

Customer: **Oasis Petroleum**
Well Name: **Kline Federal 5300 11-18 2B**
Rig #: **Nabors B27**
API #: **33-053-06243**
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **D.FOLEY / D.OGDEN**
Directional Drillers: **RPM**
Survey Corrected To: **True North**
Vertical Section Direction: **90**
Total Correction: **8.30**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
66	8298	0.50	118.60	174.00	8297.04	30.04	18.35	30.04	0.15
67	8392	0.50	108.30	159.00	8391.03	30.79	18.03	30.79	0.10
68	8486	0.50	89.90	161.00	8485.03	31.58	17.90	31.58	0.17
69	8580	0.60	102.80	167.00	8579.03	32.48	17.79	32.48	0.17
70	8674	0.70	110.80	168.00	8673.02	33.49	17.48	33.49	0.14
71	8769	0.60	115.20	170.00	8768.01	34.48	17.06	34.48	0.12
72	8863	0.60	133.30	172.00	8862.01	35.29	16.51	35.29	0.20
73	8957	0.40	114.20	172.00	8956.00	35.95	16.04	35.95	0.27
74	9051	0.40	118.30	174.00	9050.00	36.53	15.75	36.53	0.03
75	9145	0.70	97.20	176.00	9144.00	37.39	15.52	37.39	0.38
76	9240	0.20	343.00	177.00	9239.00	37.92	15.61	37.92	0.85
77	9334	0.20	335.80	177.00	9333.00	37.80	15.92	37.80	0.03
78	9428	0.40	38.40	179.00	9426.99	37.94	16.32	37.94	0.38
79	9523	0.50	27.20	177.00	9521.99	38.34	16.95	38.34	0.14
80	9617	0.60	10.90	177.00	9615.99	38.62	17.80	38.62	0.20
81	9711	0.50	2.40	181.00	9709.98	38.73	18.69	38.73	0.14
82	9805	0.50	354.20	185.00	9803.98	38.70	19.51	38.70	0.08
83	9899	0.50	330.90	185.00	9897.98	38.46	20.28	38.46	0.21
84	9993	0.60	353.10	186.00	9991.97	38.20	21.12	38.20	0.25
85	10087	0.70	0.00	188.00	10085.97	38.14	22.19	38.14	0.13
86	10171	1.00	11.90	188.00	10169.96	38.30	23.42	38.30	0.41
87	10201	1.00	4.30	168.00	10199.95	38.37	23.93	38.37	0.44
88	10232	1.50	58.70	176.00	10230.94	38.74	24.42	38.74	3.96
89	10263	4.40	88.80	176.00	10261.90	40.27	24.65	40.27	10.30
90	10295	7.40	97.40	176.00	10293.73	43.54	24.41	43.54	9.75
91	10326	11.10	97.50	177.00	10324.32	48.48	23.76	48.48	11.94
92	10357	14.50	98.30	177.00	10354.55	55.28	22.81	55.28	10.98
93	10389	17.50	100.70	177.00	10385.30	63.98	21.34	63.98	9.60
94	10420	20.90	103.40	179.00	10414.58	73.94	19.19	73.94	11.33
95	10452	24.20	104.50	179.00	10444.12	85.85	16.23	85.85	10.40
96	10483	27.20	104.60	179.00	10472.06	98.86	12.85	98.86	9.68
97	10514	30.20	104.80	181.00	10499.24	113.25	9.07	113.25	9.68
98	10546	33.30	105.00	181.00	10526.45	129.52	4.74	129.52	9.69
99	10577	36.40	105.90	181.00	10551.89	146.59	0.02	146.59	10.14
100	10609	39.70	107.80	183.00	10577.09	165.46	-5.71	165.46	10.94
101	10640	43.20	109.50	183.00	10600.32	184.90	-12.28	184.90	11.86
102	10671	46.60	110.40	183.00	10622.27	205.46	-19.75	205.46	11.16
103	10703	50.10	110.40	181.00	10643.54	227.87	-28.08	227.87	10.94
104	10734	53.80	109.50	174.00	10662.64	250.81	-36.41	250.81	12.15
105	10766	57.80	108.00	174.00	10680.62	275.87	-44.90	275.87	13.09
106	10797	62.90	106.70	176.00	10695.96	301.58	-52.93	301.58	16.85
107	10828	67.30	106.40	177.00	10709.00	328.53	-60.93	328.53	14.22
108	10860	71.50	105.70	177.00	10720.26	357.31	-69.21	357.31	13.28
109	10891	73.50	105.60	179.00	10729.58	385.78	-77.18	385.78	6.46
110	10922	76.40	104.50	179.00	10737.63	414.68	-84.96	414.68	9.96
111	10954	81.30	103.40	181.00	10743.82	445.14	-92.52	445.14	15.68
112	10985	85.90	102.10	183.00	10747.27	475.18	-99.31	475.18	15.41
113	11007	88.40	100.60	183.00	10748.37	496.72	-103.64	496.72	13.25
114	11032	88.40	101.10	230.00	10749.06	521.27	-108.34	521.27	2.00
115	11063	88.90	100.50	228.00	10749.79	551.71	-114.15	551.71	2.52
116	11094	88.80	100.70	226.00	10750.42	582.17	-119.85	582.17	0.72
117	11124	89.30	98.80	226.00	10750.91	611.73	-124.93	611.73	6.55
118	11155	88.80	97.30	224.00	10751.43	642.42	-129.27	642.42	5.10
119	11186	88.50	96.10	224.00	10752.16	673.20	-132.89	673.20	3.99
120	11217	88.70	95.40	224.00	10752.92	704.04	-135.99	704.04	2.35
121	11248	89.10	93.90	224.00	10753.51	734.93	-138.50	734.93	5.01
122	11279	91.30	93.60	226.00	10753.40	765.86	-140.53	765.86	7.16
123	11310	92.30	92.90	226.00	10752.43	796.79	-142.29	796.79	3.94
124	11341	92.10	92.80	226.00	10751.24	827.73	-143.83	827.73	0.72
125	11372	91.60	91.70	226.00	10750.24	858.69	-145.04	858.69	3.90
126	11403	90.80	90.80	226.00	10749.59	889.68	-145.72	889.68	3.88
127	11434	90.30	89.10	228.00	10749.29	920.68	-145.69	920.68	5.72
128	11465	89.60	88.10	230.00	10749.32	951.67	-144.94	951.67	3.94
129	11495	89.80	88.20	231.00	10749.48	981.65	-143.97	981.65	0.75
130	11588	89.60	87.80	233.00	10749.96	1074.59	-140.72	1074.59	0.48

Report #: **1**
Date: **5-Apr-15**



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # **8798**

SURVEY REPORT

Customer: **Oasis Petroleum**
Well Name: **Kline Federal 5300 11-18 2B**
Rig #: **Nabors B27**
API #: **33-053-06243**
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **D.FOLEY / D.OGDEN**
Directional Drillers: **RPM**
Survey Corrected To: **True North**
Vertical Section Direction: **90**
Total Correction: **8.30**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
131	11681	89.60	88.50	233.00	10750.61	1167.54	-137.72	1167.54	0.75
132	11773	90.50	88.30	237.00	10750.53	1259.50	-135.15	1259.50	1.00
133	11865	90.40	89.30	237.00	10749.81	1351.48	-133.22	1351.48	1.09
134	11960	90.90	89.50	240.00	10748.73	1446.47	-132.23	1446.47	0.57
135	12055	92.20	89.70	242.00	10746.16	1541.43	-131.57	1541.43	1.38
136	12150	91.10	89.80	242.00	10743.43	1636.39	-131.15	1636.39	1.16
137	12245	91.70	89.10	242.00	10741.11	1731.35	-130.24	1731.35	0.97
138	12340	89.80	89.20	242.00	10739.86	1826.33	-128.83	1826.33	2.00
139	12434	88.40	87.80	246.00	10741.34	1920.28	-126.37	1920.28	2.11
140	12529	88.80	87.60	248.00	10743.66	2015.18	-122.56	2015.18	0.47
141	12623	89.10	88.80	246.00	10745.38	2109.11	-119.61	2109.11	1.32
142	12719	90.90	90.30	246.00	10745.38	2205.10	-118.85	2205.10	2.44
143	12814	89.60	89.90	246.00	10744.97	2300.10	-119.02	2300.10	1.43
144	12910	90.00	89.50	248.00	10745.30	2396.10	-118.52	2396.10	0.59
145	13005	90.10	89.50	249.00	10745.22	2491.09	-117.69	2491.09	0.11
146	13101	89.80	89.30	249.00	10745.30	2587.09	-116.68	2587.09	0.38
147	13195	89.60	89.60	251.00	10745.80	2681.08	-115.78	2681.08	0.38
148	13290	89.90	89.60	253.00	10746.21	2776.08	-115.12	2776.08	0.32
149	13384	89.50	89.40	251.00	10746.70	2870.07	-114.30	2870.07	0.48
150	13480	88.90	89.00	253.00	10748.04	2966.05	-112.96	2966.05	0.75
151	13574	89.40	89.10	255.00	10749.44	3060.03	-111.40	3060.03	0.54
152	13671	89.70	88.70	257.00	10750.20	3157.01	-109.54	3157.01	0.52
153	13765	89.50	90.30	255.00	10750.86	3251.00	-108.72	3251.00	1.72
154	13860	89.70	90.20	257.00	10751.52	3346.00	-109.13	3346.00	0.24
155	13956	90.40	90.40	257.00	10751.43	3442.00	-109.63	3442.00	0.76
156	14051	90.10	89.90	258.00	10751.02	3536.99	-109.88	3536.99	0.61
157	14146	89.00	88.00	258.00	10751.77	3631.97	-108.14	3631.97	2.31
158	14240	87.40	86.80	260.00	10754.72	3725.82	-103.88	3725.82	2.13
159	14335	88.20	87.30	260.00	10758.37	3820.62	-99.00	3820.62	0.99
160	14431	89.20	87.60	262.00	10760.54	3916.50	-94.73	3916.50	1.09
161	14525	89.40	86.90	262.00	10761.69	4010.39	-90.22	4010.39	0.77
162	14620	89.20	88.60	260.00	10762.85	4105.30	-86.49	4105.30	1.80
163	14714	90.90	90.40	260.00	10762.77	4199.29	-85.67	4199.29	2.63
164	14809	91.60	92.20	260.00	10760.70	4294.24	-87.82	4294.24	2.03
165	14907	90.70	92.60	262.00	10758.73	4392.13	-91.92	4392.13	1.00
166	15003	89.90	92.90	262.00	10758.23	4488.02	-96.53	4488.02	0.89
167	15099	88.70	92.60	262.00	10759.40	4583.90	-101.13	4583.90	1.29
168	15194	88.10	92.20	264.00	10762.05	4678.78	-105.11	4678.78	0.76
169	15288	90.20	92.80	264.00	10763.45	4772.68	-109.21	4772.68	2.32
170	15382	90.50	92.90	262.00	10762.87	4866.56	-113.88	4866.56	0.34
171	15478	89.80	92.40	255.00	10762.62	4962.45	-118.32	4962.45	0.90
172	15573	90.10	91.00	257.00	10762.71	5057.41	-121.14	5057.41	1.51
173	15668	89.70	89.80	258.00	10762.87	5152.40	-121.80	5152.40	1.33
174	15763	89.00	88.40	260.00	10763.95	5247.38	-120.31	5247.38	1.65
175	15859	87.40	87.00	262.00	10766.96	5343.25	-116.46	5343.25	2.21
176	15955	88.50	88.30	260.00	10770.40	5439.11	-112.53	5439.11	1.77
177	16051	89.60	89.40	258.00	10771.99	5535.07	-110.60	5535.07	1.62
178	16148	89.80	89.20	262.00	10772.50	5632.06	-109.42	5632.06	0.29
179	16244	90.50	90.10	260.00	10772.25	5728.06	-108.83	5728.06	1.19
180	16340	92.20	91.30	262.00	10769.99	5824.02	-110.00	5824.02	2.17
181	16436	92.00	90.80	260.00	10766.47	5919.94	-111.76	5919.94	0.56
182	16532	89.90	90.00	258.00	10764.88	6015.92	-112.43	6015.92	2.34
183	16628	90.00	90.30	260.00	10764.96	6111.92	-112.68	6111.92	0.33
184	16724	88.90	90.20	262.00	10765.88	6207.91	-113.10	6207.91	1.15
185	16820	88.20	90.30	264.00	10768.31	6303.88	-113.52	6303.88	0.74
186	16916	89.50	90.80	264.00	10770.24	6399.85	-114.44	6399.85	1.45
187	17012	89.70	90.90	266.00	10770.91	6495.84	-115.87	6495.84	0.23
188	17106	90.80	91.30	266.00	10770.50	6589.82	-117.67	6589.82	1.25
189	17202	90.90	91.10	264.00	10769.07	6685.79	-119.68	6685.79	0.23
190	17299	89.80	91.30	264.00	10768.48	6782.76	-121.71	6782.76	1.15
191	17394	89.90	91.30	266.00	10768.73	6877.74	-123.87	6877.74	0.11
192	17489	91.70	91.70	266.00	10767.40	6972.69	-126.35	6972.69	1.94
Projection	17597	91.70	91.70		10764.20	7080.60	-129.56	7080.60	0.00

Report #: **2**
Date: **5-Jun-15**



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # **8983**

SURVEY REPORT

Customer: **Oasis Petroleum North America LLC**
Well Name: **Kline Federal 5300 11-18 2B**
Rig #: **Xtreme 21**
API #: **33-053-06243**
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **Sammy Hayman./ David Unger**
Directional Drillers: **RPM**
Survey Corrected To: **True North**
Vertical Section Direction: **90**
Total Correction: **8.30**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
Tie in to Gyro Surveys									
Tie In	17299	89.80	91.30	264.00	10768.48	6782.77	-121.71	6782.77	1.15
1	17394	90.00	91.40	258.00	10768.65	6877.74	-123.95	6877.74	0.24
2	17420	88.90	91.00	258.00	10768.90	6903.74	-124.49	6903.74	4.50
3	17453	87.30	90.90	258.00	10769.99	6936.71	-125.04	6936.71	4.86
4	17489	87.00	90.70	260.00	10771.78	6972.66	-125.54	6972.66	1.00
5	17584	86.00	90.30	258.00	10777.58	7067.48	-126.37	7067.48	1.13
6	17680	89.10	89.90	262.00	10781.68	7163.38	-126.54	7163.38	3.26
7	17776	88.60	89.50	262.00	10783.61	7259.36	-126.03	7259.36	0.67
8	17872	90.20	89.10	262.00	10784.61	7355.35	-124.86	7355.35	1.72
9	17969	90.80	89.10	264.00	10783.77	7452.33	-123.34	7452.33	0.62
10	18065	89.90	89.90	264.00	10783.18	7548.32	-122.50	7548.32	1.25
11	18161	89.90	90.20	264.00	10783.35	7644.32	-122.58	7644.32	0.31
12	18258	88.90	90.60	264.00	10784.36	7741.31	-123.26	7741.31	1.11
13	18354	88.70	90.70	266.00	10786.38	7837.29	-124.35	7837.29	0.23
14	18450	89.50	91.20	266.00	10787.88	7933.26	-125.94	7933.26	0.98
15	18544	90.80	91.10	267.00	10787.64	8027.24	-127.83	8027.24	1.39
16	18640	90.70	90.40	266.00	10786.38	8123.22	-129.08	8123.22	0.74
17	18735	88.40	89.70	266.00	10787.13	8218.21	-129.17	8218.21	2.53
18	18833	87.40	89.50	267.00	10790.72	8316.14	-128.48	8316.14	1.04
19	18928	88.20	90.10	266.00	10794.36	8411.07	-128.15	8411.07	1.05
20	19023	89.30	90.70	266.00	10796.44	8506.04	-128.82	8506.04	1.32
21	19120	89.90	90.70	266.00	10797.11	8603.03	-130.00	8603.03	0.62
22	19215	91.80	91.20	264.00	10795.70	8698.00	-131.57	8698.00	2.07
23	19311	93.20	90.90	266.00	10791.52	8793.89	-133.33	8793.89	1.49
24	19406	91.70	90.20	266.00	10787.46	8888.80	-134.24	8888.80	1.74
25	19502	91.00	89.70	266.00	10785.19	8984.77	-134.16	8984.77	0.90
26	19596	89.20	88.60	267.00	10785.03	9078.76	-132.76	9078.76	2.24
27	19692	89.90	88.50	266.00	10785.78	9174.72	-130.34	9174.72	0.74
28	19787	89.10	88.00	266.00	10786.61	9269.67	-127.43	9269.67	0.99
29	19882	90.30	88.40	266.00	10787.11	9364.62	-124.45	9364.62	1.33
30	19977	91.60	88.40	266.00	10785.54	9459.57	-121.80	9459.57	1.37
31	20073	92.70	88.00	264.00	10781.93	9555.45	-118.79	9555.45	1.22
32	20177	92.60	87.70	253.00	10777.13	9659.27	-114.89	9659.27	0.30
33	20273	91.70	89.80	255.00	10773.52	9755.17	-112.80	9755.17	2.38
34	20369	91.50	89.60	257.00	10770.84	9851.13	-112.29	9851.13	0.29
35	20382	91.60	89.70	257.00	10770.49	9864.13	-112.21	9864.13	1.09
Projection	20450	91.60	89.70	257.00	10768.59	9932.10	-111.86	9932.10	0.00

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
29334

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date April 9, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Suspension of Drilling

Well Name and Number Kline Federal 5300 11-18 2B				
Footages 960 F A L 318 F W L	Qtr-Qtr 11	Section 18	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie		

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum requests permission for suspension of drilling for approximately 55 days for the referenced well under NDAC 43-02-03-055. Oasis would like to suspend drilling on this well in order to drill the approved Carson SWD 5301 12-24 (well file #90329). The current rig will move to the Carson SWD pad once the vertical well bores have been drilled for all 3 wells on the Kline 11-18 pad. Oasis will return to the Kline 11-18 pad with a second rig on approximately June 6, 2015, to drill the lateral portion of the referenced well to TD.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9575	
Address 1001 Fannin St, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>Michael Kukuk</i>		Printed Name Michael Kukuk	
Title Regulatory Supervisor		Date March 26, 2015	
Email Address mkukuk@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 4/8/15	
By <i>Cost L. H. H. H.</i>	
Title Mineral Resources Permit Manager	

Holweger, Todd L.

From: Michael Kukuk <mkukuk@oasispetroleum.com>
Sent: Thursday, March 26, 2015 6:40 PM
To: Holweger, Todd L.
Cc: Regulatory; APD; Karyme Martin; Jason Swaren
Subject: SOD sundries for the Kline Federal 5300 11-18 pad
Attachments: Kline Federal 11-18 SOD sundries.pdf; ATT00001.txt

Importance: High

Good Evening Todd,

Per our conversation I have attached the SOD sundries for the 3 wells on the Kline pad. A few key points:

- 1) We will move the rig to the Carson SWD pad once we have finished drilling the vertical portions of all 3 wells on this pad.
- 2) We will finish drilling the vertical portion of the third well, the Kline Federal 5300 11-18 2B, on **April 9th**.
- 3) We will utilize a 2nd rig to drill the lateral portions of the 3 wells on this pad.
- 4) The 2nd rig is currently drilling wells on a different pad and is scheduled to reach TD on the final well in late May/early June, leaving a gap of approximately 55 days.
- 5) We will be able to return to the Kline Federal 5300 11-18 pad on or before **June 6th**.

Given the time sensitive nature of this request, expedited review of these sundries would be greatly appreciated.

Thank you for your consideration,

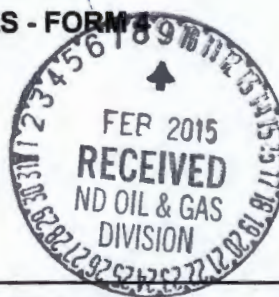
Michael P. Kukuk
Regulatory Supervisor
1001 Fannin, Suite 1500
Houston, Texas 77002
281-404-9575
281-382-5877 (cell)

mkukuk@oasispetroleum.com



**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

29334

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date March 17, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Change to Original APD

Well Name and Number Kline Federal 5300 11-18 2T2						
Footages			Qtr-Qtr	Section	Township	Range
960 F N	L	318 F W	L	LOT1	18	153 N
						100 W
Field	Pool	County				
Baker	Bakken	McKenzie				

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK**Oasis Petroleum respectfully requests approval to make the following changes to the original APD as follows:****Name Change: Kline Federal 5300 11-18 2B (previously 2T2)****Formation Change: Bakken (previously Three Forks second bench)****BHL change: 1080' FNL & 250' FEL Sec 17 T153N R100W**
(previously: 1530' FNL & 205' FEL)**Surface casing design:**

Surface Casing of 13 3/8" set at 2068' (previously 9 5/8")

Contingency Casing of 9 5/8" set at 6450'

Intermediate Casing of 7" with weight of 32 set at 11,005' (previously set at 11,111')

Production liner of 4 1/2" set from 10,446' to 20920' (previously set from 10,313' to 20,718')

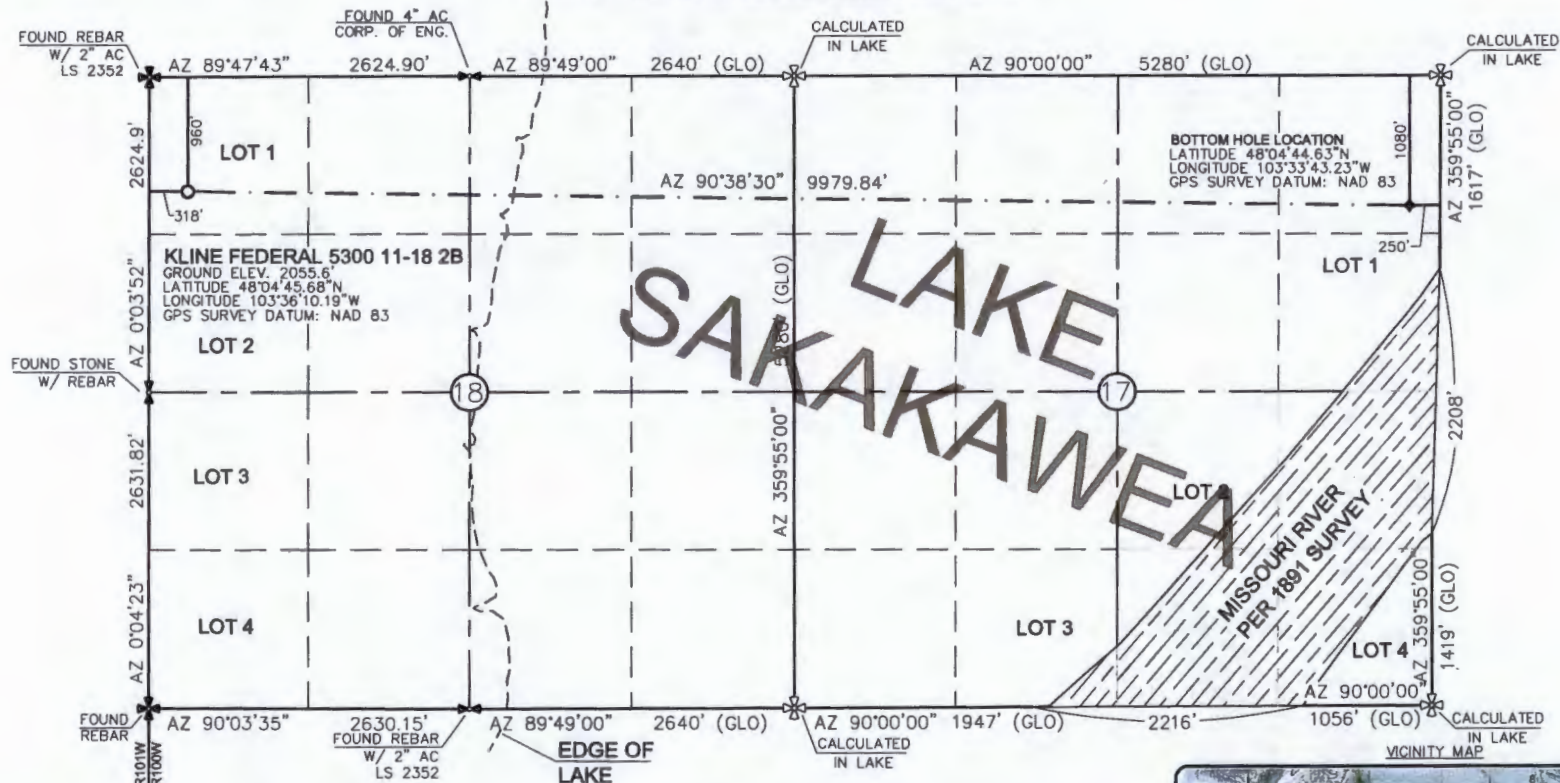
See attached supporting documents.*CC 25.00 2-20-15 KB**CC 25.00*

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9652	
Address 1001 Fannin Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>VS</i>	Printed Name Victoria Siemieniewski		
Title Regulatory Specialist	Date February 5, 2015		
Email Address vsiemieniewski@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date <i>2-17-2015</i>	
By <i>[Signature]</i>	
Title David Burns	Engineering Tech.

WELL LOCATION PLAT
OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 KLINE FEDERAL 5300 11-18 2B

960 FEET FROM NORTH LINE AND 316 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



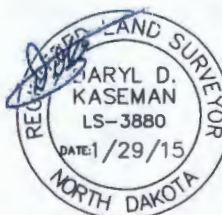
THIS DOCUMENT WAS ORIGINALLY
 ISSUED AND SEALED BY DARYL D.
 KASEMAN, PLS, REGISTRATION NUMBER
 3880 ON 1/29/15 AND THE
 ORIGINAL DOCUMENTS ARE STORED AT
 THE OFFICES OF INTERSTATE
 ENGINEERING, INC.



- ✕ - MONUMENT - RECOVERED
- ✕ - MONUMENT - NOT RECOVERED

STAKED ON 6/18/14
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'
 THIS SURVEY AND PLAT IS BEING PROVIDED AT THE
 REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I
 CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
 WORK PERFORMED BY ME OR UNDER MY SUPERVISION
 AND IS TRUE AND CORRECT TO THE BEST OF MY
 KNOWLEDGE AND BELIEF.

[Signature]
 DARYL D. KASEMAN LS-3880



© 2014, INTERSTATE ENGINEERING, INC.

Revision	Date	By	Description
REV 1	6/17/14	SKH	GROUND HILLS
REV 2	6/24/14	JAB	GROUND LATTICE
REV 3	6/24/14	JAB	GROUND LATTICE
REV 4	6/24/14	JAB	GROUND LATTICE
REV 5	6/24/14	JAB	GROUND LATTICE
REV 6	6/24/14	JAB	GROUND LATTICE
REV 7	6/24/14	JAB	GROUND LATTICE
REV 8	6/24/14	JAB	GROUND LATTICE
REV 9	6/24/14	JAB	GROUND LATTICE
REV 10	6/24/14	JAB	GROUND LATTICE
REV 11	6/24/14	JAB	GROUND LATTICE
REV 12	6/24/14	JAB	GROUND LATTICE
REV 13	6/24/14	JAB	GROUND LATTICE
REV 14	6/24/14	JAB	GROUND LATTICE
REV 15	6/24/14	JAB	GROUND LATTICE
REV 16	6/24/14	JAB	GROUND LATTICE
REV 17	6/24/14	JAB	GROUND LATTICE
REV 18	6/24/14	JAB	GROUND LATTICE
REV 19	6/24/14	JAB	GROUND LATTICE
REV 20	6/24/14	JAB	GROUND LATTICE

OASIS PETROLEUM NORTH AMERICA, LLC	WELL LOCATION PLAT
SECTION 18, T153N, R100W	
MCKENZIE COUNTY, NORTH DAKOTA	
Drawn By: JBLK	Project No.: 14-08-177
Checked By: JBLK	Date: 05-28-2014

Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Grand Forks, ND 58201
 PH: (701) 433-9917
 FX: (701) 433-9918
 www.interstateeng.com
 Our offices in Minnesota, North Dakota and South Dakota



1/8
 SHEET NO.

DRILLING PLAN										
OPERATOR	Oasis Petroleum				COUNTY/STATE	McKenzie Co. ND				
WELL NAME	Kline Federal 5300 11-18 2B				RIG	0				
WELL TYPE	Horizontal Middle Bakken									
LOCATION	NWNW 18-153N-100W				Surface Location (survey plat):	960' Int	318' Int			
EST. T.D.	20,462'				GROUND ELEV:	2053 Finished Pad Elev.		Sub Height: 25		
TOTAL LATERAL	9,457'				KB ELEV:	2078				
PROGNOSIS:	Based on 2,078' KB(est)				LOGS:	Type Interval				
MARKER	DEPTH (Surf Loc)	DATUM (Surf Loc)		OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota						
Pierre	NDIC MAP	1,988	110	CBL/GR: Above top of cement/GR to base of casing						
Greenhorn		5,023	(2,945)	MWD GR: KOP to lateral TD						
Mowry		5,103	(3,025)	DEVIATION:						
Dakota		5,469	(3,391)	Surf: 3 deg. max., 1 deg / 100'; any every 500'						
Rierdon		6,450	(4,372)	Prod: 5 deg. max., 1 deg / 100'; any every 100'						
Dunham Salt		6,785	(4,707)	DST'S:						
Dunham Salt Base		6,896	(4,818)	None planned						
Spearfish		6,993	(4,915)	CORES:						
Pine Salt		7,248	(5,170)	None planned						
Pine Salt Base		7,296	(5,218)	MUDLOGGING:						
Opeche Salt		7,341	(5,263)	Two-Man: 8,317' ~200' above the Charles (Kibbey) to Casing point; Casing point to TD						
Opeche Salt Base		7,371	(5,293)	30' samples at direction of wellsite geologist; 10' through target @ curve land						
Broom Creek (Top of Minnelusa Gp.)		7,573	(5,495)	BOP:						
Arnsden		7,653	(5,575)	11" 5000 psi blind, pipe & annular						
Tyler		7,821	(5,743)							
Otter (Base of Minnelusa Gp.)		8,012	(5,934)							
Kibbey Lime		8,367	(6,289)							
Charles Salt		8,517	(6,439)							
UB		9,141	(7,063)							
Base Last Salt		9,216	(7,138)							
Ratcliffe		9,264	(7,186)							
Mission Canyon		9,421	(7,343)							
Lodgepole		9,999	(7,921)							
Lodgepole Fracture Zone		10,173	(8,095)							
False Bakken		10,698	(8,620)							
Upper Bakken		10,708	(8,630)							
Middle Bakken		10,722	(8,644)							
Middle Bakken Sand Target		10,731	(8,653)							
Base Middle Bakken Sand Target		10,741	(8,663)							
Lower Bakken		10,766	(8,688)							
Three Forks		10,781	(8,703)							
Dip Rate:	0.3				Surface Formation: Glacial till					
Max. Anticipated BHP:	4665									
MUD:	Interval	Type	WT	Vis	WL	Remarks				
Surface:	0' -	2,068' FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks				
Intermediate:	2,068' -	11,005' Invert	9.5-10.4	40-50	30-HHlp	Circ Mud Tanks				
Lateral:	11,005' -	20,462' Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks				
CASING:	Size	Wt wpl	Hole	Depth	Cement	WOC	Remarks			
Surface:	13-3/8"	54.5#	17-1/2"	2,068'	To Surface	12	100' into Pierre			
Dakota Contingency:	9-5/8"	36#	12-1/4"	6,450'	To Surface	12	Below Dakota			
Intermediate:	7"	32#	8-3/4"	11,005'	4969	24	500' above Dakota			
Production Liner:	4.5"	13.5#	6"	20,462'	TOL @ 10,208'					50' above KOP
PROBABLE PLUGS, IF REQ'D:										
OTHER:	MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI				
Surface:	2,068	2,068	960 FNL	318 FWL	SEC 18-T153N-R100W					
KOP:	10,258'	10,258'	960 FNL	368 FWL	SEC 18-T153N-R100W		Build Rate: 12 deg /100'			
EOC:	11,005'	10,735'	1039 FNL	836 FWL	SEC 18-T153N-R100W	99.60				
Casing Point:	11,005'	10,735'	1039 FNL	836 FWL	SEC 18-T153N-R100W	99.60				
Middle Bakken Lateral TD:	20,462'	10,784'	1080 FNL	250 FEL	SEC 17-T153N-R100W	90.00				
Comments:										
Request a Sundry for an Open Hole Log Waiver										
Exception well: Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW)										
Completion Notes: 35 packers, 35 sleeves, no frac string										
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.										
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2)										
68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)										
										
Geology: M. Steed 5/12/2014					Engineering: Agonzalez 1-27-2015					

**Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 2B
Section 18 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
13-3/8"	0' - 2068'	54.5	J-55	STC	12.615"	12.459"	4100	5470	6840

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 2068'	13-3/8", 54.5#, J-55, LTC, 8rd	1130 / 1.16	2730 / 1.95	514 / 2.60

API Rating & Safety Factor

- a) Collapse based on full casing evacuation with 9 ppg fluid on backside (2068' setting depth).
- b) Burst pressure based on 13 ppg fluid with no fluid on backside (2068' setting depth).
- c) Based on string weight in 9 ppg fluid at 2068' TVD plus 100k# overpull. (Buoyed weight equals 97k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 60% excess to circulate cement back to surface.

Mix and pump the following slurry.

Pre-flush (Spacer): 20 bbls fresh water

Lead Slurry: 694 sks (358 bbls), 11.5 lb/gal, 2.97 cu. Ft./sk Varicem Cement with 0.125 il/sk Lost Circulation Additive

Tail Slurry: 300 sks (62 bbls), 13.0 lb/gal, 2.01 cu.ft./sk Varicem with .125 lb/sk Lost Circulation Agent

**Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 2B
Section 18 T153N R100W
McKenzie County, ND**

Contingency INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' - 6450'	36	HCL-80	LTC	8.835"	8.75"	5450	7270	9090

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 6450'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.17	3520 / 1.28	453 / 1.53

API Rating & Safety Factor

- a) Collapse based on full casing evacuation with 10.4 ppg fluid on backside (6450' setting depth).
- b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 15.2#/ft fracture gradient. Backup of 9 ppg fluid..
- c) Tension based on string weight in 10.4 ppg fluid at 6450' TVD plus 100k# overpull. (Buoyed weight equals 195k lbs.)

Cement volumes are based on 9-5/8" casing set in 12-1/4 " hole with 10% excess to circulate cement back to surface.

Pre-flush (Spacer): 20 bbls Chem wash

Lead Slurry: 570 sks (295 bbls), 2.90 ft³/sk, 11.5 lb/gal Conventional system with 94 lb/sk cement, 4% D079 extender, 2% D053 expanding agent, 2% CaCl₂ and 0.250 lb/sk D130 lost circulation control agent.

Tail Slurry: 605 sks (125 bbls), 1.16 ft³/sk 15.8 lb/gal Conventional system with 94 lb/sk cement, 0.25% CaCl₂, and 0.250 lb/sk lost circulation control agent

**Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 2B
Section 18 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift**	Minimum	Optimum	Max
7"	0' - 11005'	32	HCP-110	LTC	6.094"	6.000"	6730	8970	9870

**Special Drift 7" 32# to 6.0"

Interval	Length	Description	Collapse (psi) a	Burst (psi) b	Tension (1000 lbs) / c
0' - 11005'	11005'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.11*	12480 / 1.28	897 / 2.25
6785' - 9216'	2431'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.06**	12480 / 1.30	

API Rating & Safety Factor

- a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10735' TVD.
- c) Based on string weight in 10 ppg fluid, (298k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Mix and pump the following slurry

Pre-flush (Spacer): **100 bbls** Saltwater
 20 bbls Tuned Spacer III

Lead Slurry: **175 sks** (81 bbls), 11.8 ppg, 2.55 cu. ft./sk Econocem Cement with .3% Fe-2 and .25 lb/sk Lost Circulation Additive

Tail Slurry: **577 sks** (169 bbls), 14.0 ppg, 1.55 cu. ft./sk Extendcem System with .2% HR-5 Retarder and .25 lb/sk Lost Circulation Additive

**Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 2B
Section 18 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
4-1/2"	10208' - 20462'	13.5	P-110	BTC	3.920"	3.795"	2270	3020	3780

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
10208' - 20462'	10254	4-1/2", 13.5 lb, P-110, BTC	10670 / 2.00	12410 / 1.28	443 / 2.02

API Rating & Safety Factor

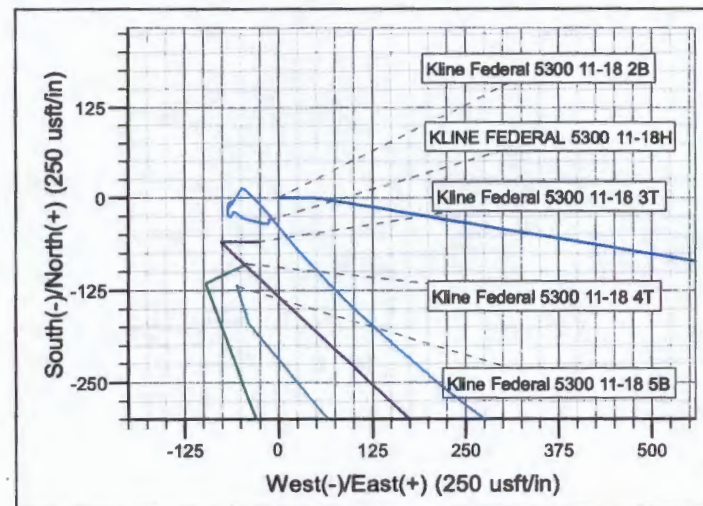
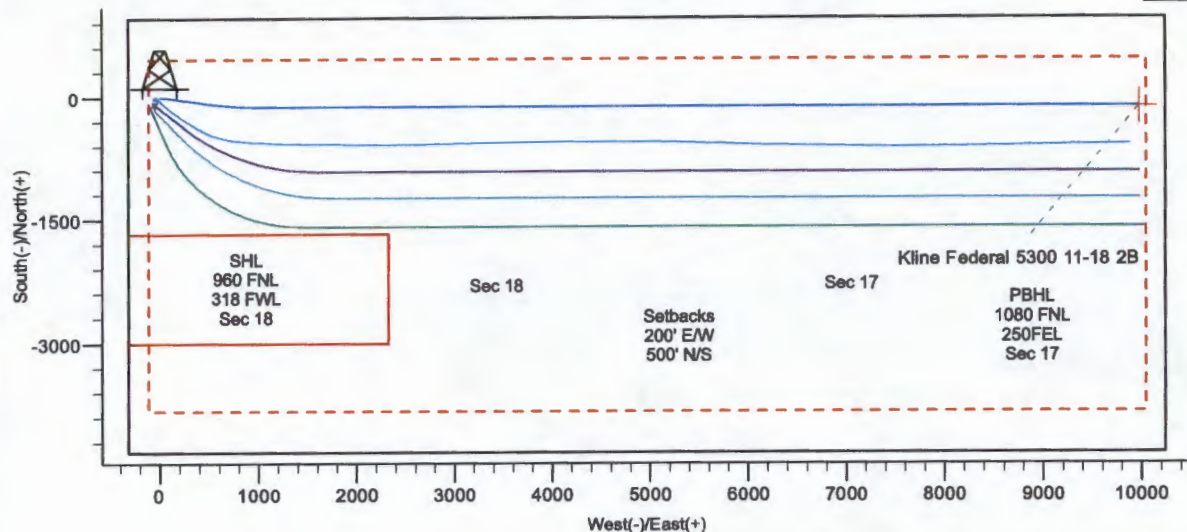
- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10784' TVD.
- b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10784' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 118k lbs.) plus 100k lbs overpull.

Project: Indian Hills
 Site: 153N-100W-17/18
 Well: Kline Federal 5300 11-18 2B
 Wellbore: Kline Federal 5300 11-18 2B
 Design: Design #5



WELL DETAILS: Kline Federal 5300 11-18 2B

Ground Level: 2053.0
 Northing 408992.29 Easting 1210243.30 Latitude 48° 4' 45.680 N Longitude 103° 36' 10.190 W



SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2500.0	0.00	0.00	2500.0	0.0	0.0	0.00	
2516.7	0.50	90.00	2516.7	0.0	0.1	3.00	
8226.7	0.50	90.00	8226.4	0.0	49.9	0.00	
8243.3	0.00	0.00	8243.1	0.0	50.0	3.00	
10257.7	0.00	0.00	10257.5	0.0	50.0	0.00	
11005.2	89.70	99.60	10735.0	-79.2	518.3	12.00	
11485.0	89.70	90.00	10737.5	-119.3	995.8	2.00	
20462.3	89.70	90.00	10784.0	-120.0	9973.0	0.00	Kline Federal 5300 11-18 2B

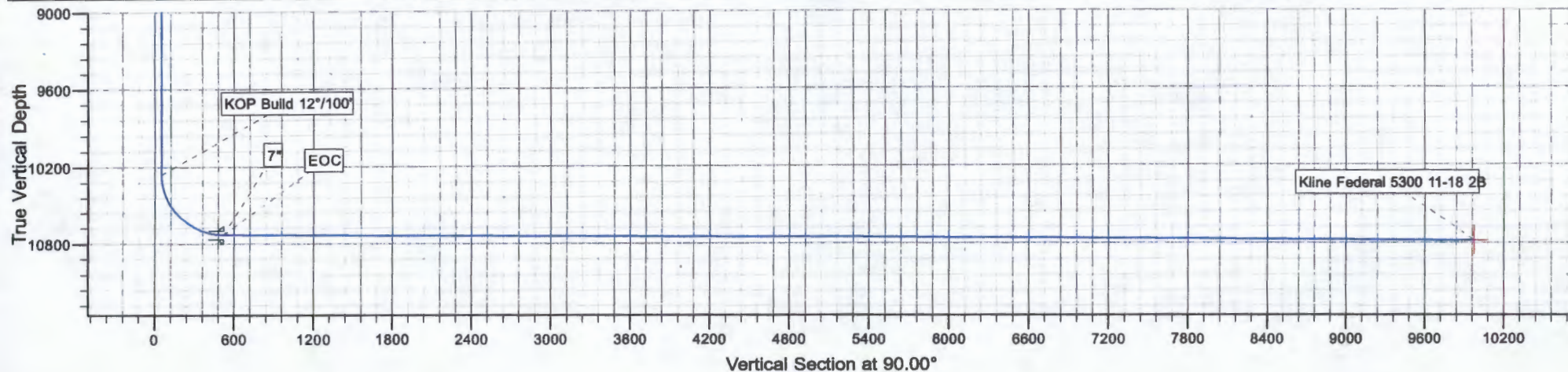
CASING DETAILS

TVD	MD	Name	Size
2068.0	2068.0	13 3/8"	13.375
6449.9	6450.0	9 5/8"	9.625
10735.0	11006.0	7"	7.000



Azimuths to True North
 Magnetic North: 8.41°

Magnetic Field
 Strength: 56376.9snT
 Dip Angle: 72.99°
 Date: 4/22/2014
 Model: IGRF2015





Oasis

Indian Hills

153N-100W-17/18

Kline Federal 5300 11-18 2B

T153N R100W SEC. 18

Plan: Design #5

Standard Planning Report

27 January, 2015

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2B
Company:	Oasis	TVD Reference:	Ref KB @ 2078.0usft
Project:	Indian Hills	MD Reference:	Ref KB @ 2078.0usft
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 2B		
Design:	Design #5		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-17/18		
Site Position:		Northing:	408,962.44 usft
From:	Lat/Long	Easting:	1,210,229.18 usft
Position Uncertainty:	0.0 usft	Slot Radius:	13.200 in
		Latitude:	48° 4' 45.380 N
		Longitude:	103° 36' 10.380 W
		Grid Convergence:	-2.31 °

Well	Kline Federal 5300 11-18 2B		
Well Position	+N/-S	30.4 usft	Northing: 408,992.29 usft
	+E/-W	12.9 usft	Easting: 1,210,243.30 usft
Position Uncertainty	2.0 usft	Wellhead Elevation:	Latitude: 48° 4' 45.680 N
			Longitude: 103° 36' 10.190 W
			Ground Level: 2,053.0 usft

Wellbore	Kline Federal 5300 11-18 2B		
Magnetics	Model Name	Sample Date	Declination
			(°)
	IGRF2015	4/22/2014	8.41
			Dip Angle
			(°)
			72.99
			Field Strength
			(nT)
			56,377

Design	Design #5		
Audit Notes:			
Version:	Phase:	PROTOTYPE	Tie On Depth: 0.0
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(usft)	(usft)	(usft)
	0.0	0.0	0.0
			Direction
			(°)
			90.00

Plan Sections										
Measured	Inclination	Azimuth	Vertical	+N/-S	+E/-W	Dogleg	Build	Turn	TFO	Target
Depth	(°)	(°)	Depth	(usft)	(usft)	Rate	Rate	Rate	(°)	
(usft)			(usft)			(°/100ft)	(°/100ft)	(°/100ft)		
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,500.0	0.00	0.00	2,500.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,516.7	0.50	90.00	2,516.7	0.0	0.1	3.00	3.00	0.00	90.00	
8,226.7	0.50	90.00	8,226.4	0.0	49.9	0.00	0.00	0.00	0.00	
8,243.3	0.00	0.00	8,243.1	0.0	50.0	3.00	-3.00	0.00	180.00	
10,257.7	0.00	0.00	10,257.5	0.0	50.0	0.00	0.00	0.00	0.00	
11,005.2	89.70	99.60	10,735.0	-79.2	518.3	12.00	12.00	0.00	99.60	
11,485.0	89.70	90.00	10,737.5	-119.3	995.8	2.00	0.00	-2.00	-90.01	
20,462.3	89.70	90.00	10,784.0	-120.0	9,973.0	0.00	0.00	0.00	0.00	Kline Federal 5300 11

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2B
Company:	Oasis	TVD Reference:	Ref KB @ 2078.0usft
Project:	Indian Hills	MD Reference:	Ref KB @ 2078.0usft
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 2B		
Design:	Design #5		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
2,500.0	0.00	0.00	2,500.0	0.0	0.0	0.0	0.00	0.00	0.00
2,516.7	0.50	90.00	2,516.7	0.0	0.1	0.1	3.00	3.00	0.00
2,600.0	0.50	90.00	2,600.0	0.0	0.8	0.8	0.00	0.00	0.00
2,700.0	0.50	90.00	2,700.0	0.0	1.7	1.7	0.00	0.00	0.00
2,800.0	0.50	90.00	2,800.0	0.0	2.5	2.5	0.00	0.00	0.00
2,900.0	0.50	90.00	2,900.0	0.0	3.4	3.4	0.00	0.00	0.00
3,000.0	0.50	90.00	3,000.0	0.0	4.3	4.3	0.00	0.00	0.00
3,100.0	0.50	90.00	3,100.0	0.0	5.2	5.2	0.00	0.00	0.00
3,200.0	0.50	90.00	3,200.0	0.0	6.0	6.0	0.00	0.00	0.00
3,300.0	0.50	90.00	3,300.0	0.0	6.9	6.9	0.00	0.00	0.00
3,400.0	0.50	90.00	3,400.0	0.0	7.8	7.8	0.00	0.00	0.00
3,500.0	0.50	90.00	3,500.0	0.0	8.7	8.7	0.00	0.00	0.00
3,600.0	0.50	90.00	3,600.0	0.0	9.5	9.5	0.00	0.00	0.00
3,700.0	0.50	90.00	3,700.0	0.0	10.4	10.4	0.00	0.00	0.00
3,800.0	0.50	90.00	3,800.0	0.0	11.3	11.3	0.00	0.00	0.00
3,900.0	0.50	90.00	3,899.9	0.0	12.1	12.1	0.00	0.00	0.00
4,000.0	0.50	90.00	3,999.9	0.0	13.0	13.0	0.00	0.00	0.00
4,100.0	0.50	90.00	4,099.9	0.0	13.9	13.9	0.00	0.00	0.00
4,200.0	0.50	90.00	4,199.9	0.0	14.8	14.8	0.00	0.00	0.00
4,300.0	0.50	90.00	4,299.9	0.0	15.6	15.6	0.00	0.00	0.00
4,400.0	0.50	90.00	4,399.9	0.0	16.5	16.5	0.00	0.00	0.00
4,500.0	0.50	90.00	4,499.9	0.0	17.4	17.4	0.00	0.00	0.00
4,600.0	0.50	90.00	4,599.9	0.0	18.3	18.3	0.00	0.00	0.00
4,700.0	0.50	90.00	4,699.9	0.0	19.1	19.1	0.00	0.00	0.00
4,800.0	0.50	90.00	4,799.9	0.0	20.0	20.0	0.00	0.00	0.00
4,900.0	0.50	90.00	4,899.9	0.0	20.9	20.9	0.00	0.00	0.00
5,000.0	0.50	90.00	4,999.9	0.0	21.7	21.7	0.00	0.00	0.00
5,100.0	0.50	90.00	5,099.9	0.0	22.6	22.6	0.00	0.00	0.00
5,200.0	0.50	90.00	5,199.9	0.0	23.5	23.5	0.00	0.00	0.00
5,300.0	0.50	90.00	5,299.9	0.0	24.4	24.4	0.00	0.00	0.00
5,400.0	0.50	90.00	5,399.9	0.0	25.2	25.2	0.00	0.00	0.00
5,500.0	0.50	90.00	5,499.9	0.0	26.1	26.1	0.00	0.00	0.00
5,600.0	0.50	90.00	5,599.9	0.0	27.0	27.0	0.00	0.00	0.00
5,700.0	0.50	90.00	5,699.9	0.0	27.9	27.9	0.00	0.00	0.00
5,800.0	0.50	90.00	5,799.9	0.0	28.7	28.7	0.00	0.00	0.00
5,900.0	0.50	90.00	5,899.9	0.0	29.6	29.6	0.00	0.00	0.00
6,000.0	0.50	90.00	5,999.9	0.0	30.5	30.5	0.00	0.00	0.00
6,100.0	0.50	90.00	6,099.9	0.0	31.3	31.3	0.00	0.00	0.00
6,200.0	0.50	90.00	6,199.9	0.0	32.2	32.2	0.00	0.00	0.00
6,300.0	0.50	90.00	6,299.9	0.0	33.1	33.1	0.00	0.00	0.00
6,400.0	0.50	90.00	6,399.9	0.0	34.0	34.0	0.00	0.00	0.00
6,450.0	0.50	90.00	6,449.8	0.0	34.4	34.4	0.00	0.00	0.00
9 5/8"									
6,500.0	0.50	90.00	6,499.8	0.0	34.8	34.8	0.00	0.00	0.00
6,600.0	0.50	90.00	6,599.8	0.0	35.7	35.7	0.00	0.00	0.00
6,700.0	0.50	90.00	6,699.8	0.0	36.6	36.6	0.00	0.00	0.00
6,800.0	0.50	90.00	6,799.8	0.0	37.5	37.5	0.00	0.00	0.00
6,900.0	0.50	90.00	6,899.8	0.0	38.3	38.3	0.00	0.00	0.00
7,000.0	0.50	90.00	6,999.8	0.0	39.2	39.2	0.00	0.00	0.00
7,100.0	0.50	90.00	7,099.8	0.0	40.1	40.1	0.00	0.00	0.00
7,200.0	0.50	90.00	7,199.8	0.0	40.9	40.9	0.00	0.00	0.00
7,300.0	0.50	90.00	7,299.8	0.0	41.8	41.8	0.00	0.00	0.00
7,400.0	0.50	90.00	7,399.8	0.0	42.7	42.7	0.00	0.00	0.00
7,500.0	0.50	90.00	7,499.8	0.0	43.8	43.8	0.00	0.00	0.00
7,600.0	0.50	90.00	7,599.8	0.0	44.4	44.4	0.00	0.00	0.00

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2B
Company:	Oasis	TVD Reference:	Ref KB @ 2078.0usft
Project:	Indian Hills	MD Reference:	Ref KB @ 2078.0usft
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 2B		
Design:	Design #5		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
7,700.0	0.50	90.00	7,699.8	0.0	45.3	45.3	0.00	0.00	0.00
7,800.0	0.50	90.00	7,799.8	0.0	46.2	46.2	0.00	0.00	0.00
7,900.0	0.50	90.00	7,899.8	0.0	47.1	47.1	0.00	0.00	0.00
8,000.0	0.50	90.00	7,999.8	0.0	47.9	47.9	0.00	0.00	0.00
8,100.0	0.50	90.00	8,099.8	0.0	48.8	48.8	0.00	0.00	0.00
8,200.0	0.50	90.00	8,199.8	0.0	49.7	49.7	0.00	0.00	0.00
8,226.7	0.50	90.00	8,226.4	0.0	49.9	49.9	0.00	0.00	0.00
8,243.3	0.00	0.00	8,243.1	0.0	50.0	50.0	3.00	-3.00	0.00
8,300.0	0.00	0.00	8,299.8	0.0	50.0	50.0	0.00	0.00	0.00
8,400.0	0.00	0.00	8,399.8	0.0	50.0	50.0	0.00	0.00	0.00
8,500.0	0.00	0.00	8,499.8	0.0	50.0	50.0	0.00	0.00	0.00
8,600.0	0.00	0.00	8,599.8	0.0	50.0	50.0	0.00	0.00	0.00
8,700.0	0.00	0.00	8,699.8	0.0	50.0	50.0	0.00	0.00	0.00
8,800.0	0.00	0.00	8,799.8	0.0	50.0	50.0	0.00	0.00	0.00
8,900.0	0.00	0.00	8,899.8	0.0	50.0	50.0	0.00	0.00	0.00
9,000.0	0.00	0.00	8,999.8	0.0	50.0	50.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	0.0	50.0	50.0	0.00	0.00	0.00
9,200.0	0.00	0.00	9,199.8	0.0	50.0	50.0	0.00	0.00	0.00
9,300.0	0.00	0.00	9,299.8	0.0	50.0	50.0	0.00	0.00	0.00
9,400.0	0.00	0.00	9,399.8	0.0	50.0	50.0	0.00	0.00	0.00
9,500.0	0.00	0.00	9,499.8	0.0	50.0	50.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	0.0	50.0	50.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	0.0	50.0	50.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	0.0	50.0	50.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.8	0.0	50.0	50.0	0.00	0.00	0.00
10,000.0	0.00	0.00	9,999.8	0.0	50.0	50.0	0.00	0.00	0.00
10,100.0	0.00	0.00	10,099.8	0.0	50.0	50.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.8	0.0	50.0	50.0	0.00	0.00	0.00
10,257.7	0.00	0.00	10,257.5	0.0	50.0	50.0	0.00	0.00	0.00
KOP Build 12°/100'									
10,275.0	2.07	99.60	10,274.8	-0.1	50.3	50.3	11.98	11.98	0.00
10,300.0	5.07	99.60	10,299.7	-0.3	51.8	51.8	12.00	12.00	0.00
10,325.0	8.07	99.60	10,324.8	-0.8	54.6	54.6	12.00	12.00	0.00
10,350.0	11.07	99.60	10,349.2	-1.5	58.7	58.7	12.00	12.00	0.00
10,375.0	14.07	99.60	10,373.6	-2.4	64.1	64.1	12.00	12.00	0.00
10,400.0	17.07	99.60	10,397.7	-3.5	70.7	70.7	12.00	12.00	0.00
10,425.0	20.07	99.60	10,421.4	-4.8	78.6	78.6	12.00	12.00	0.00
10,450.0	23.07	99.60	10,444.6	-6.4	87.6	87.6	12.00	12.00	0.00
10,475.0	26.07	99.60	10,467.4	-8.1	97.9	97.9	12.00	12.00	0.00
10,500.0	29.07	99.60	10,489.5	-10.0	109.3	109.3	12.00	12.00	0.00
10,525.0	32.07	99.60	10,511.0	-12.2	121.8	121.8	12.00	12.00	0.00
10,550.0	35.07	99.60	10,531.9	-14.5	135.5	135.5	12.00	12.00	0.00
10,575.0	38.07	99.60	10,551.9	-16.9	150.1	150.1	12.00	12.00	0.00
10,600.0	41.07	99.60	10,571.2	-19.6	165.8	165.8	12.00	12.00	0.00
10,625.0	44.07	99.60	10,589.6	-22.4	182.5	182.5	12.00	12.00	0.00
10,650.0	47.07	99.60	10,607.1	-25.4	200.1	200.1	12.00	12.00	0.00
10,675.0	50.07	99.60	10,623.7	-28.5	218.6	218.6	12.00	12.00	0.00
10,700.0	53.07	99.60	10,639.2	-31.8	237.9	237.9	12.00	12.00	0.00
10,725.0	56.07	99.60	10,653.7	-35.2	258.0	258.0	12.00	12.00	0.00
10,750.0	59.07	99.60	10,667.1	-38.7	278.8	278.8	12.00	12.00	0.00
10,775.0	62.07	99.60	10,679.4	-42.3	300.3	300.3	12.00	12.00	0.00
10,800.0	65.07	99.60	10,690.5	-46.1	322.3	322.3	12.00	12.00	0.00
10,825.0	68.07	99.60	10,700.4	-49.9	344.9	344.9	12.00	12.00	0.00
10,850.0	71.07	99.60	10,709.2	-53.8	368.0	368.0	12.00	12.00	0.00
10,875.0	74.07	99.60	10,716.6	-57.8	391.6	391.6	12.00	12.00	0.00

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2B
Company:	Oasis	TVD Reference:	Ref KB @ 2078.0usft
Project:	Indian Hills	MD Reference:	Ref KB @ 2078.0usft
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 2B		
Design:	Design #5		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N-S (usft)	+E-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,900.0	77.07	99.60	10,722.9	-61.8	415.4	415.4	12.00	12.00	0.00
10,925.0	80.07	99.60	10,727.8	-65.9	439.6	439.6	12.00	12.00	0.00
10,950.0	83.07	99.60	10,731.5	-70.0	464.0	464.0	12.00	12.00	0.00
10,975.0	86.07	99.60	10,733.9	-74.2	488.5	488.5	12.00	12.00	0.00
11,000.0	89.07	99.60	10,734.9	-78.3	513.1	513.1	12.00	12.00	0.00
11,005.2	89.70	99.60	10,735.0	-79.2	518.3	518.3	12.00	12.00	0.00
EOC									
11,006.0	89.70	99.58	10,735.0	-79.3	519.0	519.0	1.97	0.40	-1.93
T									
11,100.0	89.70	97.70	10,735.5	-93.5	612.0	812.0	2.00	0.00	-2.00
11,200.0	89.70	95.70	10,736.0	-105.1	711.3	711.3	2.00	0.00	-2.00
11,300.0	89.70	93.70	10,736.5	-113.3	810.9	810.9	2.00	0.00	-2.00
11,400.0	89.70	91.70	10,737.0	-118.1	910.8	910.8	2.00	0.00	-2.00
11,485.0	89.70	90.00	10,737.5	-119.3	995.8	995.8	2.00	0.00	-2.00
11,500.0	89.70	90.00	10,737.6	-119.3	1,010.8	1,010.8	0.00	0.00	0.00
11,600.0	89.70	90.00	10,738.1	-119.3	1,110.8	1,110.8	0.00	0.00	0.00
11,700.0	89.70	90.00	10,738.6	-119.3	1,210.8	1,210.8	0.00	0.00	0.00
11,800.0	89.70	90.00	10,739.1	-119.3	1,310.8	1,310.8	0.00	0.00	0.00
11,900.0	89.70	90.00	10,739.6	-119.4	1,410.8	1,410.8	0.00	0.00	0.00
12,000.0	89.70	90.00	10,740.1	-119.4	1,510.8	1,510.8	0.00	0.00	0.00
12,100.0	89.70	90.00	10,740.7	-119.4	1,610.8	1,610.8	0.00	0.00	0.00
12,200.0	89.70	90.00	10,741.2	-119.4	1,710.8	1,710.8	0.00	0.00	0.00
12,300.0	89.70	90.00	10,741.7	-119.4	1,810.8	1,810.8	0.00	0.00	0.00
12,400.0	89.70	90.00	10,742.2	-119.4	1,910.8	1,910.8	0.00	0.00	0.00
12,500.0	89.70	90.00	10,742.7	-119.4	2,010.8	2,010.8	0.00	0.00	0.00
12,600.0	89.70	90.00	10,743.3	-119.4	2,110.8	2,110.8	0.00	0.00	0.00
12,700.0	89.70	90.00	10,743.8	-119.4	2,210.8	2,210.8	0.00	0.00	0.00
12,800.0	89.70	90.00	10,744.3	-119.4	2,310.8	2,310.8	0.00	0.00	0.00
12,900.0	89.70	90.00	10,744.8	-119.4	2,410.8	2,410.8	0.00	0.00	0.00
13,000.0	89.70	90.00	10,745.3	-119.4	2,510.8	2,510.8	0.00	0.00	0.00
13,100.0	89.70	90.00	10,745.8	-119.4	2,610.8	2,610.8	0.00	0.00	0.00
13,200.0	89.70	90.00	10,746.4	-119.5	2,710.8	2,710.8	0.00	0.00	0.00
13,300.0	89.70	90.00	10,746.9	-119.5	2,810.8	2,810.8	0.00	0.00	0.00
13,400.0	89.70	90.00	10,747.4	-119.5	2,910.8	2,910.8	0.00	0.00	0.00
13,500.0	89.70	90.00	10,747.9	-119.5	3,010.8	3,010.8	0.00	0.00	0.00
13,600.0	89.70	90.00	10,748.4	-119.5	3,110.8	3,110.8	0.00	0.00	0.00
13,700.0	89.70	90.00	10,749.0	-119.5	3,210.8	3,210.8	0.00	0.00	0.00
13,800.0	89.70	90.00	10,749.5	-119.5	3,310.8	3,310.8	0.00	0.00	0.00
13,900.0	89.70	90.00	10,750.0	-119.5	3,410.8	3,410.8	0.00	0.00	0.00
14,000.0	89.70	90.00	10,750.5	-119.5	3,510.8	3,510.8	0.00	0.00	0.00
14,100.0	89.70	90.00	10,751.0	-119.5	3,610.8	3,610.8	0.00	0.00	0.00
14,200.0	89.70	90.00	10,751.5	-119.5	3,710.8	3,710.8	0.00	0.00	0.00
14,300.0	89.70	90.00	10,752.1	-119.5	3,810.8	3,810.8	0.00	0.00	0.00
14,400.0	89.70	90.00	10,752.6	-119.5	3,910.8	3,910.8	0.00	0.00	0.00
14,500.0	89.70	90.00	10,753.1	-119.6	4,010.8	4,010.8	0.00	0.00	0.00
14,600.0	89.70	90.00	10,753.6	-119.6	4,110.8	4,110.8	0.00	0.00	0.00
14,700.0	89.70	90.00	10,754.1	-119.6	4,210.8	4,210.8	0.00	0.00	0.00
14,800.0	89.70	90.00	10,754.7	-119.6	4,310.8	4,310.8	0.00	0.00	0.00
14,900.0	89.70	90.00	10,755.2	-119.6	4,410.8	4,410.8	0.00	0.00	0.00
15,000.0	89.70	90.00	10,755.7	-119.6	4,510.8	4,510.8	0.00	0.00	0.00
15,100.0	89.70	90.00	10,756.2	-119.6	4,610.8	4,610.8	0.00	0.00	0.00
15,200.0	89.70	90.00	10,756.7	-119.6	4,710.8	4,710.8	0.00	0.00	0.00
15,300.0	89.70	90.00	10,757.2	-119.6	4,810.8	4,810.8	0.00	0.00	0.00
15,400.0	89.70	90.00	10,757.8	-119.6	4,910.8	4,910.8	0.00	0.00	0.00

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2B
Company:	Oasis	TVD Reference:	Ref KB @ 2078.0usft
Project:	Indian Hills	MD Reference:	Ref KB @ 2078.0usft
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 2B		
Design:	Design #5		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
15,500.0	89.70	90.00	10,758.3	-119.6	5,010.8	5,010.8	0.00	0.00	0.00
15,600.0	89.70	90.00	10,758.8	-119.6	5,110.8	5,110.8	0.00	0.00	0.00
15,700.0	89.70	90.00	10,759.3	-119.6	5,210.8	5,210.8	0.00	0.00	0.00
15,800.0	89.70	90.00	10,759.8	-119.6	5,310.8	5,310.8	0.00	0.00	0.00
15,900.0	89.70	90.00	10,760.4	-119.7	5,410.8	5,410.8	0.00	0.00	0.00
16,000.0	89.70	90.00	10,760.9	-119.7	5,510.7	5,510.7	0.00	0.00	0.00
16,100.0	89.70	90.00	10,761.4	-119.7	5,610.7	5,610.7	0.00	0.00	0.00
16,200.0	89.70	90.00	10,761.9	-119.7	5,710.7	5,710.7	0.00	0.00	0.00
16,300.0	89.70	90.00	10,762.4	-119.7	5,810.7	5,810.7	0.00	0.00	0.00
16,400.0	89.70	90.00	10,762.9	-119.7	5,910.7	5,910.7	0.00	0.00	0.00
16,500.0	89.70	90.00	10,763.5	-119.7	6,010.7	6,010.7	0.00	0.00	0.00
16,600.0	89.70	90.00	10,764.0	-119.7	6,110.7	6,110.7	0.00	0.00	0.00
16,700.0	89.70	90.00	10,764.5	-119.7	6,210.7	6,210.7	0.00	0.00	0.00
16,800.0	89.70	90.00	10,765.0	-119.7	6,310.7	6,310.7	0.00	0.00	0.00
16,900.0	89.70	90.00	10,765.5	-119.7	6,410.7	6,410.7	0.00	0.00	0.00
17,000.0	89.70	90.00	10,766.1	-119.7	6,510.7	6,510.7	0.00	0.00	0.00
17,100.0	89.70	90.00	10,766.6	-119.7	6,610.7	6,610.7	0.00	0.00	0.00
17,200.0	89.70	90.00	10,767.1	-119.8	6,710.7	6,710.7	0.00	0.00	0.00
17,300.0	89.70	90.00	10,767.6	-119.8	6,810.7	6,810.7	0.00	0.00	0.00
17,400.0	89.70	90.00	10,768.1	-119.8	6,910.7	6,910.7	0.00	0.00	0.00
17,500.0	89.70	90.00	10,768.6	-119.8	7,010.7	7,010.7	0.00	0.00	0.00
17,600.0	89.70	90.00	10,769.2	-119.8	7,110.7	7,110.7	0.00	0.00	0.00
17,700.0	89.70	90.00	10,769.7	-119.8	7,210.7	7,210.7	0.00	0.00	0.00
17,800.0	89.70	90.00	10,770.2	-119.8	7,310.7	7,310.7	0.00	0.00	0.00
17,900.0	89.70	90.00	10,770.7	-119.8	7,410.7	7,410.7	0.00	0.00	0.00
18,000.0	89.70	90.00	10,771.2	-119.8	7,510.7	7,510.7	0.00	0.00	0.00
18,100.0	89.70	90.00	10,771.8	-119.8	7,610.7	7,610.7	0.00	0.00	0.00
18,200.0	89.70	90.00	10,772.3	-119.8	7,710.7	7,710.7	0.00	0.00	0.00
18,300.0	89.70	90.00	10,772.8	-119.8	7,810.7	7,810.7	0.00	0.00	0.00
18,400.0	89.70	90.00	10,773.3	-119.8	7,910.7	7,910.7	0.00	0.00	0.00
18,500.0	89.70	90.00	10,773.8	-119.8	8,010.7	8,010.7	0.00	0.00	0.00
18,600.0	89.70	90.00	10,774.3	-119.9	8,110.7	8,110.7	0.00	0.00	0.00
18,700.0	89.70	90.00	10,774.9	-119.9	8,210.7	8,210.7	0.00	0.00	0.00
18,800.0	89.70	90.00	10,775.4	-119.9	8,310.7	8,310.7	0.00	0.00	0.00
18,900.0	89.70	90.00	10,775.9	-119.9	8,410.7	8,410.7	0.00	0.00	0.00
19,000.0	89.70	90.00	10,776.4	-119.9	8,510.7	8,510.7	0.00	0.00	0.00
19,100.0	89.70	90.00	10,776.9	-119.9	8,610.7	8,610.7	0.00	0.00	0.00
19,200.0	89.70	90.00	10,777.5	-119.9	8,710.7	8,710.7	0.00	0.00	0.00
19,300.0	89.70	90.00	10,778.0	-119.9	8,810.7	8,810.7	0.00	0.00	0.00
19,400.0	89.70	90.00	10,778.5	-119.9	8,910.7	8,910.7	0.00	0.00	0.00
19,500.0	89.70	90.00	10,779.0	-119.9	9,010.7	9,010.7	0.00	0.00	0.00
19,600.0	89.70	90.00	10,779.5	-119.9	9,110.7	9,110.7	0.00	0.00	0.00
19,700.0	89.70	90.00	10,780.0	-119.9	9,210.7	9,210.7	0.00	0.00	0.00
19,800.0	89.70	90.00	10,780.6	-119.9	9,310.7	9,310.7	0.00	0.00	0.00
19,900.0	89.70	90.00	10,781.1	-120.0	9,410.7	9,410.7	0.00	0.00	0.00
20,000.0	89.70	90.00	10,781.6	-120.0	9,510.7	9,510.7	0.00	0.00	0.00
20,100.0	89.70	90.00	10,782.1	-120.0	9,610.7	9,610.7	0.00	0.00	0.00
20,200.0	89.70	90.00	10,782.6	-120.0	9,710.7	9,710.7	0.00	0.00	0.00
20,300.0	89.70	90.00	10,783.2	-120.0	9,810.7	9,810.7	0.00	0.00	0.00
20,400.0	89.70	90.00	10,783.7	-120.0	9,910.7	9,910.7	0.00	0.00	0.00
20,462.3	89.70	90.00	10,784.0	-120.0	9,973.0	9,973.0	0.00	0.00	0.00
Kline Federal 5300 11-18 2B									

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2B
Company:	Oasis	TVD Reference:	Ref KB @ 2078.0usft
Project:	Indian Hills	MD Reference:	Ref KB @ 2078.0usft
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 2B		
Design:	Design #5		

Design Targets

Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
- hit/miss target									
- Shape									
Kline Federal 5300 11-11	0.00	0.00	10,784.0	-120.0	9,973.0	408,470.60	1,220,203.37	48° 4' 44.470 N	103° 33' 43.314 W
- plan hits target center									
- Point									

Casing Points

Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (in)	Hole Diameter (in)
2,068.0	2,068.0	13 3/8"	13.375	17.500
6,450.0	6,449.8	9 5/8"	9.625	12.250
11,006.0	10,735.0	7"	7.000	8.750

Formations

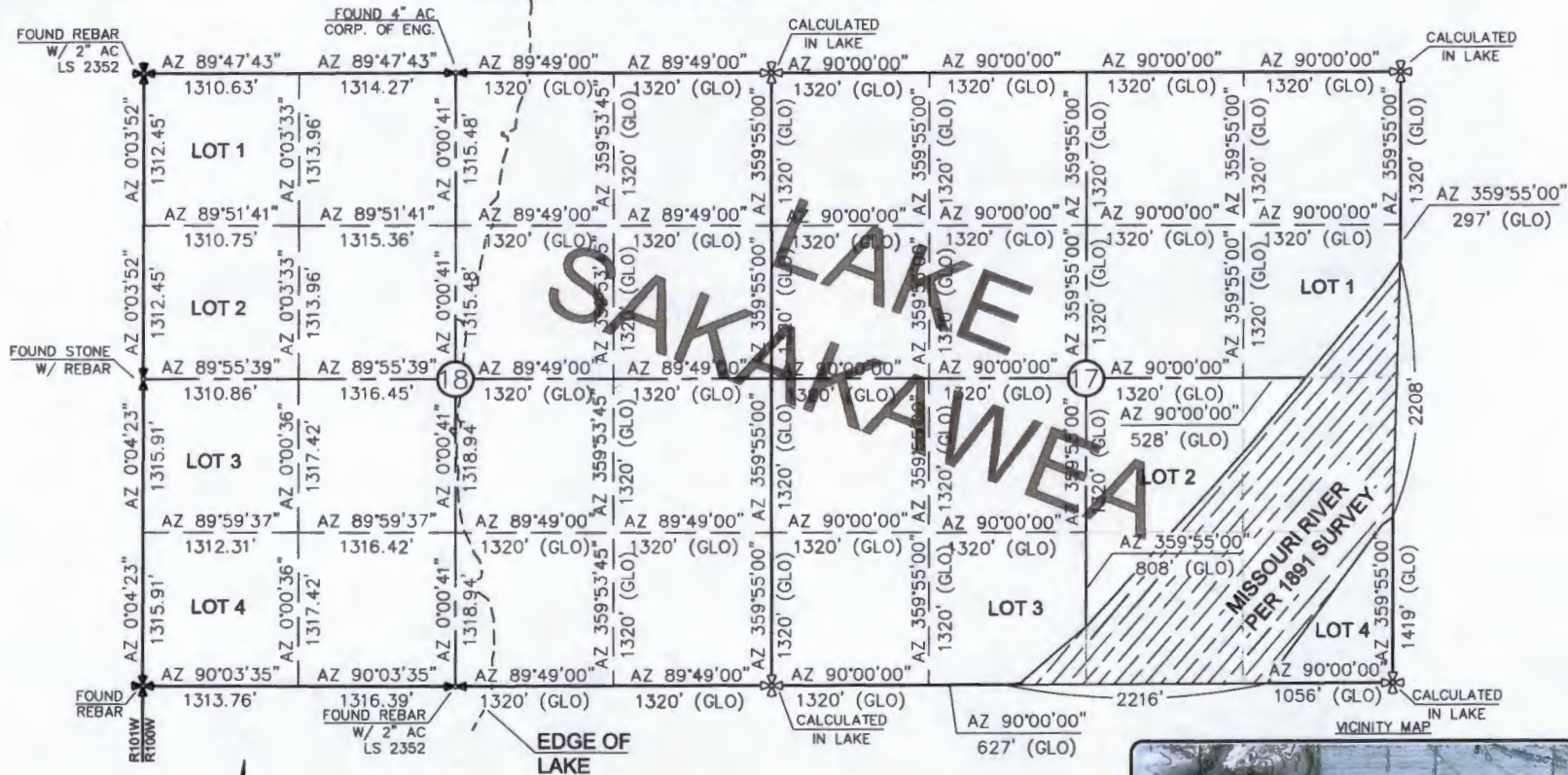
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)
1,968.0	1,968.0	Pierre			
5,023.1	5,023.0	Greenhorn			
5,103.1	5,103.0	Mowry			
5,469.1	5,469.0	Dakota			
6,450.1	6,450.0	Rierdon			
6,785.2	6,785.0	Dunham Salt			
6,896.2	6,896.0	Dunham Salt Base			
6,993.2	6,993.0	Spearfish			
7,248.2	7,248.0	Pine Salt			
7,296.2	7,296.0	Pine Salt Base			
7,341.2	7,341.0	Opeche Salt			
7,371.2	7,371.0	Opeche Salt Base			
7,573.2	7,573.0	Broom Creek (Top of Minnelusa Gp.)			
7,653.2	7,653.0	Amsden			
7,821.2	7,821.0	Tyler			
8,012.2	8,012.0	Otter (Base of Minnelusa Gp.)			
8,367.2	8,367.0	Kibbey Lime			
8,517.2	8,517.0	Charles Salt			
9,141.2	9,141.0	UB			
9,216.2	9,216.0	Base Last Salt			
9,264.2	9,264.0	Ratcliffe			
9,421.2	9,421.0	Mission Canyon			
9,999.2	9,999.0	Lodgepole			
10,173.2	10,173.0	Lodgepole Fracture Zone			
10,818.6	10,698.0	False Bakken			
10,846.5	10,708.0	Upper Bakken			
10,896.2	10,722.0	Middle Bakken			
10,946.1	10,731.0	Middle Bakken Sand Target			
12,165.5	10,741.0	Base Middle Bakken Sand Target			
16,989.2	10,766.0	Lower Bakken			
19,883.5	10,761.0	Three Forks			

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2B
Company:	Oasis	TVD Reference:	Ref KB @ 2078.0usft
Project:	Indian Hills	MD Reference:	Ref KB @ 2078.0usft
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 2B		
Design:	Design #5		

Plan Annotations				
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates		Comment
		+N/-S (usft)	+E/-W (usft)	
10,257.7	10,257.5	0.0	50.0	KOP Build 12°/100'
11,005.2	10,735.0	-79.2	518.3	EOC

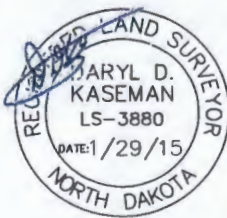
SECTION BREAKDOWN
OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 11-18 2B"
 980 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
 SECTIONS 17 & 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- ⊕ - MONUMENT - RECOVERED
- ⊗ - MONUMENT - NOT RECOVERED

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 1/29/15, AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY 0°03'.



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Revised	Date	By	Description
REV 1	1/29/15	JK	HOOD HILLS
REV 2	1/29/15	JK	REMOVED LATITUDE
REV 3	1/29/15	JK	ADDED DISTANCE RT TO PAD
REV 4	1/29/15	JK	CHANGED NELL NAME & IN
REV 5	1/27/16	JK	CHANGED NELL NAME & IN

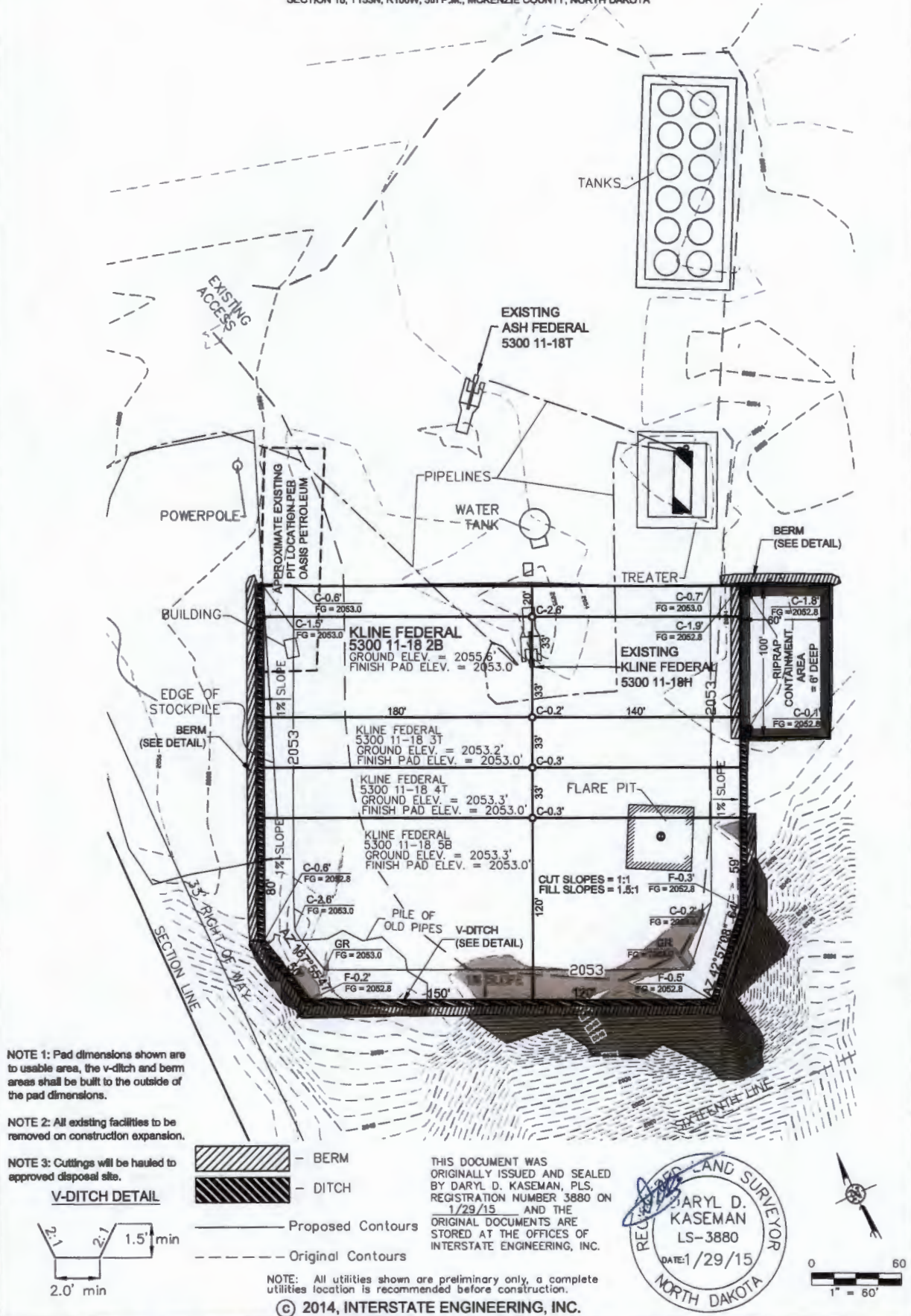
OASIS PETROLEUM NORTH AMERICA, LLC SECTION BREAKDOWN SECTIONS 17 & 18, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA	Drawn By: BAKL Project No.: 15142-177 Checked By: JDSK Date: 1/29/15
--	---

Interstate Engineering, Inc.
 P.O. Box 644
 428 East Main Street
 Sidney, Montana 59270
 Tel: (406) 433-9111
 Fax: (406) 433-9118
 www.interstateeng.com
 Other offices in Minnesota, Iowa, Illinois and South Dakota

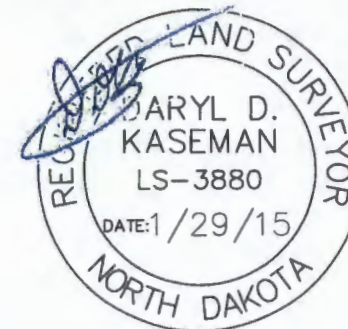


2/8
 SHEET NO.

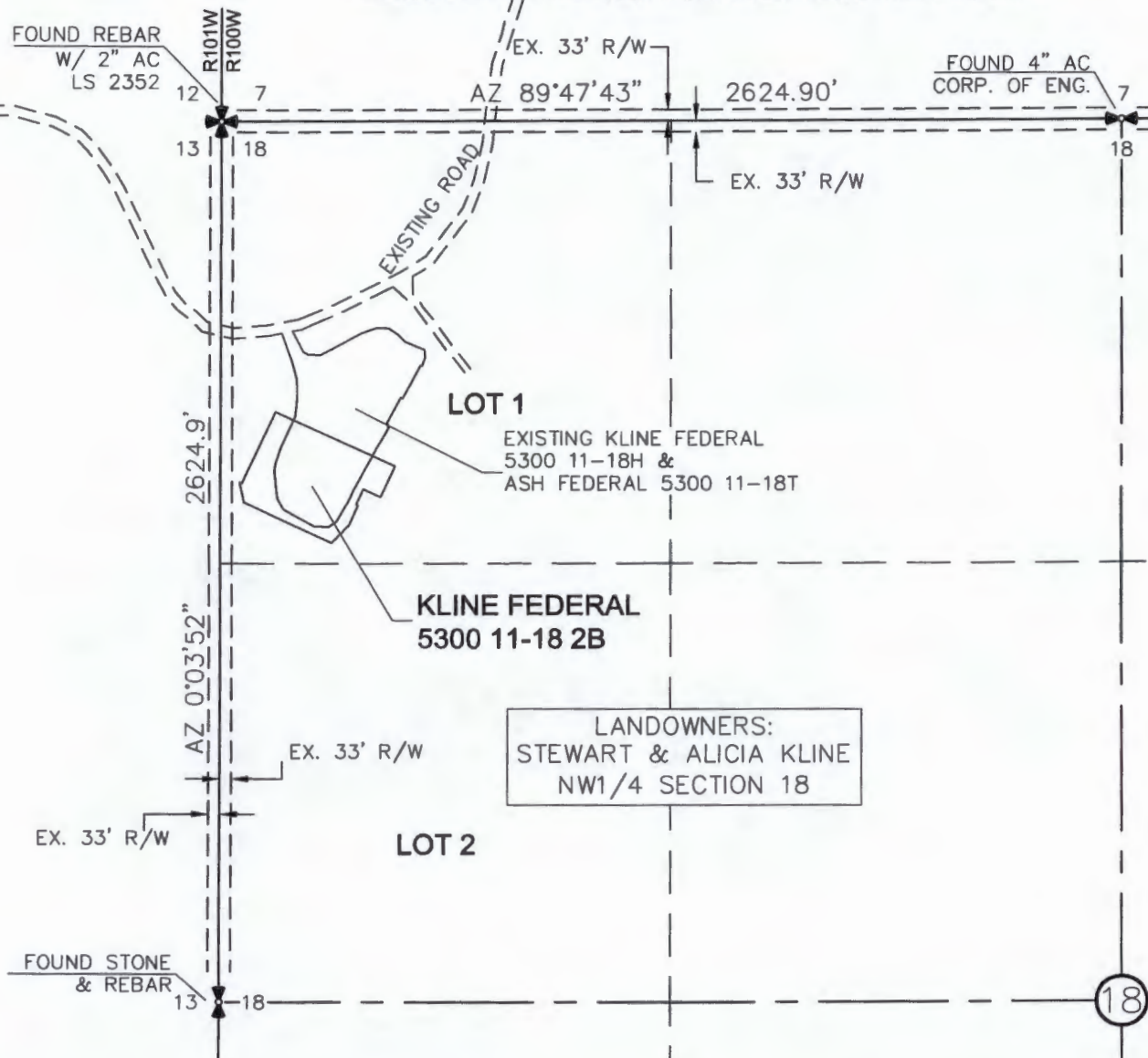
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 6300 11-18 25"
960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



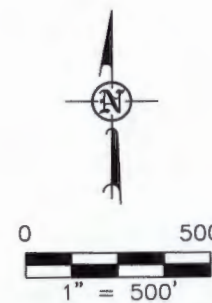
ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 11-18 2B"
 960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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 NUMBER 3880 ON 1/29/15
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LANDOWNERS:
 STEWART & ALICIA KLINE
 NW1/4 SECTION 18



NOTE: All utilities shown are preliminary only, a complete
 utilities location is recommended before construction.

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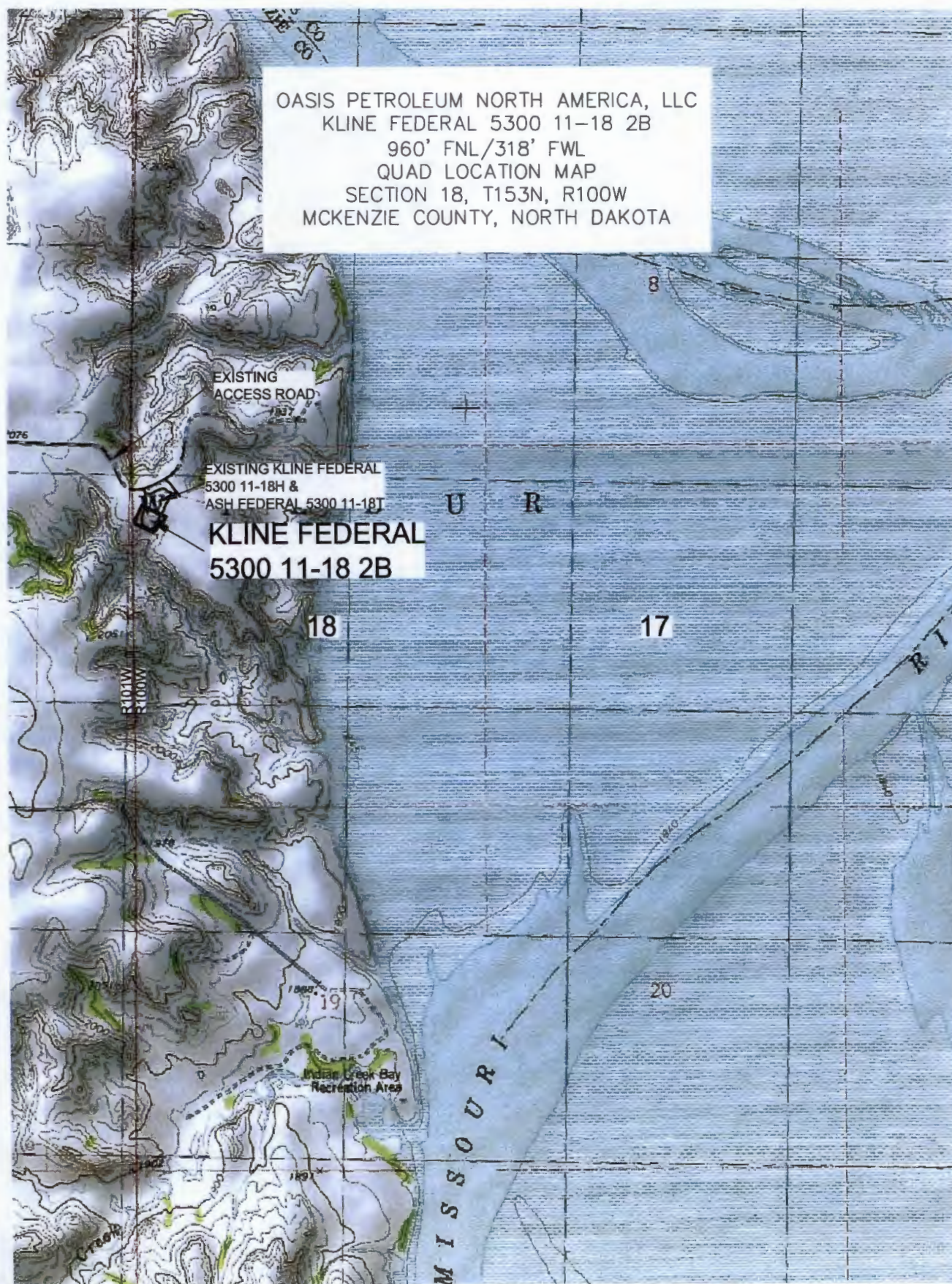
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 Fax: (406) 433-9618
 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 ACCESS APPROACH
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.J.H.
 Checked By: D.D.K.
 Project No.: S14-09-127
 Date: APRIL, 2014

Revision No.	Date	By	Description
REV 1	6/16/14	B.H.	MOVED WELLS
REV 2	6/24/14	LS	REVISED LATITUDE
REV 3	6/29/14	LS	ADDED EXISTING RT TO PAD LAYOUT
REV 4	12/29/14	LS	CHANGED WELL NAME & BH
REV 5	1/27/15	B.H.	CHANGED WELL NAME & BH

OASIS PETROLEUM NORTH AMERICA, LLC
 KLINE FEDERAL 5300 11-18 2B
 960' FNL/318' FWL
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



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 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

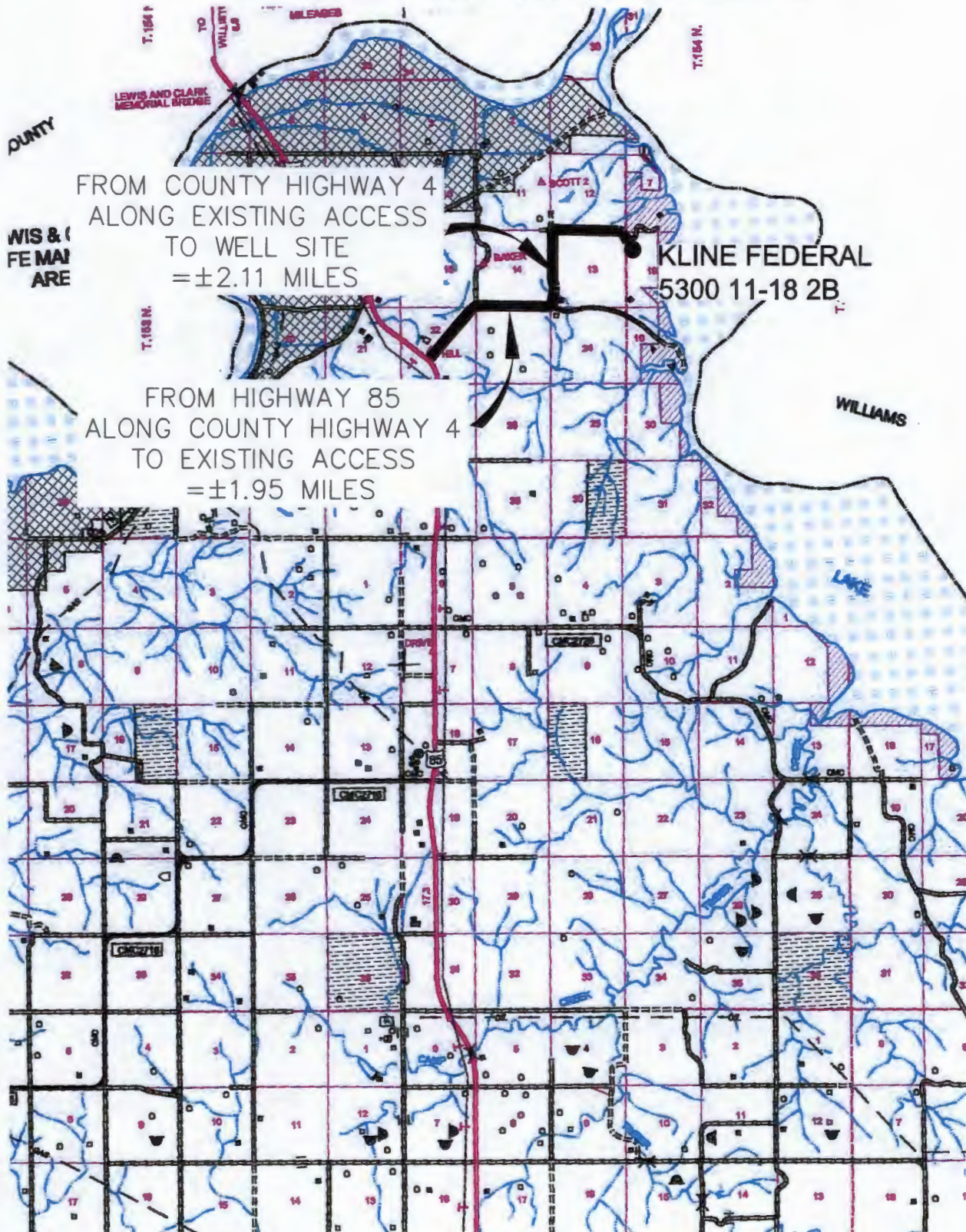
Drawn By: B.H.H. Project No.: S14-09-127
 Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	6/16/14	BH	MOVED WELLS
REV 2	6/24/14	JIS	REVISED LATITUDE
REV 3	8/29/14	JIS	ADDED EXISTING PIT TO PAD LAYOUT
REV 4	2/20/14	JIS	CHANGED WELL NAME & BH
REV 5	1/27/15	BH	CHANGED WELL NAMES & BH

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 2B"

960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127
Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS
REV 2	6/24/14	JLS	REVISED LATITUDE
REV 3	8/29/14	JLS	ADDED EXISTING PIT TO PAD LAYOUT
REV 4	12/30/14	JLS	CHANGED WELL NAME & SH
REV 5	1/27/15	BHH	CHANGED WELL NAMES & SH

CROSS SECTIONS

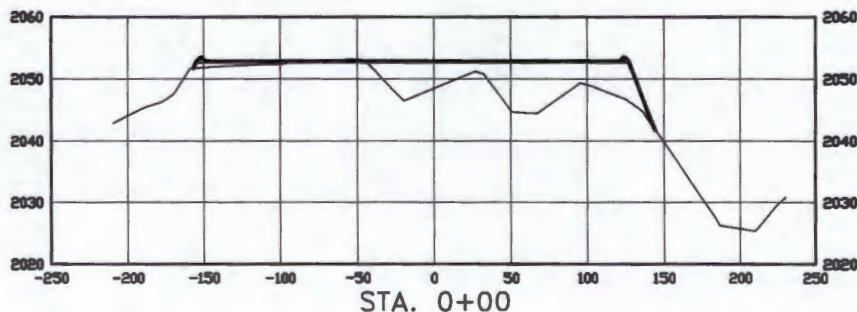
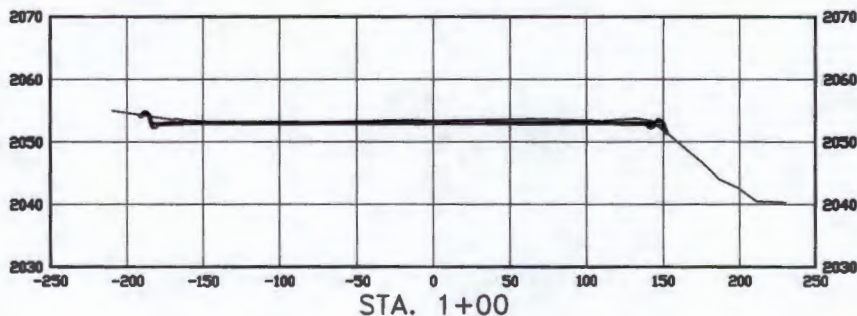
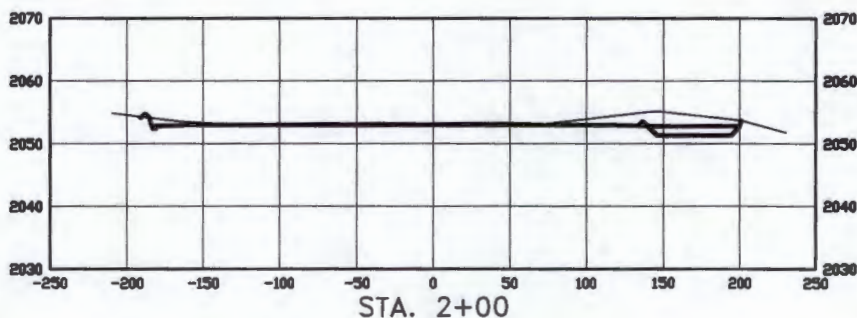
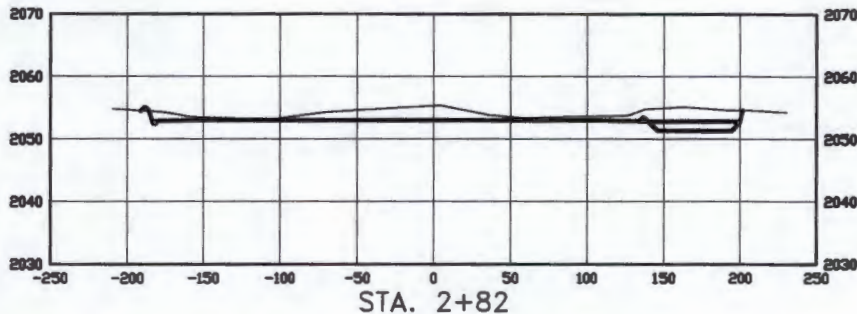
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"KLINE FEDERAL 5300 11-18 2B"

960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS
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P.L.S., REGISTRATION NUMBER
3880 ON 1/29/15 AND THE
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SCALE
HORIZ 1"=120'
VERT 1"=30'

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SHEET NO.



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OASIS PETROLEUM NORTH AMERICA, LLC
PAD CROSS SECTIONS
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: B.H.H. Project No.: S14-09-127
Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	6/16/14	BH	MOVED WELLS
REV 2	6/24/14	JAS	REVISED LATITUDE
REV 3	8/29/14	JAS	ADDED EXISTING PNT TO PAD LAYOUT
REV 4	12/30/14	JAS	CHANGED WELL NAME & BH
REV 5	1/27/15	BH	CHANGED WELL NAMES & BH

WELL LOCATION SITE QUANTITIES

OASIS PETROLEUM NORTH AMERICA, LLC

1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"KLINE FEDERAL 5300 11-18 2B"

960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

WELL SITE ELEVATION

2055.6

WELL PAD ELEVATION

2053.0

EXCAVATION

1,906

PLUS PIT

0

1,906

EMBANKMENT

869

PLUS SHRINKAGE (30%)

261

1,130

STOCKPILE PIT

0

STOCKPILE TOP SOIL (6")

1,934

BERMS

883 LF = 286 CY

DITCHES

727 LF = 111 CY

CONTAINMENT AREA

1,112 CY

ADDITIONAL MATERIAL NEEDED

221

DISTURBED AREA FROM PAD

2.40 ACRES

NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)

CUT END SLOPES AT 1:1

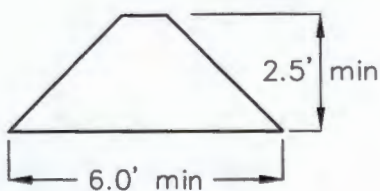
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION

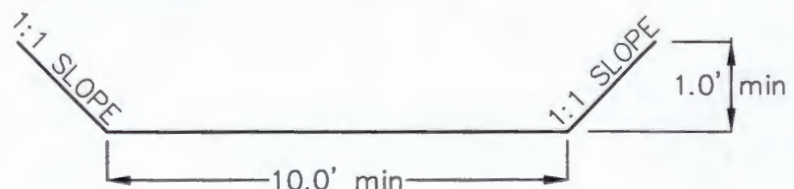
960' FNL

318' FWL

BERM DETAIL



DIVERSION DITCH DETAIL



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OASIS PETROLEUM NORTH AMERICA, LLC
QUANTITIES
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127
Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	8/18/14	BH	MOVED WELLS
REV 2	8/24/14	JLS	REVISED LATITUDE
REV 3	8/29/14	JLS	ADDED EXISTING PIT TO PAD LAYOUT
REV 4	12/30/14	JLS	CHANGED WELL NAME & PI
REV 5	1/27/15	BH	CHANGED WELL NAMES & PI



Oil and Gas Division 29334

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

BRANDI TERRY

Date: 9/15/2014

OASIS PETROLEUM NORTH AMERICA LLC

1001 FANNIN STE 1500

HOUSTON, TX 77002 USA

RE: CORES AND SAMPLES

Well Name: **KLINE FEDERAL 5300 11-18 2T2** Well File No.: **29334**

Location: **LOT1 18-153-100** County: **MCKENZIE**

Permit Type: **Development - HORIZONTAL**

Field: **BAKER** Target Horizon: **THREE FORKS B2**

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

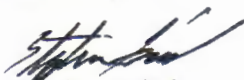
- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.
- 2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:
Base of the Last Charles Salt
 - Sample cuttings shall be collected at:
 - o 30' maximum intervals through all vertical and build sections.
 - o 100' maximum intervals through any horizontal sections.
 - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
 - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.
- 3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

**ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202**

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely


Stephen Fried
Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.
29334

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date July 2, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☒ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Waiver to rule Rule 43-02-03-31

Well Name and Number Kline Federal 5300 11-18 2T2					
Footages		Qtr-Qtr	Section	Township	Range
960 F N L 318 F W L		LOT 1	18	153 N	100 W
Field		Pool		County	
		Bakken		McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

The Oasis Petroleum/ Kline Federal 5300 11-18H located within a mile of the subject well

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Approved Per #20275

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9563	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Heather McCowan	
Title Regulatory Assistant		Date July 2, 2014	
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 9/8/2014	
By Matthew Messana	
Title Engineering Tech	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.
29334

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☐ Report of Work Done	Date Work Completed
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☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Suspension of Drilling

Well Name and Number KLINE FEDERAL 5300 11-18 2T2					
Footages	Qtr-Qtr	Section	Township	Range	
960 F N L 318 F WL	LOT 1	18	153 N	100 W	
Field BAKER	Pool BAKKEN	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Advanced Energy Services			
Address	City	State	Zip Code

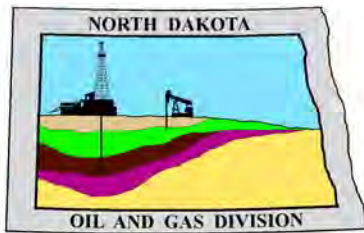
DETAILS OF WORK

Oasis Petroleum North America LLC requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Oasis Petroleum North America LLC intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Oasis Petroleum North America LLC understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Oasis Petroleum North America LL will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Notify NDIC Inspector Richard Dunn at (701)770-3554 with spud and TD info

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9563	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Heather McCowan		
Title Regulatory Assistant	Date July 2, 2014		
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 9/8/2014	
By Matthew Messana	
Title Engineering Tech	



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

September 8, 2014

Heather McCowan
Regulatory Assistant
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 1500
Houston, TX 77002

**RE: HORIZONTAL WELL
KLINE FEDERAL 5300 11-18 2T2
LOT1 Section 18-153N-100W
McKenzie County
Well File # 29334**

Dear Heather:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **500' setback** from the north & south boundaries and **200' setback** from the east & west boundaries within the 1280 acre spacing unit consisting of Sections 18 & 17, T153N R100W.

PERMIT STIPULATIONS: Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. In addition, OASIS PETROLEUM NORTH AMERICA LLC must take into consideration NDAC 43-02-03-28 (Safety Regulation) when contemplating simultaneous operations on the above captioned location. Pursuant to NDAC 43-02-03-28 (Safety Regulation) "No boiler, portable electric lighting generator, or treater shall be placed nearer than 150 feet to any producing well or oil tank." Lastly, OASIS PETROLEUM NORTH AMERICA LLC must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. Based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 10028' E.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

Matt Messana
Engineering Technician



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 10 / 1 / 2013	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC			Telephone Number 281-404-9563
Address 1001 Fannin Suite 1500		City Houston	State TX Zip Code 77002

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name KLINE FEDERAL				Well Number 5300 11-18 2T2			
Surface Footages 960 F N L 318 F W L		Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Footages 1198 F N L 782 F W L		Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 238 S From WH 464 E From WH		Azimuth 120 °	Longstring Total Depth 11111 Feet MD 10840 Feet TVD				
Bottom Hole Footages From Nearest Section Line 1530 F N L 205 F E L		Qtr-Qtr SENE	Section 17	Township 153 N	Range 100 W	County McKenzie	
Bottom Hole Coordinates From Well Head 570 S From WH 10023 E From WH		KOP Lateral 1 10363 Feet MD	Azimuth Lateral 1 90 °		Estimated Total Depth Lateral 1 20718 Feet MD 10890 Feet TVD		
Latitude of Well Head 48 ° 04 ' 45.68 "		Longitude of Well Head -103 ° 36 ' 10.19 "		NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Sections 18 & 17, T153N R100W		
Ground Elevation 2056 Feet Above S.L.	Acres in Spacing/Drilling Unit 1280	Spacing/Drilling Unit Setback Requirement 500 Feet N/S 200 Feet E/W			Industrial Commission Order 23752		
North Line of Spacing/Drilling Unit 10544 Feet	South Line of Spacing/Drilling Unit 10489 Feet	East Line of Spacing/Drilling Unit 5244 Feet			West Line of Spacing/Drilling Unit 5256 Feet		
Objective Horizons Three Forks B2					Pierre Shale Top 1968		
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 2068 Feet	Cement Volume 984 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 29/32 Lb./Ft.	Longstring Total Depth 11111 Feet MD 10840 Feet TVD		Cement Volume 835 Sacks	Cement Top 3947 Feet	Top Dakota Sand 5448 Feet
Base Last Charles Salt (If Applicable) 9216 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Salt Water Gel			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor Ryan	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats

Lateral 2

KOP Lateral 2	Azimuth Lateral 2	Estimated Total Depth Lateral 2		KOP Coordinates From Well Head		
Feet MD	°	Feet MD	Feet TVD	From WH		From WH
Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head				
From WH		From WH		From WH		From WH
KOP Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
F	L			N	W	
Bottom Hole Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
F	L			N	W	

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH				
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W	County

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date 7 / 02 / 2014
ePermit	Printed Name Heather McCowan	Title Regulatory Assistant

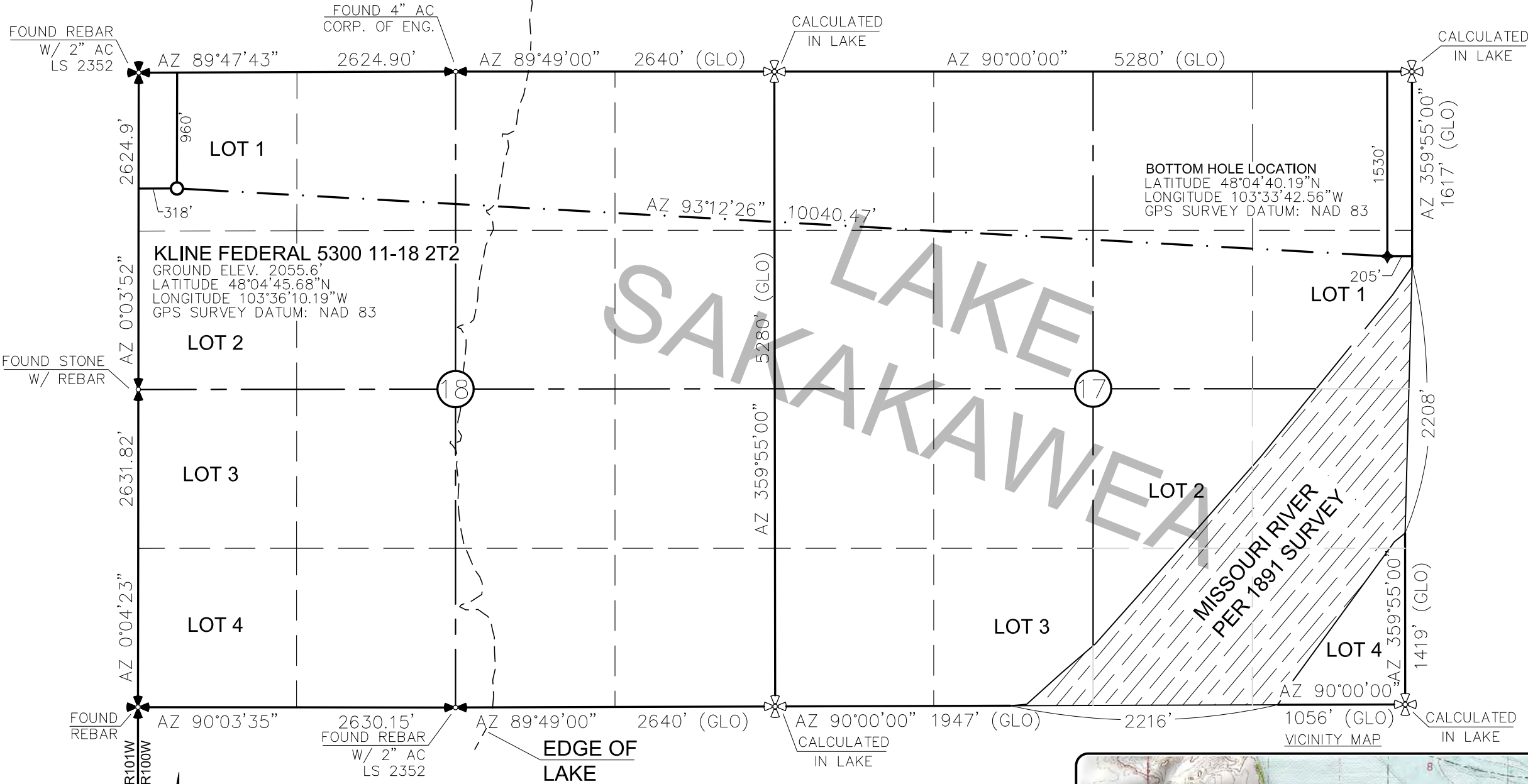
FOR STATE USE ONLY

Permit and File Number 29334	API Number 33 - 053 - 06243
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

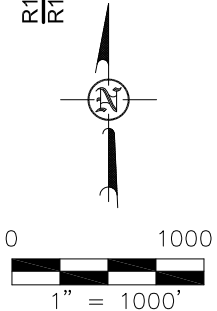
FOR STATE USE ONLY

Date Approved 9 / 8 / 2014
By Matt Messina
Title Engineering Technician

WELL LOCATION PLAT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 2T2"
960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

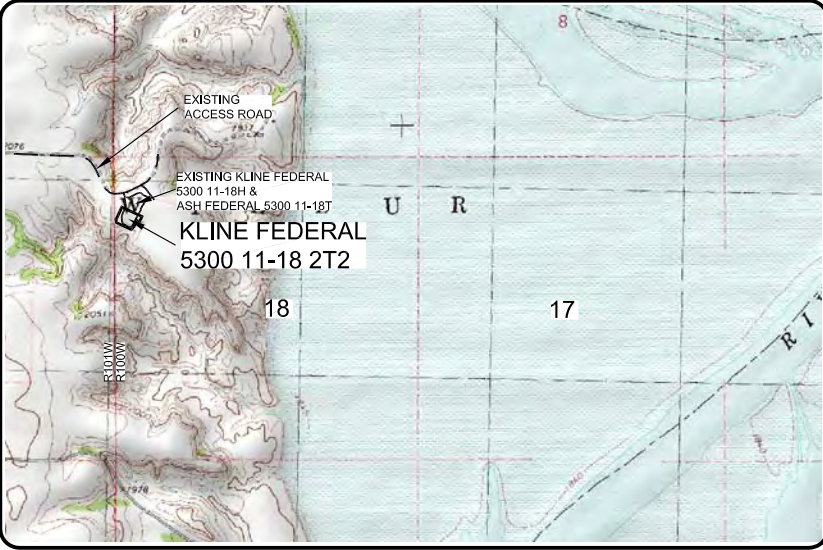
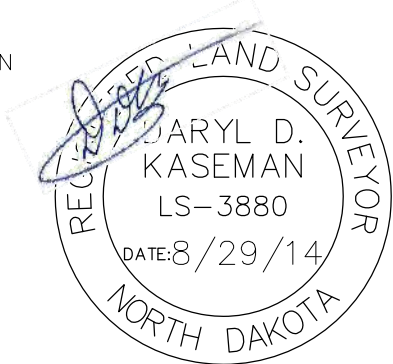


THIS DOCUMENT WAS ORIGINALLY
ISSUED AND SEALED BY DARYL D.
KASEMAN, PLS, REGISTRATION NUMBER
3880 ON 8/29/14 AND THE
ORIGINAL DOCUMENTS ARE STORED AT
THE OFFICES OF INTERSTATE
ENGINEERING, INC.



✖ - MONUMENT - RECOVERED
✖ - MONUMENT - NOT RECOVERED

[Signature]
DARYL D. KASEMAN LS-3880

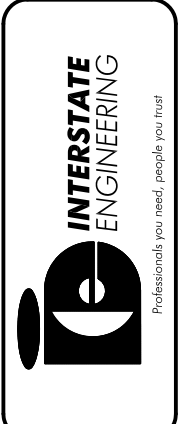


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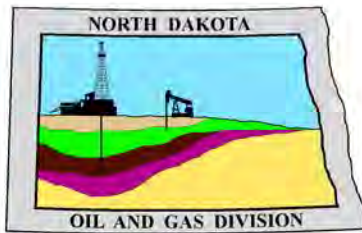
Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS
REV 2	6/24/14	JJS	REVISED LATITUDE
REV 3	8/29/14	JJS	ADDED EXISTING PIT TO PAD

OASIS PETROLEUM NORTH AMERICA, LLC WELL LOCATION PLAT SECTION 18, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA	Project No.: S14-09-127 Date: APRIL 2014
---	---

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota



1/8
SHEET NO.



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

RE: **Filter Socks and Other Filter Media**
Leakproof Container Required
Oil and Gas Wells

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

Thank you for your cooperation.

Sincerely,

Bruce E. Hicks

Assistant Director

DRILLING PLAN										
OPERATOR	Oasis Petroleum					COUNTY/STATE	Mckenzie Co., ND			
WELL NAME	Kline Federal 5300 11-18 2T2					RIG	0			
WELL TYPE	Horizontal Three Forks									
LOCATION	NWNW 18-153N-100W					Surface Location (survey plat):	960' fml	318' fwl		
EST. T.D.	20,718'					GROUND ELEV:	2052	Finished Pad Elev.	Sub Height: 25	
TOTAL LATERAL:					9,607' (est)	KB ELEV:	2077			
PROGNOSIS: Based on 2,077' KB(est)					LOGS:					
MARKER					DEPTH (Surf Loc)	DATUM (Surf Loc)	Type Interval			
Pierre					NDIC MAP	1,967	110'	OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota		
Greenhorn						4,614	-2,537'	CBL/GR: Above top of cement/GR to base of casing		
Mowry						5,020	-2,943'	MWD GR: KOP to lateral TD		
Dakota						5,447	-3,370'	DEVIATION:		
Rierdon						6,446	-4,369'	Surf:	3 deg. max., 1 deg / 100'; srvy every 500'	
Dunham Salt						6,784	-4,707'	Prod:	5 deg. max., 1 deg / 100'; srvy every 100'	
Dunham Salt Base						6,895	-4,818'	DST'S:		
Spearfish						6,992	-4,915'	None planned		
Pine Salt						7,247	-5,170'	CORES:		
Pine Salt Base						7,295	-5,218'	None planned		
Opeche Salt						7,340	-5,263'	MUDLOGGING:		
Opeche Salt Base						7,370	-5,293'	Two-Man: 8,316'		
Broom Creek (Top of Minnelusa Gp.)						7,572	-5,495'	~200' above the Charles (Kibbey) to Casing point; Casing point to TD		
Amsden						7,652	-5,575'	30' samples at direction of wellsite geologist; 10' through target @ curve land		
Tyler						7,820	-5,743'	BOP:		
Otter (Base of Minnelusa Gp.)						8,011	-5,934'	11" 5000 psi blind, pipe & annular		
Kibbey						8,366	-6,289'			
Charles Salt						8,516	-6,439'			
UB						9,140	-7,063'			
Base Last Salt						9,215	-7,138'			
Ratcliffe						9,263	-7,186'			
Mission Canyon						9,439	-7,362'			
Lodgepole						10,001	-7,924'			
Lodgepole Fracture Zone						10,207	-8,130'			
False Bakken						10,697	-8,620'			
Upper Bakken						10,707	-8,630'			
Middle Bakken						10,721	-8,644'			
Lower Bakken						10,766	-8,689'			
Pronghorn						10,780	-8,703'			
Three Forks 1st Bench						10,792	-8,715'			
Three Forks 1st Bench Claystone						10,815	-8,738'			
Three Forks 2nd Bench						10,824	-8,747'			
Three Forks 2nd Bench Claystone						10,852	-8,775'			
Three Forks 3rd Bench						10,872	-8,795'			
Dip Rate: -0.3										
Max. Anticipated BHP:					4702	Surface Formation: Glacial till				
MUD:		Interval	Type	WT	Vis	WL	Remarks			
Surface:		0' -	2,068'	FW	8.4-9.0	28-32	NC	Circ Mud Tanks		
Intermediate:		2,068' -	11,111'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks		
Lateral:		11,111' -	20,718'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks		
CASING:		Size	Wt ppf	Hole	Depth	Cement	WOC	Remarks		
Surface:		13-3/8"	54.5#	17-1/2"	2,068'	To Surface	12	100' into Pierre		
Inermediate (Dakota):		9-5/8"	40#	12-1/4"	6,100'	To Surface	24	Set Casing across Dakota		
Intermediate:		7"	29/32#	8-3/4"	11,111'	To Surface	24	1500' above Dakota		
Production Liner:		4-1/2"	13.5#	6"	20,718'	TOL @ 10,313'		50' above KOP		
PROBABLE PLUGS, IF REQ'D:										
OTHER:		MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI			
Surface:		2,068	2,068	960' FNL	318' FWL	SEC 18-T153N-R100W	Survey Company:			
KOP:		10,363'	10,363'	960' FNL	370' FWL	SEC 18-T153N-R100W	Build Rate:	12 deg /100'		
EOC:		11,111'	10,840'	1198' FNL	781' FWL	SEC 18-T153N-R100W	120.00			
Casing Point:		11,111'	10,840'	1198' FNL	781' FWL	SEC 18-T153N-R100W	120.00			
Three Forks Lateral TD:		20,718'	10,890'	1530' FNL	205' FEL	SEC 17-T153N-R100W	90.00			
Comments:										
Request Sundry to Waive Open Hole Logs										
Exception well: Oasis Petroleum's Kline 5300 11-18H										
Completion Notes: 35 packers, 35 sleeves, no frac string										
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.										
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2)										
68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)										
OASIS PETROLEUM										
Geology: M.Steed 4/9/2014					Engineering: hlbader rpm 5/29/14					

**Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 2T2
Sec. 18 T153N R100W
McKenzie County, North Dakota**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
13-3/8"	0' to 2,068'	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 2,068'	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 1.17	2730 / 2.82	514 / 2.61

API Rating & Safety Factor

- a) Collapse pressure based on full casing evacuation with 9 ppg fluid on backside (2068' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2068' setting depth).
- c) Based on string weight in 9 ppg fluid at 2068' TVD plus 100k# overpull. (Buoyed weight equals 97k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): 20 bbls fresh water

Lead Slurry: 635 sks (328 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCL₂, 4% D079 Extender, and 2% D053 Expanding Agent.

Tail Slurry: 349 sks (72 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25 lb/sk Lost Circulation Control Agent, and .25% CaCL₂.

**Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 2T2
Sec. 18 T153N R100W
McKenzie County, North Dakota**

CONTINGENCY INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' - 6101'	40	HCL-80	LTC	8.835"	8.75"***	5,450	7,270	9,090

**Special Drift

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' - 6101'	9-5/8", 40#, HCL-80, LTC, 8rd	4230 / 5.33	5750 / 1.23	837 / 2.73

API Rating & Safety Factor

- a. Collapse pressure based on 11.5ppg fluid on backside and 9ppg fluid inside of casing.
- b. Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
- c. Yield based on string weight in 10 ppg fluid, (207k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 9-5/8" casing set in a 12-1/4" hole with **10%** excess in OH and **0%** excess inside surface casing. TOC at surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **598 sks (309 bbls)** Conventional system with 75 lb/sk cement, 0.5lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl₂, 0.2% anti foam, and 0.4% fluid loss

Tail Slurry: **349 sks (72 bbls)** Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti foam, and 0.1% retarder

**Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 2T2
Sec. 18 T153N R100W
McKenzie County, North Dakota**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' - 6634'	29	P-110	LTC	6.184"	6.059"	5980	7970	8770
7"	6634' - 10363'	32	HCP-110	LTC	6.094"	6.000"	6730	8970	9870
7"	10363' - 11111'	29	P-110	LTC	6.184"	6.059"	5980	7970	8770

**Special Drift

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' - 6634'	7", 29#, P-110, LTC, 8rd	8530 / 2.47*	11220 / 1.19	797 / 2.08
6634' - 10363'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.19*	12460 / 1.29	
6634' - 10363'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.28**	12460 / 1.29	
10363' - 11111'	7", 29#, P-110, LTC, 8rd	8530 / 1.51*	11220 / 1.15	

API Rating & Safety Factor

- a. *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b. Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,840' TVD.
- c. Based on string weight in 10 ppg fluid, (301k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls Saltwater**
40 bbls Weighted MudPush Express

Lead Slurry: **207 sks (81 bbls)** 2.21 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 Extender, 3.0% KCl, 3.0% D154 Extender, 0.3% D208 Viscosifier, 0.07% Retarder, 0.2% Anti Foam, 0.5lb/sk D130 LCM

Tail Slurry: **628 sks (172 bbls)** 1.54 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% Retarder, 0.2% Fluid Loss, 0.2% Anti Foam, 0.5 lb/sk LCM

**Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 2T2
Sec. 18 T153N R100W
McKenzie County, North Dakota**

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Torque
4-1/2"	10,313' – 20,718'	13.5	P-110	BTC	3.92"	3.795"	4,500

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
10,313' – 20,718'	4-1/2", 13.5 lb, P-110, BTC, 8rd	10680 / 1.99	12410 / 1.28	443 / 1.86

API Rating & Safety Factor

- a) Collapse pressure based on full casing evacuation with 9.5 ppg fluid on backside @ 10890' TVD.
- b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10890' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 126k lbs.) plus 100k lbs overpull.

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel)
68476-34-6 (Primary Name: Fuels, diesel, No. 2)
68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)
8008-20-6 (Primary Name: Kerosene)



Company: Oasis Petroleum
Field: Indian Hills
Location: 153N-100W-17/18
Well: Kline Federal 5300 11-18 2T2
Wellbore #1

Plan: Design #6 (Kline Federal 5300 11-18 2T2/Wellbore #1)



SECTION DETAILS

Sec	MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	TFace	VSec	Target
1	0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.0	
2	2500.0	0.00	0.00	2500.0	0.0	0.0	0.00	0.00	0.0	
3	2650.0	3.00	90.00	2649.9	0.0	3.9	2.00	90.00	3.9	
4	3501.2	3.00	90.00	3500.0	0.0	48.5	0.00	0.00	48.4	
5	3651.2	0.00	0.00	3649.9	0.0	52.4	2.00	-180.00	52.3	
6	10363.3	0.00	0.00	10362.0	0.0	52.4	0.00	0.00	52.3	
7	11110.8	89.70	120.00	10839.5	-237.5	463.7	12.00	120.00	476.5	
8	11125.8	89.70	120.00	10839.5	-245.0	476.7	0.00	0.00	489.9	
9	12110.4	89.70	90.46	10844.8	-500.8	1416.3	3.00	-90.08	1442.4	PBHL Kline Federal 5300 11-18 2T2
10	20717.6	89.70	90.46	10889.9	-570.0	10023.0	0.00	0.00	10039.2	PBHL Kline Federal 5300 11-18 2T2

WELL DETAILS: Kline Federal 5300 11-18 2T2

+N/-S	+E/-W	Northing	Ground Level: Easting	2053.0	Latitude	Longitude	Slot
0.0	0.0	408992.29	1210243.30		48° 4' 45.680 N	103° 36' 10.190 W	

WELLBORE TARGET DETAILS

Name	TVD	+N/-S	+E/-W	Latitude	Longitude	Shape
PBHL Kline Federal 5300 11-18 2T2	10890.0	-570.0	10023.0	48° 4' 40.028 N	103° 33' 42.582 W	Point

ANNOTATIONS

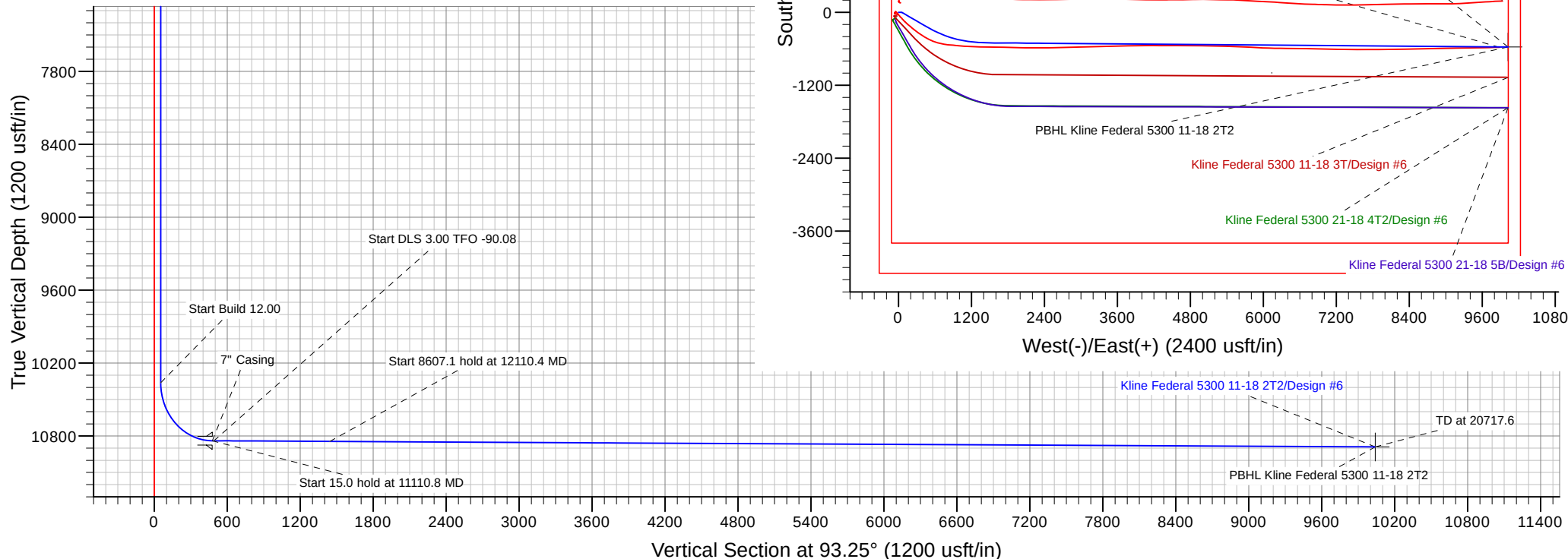
TVD	MD	Annotation
2500.0	2500.0	Start Build 2.00
2649.9	2650.0	Start 851.2 hold at 2650.0 MD
3500.0	3501.2	Start DLS 2.00 TFO -180.00
3649.9	3651.2	Start 6712.1 hold at 3651.2 MD
10362.0	10363.3	Start Build 12.00
10839.5	11110.8	Start 15.0 hold at 11110.8 MD
10839.5	11125.8	Start DLS 3.00 TFO -90.08
10844.8	12110.4	Start 8607.1 hold at 12110.4 MD
10889.9	20717.6	TD at 20717.6

CASING DETAILS

TVD	MD	Name	Size
2068.0	2068.0	13 3/8" Casing	13-3/8
10839.5	11111.0	7" Casing	7
6100.0	6101.3	9 5/8" Casing	9-5/8

Plan: Design #6 (Kline Federal 5300 11-18 2T2/Wellbore #1)

Created By: M. Loucks Date: 15:24, June 30 2014





Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 11-18 2T2

Wellbore #1

Plan: Design #6

Standard Planning Report

30 June, 2014

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-17/18			
Site Position:		Northing:	408,992.30 usft	Latitude: 48° 4' 45.680 N
From:	Lat/Long	Easting:	1,210,243.30 usft	Longitude: 103° 36' 10.190 W
Position Uncertainty:	0.0 usft	Slot Radius:	13-3/16 "	Grid Convergence: -2.31 °

Well	Kline Federal 5300 11-18 2T2			
Well Position	+N/-S	0.0 usft	Northing:	408,992.29 usft
	+E/-W	0.0 usft	Easting:	1,210,243.30 usft
Position Uncertainty		0.0 usft	Wellhead Elevation:	Ground Level: 2,053.0 usft

Wellbore	Wellbore #1				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	04/22/14	8.14	72.97	56,479

Design	Design #6			
Audit Notes:				
Version:	Phase:	PLAN	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	93.25

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,500.0	0.00	0.00	2,500.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,650.0	3.00	90.00	2,649.9	0.0	3.9	2.00	2.00	0.00	90.00	
3,501.2	3.00	90.00	3,500.0	0.0	48.5	0.00	0.00	0.00	0.00	
3,651.2	0.00	0.00	3,649.9	0.0	52.4	2.00	-2.00	-60.00	-180.00	
10,363.3	0.00	0.00	10,362.0	0.0	52.4	0.00	0.00	0.00	0.00	
11,110.8	89.70	120.00	10,839.5	-237.5	463.7	12.00	12.00	0.00	120.00	
11,125.8	89.70	120.00	10,839.5	-245.0	476.7	0.00	0.00	0.00	0.00	
12,110.4	89.70	90.46	10,844.8	-500.8	1,416.3	3.00	0.00	-3.00	-90.08	PBHL Kline Federal 5
20,717.6	89.70	90.46	10,889.9	-570.0	10,023.0	0.00	0.00	0.00	0.00	PBHL Kline Federal 5

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	0.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.00	0.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.00	0.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.00	0.00	400.0	0.0	0.0	0.0	0.00	0.00	0.00
500.0	0.00	0.00	500.0	0.0	0.0	0.0	0.00	0.00	0.00
600.0	0.00	0.00	600.0	0.0	0.0	0.0	0.00	0.00	0.00
700.0	0.00	0.00	700.0	0.0	0.0	0.0	0.00	0.00	0.00
800.0	0.00	0.00	800.0	0.0	0.0	0.0	0.00	0.00	0.00
900.0	0.00	0.00	900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,000.0	0.00	0.00	1,000.0	0.0	0.0	0.0	0.00	0.00	0.00
1,100.0	0.00	0.00	1,100.0	0.0	0.0	0.0	0.00	0.00	0.00
1,200.0	0.00	0.00	1,200.0	0.0	0.0	0.0	0.00	0.00	0.00
1,300.0	0.00	0.00	1,300.0	0.0	0.0	0.0	0.00	0.00	0.00
1,400.0	0.00	0.00	1,400.0	0.0	0.0	0.0	0.00	0.00	0.00
1,500.0	0.00	0.00	1,500.0	0.0	0.0	0.0	0.00	0.00	0.00
1,600.0	0.00	0.00	1,600.0	0.0	0.0	0.0	0.00	0.00	0.00
1,700.0	0.00	0.00	1,700.0	0.0	0.0	0.0	0.00	0.00	0.00
1,800.0	0.00	0.00	1,800.0	0.0	0.0	0.0	0.00	0.00	0.00
1,900.0	0.00	0.00	1,900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,968.0	0.00	0.00	1,968.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,000.0	0.00	0.00	2,000.0	0.0	0.0	0.0	0.00	0.00	0.00
2,068.0	0.00	0.00	2,068.0	0.0	0.0	0.0	0.00	0.00	0.00
13 3/8" Casing									
2,100.0	0.00	0.00	2,100.0	0.0	0.0	0.0	0.00	0.00	0.00
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.0	0.00	0.00	0.00
2,300.0	0.00	0.00	2,300.0	0.0	0.0	0.0	0.00	0.00	0.00
2,400.0	0.00	0.00	2,400.0	0.0	0.0	0.0	0.00	0.00	0.00
2,500.0	0.00	0.00	2,500.0	0.0	0.0	0.0	0.00	0.00	0.00
Start Build 2.00									
2,600.0	2.00	90.00	2,600.0	0.0	1.7	1.7	2.00	2.00	0.00
2,650.0	3.00	90.00	2,649.9	0.0	3.9	3.9	2.00	2.00	0.00
Start 851.2 hold at 2650.0 MD									
2,700.0	3.00	90.00	2,699.9	0.0	6.5	6.5	0.00	0.00	0.00
2,800.0	3.00	90.00	2,799.7	0.0	11.8	11.8	0.00	0.00	0.00
2,900.0	3.00	90.00	2,899.6	0.0	17.0	17.0	0.00	0.00	0.00
3,000.0	3.00	90.00	2,999.5	0.0	22.2	22.2	0.00	0.00	0.00
3,100.0	3.00	90.00	3,099.3	0.0	27.5	27.4	0.00	0.00	0.00
3,200.0	3.00	90.00	3,199.2	0.0	32.7	32.7	0.00	0.00	0.00
3,300.0	3.00	90.00	3,299.0	0.0	37.9	37.9	0.00	0.00	0.00
3,400.0	3.00	90.00	3,398.9	0.0	43.2	43.1	0.00	0.00	0.00
3,501.2	3.00	90.00	3,500.0	0.0	48.5	48.4	0.00	0.00	0.00
Start DLS 2.00 TFO -180.00									
3,600.0	1.02	90.00	3,598.7	0.0	51.9	51.9	2.00	-2.00	0.00
3,651.2	0.00	0.00	3,649.9	0.0	52.4	52.3	2.00	-2.00	-175.66
Start 6712.1 hold at 3651.2 MD									
3,700.0	0.00	0.00	3,698.7	0.0	52.4	52.3	0.00	0.00	0.00
3,800.0	0.00	0.00	3,798.7	0.0	52.4	52.3	0.00	0.00	0.00
3,900.0	0.00	0.00	3,898.7	0.0	52.4	52.3	0.00	0.00	0.00
4,000.0	0.00	0.00	3,998.7	0.0	52.4	52.3	0.00	0.00	0.00
4,100.0	0.00	0.00	4,098.7	0.0	52.4	52.3	0.00	0.00	0.00
4,200.0	0.00	0.00	4,198.7	0.0	52.4	52.3	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
4,300.0	0.00	0.00	4,298.7	0.0	52.4	52.3	0.00	0.00	0.00
4,400.0	0.00	0.00	4,398.7	0.0	52.4	52.3	0.00	0.00	0.00
4,500.0	0.00	0.00	4,498.7	0.0	52.4	52.3	0.00	0.00	0.00
4,600.0	0.00	0.00	4,598.7	0.0	52.4	52.3	0.00	0.00	0.00
4,616.3	0.00	0.00	4,615.0	0.0	52.4	52.3	0.00	0.00	0.00
Greenhorn									
4,700.0	0.00	0.00	4,698.7	0.0	52.4	52.3	0.00	0.00	0.00
4,800.0	0.00	0.00	4,798.7	0.0	52.4	52.3	0.00	0.00	0.00
4,900.0	0.00	0.00	4,898.7	0.0	52.4	52.3	0.00	0.00	0.00
5,000.0	0.00	0.00	4,998.7	0.0	52.4	52.3	0.00	0.00	0.00
5,022.3	0.00	0.00	5,021.0	0.0	52.4	52.3	0.00	0.00	0.00
Mowry									
5,100.0	0.00	0.00	5,098.7	0.0	52.4	52.3	0.00	0.00	0.00
5,200.0	0.00	0.00	5,198.7	0.0	52.4	52.3	0.00	0.00	0.00
5,300.0	0.00	0.00	5,298.7	0.0	52.4	52.3	0.00	0.00	0.00
5,400.0	0.00	0.00	5,398.7	0.0	52.4	52.3	0.00	0.00	0.00
5,449.3	0.00	0.00	5,448.0	0.0	52.4	52.3	0.00	0.00	0.00
Dakota									
5,500.0	0.00	0.00	5,498.7	0.0	52.4	52.3	0.00	0.00	0.00
5,600.0	0.00	0.00	5,598.7	0.0	52.4	52.3	0.00	0.00	0.00
5,700.0	0.00	0.00	5,698.7	0.0	52.4	52.3	0.00	0.00	0.00
5,800.0	0.00	0.00	5,798.7	0.0	52.4	52.3	0.00	0.00	0.00
5,900.0	0.00	0.00	5,898.7	0.0	52.4	52.3	0.00	0.00	0.00
6,000.0	0.00	0.00	5,998.7	0.0	52.4	52.3	0.00	0.00	0.00
6,100.0	0.00	0.00	6,098.7	0.0	52.4	52.3	0.00	0.00	0.00
6,101.3	0.00	0.00	6,100.0	0.0	52.4	52.3	0.00	0.00	0.00
9 5/8" Casing									
6,200.0	0.00	0.00	6,198.7	0.0	52.4	52.3	0.00	0.00	0.00
6,300.0	0.00	0.00	6,298.7	0.0	52.4	52.3	0.00	0.00	0.00
6,400.0	0.00	0.00	6,398.7	0.0	52.4	52.3	0.00	0.00	0.00
6,448.3	0.00	0.00	6,447.0	0.0	52.4	52.3	0.00	0.00	0.00
Rierdon									
6,500.0	0.00	0.00	6,498.7	0.0	52.4	52.3	0.00	0.00	0.00
6,600.0	0.00	0.00	6,598.7	0.0	52.4	52.3	0.00	0.00	0.00
6,700.0	0.00	0.00	6,698.7	0.0	52.4	52.3	0.00	0.00	0.00
6,786.3	0.00	0.00	6,785.0	0.0	52.4	52.3	0.00	0.00	0.00
Dunham Salt									
6,800.0	0.00	0.00	6,798.7	0.0	52.4	52.3	0.00	0.00	0.00
6,897.3	0.00	0.00	6,896.0	0.0	52.4	52.3	0.00	0.00	0.00
Dunham Salt Base									
6,900.0	0.00	0.00	6,898.7	0.0	52.4	52.3	0.00	0.00	0.00
6,994.3	0.00	0.00	6,993.0	0.0	52.4	52.3	0.00	0.00	0.00
Spearfish									
7,000.0	0.00	0.00	6,998.7	0.0	52.4	52.3	0.00	0.00	0.00
7,100.0	0.00	0.00	7,098.7	0.0	52.4	52.3	0.00	0.00	0.00
7,200.0	0.00	0.00	7,198.7	0.0	52.4	52.3	0.00	0.00	0.00
7,249.3	0.00	0.00	7,248.0	0.0	52.4	52.3	0.00	0.00	0.00
Pine Salt									
7,297.3	0.00	0.00	7,296.0	0.0	52.4	52.3	0.00	0.00	0.00
Pine Salt Base									
7,300.0	0.00	0.00	7,298.7	0.0	52.4	52.3	0.00	0.00	0.00
7,342.3	0.00	0.00	7,341.0	0.0	52.4	52.3	0.00	0.00	0.00
Opeche Salt									

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Project:	Indian Hills	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
7,372.3	0.00	0.00	7,371.0	0.0	52.4	52.3	0.00	0.00	0.00
Opeche Salt Base									
7,400.0	0.00	0.00	7,398.7	0.0	52.4	52.3	0.00	0.00	0.00
7,500.0	0.00	0.00	7,498.7	0.0	52.4	52.3	0.00	0.00	0.00
7,574.3	0.00	0.00	7,573.0	0.0	52.4	52.3	0.00	0.00	0.00
Broom Creek (Top of Minnelusa Gp.)									
7,600.0	0.00	0.00	7,598.7	0.0	52.4	52.3	0.00	0.00	0.00
7,654.3	0.00	0.00	7,653.0	0.0	52.4	52.3	0.00	0.00	0.00
Amsden									
7,700.0	0.00	0.00	7,698.7	0.0	52.4	52.3	0.00	0.00	0.00
7,800.0	0.00	0.00	7,798.7	0.0	52.4	52.3	0.00	0.00	0.00
7,822.3	0.00	0.00	7,821.0	0.0	52.4	52.3	0.00	0.00	0.00
Tyler									
7,900.0	0.00	0.00	7,898.7	0.0	52.4	52.3	0.00	0.00	0.00
8,000.0	0.00	0.00	7,998.7	0.0	52.4	52.3	0.00	0.00	0.00
8,013.3	0.00	0.00	8,012.0	0.0	52.4	52.3	0.00	0.00	0.00
Otter (Base of Minnelusa Gp.)									
8,100.0	0.00	0.00	8,098.7	0.0	52.4	52.3	0.00	0.00	0.00
8,200.0	0.00	0.00	8,198.7	0.0	52.4	52.3	0.00	0.00	0.00
8,300.0	0.00	0.00	8,298.7	0.0	52.4	52.3	0.00	0.00	0.00
8,368.3	0.00	0.00	8,367.0	0.0	52.4	52.3	0.00	0.00	0.00
Kibbey									
8,400.0	0.00	0.00	8,398.7	0.0	52.4	52.3	0.00	0.00	0.00
8,500.0	0.00	0.00	8,498.7	0.0	52.4	52.3	0.00	0.00	0.00
8,518.3	0.00	0.00	8,517.0	0.0	52.4	52.3	0.00	0.00	0.00
Charles Salt									
8,600.0	0.00	0.00	8,598.7	0.0	52.4	52.3	0.00	0.00	0.00
8,700.0	0.00	0.00	8,698.7	0.0	52.4	52.3	0.00	0.00	0.00
8,800.0	0.00	0.00	8,798.7	0.0	52.4	52.3	0.00	0.00	0.00
8,900.0	0.00	0.00	8,898.7	0.0	52.4	52.3	0.00	0.00	0.00
9,000.0	0.00	0.00	8,998.7	0.0	52.4	52.3	0.00	0.00	0.00
9,100.0	0.00	0.00	9,098.7	0.0	52.4	52.3	0.00	0.00	0.00
9,142.3	0.00	0.00	9,141.0	0.0	52.4	52.3	0.00	0.00	0.00
UB									
9,200.0	0.00	0.00	9,198.7	0.0	52.4	52.3	0.00	0.00	0.00
9,217.3	0.00	0.00	9,216.0	0.0	52.4	52.3	0.00	0.00	0.00
Base Last Salt									
9,265.3	0.00	0.00	9,264.0	0.0	52.4	52.3	0.00	0.00	0.00
Ratcliffe									
9,300.0	0.00	0.00	9,298.7	0.0	52.4	52.3	0.00	0.00	0.00
9,400.0	0.00	0.00	9,398.7	0.0	52.4	52.3	0.00	0.00	0.00
9,441.3	0.00	0.00	9,440.0	0.0	52.4	52.3	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,498.7	0.0	52.4	52.3	0.00	0.00	0.00
9,600.0	0.00	0.00	9,598.7	0.0	52.4	52.3	0.00	0.00	0.00
9,700.0	0.00	0.00	9,698.7	0.0	52.4	52.3	0.00	0.00	0.00
9,800.0	0.00	0.00	9,798.7	0.0	52.4	52.3	0.00	0.00	0.00
9,900.0	0.00	0.00	9,898.7	0.0	52.4	52.3	0.00	0.00	0.00
10,000.0	0.00	0.00	9,998.7	0.0	52.4	52.3	0.00	0.00	0.00
10,003.3	0.00	0.00	10,002.0	0.0	52.4	52.3	0.00	0.00	0.00
Lodgepole									
10,100.0	0.00	0.00	10,098.7	0.0	52.4	52.3	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,200.0	0.00	0.00	10,198.7	0.0	52.4	52.3	0.00	0.00	0.00
10,209.3	0.00	0.00	10,208.0	0.0	52.4	52.3	0.00	0.00	0.00
Lodgepole Fracture Zone									
10,300.0	0.00	0.00	10,298.7	0.0	52.4	52.3	0.00	0.00	0.00
10,363.3	0.00	0.00	10,362.0	0.0	52.4	52.3	0.00	0.00	0.00
Start Build 12.00									
10,375.0	1.40	120.00	10,373.7	-0.1	52.5	52.4	12.00	12.00	1,025.96
10,400.0	4.40	120.00	10,398.7	-0.7	53.6	53.6	12.00	12.00	0.00
10,425.0	7.40	120.00	10,423.5	-2.0	55.8	55.9	12.00	12.00	0.00
10,450.0	10.40	120.00	10,448.2	-3.9	59.2	59.3	12.00	12.00	0.00
10,475.0	13.40	120.00	10,472.7	-6.5	63.7	63.9	12.00	12.00	0.00
10,500.0	16.40	120.00	10,496.8	-9.7	69.2	69.7	12.00	12.00	0.00
10,525.0	19.40	120.00	10,520.6	-13.6	75.9	76.5	12.00	12.00	0.00
10,550.0	22.40	120.00	10,544.0	-18.0	83.6	84.5	12.00	12.00	0.00
10,575.0	25.40	120.00	10,566.8	-23.1	92.4	93.5	12.00	12.00	0.00
10,600.0	28.40	120.00	10,589.1	-28.7	102.2	103.6	12.00	12.00	0.00
10,625.0	31.40	120.00	10,610.8	-35.0	113.0	114.8	12.00	12.00	0.00
10,650.0	34.40	120.00	10,631.8	-41.8	124.7	126.9	12.00	12.00	0.00
10,675.0	37.40	120.00	10,652.0	-49.1	137.4	140.0	12.00	12.00	0.00
10,700.0	40.40	120.00	10,671.5	-56.9	151.0	154.0	12.00	12.00	0.00
10,725.0	43.40	120.00	10,690.1	-65.3	165.5	168.9	12.00	12.00	0.00
10,736.0	44.73	120.00	10,698.0	-69.1	172.1	175.8	12.00	12.00	0.00
False Bakken									
10,750.0	46.40	120.00	10,707.8	-74.1	180.8	184.7	12.00	12.00	0.00
10,750.3	46.44	120.00	10,708.0	-74.2	181.0	184.9	12.00	12.00	0.00
Upper Bakken									
10,771.1	48.94	120.00	10,722.0	-81.9	194.3	198.6	12.00	12.00	0.00
Middle Bakken									
10,775.0	49.40	120.00	10,724.5	-83.4	196.8	201.2	12.00	12.00	0.00
10,800.0	52.40	120.00	10,740.3	-93.1	213.6	218.6	12.00	12.00	0.00
10,825.0	55.40	120.00	10,755.0	-103.2	231.1	236.6	12.00	12.00	0.00
10,846.8	58.02	120.00	10,767.0	-112.3	246.9	252.9	12.00	12.00	0.00
Lower Bakken									
10,850.0	58.40	120.00	10,768.7	-113.7	249.3	255.3	12.00	12.00	0.00
10,874.5	61.35	120.00	10,781.0	-124.3	267.6	274.3	12.00	12.00	0.00
Pronghorn									
10,875.0	61.40	120.00	10,781.2	-124.5	268.0	274.6	12.00	12.00	0.00
10,900.0	64.40	120.00	10,792.6	-135.6	287.3	294.5	12.00	12.00	0.00
10,900.9	64.51	120.00	10,793.0	-136.0	288.0	295.2	12.00	12.00	0.00
Three Forks 1st Bench									
10,925.0	67.40	120.00	10,802.8	-147.0	307.0	314.9	12.00	12.00	0.00
10,950.0	70.40	120.00	10,811.8	-158.7	327.2	335.7	12.00	12.00	0.00
10,963.0	71.96	120.00	10,816.0	-164.8	337.9	346.7	12.00	12.00	0.00
Three Forks 1st Bench Claystone									
10,975.0	73.40	120.00	10,819.6	-170.5	347.8	356.9	12.00	12.00	0.00
10,995.5	75.86	120.00	10,825.0	-180.4	364.9	374.5	12.00	12.00	0.00
Three Forks 2nd Bench									
11,000.0	76.40	120.00	10,826.1	-182.6	368.7	378.5	12.00	12.00	0.00
11,025.0	79.40	120.00	10,831.3	-194.8	389.9	400.3	12.00	12.00	0.00
11,050.0	82.40	120.00	10,835.3	-207.2	411.2	422.3	12.00	12.00	0.00
11,075.0	85.40	120.00	10,837.9	-219.6	432.8	444.5	12.00	12.00	0.00
11,100.0	88.40	120.00	10,839.3	-232.1	454.4	466.8	12.00	12.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
11,110.8	89.70	120.00	10,839.5	-237.5	463.7	476.5	12.00	12.00	0.00
Start 15.0 hold at 11110.8 MD									
11,111.0	89.70	120.00	10,839.5	-237.6	463.9	476.6	0.00	0.00	0.00
7" Casing									
11,125.8	89.70	120.00	10,839.5	-245.0	476.7	489.9	0.00	0.00	0.00
Start DLS 3.00 TFO -90.08									
11,200.0	89.70	117.77	10,839.9	-280.8	541.7	556.8	3.00	0.00	-3.00
11,300.0	89.69	114.77	10,840.5	-325.1	631.3	648.8	3.00	0.00	-3.00
11,400.0	89.69	111.77	10,841.0	-364.6	723.2	742.7	3.00	0.00	-3.00
11,500.0	89.69	108.77	10,841.5	-399.2	817.0	838.3	3.00	0.00	-3.00
11,600.0	89.69	105.77	10,842.1	-428.9	912.5	935.3	3.00	0.00	-3.00
11,700.0	89.69	102.77	10,842.6	-453.6	1,009.4	1,033.5	3.00	0.00	-3.00
11,800.0	89.69	99.77	10,843.2	-473.1	1,107.4	1,132.5	3.00	0.00	-3.00
11,900.0	89.69	96.77	10,843.7	-487.5	1,206.4	1,232.1	3.00	0.00	-3.00
12,000.0	89.70	93.77	10,844.2	-496.7	1,305.9	1,332.0	3.00	0.00	-3.00
12,100.0	89.70	90.77	10,844.8	-500.7	1,405.8	1,432.0	3.00	0.00	-3.00
12,110.4	89.70	90.46	10,844.8	-500.8	1,416.3	1,442.4	3.00	0.00	-3.00
Start 8607.1 hold at 12110.4 MD									
12,200.0	89.70	90.46	10,845.3	-501.5	1,505.8	1,531.9	0.00	0.00	0.00
12,300.0	89.70	90.46	10,845.8	-502.3	1,605.8	1,631.8	0.00	0.00	0.00
12,400.0	89.70	90.46	10,846.3	-503.1	1,705.8	1,731.6	0.00	0.00	0.00
12,500.0	89.70	90.46	10,846.8	-503.9	1,805.8	1,831.5	0.00	0.00	0.00
12,600.0	89.70	90.46	10,847.4	-504.7	1,905.8	1,931.4	0.00	0.00	0.00
12,700.0	89.70	90.46	10,847.9	-505.5	2,005.8	2,031.3	0.00	0.00	0.00
12,800.0	89.70	90.46	10,848.4	-506.3	2,105.8	2,131.2	0.00	0.00	0.00
12,900.0	89.70	90.46	10,848.9	-507.1	2,205.8	2,231.0	0.00	0.00	0.00
13,000.0	89.70	90.46	10,849.5	-507.9	2,305.8	2,330.9	0.00	0.00	0.00
13,100.0	89.70	90.46	10,850.0	-508.7	2,405.8	2,430.8	0.00	0.00	0.00
13,200.0	89.70	90.46	10,850.5	-509.5	2,505.8	2,530.7	0.00	0.00	0.00
13,300.0	89.70	90.46	10,851.0	-510.4	2,605.8	2,630.6	0.00	0.00	0.00
13,400.0	89.70	90.46	10,851.6	-511.2	2,705.8	2,730.4	0.00	0.00	0.00
13,500.0	89.70	90.46	10,852.1	-512.0	2,805.8	2,830.3	0.00	0.00	0.00
13,600.0	89.70	90.46	10,852.6	-512.8	2,905.8	2,930.2	0.00	0.00	0.00
13,674.9	89.70	90.46	10,853.0	-513.4	2,980.7	3,005.1	0.00	0.00	0.00
Three Forks 2nd Bench Claystone									
13,700.0	89.70	90.46	10,853.1	-513.6	3,005.8	3,030.1	0.00	0.00	0.00
13,800.0	89.70	90.46	10,853.7	-514.4	3,105.8	3,130.0	0.00	0.00	0.00
13,900.0	89.70	90.46	10,854.2	-515.2	3,205.8	3,229.8	0.00	0.00	0.00
14,000.0	89.70	90.46	10,854.7	-516.0	3,305.8	3,329.7	0.00	0.00	0.00
14,100.0	89.70	90.46	10,855.2	-516.8	3,405.7	3,429.6	0.00	0.00	0.00
14,200.0	89.70	90.46	10,855.7	-517.6	3,505.7	3,529.5	0.00	0.00	0.00
14,300.0	89.70	90.46	10,856.3	-518.4	3,605.7	3,629.4	0.00	0.00	0.00
14,400.0	89.70	90.46	10,856.8	-519.2	3,705.7	3,729.2	0.00	0.00	0.00
14,500.0	89.70	90.46	10,857.3	-520.0	3,805.7	3,829.1	0.00	0.00	0.00
14,600.0	89.70	90.46	10,857.8	-520.8	3,905.7	3,929.0	0.00	0.00	0.00
14,700.0	89.70	90.46	10,858.4	-521.6	4,005.7	4,028.9	0.00	0.00	0.00
14,800.0	89.70	90.46	10,858.9	-522.4	4,105.7	4,128.8	0.00	0.00	0.00
14,900.0	89.70	90.46	10,859.4	-523.2	4,205.7	4,228.6	0.00	0.00	0.00
15,000.0	89.70	90.46	10,859.9	-524.0	4,305.7	4,328.5	0.00	0.00	0.00
15,100.0	89.70	90.46	10,860.5	-524.8	4,405.7	4,428.4	0.00	0.00	0.00
15,200.0	89.70	90.46	10,861.0	-525.6	4,505.7	4,528.3	0.00	0.00	0.00
15,300.0	89.70	90.46	10,861.5	-526.4	4,605.7	4,628.2	0.00	0.00	0.00
15,400.0	89.70	90.46	10,862.0	-527.2	4,705.7	4,728.0	0.00	0.00	0.00
15,500.0	89.70	90.46	10,862.6	-528.0	4,805.7	4,827.9	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
15,600.0	89.70	90.46	10,863.1	-528.8	4,905.7	4,927.8	0.00	0.00	0.00
15,700.0	89.70	90.46	10,863.6	-529.7	5,005.7	5,027.7	0.00	0.00	0.00
15,800.0	89.70	90.46	10,864.1	-530.5	5,105.7	5,127.5	0.00	0.00	0.00
15,900.0	89.70	90.46	10,864.7	-531.3	5,205.7	5,227.4	0.00	0.00	0.00
16,000.0	89.70	90.46	10,865.2	-532.1	5,305.7	5,327.3	0.00	0.00	0.00
16,100.0	89.70	90.46	10,865.7	-532.9	5,405.7	5,427.2	0.00	0.00	0.00
16,200.0	89.70	90.46	10,866.2	-533.7	5,505.6	5,527.1	0.00	0.00	0.00
16,300.0	89.70	90.46	10,866.7	-534.5	5,605.6	5,626.9	0.00	0.00	0.00
16,400.0	89.70	90.46	10,867.3	-535.3	5,705.6	5,726.8	0.00	0.00	0.00
16,500.0	89.70	90.46	10,867.8	-536.1	5,805.6	5,826.7	0.00	0.00	0.00
16,600.0	89.70	90.46	10,868.3	-536.9	5,905.6	5,926.6	0.00	0.00	0.00
16,700.0	89.70	90.46	10,868.8	-537.7	6,005.6	6,026.5	0.00	0.00	0.00
16,800.0	89.70	90.46	10,869.4	-538.5	6,105.6	6,126.3	0.00	0.00	0.00
16,900.0	89.70	90.46	10,869.9	-539.3	6,205.6	6,226.2	0.00	0.00	0.00
17,000.0	89.70	90.46	10,870.4	-540.1	6,305.6	6,326.1	0.00	0.00	0.00
17,100.0	89.70	90.46	10,870.9	-540.9	6,405.6	6,426.0	0.00	0.00	0.00
17,200.0	89.70	90.46	10,871.5	-541.7	6,505.6	6,525.9	0.00	0.00	0.00
17,300.0	89.70	90.46	10,872.0	-542.5	6,605.6	6,625.7	0.00	0.00	0.00
17,400.0	89.70	90.46	10,872.5	-543.3	6,705.6	6,725.6	0.00	0.00	0.00
17,494.7	89.70	90.46	10,873.0	-544.1	6,800.3	6,820.2	0.00	0.00	0.00
Three Forks 3rd Bench									
17,500.0	89.70	90.46	10,873.0	-544.1	6,805.6	6,825.5	0.00	0.00	0.00
17,600.0	89.70	90.46	10,873.6	-544.9	6,905.6	6,925.4	0.00	0.00	0.00
17,700.0	89.70	90.46	10,874.1	-545.7	7,005.6	7,025.3	0.00	0.00	0.00
17,800.0	89.70	90.46	10,874.6	-546.5	7,105.6	7,125.1	0.00	0.00	0.00
17,900.0	89.70	90.46	10,875.1	-547.3	7,205.6	7,225.0	0.00	0.00	0.00
18,000.0	89.70	90.46	10,875.6	-548.1	7,305.6	7,324.9	0.00	0.00	0.00
18,100.0	89.70	90.46	10,876.2	-549.0	7,405.6	7,424.8	0.00	0.00	0.00
18,200.0	89.70	90.46	10,876.7	-549.8	7,505.6	7,524.7	0.00	0.00	0.00
18,300.0	89.70	90.46	10,877.2	-550.6	7,605.6	7,624.5	0.00	0.00	0.00
18,400.0	89.70	90.46	10,877.7	-551.4	7,705.5	7,724.4	0.00	0.00	0.00
18,500.0	89.70	90.46	10,878.3	-552.2	7,805.5	7,824.3	0.00	0.00	0.00
18,600.0	89.70	90.46	10,878.8	-553.0	7,905.5	7,924.2	0.00	0.00	0.00
18,700.0	89.70	90.46	10,879.3	-553.8	8,005.5	8,024.1	0.00	0.00	0.00
18,800.0	89.70	90.46	10,879.8	-554.6	8,105.5	8,123.9	0.00	0.00	0.00
18,900.0	89.70	90.46	10,880.4	-555.4	8,205.5	8,223.8	0.00	0.00	0.00
19,000.0	89.70	90.46	10,880.9	-556.2	8,305.5	8,323.7	0.00	0.00	0.00
19,100.0	89.70	90.46	10,881.4	-557.0	8,405.5	8,423.6	0.00	0.00	0.00
19,200.0	89.70	90.46	10,881.9	-557.8	8,505.5	8,523.5	0.00	0.00	0.00
19,300.0	89.70	90.46	10,882.5	-558.6	8,605.5	8,623.3	0.00	0.00	0.00
19,400.0	89.70	90.46	10,883.0	-559.4	8,705.5	8,723.2	0.00	0.00	0.00
19,500.0	89.70	90.46	10,883.5	-560.2	8,805.5	8,823.1	0.00	0.00	0.00
19,600.0	89.70	90.46	10,884.0	-561.0	8,905.5	8,923.0	0.00	0.00	0.00
19,700.0	89.70	90.46	10,884.5	-561.8	9,005.5	9,022.9	0.00	0.00	0.00
19,800.0	89.70	90.46	10,885.1	-562.6	9,105.5	9,122.7	0.00	0.00	0.00
19,900.0	89.70	90.46	10,885.6	-563.4	9,205.5	9,222.6	0.00	0.00	0.00
20,000.0	89.70	90.46	10,886.1	-564.2	9,305.5	9,322.5	0.00	0.00	0.00
20,100.0	89.70	90.46	10,886.6	-565.0	9,405.5	9,422.4	0.00	0.00	0.00
20,200.0	89.70	90.46	10,887.2	-565.8	9,505.5	9,522.3	0.00	0.00	0.00
20,300.0	89.70	90.46	10,887.7	-566.6	9,605.5	9,622.1	0.00	0.00	0.00
20,400.0	89.70	90.46	10,888.2	-567.4	9,705.5	9,722.0	0.00	0.00	0.00
20,500.0	89.70	90.46	10,888.7	-568.3	9,805.5	9,821.9	0.00	0.00	0.00
20,600.0	89.70	90.46	10,889.3	-569.1	9,905.4	9,921.8	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
20,700.0	89.70	90.46	10,889.8	-569.9	10,005.4	10,021.7	0.00	0.00	0.00
20,717.6	89.70	90.46	10,889.9	-570.0	10,023.0	10,039.2	0.00	0.00	0.00
TD at 20717.6									

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
- hit/miss target									
- Shape									
PBHL Kline Federal 530	0.00	0.00	10,890.0	-570.0	10,023.0	408,018.95	1,220,235.20	48° 4' 40.028 N	103° 33' 42.582 W
- plan misses target center by 0.1usft at 20717.6usft MD (10889.9 TVD, -570.0 N, 10023.0 E)									
- Point									

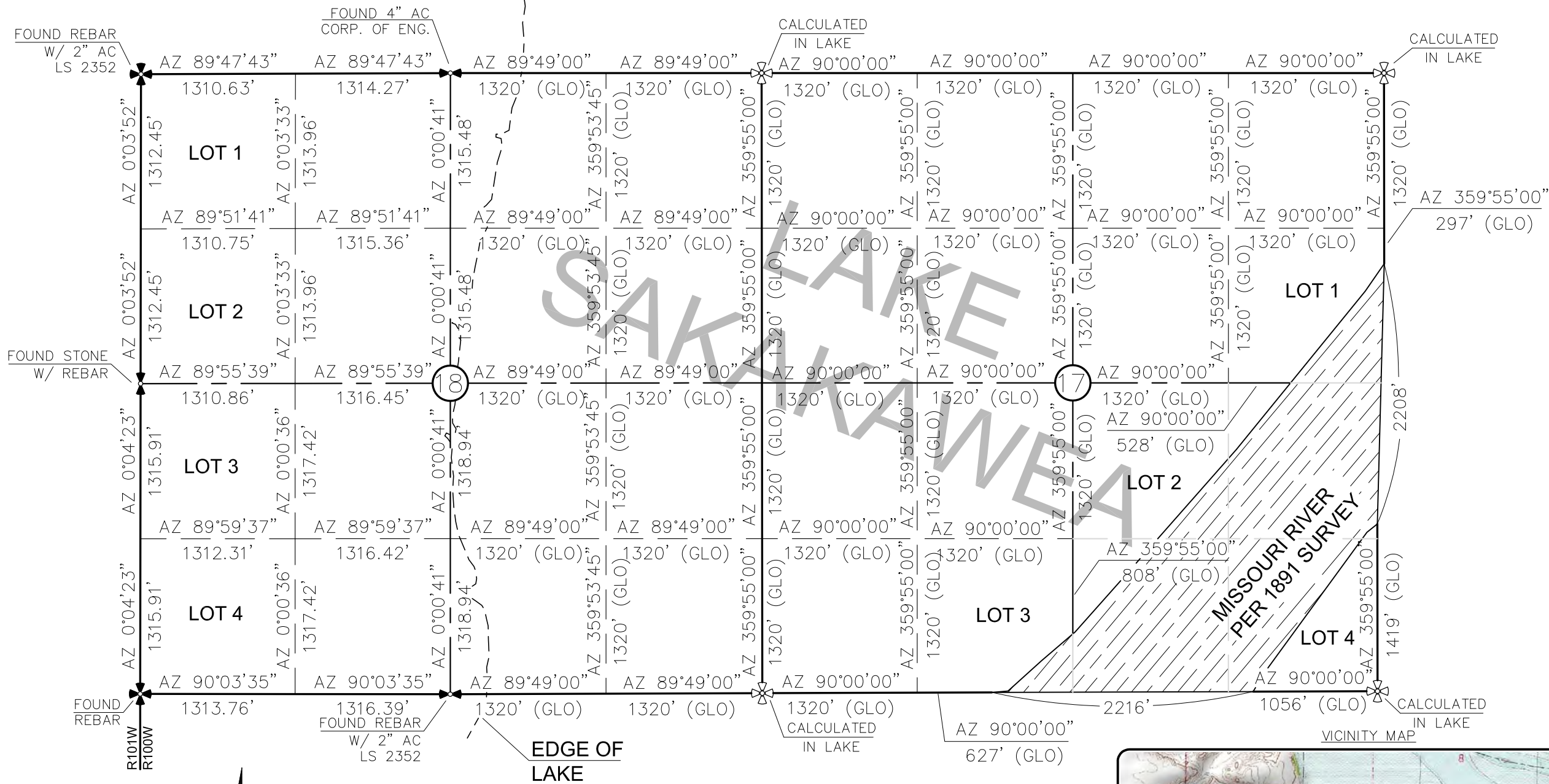
Casing Points					
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")	
2,068.0	2,068.0	13 3/8" Casing	13-3/8	17-1/2	
6,101.3	6,100.0	9 5/8" Casing	9-5/8	12-1/4	
11,111.0	10,839.5	7" Casing	7	8-3/4	

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,968.0	1,968.0	Pierre				
4,616.3	4,615.0	Greenhorn				
5,022.3	5,021.0	Mowry				
5,449.3	5,448.0	Dakota				
6,448.3	6,447.0	Rierdon				
6,786.3	6,785.0	Dunham Salt				
6,897.3	6,896.0	Dunham Salt Base				
6,994.3	6,993.0	Spearfish				
7,249.3	7,248.0	Pine Salt				
7,297.3	7,296.0	Pine Salt Base				
7,342.3	7,341.0	Opeche Salt				
7,372.3	7,371.0	Opeche Salt Base				
7,574.3	7,573.0	Broom Creek (Top of Minnelusa Gp.)				
7,654.3	7,653.0	Amsden				
7,822.3	7,821.0	Tyler				
8,013.3	8,012.0	Otter (Base of Minnelusa Gp.)				
8,368.3	8,367.0	Kibbey				
8,518.3	8,517.0	Charles Salt				
9,142.3	9,141.0	UB				
9,217.3	9,216.0	Base Last Salt				
9,265.3	9,264.0	Ratcliffe				
9,441.3	9,440.0	Mission Canyon				
10,003.3	10,002.0	Lodgepole				
10,209.3	10,208.0	Lodgepole Fracture Zone				
10,736.0	10,698.0	False Bakken				
10,750.3	10,708.0	Upper Bakken				
10,771.1	10,722.0	Middle Bakken				
10,846.8	10,767.0	Lower Bakken				
10,874.5	10,781.0	Pronghorn				
10,900.9	10,793.0	Three Forks 1st Bench				
10,963.0	10,816.0	Three Forks 1st Bench Claystone				
10,995.5	10,825.0	Three Forks 2nd Bench				
13,674.9	10,853.0	Three Forks 2nd Bench Claystone				
17,494.7	10,873.0	Three Forks 3rd Bench				

Plan Annotations					
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates			
		+N/-S (usft)	+E/-W (usft)	Comment	
2,500.0	2,500.0	0.0	0.0	Start Build 2.00	
2,650.0	2,649.9	0.0	3.9	Start 851.2 hold at 2650.0 MD	
3,501.2	3,500.0	0.0	48.5	Start DLS 2.00 TFO -180.00	
3,651.2	3,649.9	0.0	52.4	Start 6712.1 hold at 3651.2 MD	
10,363.3	10,362.0	0.0	52.4	Start Build 12.00	
11,110.8	10,839.5	-237.5	463.7	Start 15.0 hold at 11110.8 MD	
11,125.8	10,839.5	-245.0	476.7	Start DLS 3.00 TFO -90.08	
12,110.4	10,844.8	-500.8	1,416.3	Start 8607.1 hold at 12110.4 MD	
20,717.6	10,889.9	-570.0	10,023.0	TD at 20717.6	

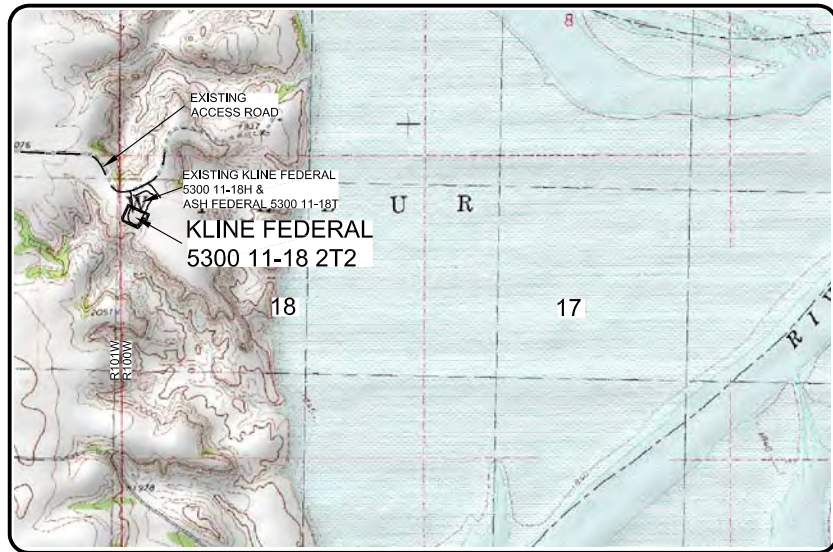
SECTION BREAKDOWN
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 2T2"
960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTIONS 17 & 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- ⊕ — MONUMENT — RECOVERED
⊗ — MONUMENT — NOT RECOVERED

THIS DOCUMENT WAS ORIGINALLY
ISSUED AND SEALED BY DARYL D.
KASEMAN, PLS, REGISTRATION NUMBER
3880 ON 8/29/14 AND THE
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ALL AZIMUTHS ARE BASED ON G.P.S.
OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
WAS 1891. THE CORNERS FOUND ARE AS
INDICATED AND ALL OTHERS ARE COMPUTED FROM
THOSE CORNERS FOUND AND BASED ON G.L.O.
DATA. THE MAPPING ANGLE FOR THIS AREA IS
APPROXIMATELY 0°03'.

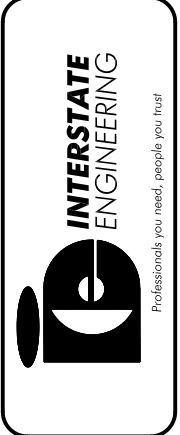


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Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS
REV 2	6/24/14	JJS	REVISED LATITUDE
REV 3	8/29/14	JJS	ADDED EXISTING PIT TO PAD

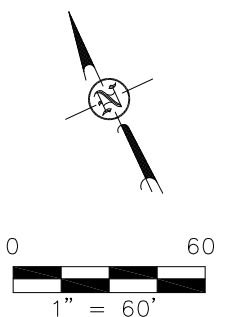
OASIS PETROLEUM NORTH AMERICA, LLC SECTION BREAKDOWN SECTIONS 17 & 18, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA	Project No.: S144-09-127 Date: APRIL 2014
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SHEET NO.

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 2T2"
960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



Q:\2014\S14-09-127 Oasis Petroleum - 1st Well of 4 on Quad Pad for Kline

WELL LOCATION SITE QUANTITIES
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 2T2"
960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

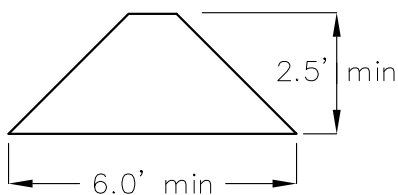
WELL SITE ELEVATION	2055.6
WELL PAD ELEVATION	2053.0
EXCAVATION	1,906
PLUS PIT	0
	1,906
EMBANKMENT	869
PLUS SHRINKAGE (30%)	261
	1,130
STOCKPILE PIT	0
STOCKPILE TOP SOIL (6")	1,934
BERMS	883 LF = 286 CY
DITCHES	727 LF = 111 CY
DETENTION AREA	1,112 CY
ADDITIONAL MATERIAL NEEDED	221
DISTURBED AREA FROM PAD	2.40 ACRES

NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
CUT END SLOPES AT 1:1
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION

960' FNL
318' FWL

BERM DETAIL



DITCH DETAIL



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OASIS PETROLEUM NORTH AMERICA, LLC
QUANTITIES
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.J.H. Project No.: S14-09-127
Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS
REV 2	6/24/14	JJS	REVISED LATITUDE
REV 3	8/29/14	JJS	ADDED EXISTING PIT TO PAD LAYOUT

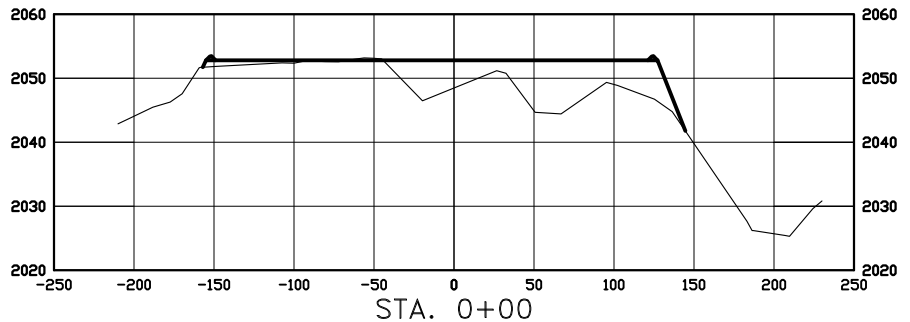
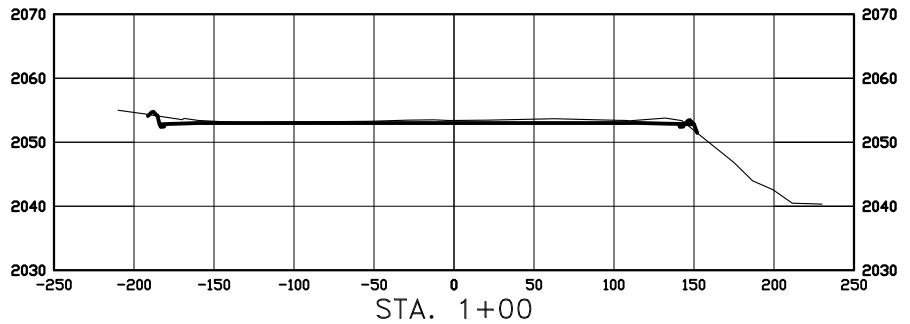
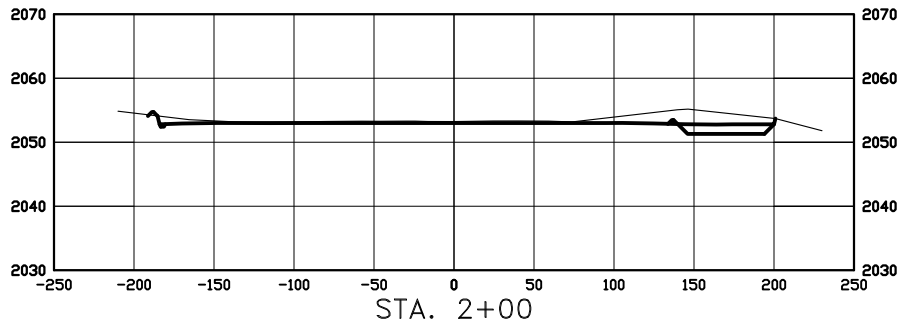
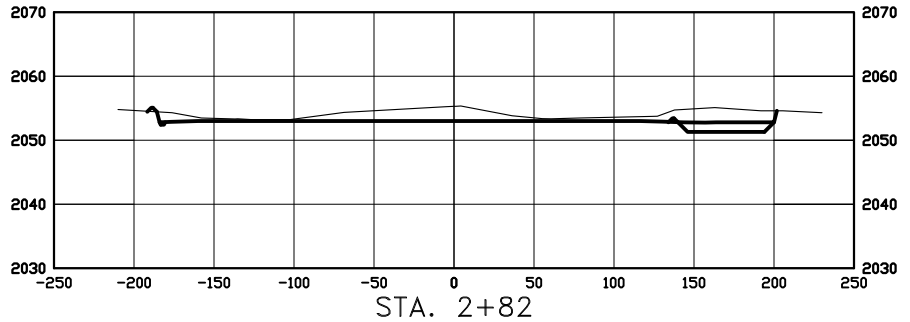
CROSS SECTIONS

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"KLINE FEDERAL 5300 11-18 2T2"

960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE

SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS
ORIGINALLY ISSUED AND
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PLS, REGISTRATION NUMBER
3880 ON 8/29/14 AND THE
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SCALE
HORIZ 1"=120'
VERT 1"=30'

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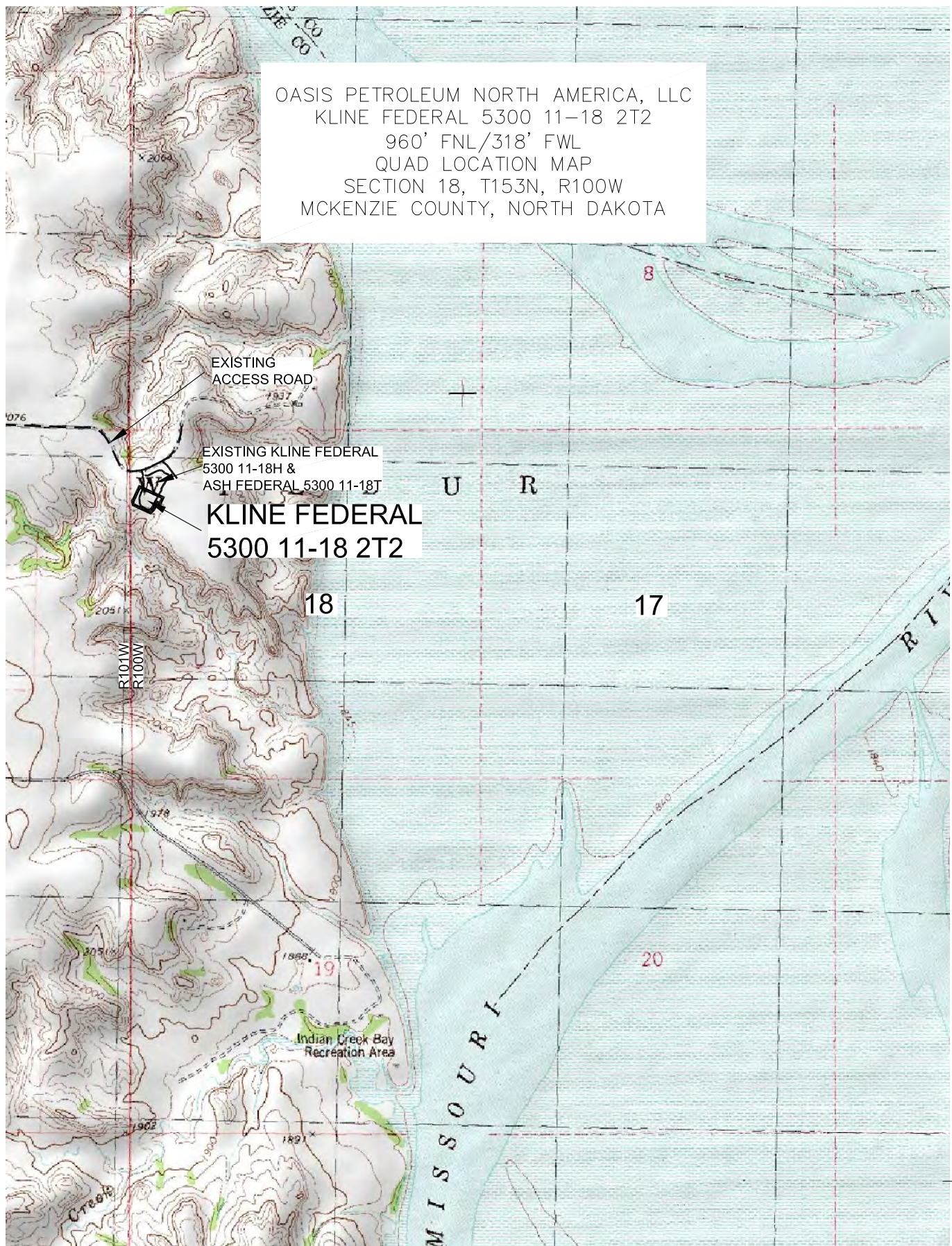


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OASIS PETROLEUM NORTH AMERICA, LLC
PAD CROSS SECTIONS
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: B.J.H. Project No.: S14-09-127
Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS
REV 2	6/24/14	JJS	REVISED LATITUDE
REV 3	8/29/14	JJS	ADDED EXISTING PIT TO PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
 KLINE FEDERAL 5300 11-18 2T2
 960' FNL/318' FWL
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



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OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127
 Checked By: D.D.K. Date: APRIL 2014

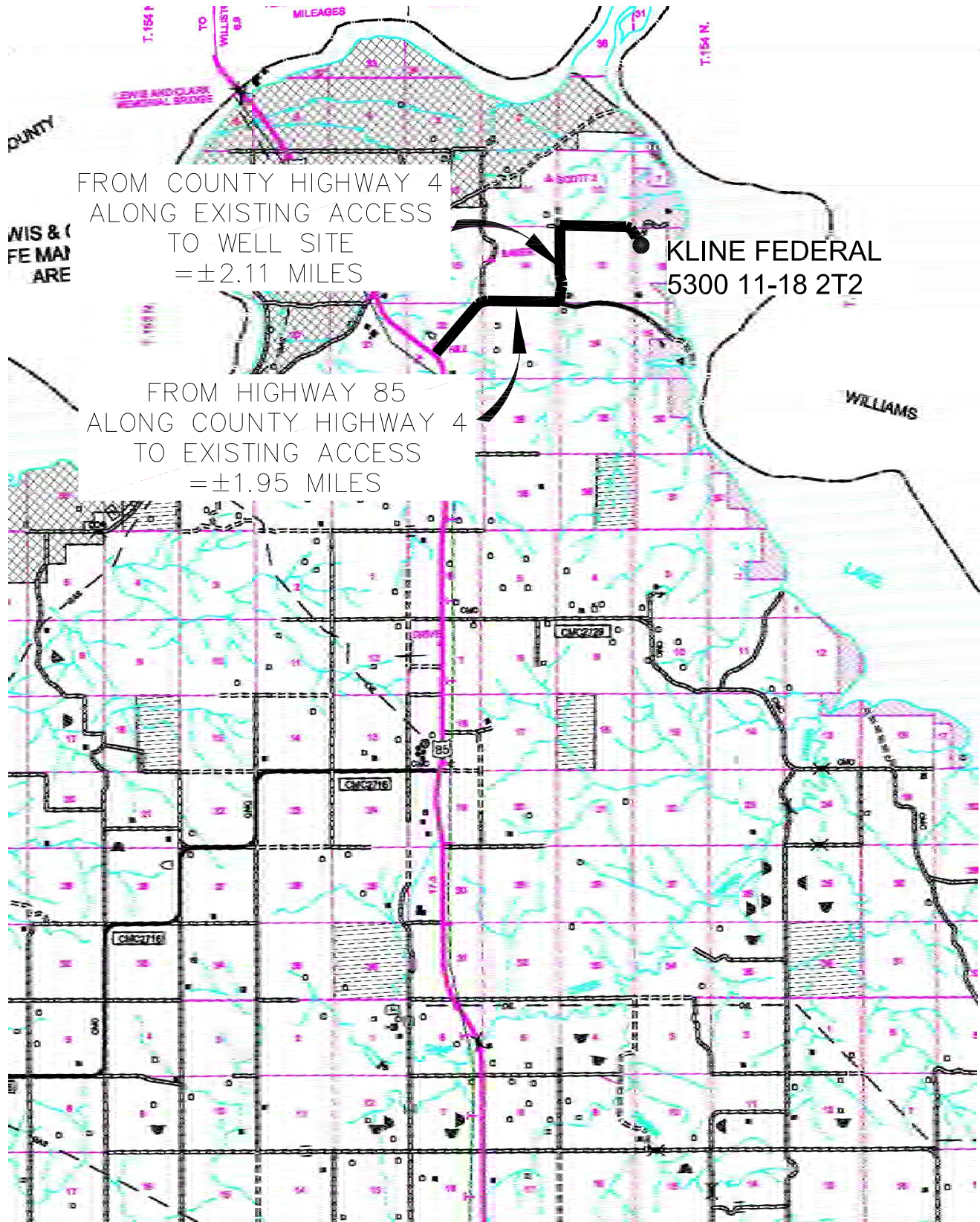
Revision No.	Date	By	Description
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REV 2	6/24/14	JJS	REVISED LATITUDE
REV 3	8/29/14	JJS	ADDED EXISTING PIT TO PAD LAYOUT

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"KLINE FEDERAL 5300 11-18 2T2"

960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127
Checked By: D.D.K. Date: APRIL 2014

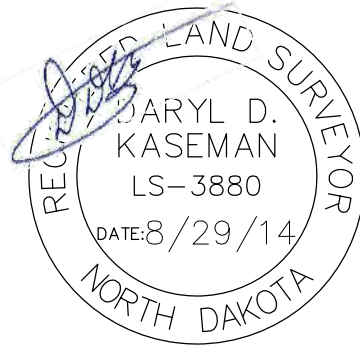
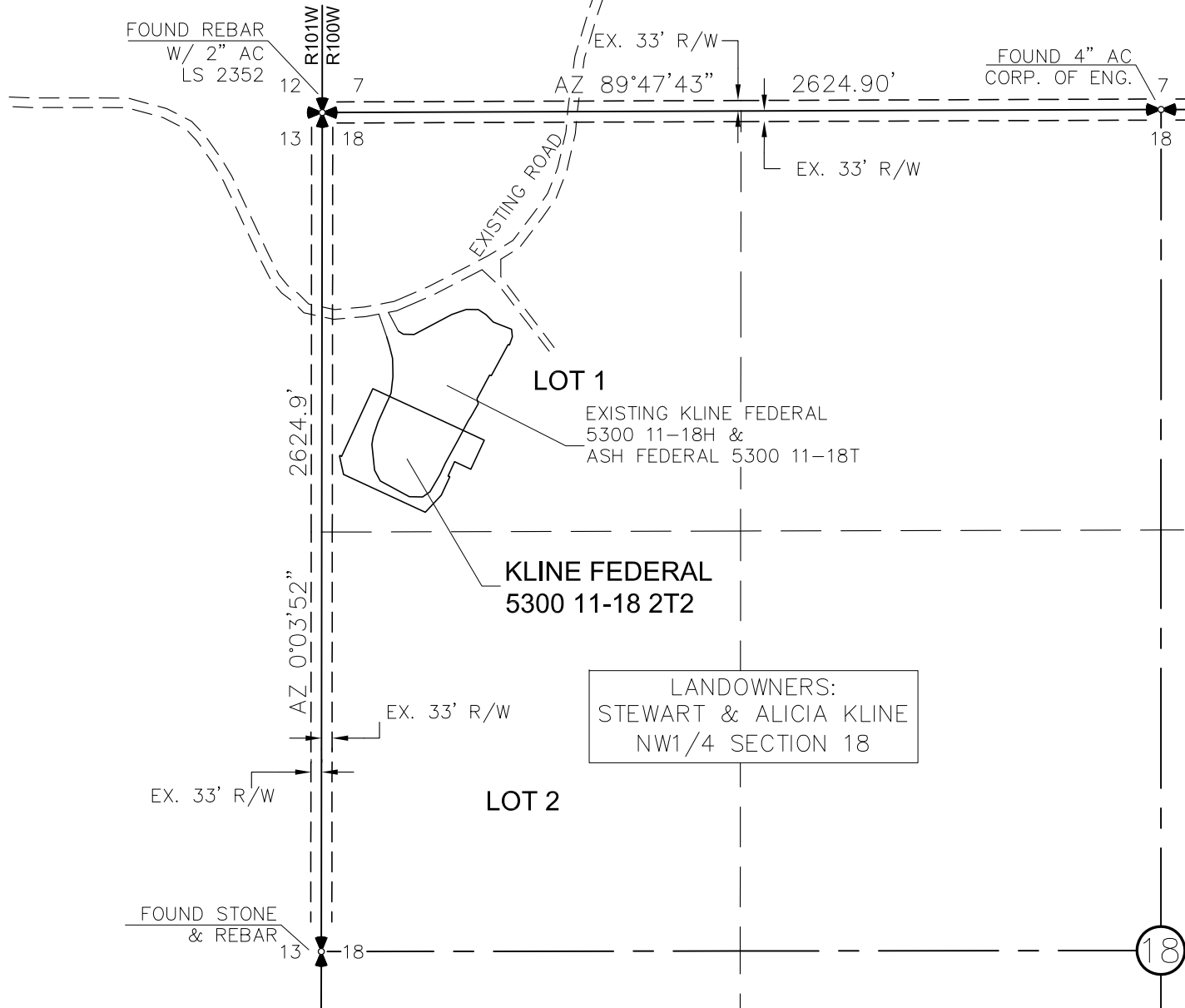
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REV 2	6/24/14	JJS	REVISED LATITUDE
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ACCESS APPROACH

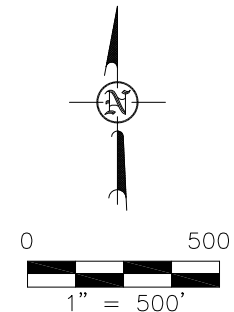
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"KLINE FEDERAL 5300 11-18 2T2"

960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS
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NUMBER 3880 ON 8/29/14
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NOTE: All utilities shown are preliminary only, a complete
utilities location is recommended before construction.

Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS
REV 2	6/24/14	JS	REVISED LATITUDE
REV 3	8/29/14	JS	ADDED EXISTING PIT TO PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
ACCESS APPROACH
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: B.J.H.
Checked By: D.J.K.
Project No.: S14-004127
Date: APRIL 2014

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GAS CAPTURE PLAN AFFIDAVIT

STATE OF TEXAS §
 §
COUNTY OF HARRIS §

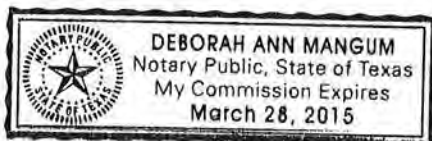
Robert Eason, being duly sworn, states as follows:

1. He is employed by Oasis Petroleum North America LLC ("Oasis") as Marketing Manager, is over the age of 21 and has personal knowledge of the matters set forth in this affidavit.
2. This affidavit is submitted in conjunction with the Application for Permit to Drill for the Kline Federal 5300 11-18 2T2 well, with a surface location in the NW NW of Section 18, Township 153 North, Range 100 West, McKenzie County, North Dakota (the "Well").
3. Oasis currently anticipates that gas to be produced from the Well will be gathered by Hiland Partners (the "Gathering Company"). Oasis has advised the Gathering Company of its intent to drill the Well and has advised the Gathering Company that it currently anticipates that the Well will be completed in May 2015, with an initial gas production rate of approximately 579 mcf/day.



Robert Eason
Marketing Manager

Subscribed and sworn to before me this 24 day of June, 2014.





Notary Public in and for the State of Texas
My Commission expires: 3-28-2015

GAS CAPTURE PLAN – OASIS PETROLEUM**Kline Federal 5300 11-18 2T2**

Section 18-T153N-R100W
Baker Field
McKenzie County, North Dakota

Anticipated first flow date May-15
Gas Gatherer: Hiland Partners
Gas to be processed at*: Hiland Operated Watford City Plant

Maximum Daily Capacity of Existing Gas Line*:	55,000 MCFD
Current Throughput of Existing Gas Line*:	33,000 MCFD
Anticipated Daily Capacity of Existing Gas Line at Date of First Gas Sales*:	66,000 MCFD
Anticipated Throughput of Existing Gas Line at Date of First Gas Sales*:	65,000 MCFD
Gas Gatherer's Issues or Expansion Plans for the Area*:	Line looping and compression

Map: Attached
Affidavit: Attached

*Provided by Gatherer

Flowback Strategy

Total Number of Wells at Location: 10

Multi-Well Start-up Plan: Initial production from the 1st new well at the CTB is anticipated in May 2015 with each following well making 1st production approximately every 5th day thereafter

Estimated Flow Rate:	Kline Federal 5300 11-18 2T2 (well)		Kline DSU (10 wells)	
	MCFD	BOPD	MCFD	BOPD
30 Days:	579	643	5,100	5,646
60 Days:	497	552	6,302	6,982
180 Days:	297	330	3,543	3,918

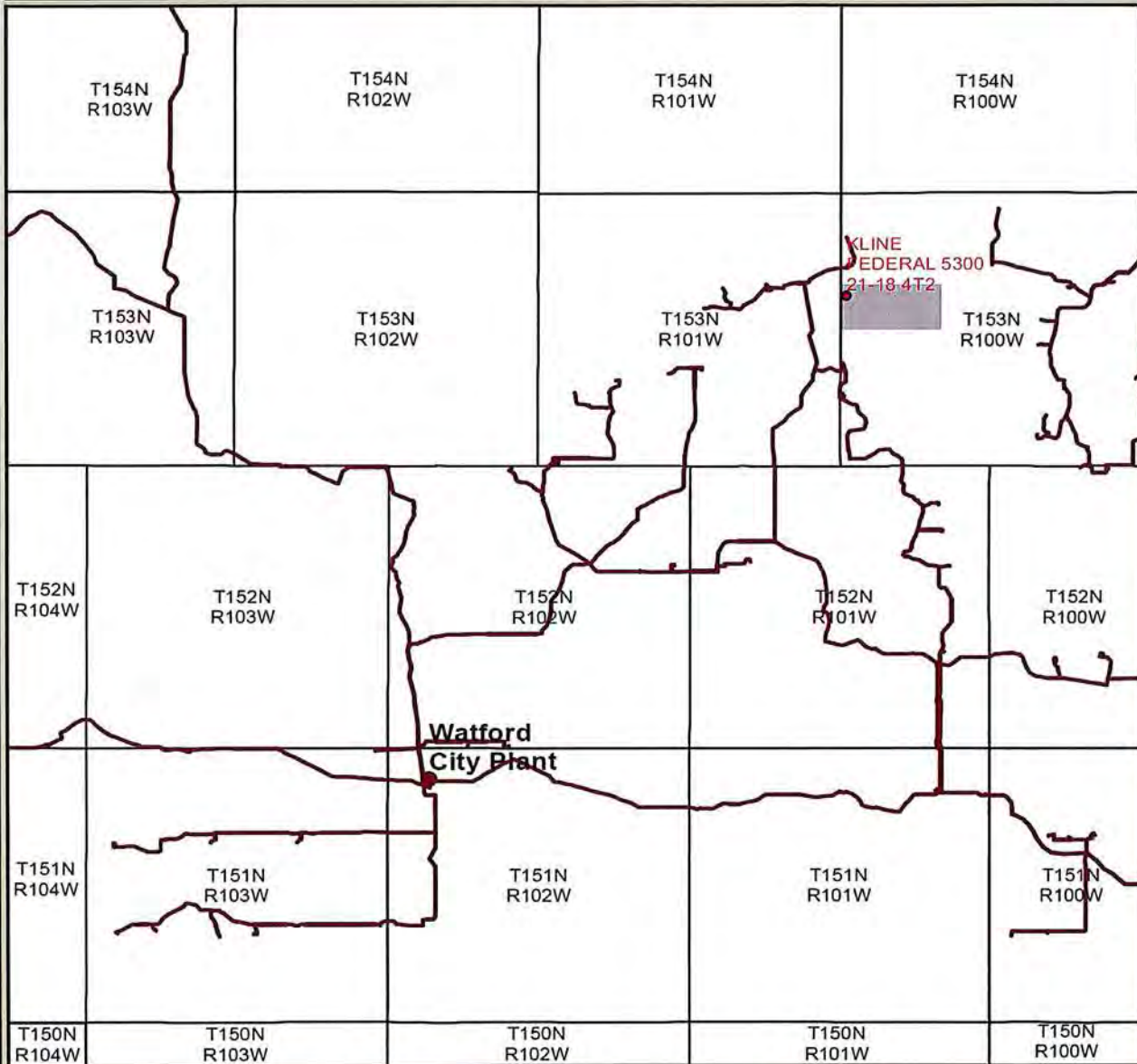
Oasis Flaring Percentage

	Statewide	Baker Field
Oasis % of Gas Flared:	12%	6%

Average over the last 6 months

Alternatives to Flaring

Gas Capture Plan - Overview
KLINE FEDERAL 5300 21-18 4T2
Section 18 T153N R100W
McKenzie County, North Dakota

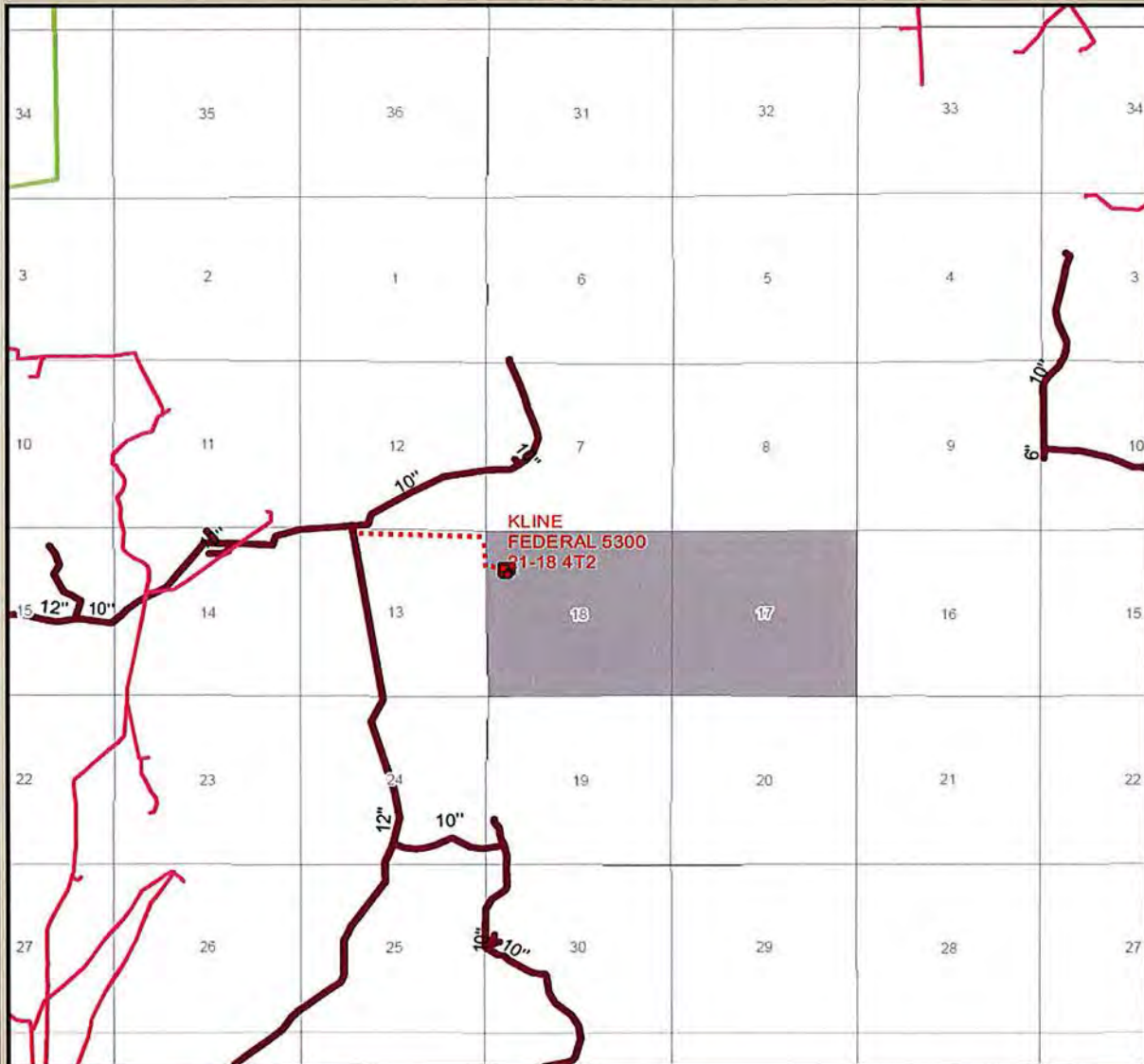


- Proposed Well
- Hiland Gas Line
- Processing Plant

Gas Gatherer: Hiland Partners, LP
Gas to be processed at: Watford City Plant



Gas Capture Plan
KLINE FEDERAL 5300 21-18 4T2
Section 18 T153N R100W
McKenzie County, North Dakota



Gas Gatherer: Hiland Partners, LP
 Gas to be processed at: Watford City Plant

- Proposed Well
- ▭ Proposed CTB
- Hiland Gas Line
- Oneok Gas Line
- Williston Basin Interstate





8/20/2014

Mineral Resources Permit Manager
North Dakota Industrial Commission
600 East Boulevard Avenue Dept. 405
Bismarck, ND 58505-0840

RE: Kline Federal 5300 11-18 2T2
Kline Federal 5300 11-18 3T
Kline Federal 5300 11-18 4T2
Kline Federal 5300 11-18 5B
Request for a legal street address

Dear NDIC:

Oasis Petroleum has requested a physical street address for the Kline Federal 5300 11-18 2T2, Kline Federal 5300 11-18 3T, Kline Federal 5300 11-18 4T2 and Kline Federal 5300 11-18 5B. The request was made to Aaron Chisolm (address@co.mckenzie.nd.us) in McKenzie County. Upon receiving a legal street address, Oasis will submit the address to the NDIC on a Sundry Notice (form 4) pursuant to 43-02-03-28.

Thank you for your consideration.

Respectfully,

A handwritten signature in blue ink that reads "Heather McCowan". The signature is fluid and cursive, with the first name "Heather" and last name "McCowan" clearly distinguishable.

Heather McCowan
Regulatory Assistant
Oasis Petroleum North America, LLC

Hello Taylor,

They will be hauled to the JMAC Resources Disposal
5009 139th Ave NW, Williston, ND 58801
(701) 774-8511

Thanks,

Heather McCowan

Regulatory Assistant | 1001 Fannin, Suite 1500, Houston, Texas 77002 | 281-404-9563 Direct |
hmccowan@oasispetroleum.com



From: Roth, Taylor J. [<mailto:tjroth@nd.gov>]
Sent: Wednesday, August 20, 2014 9:59 AM
To: Heather McCowan
Subject: RE: Kline Federal pad

Heather,

What will Oasis be doing with the cuttings on this pad?

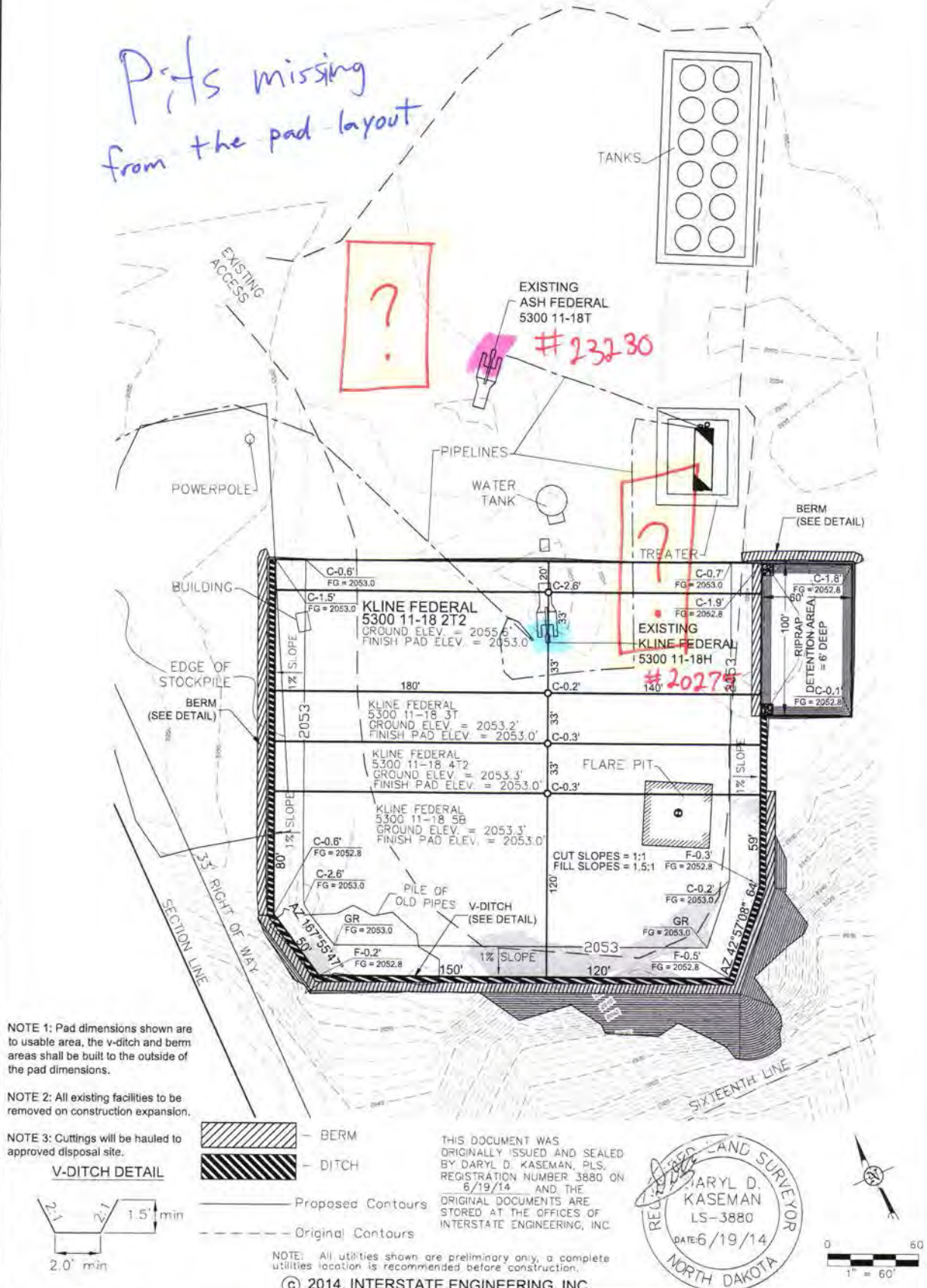
Thank you very much,

Taylor J. Roth
Survey & Permitting Technician
NDIC, Dept. Mineral Resources
Oil and Gas Division
701-328-1720 (direct)
tjroth@nd.gov



PAD LAYOUT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 11-18 2T2"
 960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

*Pits missing
from the pad layout*



3/8
SHEET NO.



Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 PH (406) 433-5617
 FAX (406) 433-5618
 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 PAD LAYOUT
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: S.B.H. Project No.: 314-09-127
 Checked By: S.B.H. Date: APRIL 2014

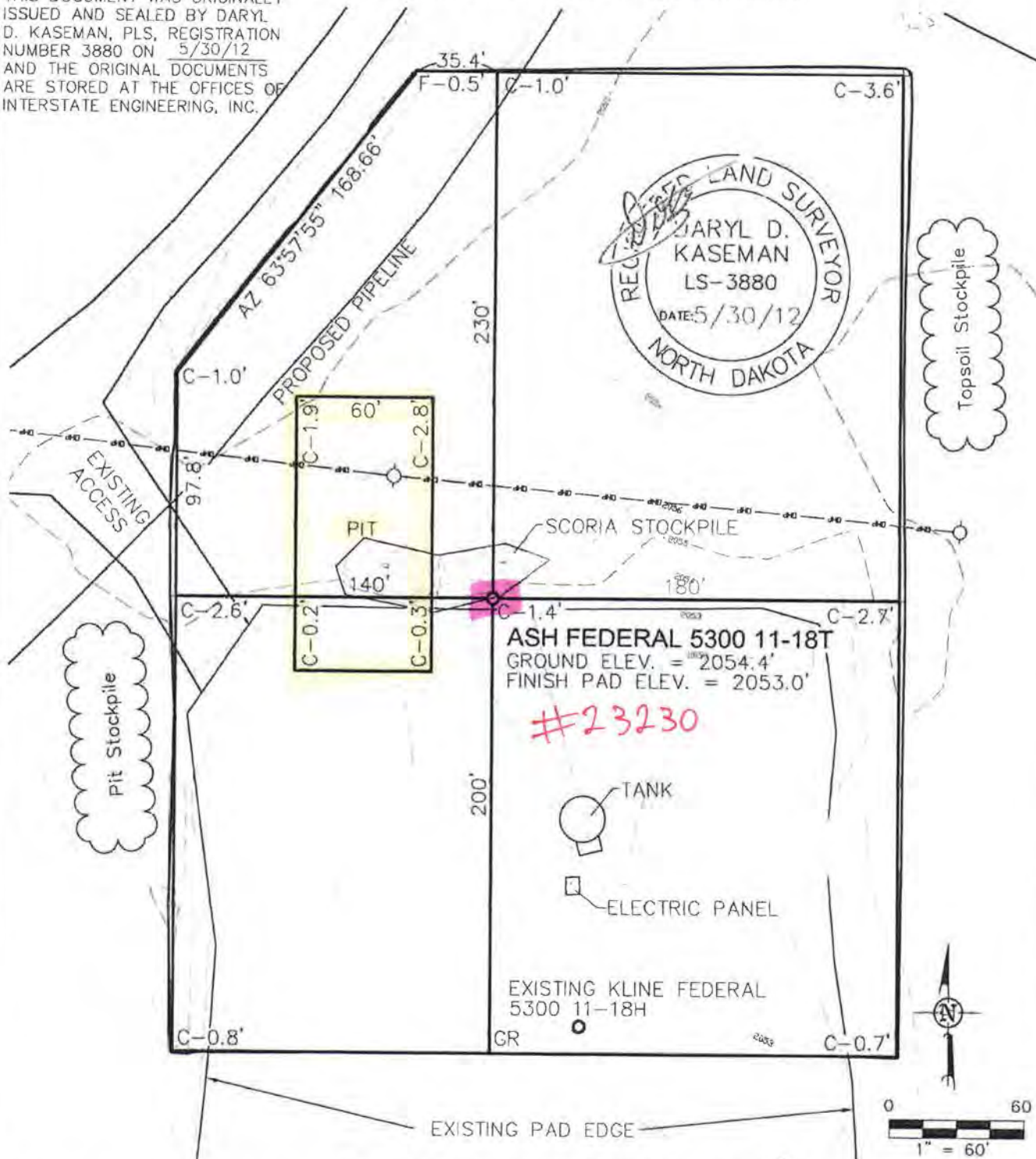
Revision No.	Date	By	Description

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"ASH FEDERAL 5300 11-18T"

800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY
ISSUED AND SEALED BY DARYL
D. KASEMAN, PLS, REGISTRATION
NUMBER 3880 ON 5/30/12
AND THE ORIGINAL DOCUMENTS
ARE STORED AT THE OFFICES OF
INTERSTATE ENGINEERING, INC.



ASH FEDERAL 5300 11-18T
GROUND ELEV. = 2054.4'
FINISH PAD ELEV. = 2053.0'

#23230

EXISTING KLINE FEDERAL
5300 11-18H

EXISTING PAD EDGE

NOTE: All utilities shown are preliminary only, a complete
utilities location is recommended before construction.

© 2012, INTERSTATE ENGINEERING, INC.

3/8

**INTERSTATE
ENGINEERING**
Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengi.com

Other (Not in Montana, North Dakota and South Dakota)

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: BJHJ Project No: S1200-145
Checked By: D.J.K. Date: MAY 2012

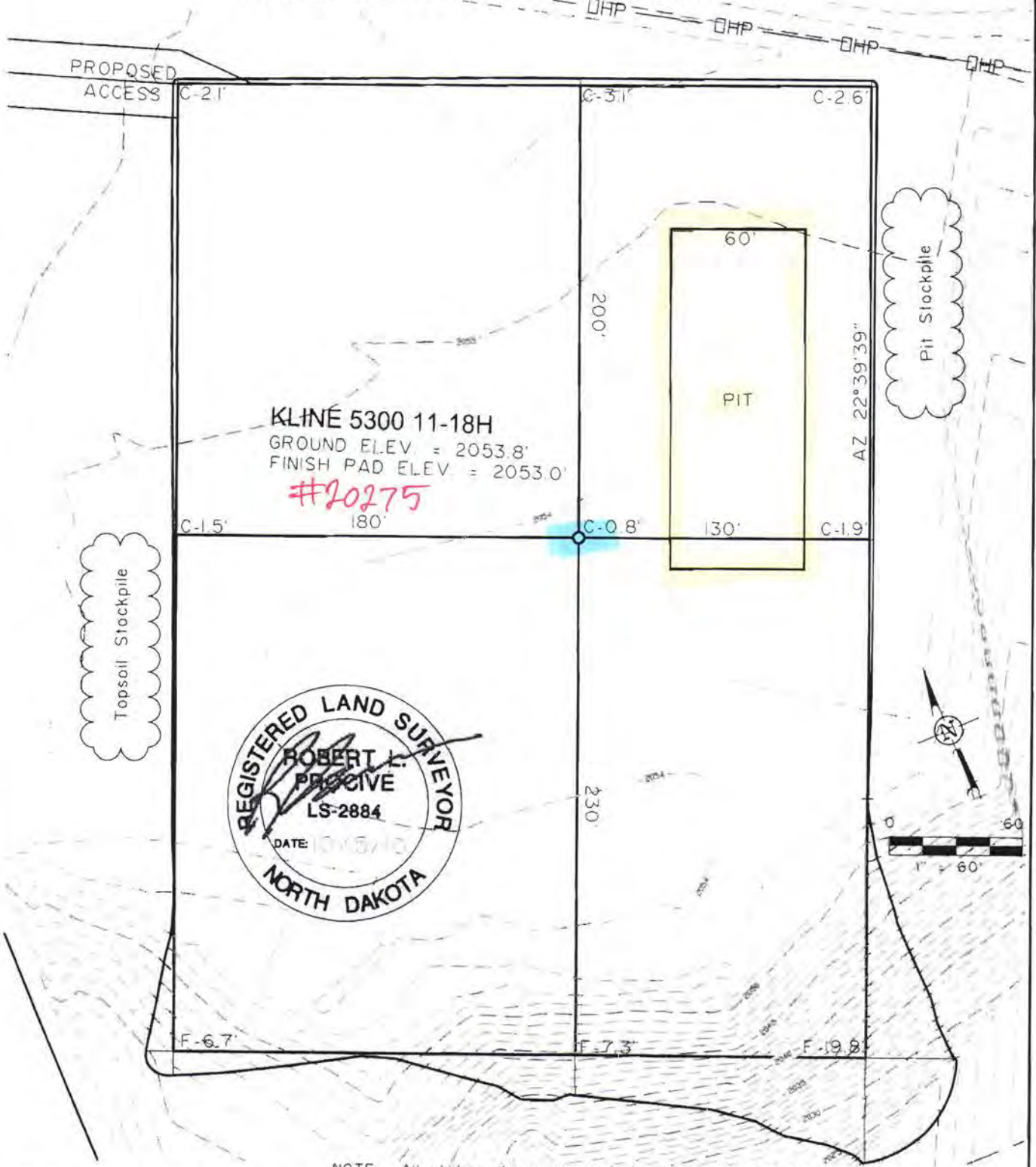
Revised	Date	By	Description

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 202 HOUSTON, TX 77002

"KLINE 5300 11-18H"

990 FEET FROM NORTH LINE AND 305 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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3

SHEET NO.



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Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 548
425 East Main Street
Sikeston, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengl.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S10-9-190
Checked By: A.J.H./R.L.P. Date: OCT. 2010

Revision	Date	By	Description



Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 11-18 2T2

Wellbore #1

Design #6

Anticollision Report

30 June, 2014



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Reference	Design #6		
Filter type:	NO GLOBAL FILTER: Using user defined selection & filtering criteria		
Interpolation Method:	Stations	Error Model:	ISCWSA
Depth Range:	Unlimited	Scan Method:	Closest Approach 3D
Results Limited by:	Maximum center-center distance of 2,000.0 usft	Error Surface:	Elliptical Conic
Warning Levels Evaluated at:	2.00 Sigma	Casing Method:	Not applied

Survey Tool Program		Date	06/30/14		
From (usft)	To (usft)	Survey (Wellbore)	Tool Name	Description	
0.0	20,717.6	Design #6 (Wellbore #1)	MWD	MWD - Standard	

Summary						
Site Name	Reference Measured Depth (usft)	Offset Measured Depth (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Separation Factor	Warning
Offset Well - Wellbore - Design						
153N-100W-17/18						
Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1	2,845.8	2,817.8	155.5	148.0	20.755	CC
Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1	2,900.0	2,871.7	155.6	147.9	20.139	ES
Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1	20,600.0	20,419.4	755.3	220.7	1.413	Level 3, SF
Kline Federal 5300 11-18 3T - Wellbore #1 - Design #6	2,500.0	2,500.0	65.7	54.7	5.993	CC
Kline Federal 5300 11-18 3T - Wellbore #1 - Design #6	20,717.6	21,035.1	500.4	-84.1	0.856	Level 1, ES, SF
Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1	2,132.6	2,108.5	33.1	28.4	7.073	CC
Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1	18,241.7	18,315.6	95.3	-200.2	0.323	Level 1, SF
Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1	18,300.0	18,372.4	95.8	-200.5	0.323	Level 1, ES
Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6	2,500.0	2,500.0	99.2	88.3	9.057	CC, ES
Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6	20,717.6	21,371.1	1,000.2	415.8	1.711	SF
Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6	2,500.0	2,500.0	131.6	120.7	12.011	CC, ES
Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6	20,717.6	21,195.2	1,003.3	421.8	1.725	SF

Offset Design		153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1											Offset Site Error:		0.0 usft		
Survey Program:		2261-MWD, 13302-MWD											Offset Well Error:		0.0 usft		
Reference		Offset		Semi Major Axis			Distance							Warning			
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor					
							+N/-S (usft)	+E/-W (usft)									
	300.0	300.0	2,180.0	2,183.0	0.5	0.0	11.34	159.1	31.9	1,913.9	1,913.4	0.53	3,583.932				
	400.0	400.0	2,180.0	2,183.0	0.8	0.0	11.34	159.1	31.9	1,814.3	1,813.5	0.76	2,391.017				
	500.0	500.0	2,180.0	2,183.0	1.0	0.0	11.34	159.1	31.9	1,714.7	1,713.7	0.98	1,743.369				
	600.0	600.0	2,180.0	2,183.0	1.2	0.0	11.34	159.1	31.9	1,615.2	1,614.0	1.21	1,336.709				
	700.0	700.0	2,180.0	2,183.0	1.4	0.0	11.34	159.1	31.9	1,515.7	1,514.3	1.43	1,057.655				
	800.0	800.0	2,180.0	2,183.0	1.7	0.0	11.34	159.1	31.9	1,416.3	1,414.7	1.66	854.313				
	900.0	900.0	2,180.0	2,183.0	1.9	0.0	11.34	159.1	31.9	1,317.0	1,315.1	1.88	699.575				
	1,000.0	1,000.0	2,180.0	2,183.0	2.1	0.0	11.34	159.1	31.9	1,217.9	1,215.7	2.11	577.900				
	1,100.0	1,100.0	2,180.0	2,183.0	2.3	0.0	11.34	159.1	31.9	1,118.8	1,116.5	2.33	479.741				
	1,200.0	1,200.0	2,180.0	2,183.0	2.6	0.0	11.34	159.1	31.9	1,020.0	1,017.4	2.56	398.913				
	1,300.0	1,300.0	2,180.0	2,183.0	2.8	0.0	11.34	159.1	31.9	921.4	918.6	2.78	331.237				
	1,400.0	1,400.0	2,180.0	2,183.0	3.0	0.0	11.34	159.1	31.9	823.1	820.1	3.01	273.794				
	1,500.0	1,500.0	2,180.0	2,183.0	3.2	0.0	11.34	159.1	31.9	725.4	722.1	3.23	224.491				
	1,600.0	1,600.0	2,180.0	2,183.0	3.5	0.0	11.34	159.1	31.9	628.3	624.9	3.46	181.803				
	1,700.0	1,700.0	2,180.0	2,183.0	3.7	0.0	11.34	159.1	31.9	532.3	528.6	3.68	144.625				
	1,800.0	1,800.0	2,180.0	2,183.0	3.9	0.0	11.34	159.1	31.9	438.1	434.2	3.91	112.187				

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1													Offset Site Error:	0.0 usft
Survey Program: 2261-MWD, 13302-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
1,900.0	1,900.0	2,180.0	2,183.0	4.1	0.0	11.34	159.1	31.9	347.2	343.1	4.13	84.070		
2,000.0	2,000.0	2,180.0	2,183.0	4.4	0.0	11.34	159.1	31.9	263.0	258.6	4.36	60.391		
2,100.0	2,100.0	2,180.0	2,183.0	4.6	0.0	11.34	159.1	31.9	194.3	189.8	4.58	42.436		
2,200.0	2,200.0	2,180.0	2,183.0	4.8	0.0	11.34	159.1	31.9	162.4	157.6	4.80	33.800		
2,300.0	2,300.0	2,276.1	2,279.1	5.0	0.1	11.47	157.9	32.0	161.2	156.0	5.15	31.270		
2,400.0	2,400.0	2,375.4	2,378.3	5.3	0.3	11.60	156.2	32.1	159.5	153.9	5.59	28.549		
2,500.0	2,500.0	2,474.6	2,477.5	5.5	0.5	11.55	155.0	31.7	158.2	152.2	6.02	26.270		
2,600.0	2,600.0	2,573.5	2,576.5	5.7	0.8	-79.29	154.6	31.0	157.3	150.9	6.45	24.406		
2,650.0	2,649.9	2,623.3	2,626.3	5.8	0.9	-80.27	154.5	30.5	156.8	150.1	6.66	23.558		
2,700.0	2,699.9	2,673.2	2,676.1	5.9	1.0	-81.48	154.6	29.8	156.3	149.4	6.87	22.755		
2,800.0	2,799.7	2,772.5	2,775.5	6.1	1.2	-83.99	154.7	28.1	155.6	148.3	7.30	21.326		
2,845.8	2,845.4	2,817.8	2,820.8	6.2	1.3	-85.17	154.9	27.3	155.5	148.0	7.49	20.755 CC		
2,900.0	2,899.6	2,871.7	2,874.6	6.3	1.4	-86.58	155.3	26.3	155.6	147.9	7.73	20.139 ES		
3,000.0	2,999.5	2,970.8	2,973.7	6.5	1.6	-89.21	156.2	24.3	156.2	148.1	8.16	19.144		
3,100.0	3,099.3	3,070.2	3,073.0	6.7	1.9	-91.90	157.5	22.1	157.6	149.0	8.59	18.342		
3,200.0	3,199.2	3,170.0	3,172.8	7.0	2.1	-94.47	158.9	20.2	159.4	150.4	9.01	17.692		
3,300.0	3,299.0	3,270.0	3,272.8	7.2	2.2	-96.92	160.1	18.4	161.3	151.9	9.42	17.130		
3,400.0	3,398.9	3,369.4	3,372.2	7.4	2.5	-99.25	161.4	16.7	163.6	153.7	9.84	16.625		
3,501.2	3,500.0	3,470.6	3,473.3	7.6	2.7	-101.52	162.8	15.1	166.2	155.9	10.27	16.179		
3,600.0	3,598.7	3,569.4	3,572.1	7.8	2.9	-103.15	164.0	13.6	168.5	157.8	10.68	15.781		
3,651.2	3,649.9	3,620.8	3,623.6	7.9	3.0	-13.52	164.6	12.8	169.3	158.5	10.84	15.617		
3,700.0	3,698.7	3,669.8	3,672.5	8.0	3.1	-13.71	165.1	12.1	170.0	158.9	11.05	15.389		
3,800.0	3,798.7	3,769.5	3,772.3	8.2	3.3	-14.08	166.2	10.7	171.3	159.9	11.48	14.928		
3,900.0	3,898.7	3,869.6	3,872.3	8.5	3.5	-14.41	167.3	9.4	172.8	160.9	11.92	14.500		
4,000.0	3,998.7	3,970.0	3,972.7	8.7	3.7	-14.69	168.3	8.3	174.0	161.7	12.35	14.089		
4,100.0	4,098.7	4,069.9	4,072.6	8.9	3.9	-14.93	169.3	7.3	175.2	162.4	12.79	13.702		
4,200.0	4,198.7	4,169.8	4,172.5	9.1	4.1	-15.15	170.3	6.3	176.4	163.2	13.23	13.340		
4,300.0	4,298.7	4,270.2	4,272.9	9.4	4.4	-15.33	171.3	5.5	177.6	164.0	13.66	13.000		
4,400.0	4,398.7	4,370.8	4,373.4	9.6	4.6	-15.44	172.0	4.9	178.5	164.4	14.10	12.660		
4,500.0	4,498.7	4,470.9	4,473.5	9.8	4.8	-15.52	172.6	4.5	179.1	164.6	14.53	12.328		
4,600.0	4,598.7	4,570.6	4,573.3	10.0	5.0	-15.60	173.2	4.1	179.8	164.9	14.96	12.016		
4,700.0	4,698.7	4,669.9	4,672.5	10.2	5.2	-15.69	174.0	3.5	180.8	165.4	15.40	11.741		
4,800.0	4,798.7	4,769.5	4,772.2	10.5	5.4	-15.81	175.2	2.8	182.1	166.3	15.83	11.501		
4,900.0	4,898.7	4,870.2	4,872.8	10.7	5.6	-15.78	176.4	2.5	183.3	167.1	16.27	11.267		
5,000.0	4,998.7	4,970.4	4,973.0	10.9	5.8	-15.66	177.5	2.7	184.3	167.6	16.71	11.031		
5,100.0	5,098.7	5,071.0	5,073.7	11.1	6.0	-15.44	178.4	3.1	185.0	167.9	17.14	10.794		
5,200.0	5,198.7	5,172.0	5,174.6	11.4	6.3	-15.22	178.9	3.8	185.4	167.8	17.57	10.548		
5,300.0	5,298.7	5,273.3	5,276.0	11.6	6.5	-15.18	178.6	3.9	185.0	167.0	18.00	10.280		
5,400.0	5,398.7	5,373.6	5,376.2	11.8	6.7	-15.09	177.8	4.5	184.1	165.7	18.43	9.990		
5,500.0	5,498.7	5,472.4	5,475.0	12.0	6.9	-15.04	177.4	4.8	183.7	164.8	18.86	9.739		
5,540.6	5,539.3	5,512.7	5,515.3	12.1	7.0	-15.02	177.4	4.8	183.6	164.6	19.03	9.648		
5,600.0	5,598.7	5,571.6	5,574.2	12.2	7.1	-15.00	177.5	4.9	183.7	164.4	19.29	9.525		
5,700.0	5,698.7	5,671.6	5,674.2	12.5	7.3	-15.03	177.8	4.7	184.1	164.4	19.72	9.337		
5,800.0	5,798.7	5,771.2	5,773.8	12.7	7.5	-15.19	177.9	4.1	184.4	164.2	20.15	9.150		
5,900.0	5,898.7	5,870.8	5,873.4	12.9	7.7	-15.40	178.4	3.3	185.0	164.4	20.58	8.990		
6,000.0	5,998.7	5,970.5	5,973.1	13.1	7.9	-15.52	179.0	2.7	185.8	164.8	21.02	8.839		
6,100.0	6,098.7	6,069.7	6,072.3	13.4	8.1	-15.53	180.0	2.4	186.8	165.4	21.46	8.708		
6,200.0	6,198.7	6,168.5	6,171.0	13.6	8.3	-15.47	181.6	2.2	188.4	166.5	21.89	8.608		
6,300.0	6,298.7	6,267.2	6,269.7	13.8	8.6	-15.25	183.9	2.3	190.7	168.4	22.33	8.540		
6,400.0	6,398.7	6,367.6	6,370.1	14.0	8.8	-14.95	186.6	2.6	193.2	170.4	22.77	8.486		
6,500.0	6,498.7	6,466.8	6,469.3	14.3	9.0	-14.49	189.5	3.4	195.8	172.6	23.20	8.437		
6,600.0	6,598.7	6,565.9	6,568.3	14.5	9.2	-13.89	192.8	4.7	198.8	175.1	23.64	8.406		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1													Offset Site Error: 0.0 usft
Survey Program: 2261-MWD, 13302-MWD													Offset Well Error: 0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
6,700.0	6,698.7	6,665.4	6,667.7	14.7	9.4	-13.17	196.9	6.4	202.3	178.2	24.08	8.400	
6,800.0	6,798.7	6,768.0	6,770.2	14.9	9.6	-12.85	199.8	6.8	204.9	180.4	24.52	8.359	
6,900.0	6,898.7	6,867.6	6,869.8	15.2	9.8	-12.83	201.9	6.4	207.1	182.2	24.95	8.302	
7,000.0	6,998.7	6,967.0	6,969.2	15.4	10.0	-12.70	204.5	6.3	209.7	184.3	25.38	8.262	
7,100.0	7,098.7	7,067.2	7,069.3	15.6	10.3	-12.51	207.2	6.4	212.3	186.4	25.82	8.221	
7,200.0	7,198.7	7,167.4	7,169.5	15.8	10.5	-12.33	209.7	6.6	214.8	188.5	26.25	8.181	
7,300.0	7,298.7	7,267.4	7,269.5	16.0	10.7	-12.11	212.2	6.9	217.1	190.4	26.69	8.135	
7,400.0	7,398.7	7,367.4	7,369.5	16.3	10.9	-11.83	214.8	7.4	219.5	192.4	27.13	8.093	
7,500.0	7,498.7	7,468.1	7,470.1	16.5	11.1	-11.40	217.3	8.6	221.8	194.2	27.57	8.044	
7,600.0	7,598.7	7,568.1	7,570.1	16.7	11.3	-10.91	219.7	10.1	223.8	195.8	28.00	7.992	
7,700.0	7,698.7	7,668.9	7,670.9	16.9	11.5	-10.41	221.9	11.6	225.7	197.2	28.43	7.937	
7,800.0	7,798.7	7,768.4	7,770.3	17.2	11.7	-9.88	224.1	13.4	227.5	198.6	28.87	7.881	
7,900.0	7,898.7	7,870.2	7,872.1	17.4	12.0	-9.45	225.9	14.8	229.0	199.7	29.30	7.816	
8,000.0	7,998.7	7,971.2	7,973.1	17.6	12.2	-9.15	227.0	15.9	229.9	200.2	29.73	7.733	
8,100.0	8,098.7	8,071.5	8,073.4	17.8	12.4	-8.83	227.8	17.0	230.5	200.3	30.16	7.643	
8,200.0	8,198.7	8,171.4	8,173.2	18.1	12.6	-8.53	228.6	18.1	231.1	200.5	30.59	7.556	
8,300.0	8,298.7	8,271.6	8,273.5	18.3	12.8	-8.30	229.3	18.9	231.7	200.7	31.02	7.471	
8,400.0	8,398.7	8,371.7	8,373.6	18.5	13.0	-8.11	229.9	19.6	232.2	200.8	31.45	7.384	
8,500.0	8,498.7	8,471.6	8,473.5	18.7	13.2	-7.93	230.5	20.3	232.7	200.8	31.88	7.300	
8,600.0	8,598.7	8,571.9	8,573.7	19.0	13.4	-7.69	231.1	21.2	233.2	200.9	32.31	7.217	
8,700.0	8,698.7	8,671.6	8,673.4	19.2	13.6	-7.36	231.7	22.5	233.6	200.9	32.75	7.134	
8,800.0	8,798.7	8,771.8	8,773.6	19.4	13.8	-6.99	232.4	23.9	234.2	201.0	33.19	7.057	
8,900.0	8,898.7	8,871.9	8,873.7	19.6	14.0	-6.60	233.0	25.4	234.6	201.0	33.62	6.978	
9,000.0	8,998.7	8,972.1	8,973.9	19.9	14.2	-6.27	233.6	26.7	235.0	200.9	34.05	6.901	
9,100.0	9,098.7	9,072.6	9,074.4	20.1	14.5	-5.92	233.9	28.2	235.2	200.7	34.48	6.821	
9,200.0	9,198.7	9,172.6	9,174.4	20.3	14.7	-5.56	234.2	29.6	235.3	200.4	34.91	6.741	
9,300.0	9,298.7	9,272.4	9,274.2	20.5	14.9	-5.22	234.5	31.0	235.5	200.2	35.34	6.664	
9,400.0	9,398.7	9,372.8	9,374.5	20.8	15.1	-4.89	234.8	32.3	235.7	199.9	35.77	6.588	
9,500.0	9,498.7	9,473.1	9,474.8	21.0	15.3	-4.58	234.8	33.6	235.6	199.4	36.20	6.508	
9,518.4	9,517.1	9,491.4	9,493.1	21.0	15.3	-4.52	234.9	33.8	235.6	199.3	36.28	6.494	
9,600.0	9,598.7	9,572.9	9,574.6	21.2	15.5	-4.29	235.0	34.8	235.7	199.0	36.63	6.433	
9,700.0	9,698.7	9,673.1	9,674.8	21.4	15.7	-4.02	235.0	35.9	235.6	198.5	37.06	6.357	
9,722.4	9,721.1	9,695.3	9,697.1	21.5	15.7	-3.96	235.0	36.1	235.6	198.4	37.16	6.340	
9,800.0	9,798.7	9,772.7	9,774.4	21.6	15.9	-3.73	235.1	37.1	235.6	198.1	37.50	6.284	
9,900.0	9,898.7	9,872.6	9,874.3	21.9	16.1	-3.45	235.4	38.2	235.8	197.9	37.93	6.217	
10,000.0	9,998.7	9,972.1	9,973.8	22.1	16.3	-3.25	235.6	39.0	236.0	197.7	38.36	6.153	
10,100.0	10,098.7	10,071.5	10,073.2	22.3	16.5	-3.07	236.2	39.7	236.6	197.8	38.79	6.098	
10,200.0	10,198.7	10,171.3	10,173.0	22.5	16.7	-2.87	236.9	40.5	237.2	198.0	39.23	6.048	
10,300.0	10,298.7	10,270.7	10,272.4	22.8	17.0	-2.66	237.8	41.4	238.1	198.4	39.66	6.002	
10,363.3	10,362.0	10,335.5	10,337.2	22.9	17.1	-2.45	238.4	42.2	238.6	198.7	39.94	5.975	
10,375.0	10,373.7	10,348.1	10,349.7	22.9	17.1	-122.32	238.5	42.8	238.7	198.7	40.00	5.968	
10,400.0	10,398.7	10,373.3	10,374.9	23.0	17.2	-121.93	238.5	45.3	239.4	199.3	40.09	5.972	
10,425.0	10,423.5	10,396.7	10,397.9	23.0	17.2	-121.34	238.7	49.1	240.8	200.7	40.15	5.997	
10,450.0	10,448.2	10,420.1	10,420.7	23.1	17.3	-120.51	239.1	54.7	243.1	202.9	40.20	6.046	
10,475.0	10,472.7	10,444.0	10,443.5	23.1	17.3	-119.54	239.6	61.6	246.2	205.9	40.25	6.115	
10,500.0	10,496.8	10,467.7	10,466.0	23.2	17.4	-118.64	240.2	69.0	250.0	209.7	40.30	6.204	
10,525.0	10,520.6	10,489.1	10,486.2	23.3	17.4	-117.84	240.8	76.0	254.6	214.3	40.33	6.313	
10,550.0	10,544.0	10,512.2	10,507.7	23.3	17.5	-116.99	241.8	84.4	260.2	219.8	40.38	6.443	
10,575.0	10,566.8	10,538.4	10,531.6	23.4	17.6	-116.01	242.8	95.2	266.2	225.7	40.44	6.582	
10,600.0	10,589.1	10,562.6	10,553.1	23.5	17.6	-115.02	243.6	106.2	272.6	232.1	40.50	6.731	
10,625.0	10,610.8	10,587.4	10,574.5	23.5	17.7	-113.91	244.3	118.7	279.6	239.0	40.59	6.888	
10,650.0	10,631.8	10,610.7	10,593.7	23.6	17.8	-112.67	244.9	131.9	287.1	246.4	40.71	7.053	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1													Offset Site Error: 0.0 usft
Survey Program: 2261-MWD, 13302-MWD													Offset Well Error: 0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
10,675.0	10,652.0	10,633.2	10,611.6	23.7	17.9	-111.37	245.5	145.6	295.1	254.3	40.84	7.226	
10,700.0	10,671.5	10,655.7	10,629.0	23.8	18.0	-110.09	246.1	159.8	303.7	262.7	41.00	7.407	
10,725.0	10,690.1	10,679.4	10,646.6	23.9	18.1	-108.77	246.7	175.6	312.8	271.5	41.20	7.590	
10,750.0	10,707.8	10,704.2	10,663.8	24.0	18.2	-107.32	247.1	193.4	322.1	280.7	41.45	7.772	
10,775.0	10,724.5	10,728.5	10,679.6	24.1	18.3	-105.82	247.3	211.8	331.7	290.0	41.72	7.950	
10,800.0	10,740.3	10,751.0	10,693.9	24.3	18.5	-104.42	247.4	229.3	341.6	299.6	42.02	8.131	
10,825.0	10,755.0	10,769.9	10,705.3	24.4	18.6	-103.04	247.6	244.3	352.0	309.7	42.32	8.317	
10,850.0	10,768.7	10,791.6	10,717.3	24.6	18.8	-101.56	247.9	262.4	362.8	320.2	42.67	8.503	
10,875.0	10,781.2	10,818.7	10,730.9	24.8	19.0	-100.02	247.9	285.9	373.7	330.6	43.09	8.673	
10,900.0	10,792.6	10,844.0	10,742.3	25.0	19.3	-98.50	247.5	308.4	384.6	341.1	43.53	8.836	
10,925.0	10,802.8	10,862.8	10,750.1	25.2	19.4	-97.01	247.2	325.5	395.7	351.7	43.94	9.005	
10,950.0	10,811.8	10,880.6	10,757.0	25.4	19.6	-95.53	247.0	342.0	407.1	362.8	44.36	9.178	
10,975.0	10,819.6	10,896.9	10,762.7	25.7	19.8	-94.03	247.0	357.2	418.9	374.2	44.78	9.356	
11,000.0	10,826.1	10,913.5	10,768.0	25.9	20.0	-92.53	247.2	372.9	431.2	386.0	45.21	9.537	
11,025.0	10,831.3	10,930.6	10,773.0	26.2	20.2	-91.05	247.5	389.4	443.7	398.0	45.66	9.718	
11,050.0	10,835.3	10,948.5	10,777.7	26.5	20.5	-89.61	248.0	406.6	456.4	410.3	46.12	9.897	
11,075.0	10,837.9	10,969.0	10,782.3	26.8	20.7	-88.25	248.6	426.5	469.3	422.7	46.61	10.069	
11,100.0	10,839.3	10,988.3	10,786.1	27.1	21.0	-86.92	249.2	445.5	482.2	435.1	47.11	10.237	
11,110.8	10,839.5	10,997.8	10,787.8	27.2	21.2	-86.39	249.4	454.8	487.8	440.5	47.33	10.305	
11,125.8	10,839.5	11,010.2	10,789.9	27.4	21.4	-86.71	249.8	467.0	495.5	447.9	47.68	10.394	
11,200.0	10,839.9	11,067.4	10,796.1	28.5	22.3	-87.72	251.5	523.8	533.0	483.4	49.59	10.749	
11,300.0	10,840.5	11,155.4	10,795.5	30.1	23.8	-87.84	254.9	611.7	580.7	528.0	52.66	11.028	
11,400.0	10,841.0	11,252.8	10,792.6	31.9	25.7	-87.70	257.3	709.0	622.6	566.3	56.29	11.060	
11,500.0	10,841.5	11,358.5	10,792.9	33.9	28.0	-87.84	259.0	814.7	658.7	598.2	60.46	10.894	
11,600.0	10,842.1	11,449.6	10,795.9	36.0	30.0	-88.17	259.4	905.8	688.7	624.1	64.64	10.655	
11,700.0	10,842.6	11,539.9	10,798.7	38.1	32.1	-88.43	260.8	996.0	714.8	645.8	68.97	10.364	
11,800.0	10,843.2	11,664.4	10,798.0	40.4	35.2	-88.38	261.3	1,120.5	734.9	660.7	74.23	9.901	
11,900.0	10,843.7	11,796.6	10,795.2	42.7	38.5	-88.13	255.8	1,252.5	745.2	665.4	79.81	9.337	
12,000.0	10,844.2	11,907.6	10,795.8	45.0	41.4	-88.12	247.8	1,363.2	747.1	662.2	84.87	8.803	
12,100.0	10,844.8	12,004.5	10,798.2	47.4	44.0	-88.24	240.5	1,459.8	743.5	654.0	89.46	8.311	
12,110.4	10,844.8	12,014.4	10,798.5	47.6	44.2	-88.26	239.8	1,469.7	742.9	652.9	89.93	8.260	
12,200.0	10,845.3	12,104.9	10,801.7	49.8	46.7	-88.45	233.4	1,560.0	737.1	642.6	94.60	7.793	
12,300.0	10,845.8	12,199.1	10,805.0	52.2	49.2	-88.66	226.8	1,653.8	730.9	631.2	99.69	7.332	
12,400.0	10,846.3	12,294.3	10,808.5	54.7	51.8	-88.89	221.0	1,748.8	725.5	620.6	104.88	6.918	
12,500.0	10,846.8	12,379.2	10,811.1	57.3	54.1	-89.06	216.8	1,833.5	721.3	611.5	109.83	6.568	
12,600.0	10,847.4	12,472.3	10,811.8	59.9	56.7	-89.07	213.9	1,926.6	719.0	603.9	115.05	6.249	
12,700.0	10,847.9	12,562.3	10,812.5	62.5	59.2	-89.09	211.8	2,016.6	717.5	597.3	120.22	5.968	
12,745.7	10,848.1	12,603.1	10,812.8	63.7	60.3	-89.09	211.3	2,057.4	717.3	594.7	122.59	5.852	
12,800.0	10,848.4	12,652.6	10,813.0	65.1	61.7	-89.09	211.1	2,106.8	717.5	592.1	125.43	5.720	
12,900.0	10,848.9	12,745.4	10,812.4	67.8	64.3	-89.01	211.5	2,199.7	718.7	588.0	130.74	5.498	
13,000.0	10,849.5	12,847.1	10,812.2	70.4	67.2	-88.95	212.3	2,301.3	720.4	584.1	136.32	5.285	
13,100.0	10,850.0	12,950.1	10,811.9	73.1	70.1	-88.88	212.7	2,404.3	721.6	579.6	141.98	5.083	
13,200.0	10,850.5	13,042.1	10,811.3	75.9	72.7	-88.80	213.4	2,496.3	723.1	575.8	147.33	4.908	
13,300.0	10,851.0	13,153.3	10,810.8	78.6	75.7	-88.72	214.0	2,607.5	724.6	571.4	153.17	4.731	
13,400.0	10,851.6	13,258.6	10,812.0	81.3	77.7	-88.77	213.4	2,712.8	724.8	566.8	157.96	4.589	
13,456.1	10,851.9	13,313.6	10,813.3	82.9	78.7	-88.85	212.8	2,767.8	724.6	564.2	160.47	4.516	
13,500.0	10,852.1	13,345.3	10,814.1	84.1	79.0	-88.90	212.8	2,799.5	724.9	562.9	162.02	4.474	
13,600.0	10,852.6	13,416.6	10,817.3	86.9	79.8	-89.13	215.2	2,870.7	728.9	563.3	165.59	4.402	
13,700.0	10,853.1	13,536.4	10,819.2	89.6	81.2	-89.23	219.8	2,990.4	733.6	563.8	169.79	4.321	
13,800.0	10,853.7	13,640.6	10,818.2	92.4	82.5	-89.12	221.7	3,094.5	736.2	562.3	173.91	4.233	
13,900.0	10,854.2	13,740.2	10,817.7	95.2	83.8	-89.04	223.7	3,194.1	739.0	561.0	178.07	4.150	
14,000.0	10,854.7	13,857.7	10,815.4	98.0	85.5	-88.82	224.5	3,311.6	740.7	558.1	182.57	4.057	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1													Offset Site Error:	0.0 usft
Survey Program: 2261-MWD, 13302-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
14,100.0	10,855.2	13,969.3	10,813.7	100.8	87.2	-88.64	222.6	3,423.1	739.8	552.8	187.08	3.955		
14,200.0	10,855.7	14,069.0	10,814.4	103.7	88.7	-88.65	220.6	3,522.9	738.6	547.1	191.51	3.857		
14,300.0	10,856.3	14,173.4	10,812.9	106.5	90.5	-88.49	217.5	3,627.1	736.4	540.4	196.07	3.756		
14,400.0	10,856.8	14,265.7	10,811.9	109.3	92.1	-88.37	215.7	3,719.4	735.3	534.8	200.50	3.668		
14,500.0	10,857.3	14,371.8	10,811.4	112.2	93.9	-88.28	213.1	3,825.4	733.7	528.5	205.23	3.575		
14,600.0	10,857.8	14,469.4	10,812.1	115.0	95.7	-88.29	210.6	3,923.0	731.9	522.0	209.89	3.487		
14,700.0	10,858.4	14,563.2	10,813.8	117.9	97.5	-88.38	209.1	4,016.8	731.0	516.5	214.53	3.408		
14,800.0	10,858.9	14,661.1	10,814.7	120.7	99.4	-88.41	207.6	4,114.7	730.3	511.0	219.29	3.330		
14,831.8	10,859.1	14,689.7	10,814.6	121.6	100.0	-88.40	207.3	4,143.3	730.3	509.5	220.77	3.308		
14,900.0	10,859.4	14,757.4	10,813.8	123.6	101.3	-88.31	206.8	4,211.0	730.3	506.2	224.06	3.259		
15,000.0	10,859.9	14,854.1	10,812.2	126.4	103.3	-88.14	206.4	4,307.7	730.9	502.0	228.87	3.193		
15,100.0	10,860.5	14,942.1	10,813.8	129.3	105.1	-88.23	206.8	4,395.6	732.0	498.4	233.57	3.134		
15,200.0	10,861.0	15,040.9	10,816.8	132.1	107.1	-88.43	208.8	4,494.4	734.7	496.2	238.54	3.080		
15,300.0	10,861.5	15,151.6	10,818.3	135.0	109.5	-88.50	209.5	4,605.0	736.2	492.4	243.80	3.020		
15,400.0	10,862.0	15,248.7	10,817.4	137.9	111.6	-88.40	209.4	4,702.2	736.9	488.1	248.78	2.962		
15,500.0	10,862.6	15,333.0	10,816.9	140.7	113.5	-88.33	210.4	4,786.4	739.0	485.5	253.52	2.915		
15,600.0	10,863.1	15,438.5	10,816.5	143.6	115.8	-88.26	213.7	4,891.9	743.1	484.3	258.75	2.872		
15,700.0	10,863.6	15,565.3	10,816.8	146.5	118.7	-88.24	213.9	5,018.7	744.1	479.5	264.52	2.813		
15,800.0	10,864.1	15,668.9	10,817.9	149.4	121.1	-88.28	211.9	5,122.2	742.9	473.1	269.80	2.753		
15,900.0	10,864.7	15,766.5	10,817.8	152.3	123.4	-88.23	210.0	5,219.8	741.8	466.8	274.97	2.698		
16,000.0	10,865.2	15,865.0	10,819.1	155.1	125.7	-88.29	208.1	5,318.3	740.6	460.4	280.19	2.643		
16,100.0	10,865.7	15,961.6	10,821.3	158.0	128.0	-88.42	206.8	5,414.9	740.0	454.6	285.40	2.593		
16,200.0	10,866.2	16,090.1	10,825.3	160.9	131.1	-88.67	204.2	5,543.3	739.0	447.6	291.42	2.536		
16,300.0	10,866.7	16,200.4	10,823.8	163.8	133.8	-88.50	197.6	5,653.3	733.8	436.9	296.98	2.471		
16,400.0	10,867.3	16,298.4	10,819.8	166.7	136.2	-88.13	190.9	5,751.0	728.0	425.7	302.23	2.409		
16,500.0	10,867.8	16,393.8	10,819.6	169.6	138.6	-88.06	185.3	5,846.3	722.9	415.5	307.47	2.351		
16,600.0	10,868.3	16,492.4	10,823.5	172.5	141.0	-88.32	180.2	5,944.6	718.4	405.5	312.86	2.296		
16,700.0	10,868.8	16,582.6	10,826.2	175.4	143.3	-88.49	176.0	6,034.7	714.5	396.5	318.05	2.247		
16,800.0	10,869.4	16,699.2	10,828.5	178.3	146.2	-88.62	170.2	6,151.1	710.4	386.5	323.90	2.193		
16,900.0	10,869.9	16,797.8	10,827.0	181.2	148.7	-88.44	163.8	6,249.5	704.7	375.4	329.29	2.140		
17,000.0	10,870.4	16,905.5	10,825.7	184.1	151.5	-88.28	156.1	6,356.9	698.4	363.4	334.92	2.085		
17,100.0	10,870.9	17,001.6	10,826.7	187.0	153.9	-88.30	149.0	6,452.7	691.8	351.5	340.29	2.033		
17,200.0	10,871.5	17,093.0	10,827.0	189.9	156.3	-88.28	143.5	6,543.9	686.6	341.1	345.55	1.987		
17,300.0	10,872.0	17,185.6	10,827.1	192.8	158.7	-88.23	138.7	6,636.5	682.3	331.4	350.85	1.945		
17,400.0	10,872.5	17,280.8	10,829.1	195.7	161.2	-88.35	135.0	6,731.5	679.1	322.8	356.25	1.906		
17,500.0	10,873.0	17,374.0	10,829.8	198.6	163.6	-88.36	131.8	6,824.7	676.4	314.8	361.59	1.871		
17,600.0	10,873.6	17,478.2	10,831.0	201.5	166.3	-88.41	128.6	6,928.8	674.2	306.9	367.24	1.836		
17,700.0	10,874.1	17,568.5	10,831.4	204.4	168.7	-88.41	126.0	7,019.1	672.2	299.6	372.53	1.804		
17,800.0	10,874.6	17,667.5	10,832.1	207.3	171.3	-88.42	124.3	7,118.1	671.2	293.1	378.05	1.775		
17,900.0	10,875.1	17,764.5	10,831.1	210.2	173.9	-88.28	122.4	7,215.0	670.1	286.6	383.51	1.747		
18,000.0	10,875.6	17,863.1	10,831.3	213.1	176.5	-88.25	121.3	7,313.6	669.8	280.8	389.03	1.722		
18,088.9	10,876.1	17,949.5	10,830.2	215.7	178.8	-88.13	120.3	7,400.0	669.5	275.6	393.90	1.700		
18,100.0	10,876.2	17,959.6	10,830.1	216.0	179.1	-88.11	120.2	7,410.1	669.5	275.1	394.49	1.697		
18,200.0	10,876.7	18,053.9	10,829.1	218.9	181.6	-87.99	120.1	7,504.4	670.3	270.4	399.90	1.676		
18,300.0	10,877.2	18,151.0	10,828.4	221.8	184.2	-87.89	120.2	7,601.5	671.2	265.8	405.38	1.656		
18,400.0	10,877.7	18,240.3	10,827.2	224.7	186.6	-87.75	121.5	7,690.8	673.5	262.8	410.66	1.640		
18,500.0	10,878.3	18,333.0	10,827.8	227.6	189.1	-87.77	123.8	7,783.4	676.8	260.8	416.06	1.627		
18,600.0	10,878.8	18,420.8	10,827.5	230.5	191.4	-87.72	127.1	7,871.2	681.5	260.2	421.32	1.618		
18,700.0	10,879.3	18,527.5	10,827.7	233.4	194.3	-87.71	132.6	7,977.7	687.4	260.4	427.10	1.610		
18,800.0	10,879.8	18,645.1	10,830.1	236.4	197.5	-87.88	136.0	8,095.2	691.2	257.9	433.23	1.595		
18,900.0	10,880.4	18,749.9	10,832.2	239.3	200.3	-88.01	137.3	8,200.0	693.1	254.1	439.02	1.579		
19,000.0	10,880.9	18,847.2	10,834.9	242.2	203.0	-88.20	138.2	8,297.3	694.8	250.2	444.63	1.563		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1												Offset Site Error:	0.0 usft
Survey Program: 2261-MWD, 13302-MWD												Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
19,100.0	10,881.4	18,942.7	10,837.7	245.1	205.6	-88.39	139.7	8,392.7	697.1	246.9	450.18	1.548	
19,200.0	10,881.9	19,048.8	10,841.4	248.0	208.5	-88.65	141.3	8,498.8	699.3	243.3	456.04	1.533	
19,300.0	10,882.5	19,158.6	10,844.7	250.9	211.5	-88.87	141.5	8,608.5	700.3	238.3	462.00	1.516	
19,400.0	10,883.0	19,258.1	10,846.2	253.8	214.2	-88.95	141.1	8,708.0	700.6	233.0	467.66	1.498 Level 3	
19,500.0	10,883.5	19,356.1	10,846.2	256.7	216.9	-88.91	140.8	8,806.0	701.1	227.9	473.27	1.481 Level 3	
19,600.0	10,884.0	19,454.3	10,845.4	259.7	219.6	-88.81	140.8	8,904.2	702.0	223.1	478.88	1.466 Level 3	
19,700.0	10,884.5	19,555.0	10,843.7	262.6	222.4	-88.63	140.7	9,004.9	702.7	218.2	484.55	1.450 Level 3	
19,800.0	10,885.1	19,644.0	10,841.9	265.5	224.9	-88.44	141.1	9,093.9	704.0	214.1	489.89	1.437 Level 3	
19,900.0	10,885.6	19,730.6	10,840.8	268.4	227.3	-88.33	142.9	9,180.4	707.1	211.9	495.17	1.428 Level 3	
20,000.0	10,886.1	19,815.9	10,840.3	271.3	229.6	-88.26	146.7	9,265.6	712.4	212.0	500.43	1.424 Level 3	
20,100.0	10,886.6	19,905.8	10,840.7	274.2	232.1	-88.27	152.6	9,355.4	719.8	213.9	505.82	1.423 Level 3	
20,200.0	10,887.2	20,007.7	10,840.8	277.2	234.9	-88.26	159.6	9,457.0	727.4	215.9	511.54	1.422 Level 3	
20,300.0	10,887.7	20,112.7	10,841.0	280.1	237.8	-88.25	166.3	9,561.8	734.6	217.2	517.35	1.420 Level 3	
20,400.0	10,888.2	20,212.7	10,841.0	283.0	240.5	-88.22	172.4	9,661.6	741.5	218.4	523.03	1.418 Level 3	
20,500.0	10,888.7	20,309.1	10,840.5	285.9	243.2	-88.16	178.5	9,757.8	748.7	220.1	528.60	1.416 Level 3	
20,600.0	10,889.3	20,419.4	10,838.9	288.8	246.3	-88.01	184.8	9,868.0	755.3	220.7	534.53	1.413 Level 3, SF	
20,700.0	10,889.8	20,482.0	10,837.6	291.7	248.0	-87.90	187.9	9,930.4	762.0	222.8	539.15	1.413 Level 3	
20,717.6	10,889.9	20,482.0	10,837.6	292.3	248.0	-87.90	187.9	9,930.4	764.0	224.4	539.67	1.416 Level 3	

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18 3T - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
0.0	0.0	0.0	0.0	0.0	0.0	-155.57	-59.8	-27.2	65.7					
100.0	100.0	100.0	100.0	0.1	0.1	-155.57	-59.8	-27.2	65.7	65.5	0.17	389.542		
200.0	200.0	200.0	200.0	0.3	0.3	-155.57	-59.8	-27.2	65.7	65.0	0.62	106.239		
300.0	300.0	300.0	300.0	0.5	0.5	-155.57	-59.8	-27.2	65.7	64.6	1.07	61.507		
400.0	400.0	400.0	400.0	0.8	0.8	-155.57	-59.8	-27.2	65.7	64.1	1.52	43.283		
500.0	500.0	500.0	500.0	1.0	1.0	-155.57	-59.8	-27.2	65.7	63.7	1.97	33.389		
600.0	600.0	600.0	600.0	1.2	1.2	-155.57	-59.8	-27.2	65.7	63.3	2.42	27.177		
700.0	700.0	700.0	700.0	1.4	1.4	-155.57	-59.8	-27.2	65.7	62.8	2.87	22.914		
800.0	800.0	800.0	800.0	1.7	1.7	-155.57	-59.8	-27.2	65.7	62.4	3.32	19.807		
900.0	900.0	900.0	900.0	1.9	1.9	-155.57	-59.8	-27.2	65.7	61.9	3.76	17.442		
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.1	-155.57	-59.8	-27.2	65.7	61.5	4.21	15.582		
1,100.0	1,100.0	1,100.0	1,100.0	2.3	2.3	-155.57	-59.8	-27.2	65.7	61.0	4.66	14.080		
1,200.0	1,200.0	1,200.0	1,200.0	2.6	2.6	-155.57	-59.8	-27.2	65.7	60.6	5.11	12.842		
1,300.0	1,300.0	1,300.0	1,300.0	2.8	2.8	-155.57	-59.8	-27.2	65.7	60.1	5.56	11.804		
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.0	-155.57	-59.8	-27.2	65.7	59.7	6.01	10.922		
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.2	-155.57	-59.8	-27.2	65.7	59.2	6.46	10.162		
1,600.0	1,600.0	1,600.0	1,600.0	3.5	3.5	-155.57	-59.8	-27.2	65.7	58.8	6.91	9.501		
1,700.0	1,700.0	1,700.0	1,700.0	3.7	3.7	-155.57	-59.8	-27.2	65.7	58.3	7.36	8.921		
1,800.0	1,800.0	1,800.0	1,800.0	3.9	3.9	-155.57	-59.8	-27.2	65.7	57.9	7.81	8.407		
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.1	-155.57	-59.8	-27.2	65.7	57.4	8.26	7.950		
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.4	-155.57	-59.8	-27.2	65.7	57.0	8.71	7.540		
2,100.0	2,100.0	2,100.0	2,100.0	4.6	4.6	-155.57	-59.8	-27.2	65.7	56.5	9.16	7.170		
2,200.0	2,200.0	2,200.0	2,200.0	4.8	4.8	-155.57	-59.8	-27.2	65.7	56.1	9.61	6.834		
2,300.0	2,300.0	2,300.0	2,300.0	5.0	5.0	-155.57	-59.8	-27.2	65.7	55.6	10.06	6.529		
2,400.0	2,400.0	2,400.0	2,400.0	5.3	5.3	-155.57	-59.8	-27.2	65.7	55.2	10.51	6.249		
2,500.0	2,500.0	2,500.0	2,500.0	5.5	5.5	-155.57	-59.8	-27.2	65.7	54.7	10.96	5.993 CC		
2,600.0	2,600.0	2,598.9	2,598.9	5.7	5.7	117.07	-59.8	-28.9	67.2	55.8	11.38	5.904		
2,650.0	2,649.9	2,648.2	2,648.1	5.8	5.8	120.18	-59.8	-31.0	69.3	57.7	11.57	5.984		
2,700.0	2,699.9	2,697.9	2,697.8	5.9	5.9	123.76	-59.8	-33.6	72.0	60.3	11.78	6.118		
2,800.0	2,799.7	2,797.3	2,797.1	6.1	6.1	130.11	-59.8	-38.8	78.4	66.2	12.18	6.433		
2,900.0	2,899.6	2,896.8	2,896.4	6.3	6.3	135.46	-59.8	-44.0	85.5	72.9	12.59	6.792		
3,000.0	2,999.5	2,996.3	2,995.7	6.5	6.5	139.96	-59.8	-49.2	93.2	80.2	12.99	7.176		
3,100.0	3,099.3	3,095.7	3,095.0	6.7	6.7	143.75	-59.8	-54.4	101.5	88.1	13.40	7.572		
3,200.0	3,199.2	3,195.2	3,194.3	7.0	6.9	146.97	-59.8	-59.6	110.1	96.3	13.81	7.970		
3,300.0	3,299.0	3,294.6	3,293.7	7.2	7.2	149.71	-59.8	-64.8	119.0	104.8	14.23	8.365		
3,400.0	3,398.9	3,394.1	3,393.0	7.4	7.4	152.06	-59.8	-70.0	128.2	113.5	14.64	8.752		
3,501.2	3,500.0	3,494.7	3,493.5	7.6	7.6	154.13	-59.8	-75.3	137.6	122.5	15.07	9.133		
3,600.0	3,598.7	3,597.5	3,596.2	7.8	7.8	155.46	-59.8	-79.1	144.0	128.5	15.48	9.301		
3,651.2	3,649.9	3,651.2	3,649.9	7.9	7.9	-114.37	-59.8	-79.6	144.9	129.2	15.71	9.220		
3,700.0	3,698.7	3,700.0	3,698.7	8.0	8.0	-114.37	-59.8	-79.6	144.9	129.0	15.92	9.103		
3,800.0	3,798.7	3,800.0	3,798.7	8.2	8.2	-114.37	-59.8	-79.6	144.9	128.5	16.36	8.856		
3,900.0	3,898.7	3,900.0	3,898.7	8.5	8.5	-114.37	-59.8	-79.6	144.9	128.1	16.80	8.622		
4,000.0	3,998.7	4,000.0	3,998.7	8.7	8.7	-114.37	-59.8	-79.6	144.9	127.6	17.25	8.400		
4,100.0	4,098.7	4,100.0	4,098.7	8.9	8.9	-114.37	-59.8	-79.6	144.9	127.2	17.69	8.189		
4,200.0	4,198.7	4,200.0	4,198.7	9.1	9.1	-114.37	-59.8	-79.6	144.9	126.7	18.14	7.988		
4,300.0	4,298.7	4,300.0	4,298.7	9.4	9.4	-114.37	-59.8	-79.6	144.9	126.3	18.58	7.797		
4,400.0	4,398.7	4,400.0	4,398.7	9.6	9.6	-114.37	-59.8	-79.6	144.9	125.8	19.03	7.614		
4,500.0	4,498.7	4,500.0	4,498.7	9.8	9.8	-114.37	-59.8	-79.6	144.9	125.4	19.47	7.440		
4,600.0	4,598.7	4,600.0	4,598.7	10.0	10.0	-114.37	-59.8	-79.6	144.9	125.0	19.92	7.274		
4,700.0	4,698.7	4,700.0	4,698.7	10.2	10.2	-114.37	-59.8	-79.6	144.9	124.5	20.36	7.114		
4,800.0	4,798.7	4,800.0	4,798.7	10.5	10.5	-114.37	-59.8	-79.6	144.9	124.1	20.81	6.962		
4,900.0	4,898.7	4,900.0	4,898.7	10.7	10.7	-114.37	-59.8	-79.6	144.9	123.6	21.26	6.816		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18 3T - Wellbore #1 - Design #6													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
5,000.0	4,998.7	5,000.0	4,998.7	10.9	10.9	-114.37	-59.8	-79.6	144.9	123.2	21.70	6.676	
5,100.0	5,098.7	5,100.0	5,098.7	11.1	11.1	-114.37	-59.8	-79.6	144.9	122.7	22.15	6.541	
5,200.0	5,198.7	5,200.0	5,198.7	11.4	11.4	-114.37	-59.8	-79.6	144.9	122.3	22.59	6.412	
5,300.0	5,298.7	5,300.0	5,298.7	11.6	11.6	-114.37	-59.8	-79.6	144.9	121.8	23.04	6.288	
5,400.0	5,398.7	5,400.0	5,398.7	11.8	11.8	-114.37	-59.8	-79.6	144.9	121.4	23.49	6.168	
5,500.0	5,498.7	5,500.0	5,498.7	12.0	12.0	-114.37	-59.8	-79.6	144.9	120.9	23.94	6.053	
5,600.0	5,598.7	5,600.0	5,598.7	12.2	12.2	-114.37	-59.8	-79.6	144.9	120.5	24.38	5.942	
5,700.0	5,698.7	5,700.0	5,698.7	12.5	12.5	-114.37	-59.8	-79.6	144.9	120.0	24.83	5.835	
5,800.0	5,798.7	5,800.0	5,798.7	12.7	12.7	-114.37	-59.8	-79.6	144.9	119.6	25.28	5.732	
5,900.0	5,898.7	5,900.0	5,898.7	12.9	12.9	-114.37	-59.8	-79.6	144.9	119.2	25.72	5.632	
6,000.0	5,998.7	6,000.0	5,998.7	13.1	13.1	-114.37	-59.8	-79.6	144.9	118.7	26.17	5.536	
6,100.0	6,098.7	6,100.0	6,098.7	13.4	13.4	-114.37	-59.8	-79.6	144.9	118.3	26.62	5.443	
6,200.0	6,198.7	6,200.0	6,198.7	13.6	13.6	-114.37	-59.8	-79.6	144.9	117.8	27.07	5.353	
6,300.0	6,298.7	6,300.0	6,298.7	13.8	13.8	-114.37	-59.8	-79.6	144.9	117.4	27.51	5.266	
6,400.0	6,398.7	6,400.0	6,398.7	14.0	14.0	-114.37	-59.8	-79.6	144.9	116.9	27.96	5.181	
6,500.0	6,498.7	6,500.0	6,498.7	14.3	14.3	-114.37	-59.8	-79.6	144.9	116.5	28.41	5.100	
6,600.0	6,598.7	6,600.0	6,598.7	14.5	14.5	-114.37	-59.8	-79.6	144.9	116.0	28.86	5.021	
6,700.0	6,698.7	6,700.0	6,698.7	14.7	14.7	-114.37	-59.8	-79.6	144.9	115.6	29.30	4.944	
6,800.0	6,798.7	6,800.0	6,798.7	14.9	14.9	-114.37	-59.8	-79.6	144.9	115.1	29.75	4.870	
6,900.0	6,898.7	6,900.0	6,898.7	15.2	15.2	-114.37	-59.8	-79.6	144.9	114.7	30.20	4.797	
7,000.0	6,998.7	7,000.0	6,998.7	15.4	15.4	-114.37	-59.8	-79.6	144.9	114.2	30.65	4.727	
7,100.0	7,098.7	7,100.0	7,098.7	15.6	15.6	-114.37	-59.8	-79.6	144.9	113.8	31.09	4.659	
7,200.0	7,198.7	7,200.0	7,198.7	15.8	15.8	-114.37	-59.8	-79.6	144.9	113.3	31.54	4.593	
7,300.0	7,298.7	7,300.0	7,298.7	16.0	16.0	-114.37	-59.8	-79.6	144.9	112.9	31.99	4.529	
7,400.0	7,398.7	7,400.0	7,398.7	16.3	16.3	-114.37	-59.8	-79.6	144.9	112.4	32.44	4.466	
7,500.0	7,498.7	7,500.0	7,498.7	16.5	16.5	-114.37	-59.8	-79.6	144.9	112.0	32.89	4.405	
7,600.0	7,598.7	7,600.0	7,598.7	16.7	16.7	-114.37	-59.8	-79.6	144.9	111.5	33.34	4.346	
7,700.0	7,698.7	7,700.0	7,698.7	16.9	16.9	-114.37	-59.8	-79.6	144.9	111.1	33.78	4.288	
7,800.0	7,798.7	7,800.0	7,798.7	17.2	17.2	-114.37	-59.8	-79.6	144.9	110.6	34.23	4.232	
7,900.0	7,898.7	7,900.0	7,898.7	17.4	17.4	-114.37	-59.8	-79.6	144.9	110.2	34.68	4.178	
8,000.0	7,998.7	8,000.0	7,998.7	17.6	17.6	-114.37	-59.8	-79.6	144.9	109.7	35.13	4.124	
8,100.0	8,098.7	8,100.0	8,098.7	17.8	17.8	-114.37	-59.8	-79.6	144.9	109.3	35.58	4.072	
8,200.0	8,198.7	8,200.0	8,198.7	18.1	18.1	-114.37	-59.8	-79.6	144.9	108.9	36.02	4.022	
8,300.0	8,298.7	8,300.0	8,298.7	18.3	18.3	-114.37	-59.8	-79.6	144.9	108.4	36.47	3.972	
8,400.0	8,398.7	8,400.0	8,398.7	18.5	18.5	-114.37	-59.8	-79.6	144.9	108.0	36.92	3.924	
8,500.0	8,498.7	8,500.0	8,498.7	18.7	18.7	-114.37	-59.8	-79.6	144.9	107.5	37.37	3.877	
8,600.0	8,598.7	8,600.0	8,598.7	19.0	19.0	-114.37	-59.8	-79.6	144.9	107.1	37.82	3.831	
8,700.0	8,698.7	8,700.0	8,698.7	19.2	19.2	-114.37	-59.8	-79.6	144.9	106.6	38.27	3.786	
8,800.0	8,798.7	8,800.0	8,798.7	19.4	19.4	-114.37	-59.8	-79.6	144.9	106.2	38.72	3.742	
8,900.0	8,898.7	8,900.0	8,898.7	19.6	19.6	-114.37	-59.8	-79.6	144.9	105.7	39.16	3.699	
9,000.0	8,998.7	9,000.0	8,998.7	19.9	19.9	-114.37	-59.8	-79.6	144.9	105.3	39.61	3.657	
9,100.0	9,098.7	9,100.0	9,098.7	20.1	20.1	-114.37	-59.8	-79.6	144.9	104.8	40.06	3.616	
9,200.0	9,198.7	9,200.0	9,198.7	20.3	20.3	-114.37	-59.8	-79.6	144.9	104.4	40.51	3.576	
9,300.0	9,298.7	9,300.0	9,298.7	20.5	20.5	-114.37	-59.8	-79.6	144.9	103.9	40.96	3.537	
9,400.0	9,398.7	9,400.0	9,398.7	20.8	20.8	-114.37	-59.8	-79.6	144.9	103.5	41.41	3.499	
9,500.0	9,498.7	9,500.0	9,498.7	21.0	21.0	-114.37	-59.8	-79.6	144.9	103.0	41.86	3.461	
9,600.0	9,598.7	9,600.0	9,598.7	21.2	21.2	-114.37	-59.8	-79.6	144.9	102.6	42.30	3.425	
9,700.0	9,698.7	9,700.0	9,698.7	21.4	21.4	-114.37	-59.8	-79.6	144.9	102.1	42.75	3.389	
9,800.0	9,798.7	9,800.0	9,798.7	21.6	21.6	-114.37	-59.8	-79.6	144.9	101.7	43.20	3.353	
9,900.0	9,898.7	9,900.0	9,898.7	21.9	21.9	-114.37	-59.8	-79.6	144.9	101.2	43.65	3.319	
10,000.0	9,998.7	10,000.0	9,998.7	22.1	22.1	-114.37	-59.8	-79.6	144.9	100.8	44.10	3.285	
10,100.0	10,098.7	10,100.0	10,098.7	22.3	22.3	-114.37	-59.8	-79.6	144.9	100.3	44.55	3.252	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18 3T - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
10,200.0	10,198.7	10,200.0	10,198.7	22.5	22.5	-114.37	-59.8	-79.6	144.9	99.9	45.00	3.220		
10,300.0	10,298.7	10,300.0	10,298.7	22.8	22.8	-114.37	-59.8	-79.6	144.9	99.4	45.45	3.188		
10,363.3	10,362.0	10,366.1	10,364.8	22.9	22.9	-114.72	-60.5	-78.9	144.6	98.9	45.72	3.163		
10,375.0	10,373.7	10,379.0	10,377.7	22.9	22.9	124.99	-61.2	-78.3	144.4	98.7	45.76	3.156		
10,400.0	10,398.7	10,406.5	10,404.9	23.0	23.0	124.24	-63.5	-76.1	144.3	98.4	45.83	3.148		
10,400.5	10,399.1	10,407.0	10,405.4	23.0	23.0	124.22	-63.6	-76.0	144.3	98.4	45.83	3.148		
10,425.0	10,423.5	10,433.8	10,431.8	23.0	23.0	123.34	-67.0	-72.8	144.4	98.5	45.87	3.148		
10,450.0	10,448.2	10,461.0	10,458.3	23.1	23.1	122.29	-71.6	-68.6	144.9	99.0	45.90	3.158		
10,475.0	10,472.7	10,488.0	10,484.1	23.1	23.1	121.11	-77.2	-63.3	145.8	99.9	45.91	3.176		
10,500.0	10,496.8	10,514.8	10,509.4	23.2	23.2	119.82	-83.8	-57.2	147.0	101.1	45.93	3.202		
10,525.0	10,520.6	10,541.4	10,533.9	23.3	23.2	118.42	-91.4	-50.1	148.7	102.7	45.94	3.236		
10,550.0	10,544.0	10,567.8	10,557.6	23.3	23.3	116.94	-99.9	-42.1	150.7	104.7	45.98	3.277		
10,575.0	10,566.8	10,594.0	10,580.4	23.4	23.3	115.38	-109.3	-33.4	153.1	107.1	46.03	3.326		
10,600.0	10,589.1	10,620.0	10,602.3	23.5	23.4	113.77	-119.5	-23.9	155.9	109.8	46.10	3.381		
10,625.0	10,610.8	10,645.7	10,623.3	23.5	23.4	112.13	-130.5	-13.7	159.1	112.9	46.21	3.443		
10,650.0	10,631.8	10,671.3	10,643.2	23.6	23.5	110.45	-142.1	-2.8	162.7	116.3	46.35	3.510		
10,675.0	10,652.0	10,696.6	10,662.1	23.7	23.5	108.77	-154.4	8.7	166.6	120.1	46.53	3.581		
10,700.0	10,671.5	10,721.7	10,680.0	23.8	23.6	107.08	-167.3	20.7	171.0	124.2	46.74	3.658		
10,725.0	10,690.1	10,746.5	10,696.7	23.9	23.7	105.41	-180.8	33.3	175.7	128.7	46.99	3.739		
10,750.0	10,707.8	10,771.2	10,712.3	24.0	23.8	103.75	-194.7	46.3	180.7	133.4	47.27	3.823		
10,775.0	10,724.5	10,795.6	10,726.8	24.1	23.9	102.12	-209.1	59.7	186.1	138.5	47.57	3.911		
10,800.0	10,740.3	10,819.9	10,740.2	24.3	24.0	100.52	-223.9	73.5	191.7	143.8	47.91	4.002		
10,825.0	10,755.0	10,843.9	10,752.4	24.4	24.1	98.95	-239.0	87.6	197.6	149.4	48.26	4.095		
10,850.0	10,768.7	10,867.8	10,763.5	24.6	24.3	97.43	-254.5	102.0	203.8	155.2	48.63	4.191		
10,875.0	10,781.2	10,891.5	10,773.4	24.8	24.4	95.94	-270.2	116.6	210.2	161.2	49.03	4.288		
10,900.0	10,792.6	10,915.0	10,782.2	25.0	24.6	94.50	-286.1	131.5	216.9	167.4	49.43	4.387		
10,925.0	10,802.8	10,938.3	10,789.9	25.2	24.8	93.11	-302.3	146.6	223.7	173.8	49.85	4.487		
10,950.0	10,811.8	10,961.6	10,796.5	25.4	24.9	91.76	-318.6	161.8	230.6	180.3	50.27	4.587		
10,975.0	10,819.6	10,984.7	10,801.9	25.7	25.1	90.46	-335.0	177.1	237.7	187.0	50.71	4.688		
11,000.0	10,826.1	11,007.6	10,806.2	25.9	25.3	89.21	-351.5	192.5	244.9	193.7	51.15	4.788		
11,025.0	10,831.3	11,030.5	10,809.5	26.2	25.6	88.00	-368.0	207.9	252.2	200.6	51.60	4.887		
11,050.0	10,835.3	11,053.3	10,811.6	26.5	25.8	86.84	-384.6	223.4	259.5	207.5	52.05	4.986		
11,075.0	10,837.9	11,076.0	10,812.6	26.8	26.0	85.72	-401.2	238.8	266.9	214.4	52.52	5.082		
11,100.0	10,839.3	11,099.5	10,812.8	27.1	26.3	84.70	-418.4	254.9	274.3	221.3	53.00	5.175		
11,110.8	10,839.5	11,111.6	10,812.9	27.2	26.4	84.36	-427.2	263.1	277.4	224.1	53.25	5.210		
11,125.8	10,839.5	11,128.4	10,813.0	27.4	26.6	84.46	-439.4	274.7	281.6	228.0	53.63	5.251		
11,200.0	10,839.9	11,212.5	10,813.4	28.5	27.7	84.86	-498.7	334.4	301.9	246.3	55.62	5.428		
11,300.0	10,840.5	11,327.8	10,814.1	30.1	29.4	85.31	-575.7	420.2	328.7	270.0	58.69	5.601		
11,400.0	10,841.0	11,445.4	10,814.7	31.9	31.4	85.69	-648.6	512.3	354.7	292.6	62.09	5.713		
11,500.0	10,841.5	11,565.1	10,815.4	33.9	33.6	86.00	-717.0	610.7	379.8	314.0	65.75	5.776		
11,600.0	10,842.1	11,687.2	10,816.1	36.0	36.1	86.28	-780.2	715.0	403.8	334.2	69.60	5.801		
11,700.0	10,842.6	11,811.5	10,816.8	38.1	38.7	86.51	-837.6	825.3	426.6	353.0	73.62	5.795		
11,800.0	10,843.2	11,938.1	10,817.5	40.4	41.5	86.71	-888.5	941.2	448.1	370.4	77.73	5.765		
11,900.0	10,843.7	12,066.9	10,818.2	42.7	44.5	86.88	-932.3	1,062.3	468.3	386.4	81.88	5.719		
12,000.0	10,844.2	12,197.9	10,819.0	45.0	47.6	87.03	-968.4	1,188.1	486.9	400.8	86.02	5.660		
12,100.0	10,844.8	12,330.9	10,819.7	47.4	50.7	87.16	-996.2	1,318.2	503.8	413.7	90.14	5.589		
12,110.4	10,844.8	12,344.9	10,819.8	47.6	51.0	87.17	-998.6	1,332.0	505.5	414.9	90.56	5.582		
12,200.0	10,845.3	12,466.2	10,820.5	49.8	53.9	87.26	-1,015.0	1,452.1	516.9	421.1	95.79	5.396		
12,300.0	10,845.8	12,603.3	10,821.2	52.2	57.1	87.31	-1,024.4	1,588.9	523.0	421.2	101.80	5.137		
12,400.0	10,846.3	12,717.6	10,821.8	54.7	59.8	87.31	-1,025.7	1,703.2	523.2	415.9	107.33	4.875		
12,500.0	10,846.8	12,817.6	10,822.3	57.3	62.2	87.31	-1,026.3	1,803.2	522.9	410.4	112.56	4.646		
12,600.0	10,847.4	12,917.6	10,822.8	59.9	64.6	87.31	-1,026.8	1,903.2	522.7	404.8	117.85	4.435		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18 3T - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)						
12,700.0	10,847.9	13,017.6	10,823.4	62.5	67.1	87.31	-1,027.3	2,003.2	522.4	399.2	123.20	4.240		
12,800.0	10,848.4	13,117.6	10,823.9	65.1	69.6	87.31	-1,027.9	2,103.2	522.1	393.5	128.59	4.060		
12,900.0	10,848.9	13,217.6	10,824.4	67.8	72.1	87.30	-1,028.4	2,203.2	521.8	387.8	134.01	3.894		
13,000.0	10,849.5	13,317.6	10,824.9	70.4	74.7	87.30	-1,028.9	2,303.2	521.6	382.1	139.47	3.740		
13,100.0	10,850.0	13,417.6	10,825.5	73.1	77.3	87.30	-1,029.5	2,403.2	521.3	376.3	144.97	3.596		
13,200.0	10,850.5	13,517.6	10,826.0	75.9	79.9	87.30	-1,030.0	2,503.2	521.0	370.5	150.49	3.462		
13,300.0	10,851.0	13,617.6	10,826.5	78.6	82.6	87.30	-1,030.5	2,603.2	520.7	364.7	156.03	3.337		
13,400.0	10,851.6	13,717.6	10,827.0	81.3	85.2	87.30	-1,031.0	2,703.2	520.5	358.9	161.60	3.221		
13,500.0	10,852.1	13,817.6	10,827.6	84.1	87.9	87.30	-1,031.6	2,803.2	520.2	353.0	167.18	3.112		
13,600.0	10,852.6	13,917.6	10,828.1	86.9	90.6	87.29	-1,032.1	2,903.1	519.9	347.1	172.79	3.009		
13,700.0	10,853.1	14,017.6	10,828.6	89.6	93.3	87.29	-1,032.6	3,003.1	519.6	341.2	178.41	2.913		
13,800.0	10,853.7	14,117.6	10,829.1	92.4	96.0	87.29	-1,033.2	3,103.1	519.4	335.3	184.04	2.822		
13,900.0	10,854.2	14,217.6	10,829.6	95.2	98.7	87.29	-1,033.7	3,203.1	519.1	329.4	189.69	2.737		
14,000.0	10,854.7	14,317.6	10,830.2	98.0	101.5	87.29	-1,034.2	3,303.1	518.8	323.5	195.35	2.656		
14,100.0	10,855.2	14,417.6	10,830.7	100.8	104.2	87.29	-1,034.7	3,403.1	518.6	317.5	201.03	2.580		
14,200.0	10,855.7	14,517.6	10,831.2	103.7	107.0	87.29	-1,035.3	3,503.1	518.3	311.6	206.71	2.507		
14,300.0	10,856.3	14,617.6	10,831.7	106.5	109.8	87.28	-1,035.8	3,603.1	518.0	305.6	212.40	2.439		
14,400.0	10,856.8	14,717.6	10,832.3	109.3	112.6	87.28	-1,036.3	3,703.1	517.7	299.6	218.10	2.374		
14,500.0	10,857.3	14,817.6	10,832.8	112.2	115.3	87.28	-1,036.9	3,803.1	517.5	293.6	223.81	2.312		
14,600.0	10,857.8	14,917.6	10,833.3	115.0	118.1	87.28	-1,037.4	3,903.1	517.2	287.7	229.52	2.253		
14,700.0	10,858.4	15,017.6	10,833.8	117.9	120.9	87.28	-1,037.9	4,003.1	516.9	281.7	235.25	2.197		
14,800.0	10,858.9	15,117.6	10,834.4	120.7	123.7	87.28	-1,038.5	4,103.1	516.6	275.7	240.98	2.144		
14,900.0	10,859.4	15,217.6	10,834.9	123.6	126.6	87.28	-1,039.0	4,203.1	516.4	269.6	246.71	2.093		
15,000.0	10,859.9	15,317.6	10,835.4	126.4	129.4	87.27	-1,039.5	4,303.1	516.1	263.6	252.45	2.044		
15,100.0	10,860.5	15,417.6	10,835.9	129.3	132.2	87.27	-1,040.0	4,403.1	515.8	257.6	258.20	1.998		
15,200.0	10,861.0	15,517.6	10,836.5	132.1	135.0	87.27	-1,040.6	4,503.1	515.5	251.6	263.95	1.953		
15,300.0	10,861.5	15,617.6	10,837.0	135.0	137.9	87.27	-1,041.1	4,603.1	515.3	245.6	269.70	1.910		
15,400.0	10,862.0	15,717.6	10,837.5	137.9	140.7	87.27	-1,041.6	4,703.1	515.0	239.5	275.46	1.870		
15,500.0	10,862.6	15,817.6	10,838.0	140.7	143.5	87.27	-1,042.2	4,803.1	514.7	233.5	281.23	1.830		
15,600.0	10,863.1	15,917.6	10,838.5	143.6	146.4	87.27	-1,042.7	4,903.1	514.4	227.4	286.99	1.792		
15,700.0	10,863.6	16,017.6	10,839.1	146.5	149.2	87.26	-1,043.2	5,003.1	514.2	221.4	292.76	1.756		
15,800.0	10,864.1	16,117.6	10,839.6	149.4	152.1	87.26	-1,043.7	5,103.1	513.9	215.3	298.54	1.721		
15,900.0	10,864.7	16,217.6	10,840.1	152.3	154.9	87.26	-1,044.3	5,203.1	513.6	209.3	304.31	1.688		
16,000.0	10,865.2	16,317.6	10,840.6	155.1	157.8	87.26	-1,044.8	5,303.1	513.3	203.2	310.09	1.655		
16,100.0	10,865.7	16,417.6	10,841.2	158.0	160.6	87.26	-1,045.3	5,403.1	513.1	197.2	315.88	1.624		
16,200.0	10,866.2	16,517.6	10,841.7	160.9	163.5	87.26	-1,045.9	5,503.1	512.8	191.1	321.66	1.594		
16,300.0	10,866.7	16,617.6	10,842.2	163.8	166.4	87.26	-1,046.4	5,603.1	512.5	185.1	327.45	1.565		
16,400.0	10,867.3	16,717.6	10,842.7	166.7	169.2	87.25	-1,046.9	5,703.1	512.2	179.0	333.24	1.537		
16,500.0	10,867.8	16,817.6	10,843.3	169.6	172.1	87.25	-1,047.5	5,803.1	512.0	172.9	339.03	1.510		
16,600.0	10,868.3	16,917.6	10,843.8	172.5	175.0	87.25	-1,048.0	5,903.1	511.7	166.9	344.83	1.484 Level 3		
16,700.0	10,868.8	17,017.6	10,844.3	175.4	177.8	87.25	-1,048.5	6,003.1	511.4	160.8	350.63	1.459 Level 3		
16,800.0	10,869.4	17,117.6	10,844.8	178.3	180.7	87.25	-1,049.0	6,103.0	511.1	154.7	356.42	1.434 Level 3		
16,900.0	10,869.9	17,217.6	10,845.4	181.2	183.6	87.25	-1,049.6	6,203.0	510.9	148.6	362.22	1.410 Level 3		
17,000.0	10,870.4	17,317.6	10,845.9	184.1	186.5	87.25	-1,050.1	6,303.0	510.6	142.6	368.03	1.387 Level 3		
17,100.0	10,870.9	17,417.6	10,846.4	187.0	189.4	87.24	-1,050.6	6,403.0	510.3	136.5	373.83	1.365 Level 3		
17,200.0	10,871.5	17,517.6	10,846.9	189.9	192.2	87.24	-1,051.2	6,503.0	510.0	130.4	379.64	1.344 Level 3		
17,300.0	10,872.0	17,617.6	10,847.4	192.8	195.1	87.24	-1,051.7	6,603.0	509.8	124.3	385.44	1.323 Level 3		
17,400.0	10,872.5	17,717.6	10,848.0	195.7	198.0	87.24	-1,052.2	6,703.0	509.5	118.2	391.25	1.302 Level 3		
17,500.0	10,873.0	17,817.6	10,848.5	198.6	200.9	87.24	-1,052.7	6,803.0	509.2	112.2	397.06	1.282 Level 3		
17,600.0	10,873.6	17,917.6	10,849.0	201.5	203.8	87.24	-1,053.3	6,903.0	508.9	106.1	402.87	1.263 Level 3		
17,700.0	10,874.1	18,017.6	10,849.5	204.4	206.7	87.23	-1,053.8	7,003.0	508.7	100.0	408.68	1.245 Level 2		
17,800.0	10,874.6	18,117.6	10,850.1	207.3	209.5	87.23	-1,054.3	7,103.0	508.4	93.9	414.50	1.227 Level 2		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18 3T - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
17,900.0	10,875.1	18,217.6	10,850.6	210.2	212.4	87.23	-1,054.9	7,203.0	508.1	87.8	420.31	1.209	Level 2	
18,000.0	10,875.6	18,317.6	10,851.1	213.1	215.3	87.23	-1,055.4	7,303.0	507.8	81.7	426.13	1.192	Level 2	
18,100.0	10,876.2	18,417.6	10,851.6	216.0	218.2	87.23	-1,055.9	7,403.0	507.6	75.6	431.94	1.175	Level 2	
18,200.0	10,876.7	18,517.6	10,852.2	218.9	221.1	87.23	-1,056.5	7,503.0	507.3	69.5	437.76	1.159	Level 2	
18,300.0	10,877.2	18,617.6	10,852.7	221.8	224.0	87.23	-1,057.0	7,603.0	507.0	63.4	443.58	1.143	Level 2	
18,400.0	10,877.7	18,717.6	10,853.2	224.7	226.9	87.22	-1,057.5	7,703.0	506.8	57.3	449.40	1.128	Level 2	
18,500.0	10,878.3	18,817.6	10,853.7	227.6	229.8	87.22	-1,058.0	7,803.0	506.5	51.3	455.22	1.113	Level 2	
18,600.0	10,878.8	18,917.6	10,854.3	230.5	232.7	87.22	-1,058.6	7,903.0	506.2	45.2	461.04	1.098	Level 2	
18,700.0	10,879.3	19,017.6	10,854.8	233.4	235.6	87.22	-1,059.1	8,003.0	505.9	39.1	466.87	1.084	Level 2	
18,800.0	10,879.8	19,117.6	10,855.3	236.4	238.5	87.22	-1,059.6	8,103.0	505.7	33.0	472.69	1.070	Level 2	
18,900.0	10,880.4	19,217.6	10,855.8	239.3	241.4	87.22	-1,060.2	8,203.0	505.4	26.9	478.51	1.056	Level 2	
19,000.0	10,880.9	19,317.6	10,856.3	242.2	244.3	87.22	-1,060.7	8,303.0	505.1	20.8	484.34	1.043	Level 2	
19,100.0	10,881.4	19,417.6	10,856.9	245.1	247.2	87.21	-1,061.2	8,403.0	504.8	14.7	490.16	1.030	Level 2	
19,200.0	10,881.9	19,517.6	10,857.4	248.0	250.1	87.21	-1,061.7	8,503.0	504.6	8.6	495.99	1.017	Level 2	
19,300.0	10,882.5	19,617.6	10,857.9	250.9	253.0	87.21	-1,062.3	8,603.0	504.3	2.5	501.82	1.005	Level 2	
19,400.0	10,883.0	19,717.6	10,858.4	253.8	255.9	87.21	-1,062.8	8,703.0	504.0	-3.6	507.64	0.993	Level 1	
19,500.0	10,883.5	19,817.6	10,859.0	256.7	258.8	87.21	-1,063.3	8,803.0	503.7	-9.7	513.47	0.981	Level 1	
19,600.0	10,884.0	19,917.6	10,859.5	259.7	261.7	87.21	-1,063.9	8,903.0	503.5	-15.8	519.30	0.969	Level 1	
19,700.0	10,884.5	20,017.6	10,860.0	262.6	264.6	87.20	-1,064.4	9,003.0	503.2	-21.9	525.13	0.958	Level 1	
19,800.0	10,885.1	20,117.6	10,860.5	265.5	267.5	87.20	-1,064.9	9,103.0	502.9	-28.1	530.96	0.947	Level 1	
19,900.0	10,885.6	20,217.6	10,861.1	268.4	270.4	87.20	-1,065.5	9,202.9	502.6	-34.2	536.79	0.936	Level 1	
20,000.0	10,886.1	20,317.6	10,861.6	271.3	273.3	87.20	-1,066.0	9,302.9	502.4	-40.3	542.62	0.926	Level 1	
20,100.0	10,886.6	20,417.6	10,862.1	274.2	276.2	87.20	-1,066.5	9,402.9	502.1	-46.4	548.45	0.915	Level 1	
20,200.0	10,887.2	20,517.6	10,862.6	277.2	279.2	87.20	-1,067.0	9,502.9	501.8	-52.5	554.28	0.905	Level 1	
20,300.0	10,887.7	20,617.6	10,863.2	280.1	282.1	87.20	-1,067.6	9,602.9	501.5	-58.6	560.12	0.895	Level 1	
20,400.0	10,888.2	20,717.6	10,863.7	283.0	285.0	87.19	-1,068.1	9,702.9	501.3	-64.7	565.95	0.886	Level 1	
20,500.0	10,888.7	20,817.6	10,864.2	285.9	287.9	87.19	-1,068.6	9,802.9	501.0	-70.8	571.78	0.876	Level 1	
20,600.0	10,889.3	20,917.6	10,864.7	288.8	290.8	87.19	-1,069.2	9,902.9	500.7	-76.9	577.62	0.867	Level 1	
20,700.0	10,889.8	21,017.6	10,865.3	291.7	293.7	87.19	-1,069.7	10,002.9	500.4	-83.0	583.45	0.858	Level 1	
20,717.6	10,889.9	21,035.1	10,865.3	292.3	294.2	87.19	-1,069.8	10,020.5	500.4	-84.1	584.47	0.856	Level 1, ES, SF	

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1													Offset Site Error:	0.0 usft
Survey Program: 2175-MWDD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
200.0	200.0	2,090.0	2,090.0	0.3	0.0	-157.01	-30.4	-12.9	1,914.3	1,914.0	0.31	6,190.021		
300.0	300.0	2,090.0	2,090.0	0.5	0.0	-157.01	-30.4	-12.9	1,814.3	1,813.8	0.53	3,397.443		
400.0	400.0	2,090.0	2,090.0	0.8	0.0	-157.01	-30.4	-12.9	1,714.3	1,713.6	0.76	2,259.292		
500.0	500.0	2,090.0	2,090.0	1.0	0.0	-157.01	-30.4	-12.9	1,614.3	1,613.4	0.98	1,641.335		
600.0	600.0	2,090.0	2,090.0	1.2	0.0	-157.01	-30.4	-12.9	1,514.4	1,513.2	1.21	1,253.280		
700.0	700.0	2,090.0	2,090.0	1.4	0.0	-157.01	-30.4	-12.9	1,414.4	1,413.0	1.43	986.953		
800.0	800.0	2,090.0	2,090.0	1.7	0.0	-157.01	-30.4	-12.9	1,314.4	1,312.8	1.66	792.843		
900.0	900.0	2,090.0	2,090.0	1.9	0.0	-157.01	-30.4	-12.9	1,214.4	1,212.6	1.88	645.086		
1,000.0	1,000.0	2,090.0	2,090.0	2.1	0.0	-157.01	-30.4	-12.9	1,114.5	1,112.4	2.11	528.850		
1,100.0	1,100.0	2,090.0	2,090.0	2.3	0.0	-157.01	-30.4	-12.9	1,014.5	1,012.2	2.33	435.023		
1,200.0	1,200.0	2,090.0	2,090.0	2.6	0.0	-157.01	-30.4	-12.9	914.6	912.0	2.56	357.696		
1,300.0	1,300.0	2,090.0	2,090.0	2.8	0.0	-157.01	-30.4	-12.9	814.7	811.9	2.78	292.870		
1,400.0	1,400.0	2,090.0	2,090.0	3.0	0.0	-157.01	-30.4	-12.9	714.8	711.8	3.01	237.744		
1,500.0	1,500.0	2,090.0	2,090.0	3.2	0.0	-157.01	-30.4	-12.9	614.9	611.7	3.23	190.296		
1,600.0	1,600.0	2,090.0	2,090.0	3.5	0.0	-157.01	-30.4	-12.9	515.1	511.6	3.46	149.035		
1,700.0	1,700.0	2,090.0	2,090.0	3.7	0.0	-157.01	-30.4	-12.9	415.3	411.6	3.68	112.835		
1,800.0	1,800.0	2,090.0	2,090.0	3.9	0.0	-157.01	-30.4	-12.9	315.7	311.8	3.91	80.843		
1,900.0	1,900.0	2,090.0	2,090.0	4.1	0.0	-157.01	-30.4	-12.9	216.5	212.4	4.13	52.426		
2,000.0	2,000.0	2,090.0	2,090.0	4.4	0.0	-157.01	-30.4	-12.9	118.7	114.3	4.36	27.253		
2,100.0	2,100.0	2,090.0	2,090.0	4.6	0.0	-157.01	-30.4	-12.9	35.9	31.3	4.58	7.833		
2,132.6	2,132.6	2,108.5	2,108.5	4.7	0.0	-157.04	-30.4	-12.9	33.1	28.4	4.67	7.073 CC		
2,200.0	2,200.0	2,175.5	2,175.5	4.8	0.1	-157.59	-31.2	-12.8	33.7	28.8	4.90	6.873		
2,300.0	2,300.0	2,275.5	2,275.5	5.0	0.3	-158.55	-32.8	-12.9	35.2	29.9	5.34	6.600		
2,400.0	2,400.0	2,375.5	2,375.5	5.3	0.5	-158.59	-34.1	-13.4	36.7	30.9	5.78	6.346		
2,500.0	2,500.0	2,475.6	2,475.6	5.5	0.7	-157.20	-34.9	-14.7	37.9	31.7	6.23	6.089		
2,600.0	2,600.0	2,575.7	2,575.6	5.7	1.0	-117.42	-35.2	-16.5	39.7	33.0	6.65	5.963		
2,650.0	2,649.9	2,625.6	2,625.5	5.8	1.1	121.61	-35.1	-17.7	41.2	34.4	6.86	6.010		
2,700.0	2,699.9	2,675.3	2,675.2	5.9	1.2	126.36	-34.8	-19.2	43.3	36.2	7.07	6.123		
2,800.0	2,799.7	2,775.0	2,774.8	6.1	1.4	135.09	-34.3	-22.7	48.7	41.2	7.49	6.500		
2,900.0	2,899.6	2,875.5	2,875.3	6.3	1.6	141.31	-33.4	-24.8	53.5	45.6	7.90	6.776		
3,000.0	2,999.5	2,975.0	2,974.8	6.5	1.8	146.23	-32.5	-26.4	58.5	50.2	8.31	7.043		
3,100.0	3,099.3	3,074.3	3,074.1	6.7	2.0	150.31	-31.9	-28.6	64.5	55.8	8.73	7.389		
3,200.0	3,199.2	3,173.7	3,173.5	7.0	2.3	153.83	-31.4	-31.4	71.4	62.2	9.15	7.801		
3,300.0	3,299.0	3,273.6	3,273.3	7.2	2.5	156.60	-31.1	-34.1	78.5	69.0	9.56	8.213		
3,400.0	3,398.9	3,373.5	3,373.1	7.4	2.7	158.89	-30.7	-36.6	85.5	75.5	9.97	8.576		
3,501.2	3,500.0	3,474.7	3,474.3	7.6	2.9	160.89	-30.2	-38.9	92.5	82.1	10.38	8.907		
3,600.0	3,598.7	3,573.2	3,572.8	7.8	3.1	162.31	-29.6	-41.0	97.6	86.8	10.79	9.041		
3,651.2	3,649.9	3,624.4	3,624.0	7.9	3.2	-107.21	-29.3	-42.2	99.0	88.0	11.04	8.973		
3,700.0	3,698.7	3,673.3	3,672.9	8.0	3.3	-106.86	-29.0	-43.2	100.0	88.7	11.24	8.892		
3,800.0	3,798.7	3,773.8	3,773.4	8.2	3.5	-106.18	-28.3	-45.1	101.6	89.9	11.67	8.703		
3,900.0	3,898.7	3,873.6	3,873.2	8.5	3.7	-105.49	-27.5	-46.7	102.9	90.8	12.10	8.503		
4,000.0	3,998.7	3,973.9	3,973.5	8.7	3.9	-104.82	-26.7	-48.4	104.3	91.7	12.53	8.320		
4,100.0	4,098.7	4,074.2	4,073.7	8.9	4.1	-104.14	-25.7	-49.7	105.3	92.3	12.96	8.123		
4,200.0	4,198.7	4,174.5	4,174.1	9.1	4.3	-103.56	-24.9	-50.7	106.1	92.7	13.39	7.920		
4,300.0	4,298.7	4,274.2	4,273.7	9.4	4.6	-103.00	-24.0	-51.7	106.8	93.0	13.82	7.729		
4,400.0	4,398.7	4,374.3	4,373.8	9.6	4.8	-102.43	-23.2	-52.8	107.7	93.5	14.26	7.556		
4,500.0	4,498.7	4,474.5	4,474.0	9.8	5.0	-101.94	-22.5	-53.7	108.5	93.8	14.69	7.387		
4,600.0	4,598.7	4,574.7	4,574.1	10.0	5.2	-101.55	-21.8	-54.4	109.1	93.9	15.12	7.213		
4,700.0	4,698.7	4,674.5	4,674.0	10.2	5.4	-101.22	-21.3	-55.2	109.7	94.1	15.55	7.052		
4,800.0	4,798.7	4,774.5	4,774.0	10.5	5.6	-100.92	-20.9	-55.9	110.3	94.3	15.98	6.902		
4,900.0	4,898.7	4,874.6	4,874.1	10.7	5.8	-100.52	-20.3	-56.6	110.9	94.5	16.41	6.757		
5,000.0	4,998.7	4,974.7	4,974.1	10.9	6.0	-100.29	-19.9	-57.2	111.4	94.5	16.85	6.612		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1													Offset Site Error: 0.0 usft
Survey Program: 2175-MWVD													Offset Well Error: 0.0 usft
Reference		Offset		Semi Major Axis			Distance						
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
5,100.0	5,098.7	5,074.4	5,073.8	11.1	6.2	-99.95	-19.4	-58.0	112.0	94.8	17.28	6.484	
5,200.0	5,198.7	5,174.4	5,173.8	11.4	6.4	-99.72	-19.1	-58.9	112.9	95.2	17.71	6.374	
5,300.0	5,298.7	5,274.5	5,274.0	11.6	6.6	-99.95	-19.6	-59.5	113.6	95.4	18.15	6.259	
5,400.0	5,398.7	5,374.6	5,374.0	11.8	6.8	-100.24	-20.3	-60.0	114.2	95.6	18.58	6.148	
5,500.0	5,498.7	5,474.0	5,473.5	12.0	7.1	-100.62	-21.2	-60.6	114.9	95.9	19.01	6.046	
5,600.0	5,598.7	5,574.1	5,573.5	12.2	7.3	-101.26	-22.7	-61.5	116.2	96.7	19.44	5.974	
5,700.0	5,698.7	5,674.4	5,673.8	12.5	7.5	-101.98	-24.3	-62.0	117.0	97.1	19.88	5.885	
5,800.0	5,798.7	5,774.4	5,773.8	12.7	7.7	-102.48	-25.4	-62.5	117.7	97.4	20.31	5.797	
5,900.0	5,898.7	5,874.2	5,873.5	12.9	7.9	-102.68	-26.0	-63.2	118.5	97.8	20.74	5.715	
6,000.0	5,998.7	5,974.1	5,973.5	13.1	8.1	-102.57	-26.0	-64.3	119.6	98.4	21.18	5.646	
6,100.0	6,098.7	6,074.1	6,073.5	13.4	8.3	-102.09	-25.3	-65.5	120.6	99.0	21.61	5.580	
6,200.0	6,198.7	6,174.7	6,174.0	13.6	8.5	-101.50	-24.2	-66.6	121.4	99.4	22.05	5.506	
6,300.0	6,298.7	6,275.3	6,274.6	13.8	8.7	-100.79	-22.8	-67.1	121.7	99.2	22.48	5.412	
6,400.0	6,398.7	6,375.5	6,374.9	14.0	9.0	-99.92	-21.0	-67.4	121.6	98.7	22.92	5.307	
6,500.0	6,498.7	6,475.7	6,475.0	14.3	9.2	-98.97	-18.9	-67.5	121.4	98.1	23.36	5.199	
6,600.0	6,598.7	6,575.8	6,575.1	14.5	9.4	-97.78	-16.4	-67.6	121.2	97.4	23.79	5.092	
6,700.0	6,698.7	6,676.0	6,675.2	14.7	9.6	-96.47	-13.6	-67.6	120.8	96.6	24.23	4.985	
6,800.0	6,798.7	6,776.5	6,775.7	14.9	9.8	-95.00	-10.5	-67.4	120.2	95.6	24.67	4.874	
6,900.0	6,898.7	6,877.2	6,876.4	15.2	10.0	-93.90	-8.1	-66.3	119.0	93.9	25.10	4.740	
7,000.0	6,998.7	6,976.8	6,976.0	15.4	10.2	-93.04	-6.2	-65.1	117.7	92.2	25.53	4.611	
7,100.0	7,098.7	7,076.5	7,075.7	15.6	10.4	-92.89	-5.9	-64.2	116.8	90.8	25.96	4.499	
7,200.0	7,198.7	7,175.9	7,175.1	15.8	10.6	-92.85	-5.8	-63.5	116.1	89.7	26.39	4.398	
7,228.5	7,227.2	7,204.1	7,203.2	15.9	10.7	-92.96	-6.0	-63.5	116.0	89.5	26.51	4.376	
7,300.0	7,298.7	7,275.2	7,274.3	16.0	10.8	-93.47	-7.0	-63.6	116.2	89.4	26.82	4.333	
7,400.0	7,398.7	7,374.9	7,374.0	16.3	11.0	-94.22	-8.6	-63.9	116.7	89.4	27.25	4.281	
7,500.0	7,498.7	7,474.7	7,473.7	16.5	11.2	-94.97	-10.2	-64.5	117.4	89.7	27.68	4.240	
7,600.0	7,598.7	7,574.6	7,573.7	16.7	11.5	-95.60	-11.5	-65.3	118.2	90.1	28.12	4.205	
7,700.0	7,698.7	7,675.0	7,674.1	16.9	11.7	-96.08	-12.6	-65.9	119.0	90.5	28.56	4.168	
7,800.0	7,798.7	7,775.5	7,774.6	17.2	11.9	-96.38	-13.2	-66.2	119.3	90.3	28.99	4.115	
7,900.0	7,898.7	7,875.8	7,874.9	17.4	12.1	-96.52	-13.5	-66.1	119.3	89.8	29.43	4.053	
8,000.0	7,998.7	7,976.2	7,975.3	17.6	12.3	-96.37	-13.2	-65.8	119.0	89.1	29.87	3.983	
8,100.0	8,098.7	8,076.1	8,075.1	17.8	12.5	-96.14	-12.7	-65.4	118.5	88.2	30.30	3.910	
8,200.0	8,198.7	8,176.3	8,175.3	18.1	12.7	-95.84	-12.0	-65.0	118.1	87.3	30.74	3.841	
8,300.0	8,298.7	8,276.8	8,275.8	18.3	12.9	-95.44	-11.1	-64.4	117.3	86.1	31.17	3.763	
8,400.0	8,398.7	8,376.8	8,375.8	18.5	13.1	-94.96	-10.1	-63.5	116.3	84.7	31.61	3.680	
8,500.0	8,498.7	8,476.7	8,475.8	18.7	13.4	-94.46	-9.0	-62.6	115.3	83.3	32.04	3.600	
8,600.0	8,598.7	8,576.8	8,575.8	19.0	13.6	-93.96	-7.9	-61.7	114.4	81.9	32.47	3.522	
8,700.0	8,698.7	8,676.4	8,675.4	19.2	13.8	-93.45	-6.8	-61.0	113.6	80.7	32.90	3.453	
8,800.0	8,798.7	8,776.5	8,775.5	19.4	14.0	-92.93	-5.8	-60.4	112.9	79.6	33.34	3.388	
8,900.0	8,898.7	8,876.6	8,875.6	19.6	14.2	-92.37	-4.6	-59.6	112.1	78.4	33.77	3.321	
9,000.0	8,998.7	8,976.6	8,975.6	19.9	14.4	-91.62	-3.2	-58.9	111.4	77.2	34.21	3.256	
9,100.0	9,098.7	9,076.8	9,075.8	20.1	14.6	-90.89	-1.7	-58.0	110.4	75.7	34.64	3.186	
9,200.0	9,198.7	9,176.7	9,175.7	20.3	14.8	-90.12	-0.2	-57.1	109.5	74.4	35.08	3.122	
9,300.0	9,298.7	9,276.7	9,275.6	20.5	15.0	-89.47	1.0	-56.1	108.5	73.0	35.51	3.057	
9,400.0	9,398.7	9,376.7	9,375.6	20.8	15.2	-88.81	2.2	-55.3	107.8	71.8	35.94	2.999	
9,500.0	9,498.7	9,476.7	9,475.6	21.0	15.4	-88.08	3.6	-54.3	106.8	70.4	36.38	2.936	
9,600.0	9,598.7	9,576.5	9,575.4	21.2	15.6	-87.35	4.9	-53.5	106.1	69.2	36.81	2.881	
9,700.0	9,698.7	9,676.4	9,675.3	21.4	15.9	-86.69	6.1	-52.8	105.4	68.2	37.24	2.831	
9,800.0	9,798.7	9,776.5	9,775.4	21.6	16.1	-86.11	7.1	-52.1	104.8	67.1	37.67	2.782	
9,900.0	9,898.7	9,876.4	9,875.2	21.9	16.3	-85.54	8.1	-51.4	104.2	66.1	38.10	2.734	
10,000.0	9,998.7	9,976.4	9,975.3	22.1	16.5	-84.90	9.2	-50.8	103.7	65.1	38.54	2.690	
10,100.0	10,098.7	10,076.5	10,075.3	22.3	16.7	-84.17	10.5	-50.1	103.1	64.1	38.97	2.644	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1													Offset Site Error:	0.0 usft
Survey Program: 2175-MWDD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
10,200.0	10,198.7	10,176.4	10,175.3	22.5	16.9	-83.36	11.8	-49.3	102.4	63.0	39.41	2.599		
10,300.0	10,298.7	10,280.4	10,279.2	22.8	17.1	-82.66	12.9	-47.7	101.1	61.2	39.85	2.536		
10,363.3	10,362.0	10,358.0	10,355.9	22.9	17.3	-86.17	6.1	-39.0	93.4	53.2	40.15	2.325		
10,375.0	10,373.7	10,371.1	10,368.6	22.9	17.3	152.57	3.7	-36.6	91.2	51.0	40.15	2.271		
10,400.0	10,398.7	10,398.5	10,394.6	23.0	17.3	149.40	-2.3	-30.8	86.7	46.5	40.20	2.158		
10,425.0	10,423.5	10,424.8	10,419.3	23.0	17.4	145.79	-9.0	-24.4	83.0	42.8	40.21	2.064		
10,450.0	10,448.2	10,450.0	10,442.5	23.1	17.4	141.86	-16.4	-17.9	80.2	40.0	40.19	1.996		
10,475.0	10,472.7	10,475.0	10,465.0	23.1	17.5	137.54	-24.6	-11.1	78.6	38.5	40.18	1.958		
10,493.4	10,490.5	10,493.2	10,481.2	23.2	17.5	134.19	-31.1	-5.9	78.3	38.1	40.18	1.949		
10,500.0	10,496.8	10,499.7	10,486.9	23.2	17.5	133.00	-33.6	-4.1	78.3	38.2	40.19	1.949		
10,525.0	10,520.6	10,524.4	10,508.5	23.3	17.6	128.55	-43.1	3.3	79.3	39.1	40.23	1.971		
10,550.0	10,544.0	10,549.8	10,530.4	23.3	17.7	124.34	-53.3	11.2	81.2	40.9	40.31	2.015		
10,575.0	10,566.8	10,576.8	10,553.1	23.4	17.7	120.16	-64.7	20.5	83.7	43.2	40.46	2.068		
10,600.0	10,589.1	10,603.4	10,574.4	23.5	17.8	116.09	-76.5	30.9	86.3	45.7	40.65	2.124		
10,625.0	10,610.8	10,629.3	10,594.2	23.5	17.9	111.96	-89.0	42.2	89.4	48.5	40.89	2.186		
10,650.0	10,631.8	10,654.6	10,612.2	23.6	18.0	107.81	-102.0	54.2	92.8	51.7	41.18	2.254		
10,675.0	10,652.0	10,679.2	10,628.7	23.7	18.1	103.77	-115.3	66.8	96.8	55.3	41.50	2.334		
10,700.0	10,671.5	10,704.2	10,644.3	23.8	18.2	99.80	-129.4	80.3	101.3	59.5	41.83	2.421		
10,725.0	10,690.1	10,729.4	10,658.9	23.9	18.4	96.05	-143.9	94.8	106.0	63.8	42.16	2.514		
10,750.0	10,707.8	10,754.3	10,672.4	24.0	18.5	92.65	-158.5	109.8	110.8	68.3	42.48	2.609		
10,775.0	10,724.5	10,778.4	10,684.2	24.1	18.7	89.42	-172.9	125.1	115.8	73.1	42.77	2.708		
10,800.0	10,740.3	10,801.4	10,694.0	24.3	18.9	86.33	-187.0	140.3	121.2	78.1	43.03	2.816		
10,825.0	10,755.0	10,825.3	10,702.8	24.4	19.1	83.16	-201.9	156.8	126.7	83.5	43.26	2.930		
10,850.0	10,768.7	10,849.1	10,710.2	24.6	19.3	80.10	-216.8	173.8	132.3	88.9	43.44	3.046		
10,875.0	10,781.2	10,872.7	10,717.0	24.8	19.5	77.53	-231.6	190.9	138.0	94.3	43.61	3.163		
10,900.0	10,792.6	10,895.7	10,722.8	25.0	19.8	75.33	-246.0	207.8	143.5	99.8	43.78	3.279		
10,925.0	10,802.8	10,919.0	10,727.6	25.2	20.1	73.24	-260.7	225.4	149.1	105.1	43.94	3.392		
10,950.0	10,811.8	10,942.4	10,730.8	25.4	20.4	71.12	-275.2	243.4	154.4	110.4	44.06	3.505		
10,975.0	10,819.6	10,965.8	10,732.7	25.7	20.7	69.04	-289.4	261.8	159.6	115.4	44.17	3.614		
11,000.0	10,826.1	10,990.4	10,734.0	25.9	21.0	67.16	-304.3	281.5	164.5	120.1	44.32	3.710		
11,025.0	10,831.3	11,015.2	10,734.9	26.2	21.4	65.71	-319.0	301.4	168.8	124.3	44.54	3.790		
11,050.0	10,835.3	11,040.2	10,735.5	26.5	21.7	64.64	-333.7	321.6	172.6	127.8	44.85	3.849		
11,075.0	10,837.9	11,066.6	10,735.9	26.8	22.1	63.94	-349.0	343.0	175.8	130.5	45.28	3.882		
11,100.0	10,839.3	11,094.0	10,736.3	27.1	22.6	63.69	-364.6	365.6	178.0	132.2	45.88	3.880		
11,110.8	10,839.5	11,105.9	10,736.4	27.2	22.7	63.73	-371.3	375.4	178.7	132.5	46.19	3.870		
11,125.8	10,839.5	11,122.5	10,736.7	27.4	23.0	63.93	-380.5	389.3	179.5	132.8	46.65	3.847		
11,200.0	10,839.9	11,206.0	10,737.0	28.5	24.5	64.34	-423.8	460.7	182.3	133.4	48.93	3.727		
11,300.0	10,840.5	11,330.7	10,737.1	30.1	26.9	63.93	-477.0	573.3	180.9	128.6	52.30	3.459		
11,400.0	10,841.0	11,451.7	10,738.7	31.9	29.4	62.45	-512.8	688.8	171.1	115.6	55.47	3.084		
11,500.0	10,841.5	11,570.1	10,739.6	33.9	31.9	58.92	-532.1	805.5	154.4	96.7	57.74	2.674		
11,600.0	10,842.1	11,665.7	10,740.1	36.0	34.0	54.96	-541.3	900.6	137.3	78.3	59.04	2.326		
11,700.0	10,842.6	11,767.5	10,739.3	38.1	36.4	50.26	-549.9	1,002.1	125.0	65.4	59.54	2.099		
11,800.0	10,843.2	11,867.8	10,739.3	40.4	38.8	46.16	-556.7	1,102.1	115.7	55.9	59.81	1.935		
11,900.0	10,843.7	11,968.1	10,739.4	42.7	41.2	43.23	-563.1	1,202.3	110.4	49.9	60.45	1.825		
12,000.0	10,844.2	12,069.2	10,739.6	45.0	43.7	41.23	-567.3	1,303.3	107.2	45.7	61.52	1.743		
12,031.4	10,844.4	12,100.3	10,739.6	45.8	44.5	40.97	-568.6	1,334.3	107.0	44.9	62.13	1.722		
12,100.0	10,844.8	12,170.8	10,739.8	47.4	46.3	41.11	-571.3	1,404.8	107.5	43.4	64.08	1.677		
12,110.4	10,844.8	12,181.8	10,739.9	47.6	46.6	41.20	-571.6	1,415.8	107.5	43.1	64.44	1.668		
12,200.0	10,845.3	12,273.5	10,741.8	49.8	48.9	41.79	-572.5	1,507.4	106.6	38.4	68.13	1.564		
12,300.0	10,845.8	12,372.7	10,744.2	52.2	51.5	42.40	-573.1	1,606.7	105.0	32.8	72.27	1.453 Level 3		
12,400.0	10,846.3	12,473.6	10,746.1	54.7	54.1	42.82	-573.8	1,707.5	104.0	27.6	76.36	1.362 Level 3		
12,487.0	10,846.8	12,558.8	10,748.2	56.9	56.4	43.57	-574.8	1,792.7	103.0	22.6	80.40	1.281 Level 3		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1													Offset Site Error:	0.0 usft
Survey Program: 2175-MWDD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
12,500.0	10,846.8	12,571.4	10,748.4	57.3	56.7	43.75	-575.1	1,805.3	103.0	21.9	81.09	1.270	Level 3	
12,600.0	10,847.4	12,669.6	10,750.2	59.9	59.4	45.43	-579.0	1,903.4	104.3	17.4	86.89	1.200	Level 2	
12,700.0	10,847.9	12,772.8	10,752.1	62.5	62.2	46.74	-581.8	2,006.5	104.8	12.3	92.52	1.132	Level 2	
12,760.8	10,848.2	12,832.6	10,753.1	64.1	63.8	47.12	-582.6	2,066.3	104.5	9.1	95.44	1.095	Level 2	
12,800.0	10,848.4	12,871.7	10,753.5	65.1	64.9	47.29	-583.2	2,105.3	104.6	7.3	97.26	1.095	Level 2	
12,900.0	10,848.9	12,972.9	10,754.6	67.8	67.7	47.60	-584.2	2,206.5	104.4	2.6	101.81	1.025	Level 2	
13,000.0	10,849.5	13,074.4	10,757.1	70.4	70.5	48.09	-584.1	2,308.1	102.4	-4.3	106.67	0.960	Level 1	
13,100.0	10,850.0	13,173.9	10,759.5	73.1	73.2	48.33	-583.5	2,407.5	100.1	-11.1	111.21	0.900	Level 1	
13,200.0	10,850.5	13,274.3	10,760.8	75.9	76.0	47.97	-582.5	2,507.9	98.2	-16.7	114.91	0.855	Level 1	
13,300.0	10,851.0	13,374.5	10,762.6	78.6	78.8	47.43	-580.5	2,608.0	95.3	-23.0	118.27	0.805	Level 1	
13,400.0	10,851.6	13,473.4	10,762.4	81.3	81.5	45.72	-578.0	2,706.9	93.4	-26.2	119.58	0.781	Level 1	
13,500.0	10,852.1	13,572.7	10,761.5	84.1	84.3	43.78	-575.8	2,806.2	92.3	-27.9	120.19	0.768	Level 1	
13,600.0	10,852.6	13,672.4	10,760.1	86.9	87.1	41.81	-574.0	2,905.9	91.9	-28.5	120.41	0.763	Level 1	
13,700.0	10,853.1	13,775.4	10,759.8	89.6	89.9	39.98	-571.8	3,008.8	90.6	-30.0	120.61	0.751	Level 1	
13,800.0	10,853.7	13,873.9	10,761.2	92.4	92.7	38.15	-568.2	3,107.3	87.1	-33.3	120.43	0.723	Level 1	
13,900.0	10,854.2	13,974.0	10,759.9	95.2	95.5	34.93	-564.3	3,207.3	85.8	-31.0	116.78	0.734	Level 1	
14,000.0	10,854.7	14,074.1	10,759.8	98.0	98.3	32.18	-560.6	3,307.3	83.8	-29.8	113.63	0.738	Level 1	
14,100.0	10,855.2	14,174.7	10,760.6	100.8	101.2	30.11	-557.8	3,407.9	81.7	-30.0	111.72	0.731	Level 1	
14,165.7	10,855.6	14,238.4	10,760.9	102.7	103.0	28.88	-556.3	3,471.5	80.7	-29.8	110.57	0.730	Level 1	
14,200.0	10,855.7	14,271.4	10,760.3	103.7	103.9	28.12	-555.8	3,504.5	81.0	-28.6	109.60	0.739	Level 1	
14,300.0	10,856.3	14,370.7	10,757.4	106.5	106.8	25.76	-554.5	3,603.8	83.2	-23.1	106.28	0.783	Level 1	
14,400.0	10,856.8	14,472.4	10,755.6	109.3	109.7	24.32	-554.1	3,705.4	84.7	-20.4	105.12	0.806	Level 1	
14,500.0	10,857.3	14,573.5	10,755.5	112.2	112.5	23.06	-553.2	3,806.5	84.6	-19.7	104.30	0.811	Level 1	
14,600.0	10,857.8	14,675.7	10,757.1	115.0	115.5	20.93	-550.2	3,908.7	82.2	-18.7	100.88	0.815	Level 1	
14,700.0	10,858.4	14,774.5	10,759.2	117.9	118.3	18.99	-547.5	4,007.5	79.5	-18.3	97.78	0.813	Level 1	
14,794.1	10,858.9	14,867.1	10,760.0	120.5	120.9	17.57	-546.1	4,100.0	78.5	-17.3	95.87	0.819	Level 1	
14,800.0	10,858.9	14,872.9	10,760.0	120.7	121.1	17.49	-546.0	4,105.8	78.5	-17.3	95.79	0.820	Level 1	
14,900.0	10,859.4	14,971.7	10,758.9	123.6	123.9	16.77	-546.3	4,204.6	79.9	-16.0	95.89	0.833	Level 1	
15,000.0	10,859.9	15,072.5	10,758.4	126.4	126.9	16.20	-546.5	4,305.4	80.7	-15.6	96.34	0.838	Level 1	
15,100.0	10,860.5	15,171.8	10,757.7	129.3	129.7	15.88	-547.2	4,404.7	81.9	-15.6	97.49	0.840	Level 1	
15,200.0	10,861.0	15,271.7	10,756.6	132.1	132.6	15.68	-548.2	4,504.6	83.5	-15.4	98.93	0.844	Level 1	
15,300.0	10,861.5	15,372.4	10,755.7	135.0	135.5	15.29	-548.8	4,605.3	84.9	-14.9	99.80	0.850	Level 1	
15,400.0	10,862.0	15,473.0	10,755.9	137.9	138.4	15.05	-549.3	4,705.9	85.0	-16.0	101.07	0.841	Level 1	
15,500.0	10,862.6	15,573.0	10,756.2	140.7	141.3	14.64	-549.6	4,805.9	85.2	-16.7	101.83	0.836	Level 1	
15,600.0	10,863.1	15,673.7	10,756.7	143.6	144.2	13.93	-549.3	4,906.6	84.9	-16.7	101.60	0.836	Level 1	
15,700.0	10,863.6	15,774.5	10,758.3	146.5	147.1	13.40	-549.0	5,007.4	83.6	-18.3	101.91	0.821	Level 1	
15,793.8	10,864.1	15,866.9	10,759.9	149.2	149.8	13.41	-549.5	5,099.8	82.4	-21.3	103.71	0.795	Level 1	
15,800.0	10,864.1	15,872.9	10,759.9	149.4	150.0	13.44	-549.6	5,105.8	82.4	-21.5	103.90	0.794	Level 1	
15,900.0	10,864.7	15,972.6	10,759.9	152.3	152.9	14.21	-551.7	5,205.4	83.3	-25.0	108.27	0.769	Level 1	
16,000.0	10,865.2	16,072.2	10,760.3	155.1	155.8	15.61	-554.7	5,305.0	84.0	-31.0	115.00	0.730	Level 1	
16,100.0	10,865.7	16,172.3	10,760.4	158.0	158.7	17.23	-558.1	5,405.1	85.1	-37.8	122.94	0.692	Level 1	
16,200.0	10,866.2	16,273.7	10,761.6	160.9	161.6	19.17	-561.7	5,506.4	85.4	-47.2	132.57	0.644	Level 1	
16,300.0	10,866.7	16,373.8	10,764.3	163.8	164.6	22.04	-566.3	5,606.3	84.7	-61.8	146.50	0.578	Level 1	
16,400.0	10,867.3	16,474.1	10,767.7	166.7	167.5	25.80	-571.8	5,706.4	84.0	-80.9	164.94	0.509	Level 1	
16,442.4	10,867.5	16,515.9	10,769.1	167.9	168.7	27.43	-574.2	5,748.1	83.8	-89.3	173.12	0.484	Level 1	
16,500.0	10,867.8	16,572.9	10,770.4	169.6	170.4	29.31	-577.3	5,805.0	84.1	-98.9	182.97	0.460	Level 1	
16,600.0	10,868.3	16,672.9	10,772.1	172.5	173.3	31.83	-581.7	5,904.9	85.0	-112.1	197.15	0.431	Level 1	
16,700.0	10,868.8	16,773.1	10,774.0	175.4	176.2	34.31	-586.0	6,004.9	85.8	-125.6	211.41	0.406	Level 1	
16,800.0	10,869.4	16,872.5	10,774.8	178.3	179.1	35.88	-589.5	6,104.3	87.1	-134.7	221.79	0.393	Level 1	
16,900.0	10,869.9	16,973.4	10,774.8	181.2	182.0	36.24	-591.4	6,205.2	88.1	-138.9	227.03	0.388	Level 1	
17,000.0	10,870.4	17,074.0	10,775.3	184.1	185.0	36.10	-592.0	6,305.7	88.0	-142.0	230.02	0.383	Level 1	
17,016.0	10,870.5	17,089.8	10,775.4	184.5	185.4	36.10	-592.1	6,321.6	88.0	-142.6	230.57	0.382	Level 1	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1													Offset Site Error:	0.0 usft
Survey Program: 2175-MWDD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
17,100.0	10,870.9	17,173.5	10,775.8	187.0	187.9	36.30	-593.2	6,405.3	88.3	-146.3	234.55	0.376	Level 1	
17,200.0	10,871.5	17,274.1	10,776.8	189.9	190.8	36.75	-594.5	6,505.9	88.2	-152.1	240.29	0.367	Level 1	
17,300.0	10,872.0	17,373.8	10,778.0	192.8	193.7	37.22	-595.7	6,605.6	87.8	-158.3	246.15	0.357	Level 1	
17,302.9	10,872.0	17,376.7	10,778.1	192.8	193.8	37.24	-595.7	6,608.4	87.8	-158.5	246.34	0.357	Level 1	
17,400.0	10,872.5	17,473.2	10,779.0	195.7	196.6	38.05	-597.7	6,704.9	88.2	-165.6	253.79	0.348	Level 1	
17,500.0	10,873.0	17,572.7	10,779.4	198.6	199.5	38.72	-599.9	6,804.4	89.2	-171.5	260.74	0.342	Level 1	
17,600.0	10,873.6	17,672.4	10,779.6	201.5	202.4	39.49	-602.6	6,904.0	90.7	-177.6	268.29	0.338	Level 1	
17,700.0	10,874.1	17,772.5	10,779.5	204.4	205.3	40.07	-605.1	7,004.2	92.3	-182.7	274.99	0.336	Level 1	
17,800.0	10,874.6	17,872.4	10,779.3	207.3	208.2	40.40	-607.2	7,104.0	93.7	-186.8	280.46	0.334	Level 1	
17,900.0	10,875.1	17,973.5	10,778.7	210.2	211.2	40.29	-608.7	7,205.1	95.0	-188.9	283.87	0.334	Level 1	
18,000.0	10,875.6	18,073.5	10,779.1	213.1	214.1	40.24	-609.6	7,305.1	95.1	-192.5	287.56	0.331	Level 1	
18,100.0	10,876.2	18,173.5	10,779.6	216.0	217.0	40.59	-611.1	7,405.0	95.6	-197.7	293.23	0.326	Level 1	
18,200.0	10,876.7	18,274.5	10,779.6	218.9	219.9	40.08	-611.3	7,506.1	95.5	-199.0	294.55	0.324	Level 1	
18,241.7	10,876.9	18,315.6	10,779.8	220.1	221.1	39.96	-611.3	7,547.1	95.3	-200.2	295.53	0.323	Level 1, SF	
18,300.0	10,877.2	18,372.4	10,779.5	221.8	222.8	39.67	-611.7	7,604.0	95.8	-200.5	296.23	0.323	Level 1, ES	
18,400.0	10,877.7	18,472.5	10,777.7	224.7	225.7	38.67	-612.2	7,704.0	97.4	-197.3	294.70	0.330	Level 1	
18,500.0	10,878.3	18,573.6	10,776.2	227.6	228.6	37.44	-611.9	7,805.1	98.3	-193.3	291.66	0.337	Level 1	
18,600.0	10,878.8	18,675.0	10,776.1	230.5	231.6	36.48	-611.2	7,906.6	97.9	-192.1	289.98	0.338	Level 1	
18,700.0	10,879.3	18,775.0	10,776.6	233.4	234.5	35.60	-610.2	8,006.5	96.8	-191.7	288.53	0.336	Level 1	
18,800.0	10,879.8	18,874.6	10,777.0	236.4	237.4	34.87	-609.5	8,106.1	96.1	-191.7	287.77	0.334	Level 1	
18,900.0	10,880.4	18,975.1	10,777.0	239.3	240.3	33.54	-608.0	8,206.6	95.2	-188.1	283.26	0.336	Level 1	
19,000.0	10,880.9	19,075.3	10,777.2	242.2	243.2	31.90	-605.8	8,306.8	93.9	-182.7	276.57	0.339	Level 1	
19,100.0	10,881.4	19,175.1	10,777.4	245.1	246.1	30.46	-604.0	8,406.5	92.8	-178.0	270.83	0.343	Level 1	
19,200.0	10,881.9	19,275.0	10,777.8	248.0	249.0	29.01	-602.2	8,506.4	91.6	-173.1	264.73	0.346	Level 1	
19,300.0	10,882.5	19,374.6	10,777.5	250.9	251.9	27.47	-600.7	8,606.0	91.2	-166.6	257.80	0.354	Level 1	
19,400.0	10,883.0	19,475.1	10,777.5	253.8	254.9	25.62	-598.5	8,706.5	90.4	-158.2	248.62	0.364	Level 1	
19,500.0	10,883.5	19,574.8	10,777.6	256.7	257.8	23.63	-596.1	8,806.2	89.4	-148.8	238.25	0.375	Level 1	
19,600.0	10,884.0	19,674.7	10,777.4	259.7	260.7	21.96	-594.3	8,906.1	89.1	-140.7	229.84	0.388	Level 1	
19,700.0	10,884.5	19,775.1	10,777.7	262.6	263.6	20.28	-592.5	9,006.5	88.4	-132.9	221.25	0.399	Level 1	
19,800.0	10,885.1	19,874.7	10,777.9	265.5	266.5	18.27	-590.1	9,106.1	87.6	-122.9	210.48	0.416	Level 1	
19,900.0	10,885.6	19,975.0	10,777.9	268.4	269.4	16.42	-588.1	9,206.3	87.2	-113.6	200.81	0.434	Level 1	
20,000.0	10,886.1	20,074.9	10,778.3	271.3	272.3	14.36	-585.7	9,306.1	86.5	-103.7	190.21	0.455	Level 1	
20,076.0	10,886.5	20,150.5	10,778.4	273.5	274.5	12.97	-584.2	9,381.8	86.3	-97.1	183.47	0.471	Level 1	
20,100.0	10,886.6	20,174.4	10,778.3	274.2	275.2	12.53	-583.8	9,405.6	86.4	-95.0	181.38	0.476	Level 1	
20,200.0	10,887.2	20,274.0	10,777.9	277.2	278.1	10.71	-582.0	9,505.2	86.8	-86.5	173.29	0.501	Level 1	
20,300.0	10,887.7	20,374.3	10,777.4	280.1	281.0	9.01	-580.3	9,605.5	87.4	-79.2	166.55	0.525	Level 1	
20,400.0	10,888.2	20,473.8	10,777.0	283.0	283.9	7.37	-578.7	9,705.0	87.9	-73.1	161.02	0.546	Level 1	
20,500.0	10,888.7	20,572.7	10,775.8	285.9	286.8	5.95	-577.5	9,803.9	89.4	-67.8	157.23	0.569	Level 1	
20,542.0	10,889.0	20,614.4	10,774.9	287.1	288.0	5.49	-577.2	9,845.6	90.5	-65.9	156.35	0.579	Level 1	
20,600.0	10,889.3	20,650.0	10,774.2	288.8	289.0	5.13	-577.0	9,881.2	94.6	-61.4	155.98	0.606	Level 1	
20,700.0	10,889.8	20,650.0	10,774.2	291.7	289.0	5.13	-577.0	9,881.2	154.6	-2.2	156.74	0.986	Level 1	
20,717.6	10,889.9	20,650.0	10,774.2	292.3	289.0	5.13	-577.0	9,881.2	169.0	12.2	156.87	1.078	Level 2	

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
0.0	0.0	0.0	0.0	0.0	0.0	-155.33	-90.2	-41.4	99.2				
100.0	100.0	100.0	100.0	0.1	0.1	-155.33	-90.2	-41.4	99.2	99.1	0.17	588.711	
200.0	200.0	200.0	200.0	0.3	0.3	-155.33	-90.2	-41.4	99.2	98.6	0.62	160.558	
300.0	300.0	300.0	300.0	0.5	0.5	-155.33	-90.2	-41.4	99.2	98.2	1.07	92.954	
400.0	400.0	400.0	400.0	0.8	0.8	-155.33	-90.2	-41.4	99.2	97.7	1.52	65.412	
500.0	500.0	500.0	500.0	1.0	1.0	-155.33	-90.2	-41.4	99.2	97.3	1.97	50.461	
600.0	600.0	600.0	600.0	1.2	1.2	-155.33	-90.2	-41.4	99.2	96.8	2.42	41.073	
700.0	700.0	700.0	700.0	1.4	1.4	-155.33	-90.2	-41.4	99.2	96.4	2.87	34.630	
800.0	800.0	800.0	800.0	1.7	1.7	-155.33	-90.2	-41.4	99.2	95.9	3.32	29.934	
900.0	900.0	900.0	900.0	1.9	1.9	-155.33	-90.2	-41.4	99.2	95.5	3.76	26.360	
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.1	-155.33	-90.2	-41.4	99.2	95.0	4.21	23.548	
1,100.0	1,100.0	1,100.0	1,100.0	2.3	2.3	-155.33	-90.2	-41.4	99.2	94.6	4.66	21.279	
1,200.0	1,200.0	1,200.0	1,200.0	2.6	2.6	-155.33	-90.2	-41.4	99.2	94.1	5.11	19.408	
1,300.0	1,300.0	1,300.0	1,300.0	2.8	2.8	-155.33	-90.2	-41.4	99.2	93.7	5.56	17.840	
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.0	-155.33	-90.2	-41.4	99.2	93.2	6.01	16.506	
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.2	-155.33	-90.2	-41.4	99.2	92.8	6.46	15.358	
1,600.0	1,600.0	1,600.0	1,600.0	3.5	3.5	-155.33	-90.2	-41.4	99.2	92.3	6.91	14.359	
1,700.0	1,700.0	1,700.0	1,700.0	3.7	3.7	-155.33	-90.2	-41.4	99.2	91.9	7.36	13.482	
1,800.0	1,800.0	1,800.0	1,800.0	3.9	3.9	-155.33	-90.2	-41.4	99.2	91.4	7.81	12.706	
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.1	-155.33	-90.2	-41.4	99.2	91.0	8.26	12.015	
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.4	-155.33	-90.2	-41.4	99.2	90.5	8.71	11.394	
2,100.0	2,100.0	2,100.0	2,100.0	4.6	4.6	-155.33	-90.2	-41.4	99.2	90.1	9.16	10.835	
2,200.0	2,200.0	2,200.0	2,200.0	4.8	4.8	-155.33	-90.2	-41.4	99.2	89.6	9.61	10.328	
2,300.0	2,300.0	2,300.0	2,300.0	5.0	5.0	-155.33	-90.2	-41.4	99.2	89.2	10.06	9.867	
2,400.0	2,400.0	2,400.0	2,400.0	5.3	5.3	-155.33	-90.2	-41.4	99.2	88.7	10.51	9.445	
2,500.0	2,500.0	2,500.0	2,500.0	5.5	5.5	-155.33	-90.2	-41.4	99.2	88.3	10.96	9.057 CC, ES	
2,600.0	2,600.0	2,597.1	2,597.1	5.7	5.7	116.04	-91.0	-42.8	101.4	90.0	11.37	8.920	
2,650.0	2,649.9	2,645.5	2,645.5	5.8	5.8	117.66	-92.0	-44.6	104.1	92.6	11.56	9.013	
2,700.0	2,699.9	2,695.1	2,695.0	5.9	5.9	119.63	-93.3	-46.9	107.6	95.9	11.75	9.161	
2,800.0	2,799.7	2,794.6	2,794.3	6.1	6.1	123.20	-95.9	-51.4	115.0	102.8	12.14	9.470	
2,900.0	2,899.6	2,894.1	2,893.7	6.3	6.2	126.34	-98.5	-55.9	122.7	110.2	12.54	9.789	
3,000.0	2,999.5	2,993.6	2,993.1	6.5	6.4	129.10	-101.1	-60.4	130.8	117.8	12.93	10.110	
3,100.0	3,099.3	3,093.1	3,092.4	6.7	6.6	131.53	-103.7	-64.9	139.1	125.7	13.34	10.429	
3,200.0	3,199.2	3,192.6	3,191.8	7.0	6.9	133.69	-106.3	-69.4	147.6	133.9	13.74	10.743	
3,300.0	3,299.0	3,292.1	3,291.1	7.2	7.1	135.61	-108.9	-73.9	156.4	142.2	14.15	11.050	
3,400.0	3,398.9	3,391.5	3,390.5	7.4	7.3	137.33	-111.6	-78.4	165.2	150.7	14.56	11.349	
3,501.2	3,500.0	3,492.3	3,491.0	7.6	7.5	138.88	-114.2	-83.0	174.4	159.4	14.98	11.642	
3,600.0	3,598.7	3,590.7	3,589.3	7.8	7.7	140.01	-116.8	-87.5	182.1	166.7	15.38	11.835	
3,651.2	3,649.9	3,641.8	3,640.4	7.9	7.8	-129.72	-118.1	-89.8	185.1	169.5	15.59	11.872	
3,700.0	3,698.7	3,690.5	3,689.0	8.0	7.9	-129.58	-119.4	-92.0	187.6	171.8	15.79	11.877	
3,800.0	3,798.7	3,790.4	3,788.7	8.2	8.1	-129.33	-122.0	-96.5	192.8	176.5	16.23	11.875	
3,900.0	3,898.7	3,896.7	3,895.0	8.5	8.4	-129.14	-124.0	-100.0	196.5	179.8	16.68	11.780	
4,000.0	3,998.7	4,000.4	3,998.7	8.7	8.6	-129.11	-124.2	-100.4	197.0	179.8	17.11	11.508	
4,100.0	4,098.7	4,100.4	4,098.7	8.9	8.8	-129.11	-124.2	-100.4	197.0	179.4	17.55	11.220	
4,200.0	4,198.7	4,200.4	4,198.7	9.1	9.0	-129.11	-124.2	-100.4	197.0	179.0	18.00	10.945	
4,300.0	4,298.7	4,300.4	4,298.7	9.4	9.2	-129.11	-124.2	-100.4	197.0	178.5	18.44	10.683	
4,400.0	4,398.7	4,400.4	4,398.7	9.6	9.4	-129.11	-124.2	-100.4	197.0	178.1	18.88	10.433	
4,500.0	4,498.7	4,500.4	4,498.7	9.8	9.7	-129.11	-124.2	-100.4	197.0	177.6	19.32	10.195	
4,600.0	4,598.7	4,600.4	4,598.7	10.0	9.9	-129.11	-124.2	-100.4	197.0	177.2	19.76	9.966	
4,700.0	4,698.7	4,700.4	4,698.7	10.2	10.1	-129.11	-124.2	-100.4	197.0	176.8	20.20	9.748	
4,800.0	4,798.7	4,800.4	4,798.7	10.5	10.3	-129.11	-124.2	-100.4	197.0	176.3	20.65	9.539	
4,900.0	4,898.7	4,900.4	4,898.7	10.7	10.5	-129.11	-124.2	-100.4	197.0	175.9	21.09	9.339	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
5,000.0	4,998.7	5,000.4	4,998.7	10.9	10.8	-129.11	-124.2	-100.4	197.0	175.4	21.53	9.146		
5,100.0	5,098.7	5,100.4	5,098.7	11.1	11.0	-129.11	-124.2	-100.4	197.0	175.0	21.98	8.962		
5,200.0	5,198.7	5,200.4	5,198.7	11.4	11.2	-129.11	-124.2	-100.4	197.0	174.5	22.42	8.784		
5,300.0	5,298.7	5,300.4	5,298.7	11.6	11.4	-129.11	-124.2	-100.4	197.0	174.1	22.87	8.614		
5,400.0	5,398.7	5,400.4	5,398.7	11.8	11.6	-129.11	-124.2	-100.4	197.0	173.6	23.31	8.449		
5,500.0	5,498.7	5,500.4	5,498.7	12.0	11.9	-129.11	-124.2	-100.4	197.0	173.2	23.75	8.291		
5,600.0	5,598.7	5,600.4	5,598.7	12.2	12.1	-129.11	-124.2	-100.4	197.0	172.8	24.20	8.139		
5,700.0	5,698.7	5,700.4	5,698.7	12.5	12.3	-129.11	-124.2	-100.4	197.0	172.3	24.64	7.992		
5,800.0	5,798.7	5,800.4	5,798.7	12.7	12.5	-129.11	-124.2	-100.4	197.0	171.9	25.09	7.850		
5,900.0	5,898.7	5,900.4	5,898.7	12.9	12.8	-129.11	-124.2	-100.4	197.0	171.4	25.53	7.713		
6,000.0	5,998.7	6,000.4	5,998.7	13.1	13.0	-129.11	-124.2	-100.4	197.0	171.0	25.98	7.581		
6,100.0	6,098.7	6,100.4	6,098.7	13.4	13.2	-129.11	-124.2	-100.4	197.0	170.5	26.42	7.453		
6,200.0	6,198.7	6,200.4	6,198.7	13.6	13.4	-129.11	-124.2	-100.4	197.0	170.1	26.87	7.330		
6,300.0	6,298.7	6,300.4	6,298.7	13.8	13.6	-129.11	-124.2	-100.4	197.0	169.6	27.32	7.210		
6,400.0	6,398.7	6,400.4	6,398.7	14.0	13.9	-129.11	-124.2	-100.4	197.0	169.2	27.76	7.094		
6,500.0	6,498.7	6,500.4	6,498.7	14.3	14.1	-129.11	-124.2	-100.4	197.0	168.7	28.21	6.982		
6,600.0	6,598.7	6,600.4	6,598.7	14.5	14.3	-129.11	-124.2	-100.4	197.0	168.3	28.65	6.874		
6,700.0	6,698.7	6,700.4	6,698.7	14.7	14.5	-129.11	-124.2	-100.4	197.0	167.9	29.10	6.768		
6,800.0	6,798.7	6,800.4	6,798.7	14.9	14.7	-129.11	-124.2	-100.4	197.0	167.4	29.55	6.666		
6,900.0	6,898.7	6,900.4	6,898.7	15.2	15.0	-129.11	-124.2	-100.4	197.0	167.0	29.99	6.567		
7,000.0	6,998.7	7,000.4	6,998.7	15.4	15.2	-129.11	-124.2	-100.4	197.0	166.5	30.44	6.470		
7,100.0	7,098.7	7,100.4	7,098.7	15.6	15.4	-129.11	-124.2	-100.4	197.0	166.1	30.89	6.377		
7,200.0	7,198.7	7,200.4	7,198.7	15.8	15.6	-129.11	-124.2	-100.4	197.0	165.6	31.33	6.286		
7,300.0	7,298.7	7,300.4	7,298.7	16.0	15.9	-129.11	-124.2	-100.4	197.0	165.2	31.78	6.198		
7,400.0	7,398.7	7,400.4	7,398.7	16.3	16.1	-129.11	-124.2	-100.4	197.0	164.7	32.23	6.112		
7,500.0	7,498.7	7,500.4	7,498.7	16.5	16.3	-129.11	-124.2	-100.4	197.0	164.3	32.67	6.028		
7,600.0	7,598.7	7,600.4	7,598.7	16.7	16.5	-129.11	-124.2	-100.4	197.0	163.8	33.12	5.947		
7,700.0	7,698.7	7,700.4	7,698.7	16.9	16.7	-129.11	-124.2	-100.4	197.0	163.4	33.57	5.867		
7,800.0	7,798.7	7,800.4	7,798.7	17.2	17.0	-129.11	-124.2	-100.4	197.0	162.9	34.01	5.790		
7,900.0	7,898.7	7,900.4	7,898.7	17.4	17.2	-129.11	-124.2	-100.4	197.0	162.5	34.46	5.715		
8,000.0	7,998.7	8,000.4	7,998.7	17.6	17.4	-129.11	-124.2	-100.4	197.0	162.0	34.91	5.642		
8,100.0	8,098.7	8,100.4	8,098.7	17.8	17.6	-129.11	-124.2	-100.4	197.0	161.6	35.36	5.571		
8,200.0	8,198.7	8,200.4	8,198.7	18.1	17.9	-129.11	-124.2	-100.4	197.0	161.2	35.80	5.501		
8,300.0	8,298.7	8,300.4	8,298.7	18.3	18.1	-129.11	-124.2	-100.4	197.0	160.7	36.25	5.433		
8,400.0	8,398.7	8,400.4	8,398.7	18.5	18.3	-129.11	-124.2	-100.4	197.0	160.3	36.70	5.367		
8,500.0	8,498.7	8,500.4	8,498.7	18.7	18.5	-129.11	-124.2	-100.4	197.0	159.8	37.15	5.302		
8,600.0	8,598.7	8,600.4	8,598.7	19.0	18.8	-129.11	-124.2	-100.4	197.0	159.4	37.59	5.239		
8,700.0	8,698.7	8,700.4	8,698.7	19.2	19.0	-129.11	-124.2	-100.4	197.0	158.9	38.04	5.177		
8,800.0	8,798.7	8,800.4	8,798.7	19.4	19.2	-129.11	-124.2	-100.4	197.0	158.5	38.49	5.117		
8,900.0	8,898.7	8,900.4	8,898.7	19.6	19.4	-129.11	-124.2	-100.4	197.0	158.0	38.94	5.058		
9,000.0	8,998.7	9,000.4	8,998.7	19.9	19.6	-129.11	-124.2	-100.4	197.0	157.6	39.38	5.001		
9,100.0	9,098.7	9,100.4	9,098.7	20.1	19.9	-129.11	-124.2	-100.4	197.0	157.1	39.83	4.945		
9,200.0	9,198.7	9,200.4	9,198.7	20.3	20.1	-129.11	-124.2	-100.4	197.0	156.7	40.28	4.890		
9,300.0	9,298.7	9,300.4	9,298.7	20.5	20.3	-129.11	-124.2	-100.4	197.0	156.2	40.73	4.836		
9,400.0	9,398.7	9,400.4	9,398.7	20.8	20.5	-129.11	-124.2	-100.4	197.0	155.8	41.18	4.783		
9,500.0	9,498.7	9,500.4	9,498.7	21.0	20.8	-129.11	-124.2	-100.4	197.0	155.3	41.62	4.732		
9,600.0	9,598.7	9,600.4	9,598.7	21.2	21.0	-129.11	-124.2	-100.4	197.0	154.9	42.07	4.681		
9,700.0	9,698.7	9,700.4	9,698.7	21.4	21.2	-129.11	-124.2	-100.4	197.0	154.4	42.52	4.632		
9,800.0	9,798.7	9,800.4	9,798.7	21.6	21.4	-129.11	-124.2	-100.4	197.0	154.0	42.97	4.584		
9,900.0	9,898.7	9,900.4	9,898.7	21.9	21.7	-129.11	-124.2	-100.4	197.0	153.5	43.42	4.537		
10,000.0	9,998.7	10,000.4	9,998.7	22.1	21.9	-129.11	-124.2	-100.4	197.0	153.1	43.86	4.490		
10,100.0	10,098.7	10,100.4	10,098.7	22.3	22.1	-129.11	-124.2	-100.4	197.0	152.6	44.31	4.445		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
10,200.0	10,198.7	10,200.4	10,198.7	22.5	22.3	-129.11	-124.2	-100.4	197.0	152.2	44.76	4.400		
10,300.0	10,298.7	10,300.4	10,298.7	22.8	22.6	-129.11	-124.2	-100.4	197.0	151.7	45.21	4.357		
10,342.2	10,340.9	10,342.6	10,340.9	22.9	22.7	-129.11	-124.2	-100.4	197.0	151.6	45.40	4.339		
10,363.3	10,362.0	10,363.7	10,362.0	22.9	22.7	-129.11	-124.2	-100.4	197.0	151.5	45.49	4.329		
10,375.0	10,373.7	10,375.0	10,373.3	22.9	22.7	110.88	-124.4	-100.4	197.0	151.5	45.55	4.325		
10,400.0	10,398.7	10,397.9	10,396.2	23.0	22.8	110.81	-125.3	-99.8	197.7	152.1	45.63	4.332		
10,425.0	10,423.5	10,421.2	10,419.3	23.0	22.8	110.67	-127.3	-98.7	199.0	153.3	45.70	4.355		
10,450.0	10,448.2	10,444.4	10,442.3	23.1	22.8	110.47	-130.2	-97.1	201.0	155.3	45.76	4.393		
10,475.0	10,472.7	10,467.5	10,465.0	23.1	22.9	110.19	-134.1	-95.0	203.7	157.9	45.81	4.447		
10,500.0	10,496.8	10,490.6	10,487.4	23.2	22.9	109.85	-138.9	-92.3	207.0	161.2	45.86	4.515		
10,525.0	10,520.6	10,513.5	10,509.4	23.3	23.0	109.45	-144.6	-89.1	211.0	165.1	45.91	4.597		
10,550.0	10,544.0	10,536.4	10,530.9	23.3	23.0	108.99	-151.2	-85.4	215.6	169.7	45.96	4.692		
10,575.0	10,566.8	10,559.0	10,551.9	23.4	23.0	108.47	-158.7	-81.3	220.9	174.9	46.01	4.801		
10,600.0	10,589.1	10,581.6	10,572.4	23.5	23.1	107.89	-167.0	-76.7	226.7	180.6	46.07	4.921		
10,625.0	10,610.8	10,603.9	10,592.2	23.5	23.1	107.27	-176.0	-71.7	233.1	187.0	46.15	5.052		
10,650.0	10,631.8	10,626.1	10,611.4	23.6	23.2	106.59	-185.7	-66.3	240.1	193.9	46.24	5.193		
10,675.0	10,652.0	10,648.1	10,629.9	23.7	23.2	105.87	-196.2	-60.6	247.6	201.3	46.34	5.343		
10,700.0	10,671.5	10,670.0	10,647.7	23.8	23.2	105.10	-207.2	-54.4	255.7	209.2	46.47	5.501		
10,725.0	10,690.1	10,691.6	10,664.7	23.9	23.3	104.29	-218.9	-47.9	264.2	217.6	46.62	5.667		
10,750.0	10,707.8	10,713.1	10,681.0	24.0	23.3	103.44	-231.2	-41.2	273.2	226.4	46.80	5.838		
10,775.0	10,724.5	10,734.4	10,696.6	24.1	23.4	102.55	-243.9	-34.1	282.6	235.6	47.00	6.013		
10,800.0	10,740.3	10,755.6	10,711.3	24.3	23.4	101.63	-257.2	-26.7	292.5	245.2	47.23	6.193		
10,825.0	10,755.0	10,776.6	10,725.3	24.4	23.5	100.68	-270.9	-19.1	302.7	255.2	47.48	6.376		
10,850.0	10,768.7	10,797.4	10,738.5	24.6	23.6	99.70	-285.0	-11.3	313.3	265.6	47.76	6.561		
10,875.0	10,781.2	10,818.2	10,750.9	24.8	23.7	98.69	-299.5	-3.3	324.3	276.2	48.06	6.747		
10,900.0	10,792.6	10,838.7	10,762.5	25.0	23.8	97.65	-314.4	5.0	335.5	287.1	48.38	6.935		
10,925.0	10,802.8	10,859.2	10,773.2	25.2	23.9	96.60	-329.7	13.5	347.0	298.3	48.72	7.122		
10,950.0	10,811.8	10,879.7	10,783.2	25.4	24.0	95.53	-345.2	22.1	358.8	309.7	49.08	7.310		
10,975.0	10,819.6	10,900.0	10,792.4	25.7	24.1	94.45	-361.1	30.9	370.8	321.3	49.46	7.497		
11,000.0	10,826.1	10,920.3	10,800.8	25.9	24.2	93.35	-377.3	39.8	383.0	333.1	49.85	7.683		
11,025.0	10,831.3	10,940.6	10,808.4	26.2	24.3	92.25	-393.7	49.0	395.3	345.1	50.25	7.868		
11,050.0	10,835.3	10,960.9	10,815.2	26.5	24.5	91.15	-410.5	58.2	407.8	357.2	50.65	8.052		
11,075.0	10,837.9	10,981.2	10,821.2	26.8	24.6	90.05	-427.4	67.6	420.5	369.4	51.06	8.234		
11,100.0	10,839.3	11,001.6	10,826.4	27.1	24.8	88.96	-444.7	77.2	433.2	381.7	51.48	8.414		
11,110.8	10,839.5	11,010.4	10,828.3	27.2	24.8	88.49	-452.2	81.4	438.7	387.0	51.66	8.491		
11,125.8	10,839.5	11,022.8	10,830.8	27.4	24.9	88.88	-462.8	87.2	446.3	394.4	51.94	8.594		
11,200.0	10,839.9	11,084.6	10,838.6	28.5	25.5	90.01	-516.4	116.9	485.7	432.4	53.33	9.107		
11,300.0	10,840.5	11,182.7	10,839.8	30.1	26.5	90.13	-601.8	165.2	542.0	486.5	55.50	9.766		
11,400.0	10,841.0	11,300.5	10,840.5	31.9	27.9	90.15	-701.2	228.5	598.4	540.3	58.05	10.308		
11,500.0	10,841.5	11,424.4	10,841.2	33.9	29.7	90.16	-801.3	301.4	653.8	592.9	60.88	10.739		
11,600.0	10,842.1	11,555.0	10,841.9	36.0	31.8	90.18	-901.4	385.2	708.0	644.1	63.95	11.071		
11,700.0	10,842.6	11,693.1	10,842.7	38.1	34.3	90.20	-1,000.9	481.0	760.7	693.5	67.23	11.315		
11,800.0	10,843.2	11,839.6	10,843.6	40.4	37.1	90.21	-1,098.4	590.1	811.5	740.8	70.75	11.470		
11,900.0	10,843.7	11,995.1	10,844.5	42.7	40.4	90.23	-1,192.6	713.9	860.0	785.6	74.47	11.549		
12,000.0	10,844.2	12,160.6	10,845.5	45.0	44.0	90.24	-1,281.4	853.5	905.8	827.3	78.43	11.548		
12,100.0	10,844.8	12,336.7	10,846.6	47.4	47.9	90.25	-1,362.2	1,009.8	948.1	865.5	82.61	11.477		
12,110.4	10,844.8	12,355.7	10,846.7	47.6	48.4	90.26	-1,370.0	1,027.1	952.3	869.3	83.06	11.466		
12,200.0	10,845.3	12,525.2	10,847.7	49.8	52.3	90.25	-1,432.3	1,184.8	984.6	895.7	88.89	11.077		
12,300.0	10,845.8	12,727.1	10,848.9	52.2	56.9	90.25	-1,488.0	1,378.8	1,011.5	915.3	96.12	10.523		
12,400.0	10,846.3	12,938.9	10,850.1	54.7	61.8	90.24	-1,524.0	1,587.4	1,027.7	923.7	103.97	9.885		
12,500.0	10,846.8	13,153.6	10,851.3	57.3	66.6	90.24	-1,536.7	1,801.6	1,032.8	920.7	112.12	9.211		
12,600.0	10,847.4	13,253.6	10,851.8	59.9	68.8	90.24	-1,537.1	1,901.6	1,032.4	915.0	117.41	8.793		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWMD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)						
12,700.0	10,847.9	13,353.6	10,852.3	62.5	71.1	90.24	-1,537.5	2,001.6	1,032.0	909.2	122.74	8.408		
12,800.0	10,848.4	13,453.6	10,852.8	65.1	73.4	90.24	-1,537.9	2,101.6	1,031.6	903.5	128.13	8.051		
12,900.0	10,848.9	13,553.6	10,853.4	67.8	75.8	90.24	-1,538.3	2,201.6	1,031.2	897.7	133.55	7.721		
13,000.0	10,849.5	13,653.6	10,853.9	70.4	78.2	90.24	-1,538.7	2,301.6	1,030.8	891.8	139.01	7.415		
13,100.0	10,850.0	13,753.6	10,854.4	73.1	80.6	90.24	-1,539.1	2,401.6	1,030.4	885.9	144.50	7.131		
13,200.0	10,850.5	13,853.6	10,854.9	75.9	83.1	90.24	-1,539.5	2,501.6	1,030.0	880.0	150.02	6.866		
13,300.0	10,851.0	13,953.6	10,855.5	78.6	85.6	90.24	-1,539.9	2,601.6	1,029.6	874.1	155.56	6.619		
13,400.0	10,851.6	14,053.6	10,856.0	81.3	88.2	90.24	-1,540.4	2,701.6	1,029.2	868.1	161.13	6.388		
13,500.0	10,852.1	14,153.6	10,856.5	84.1	90.7	90.25	-1,540.8	2,801.6	1,028.8	862.1	166.71	6.171		
13,600.0	10,852.6	14,253.6	10,857.0	86.9	93.3	90.25	-1,541.2	2,901.6	1,028.4	856.1	172.32	5.968		
13,700.0	10,853.1	14,353.6	10,857.6	89.6	95.9	90.25	-1,541.6	3,001.5	1,028.0	850.1	177.94	5.777		
13,800.0	10,853.7	14,453.6	10,858.1	92.4	98.5	90.25	-1,542.0	3,101.5	1,027.6	844.1	183.58	5.598		
13,900.0	10,854.2	14,553.6	10,858.6	95.2	101.1	90.25	-1,542.4	3,201.5	1,027.2	838.0	189.23	5.429		
14,000.0	10,854.7	14,653.6	10,859.1	98.0	103.8	90.25	-1,542.8	3,301.5	1,026.8	831.9	194.89	5.269		
14,100.0	10,855.2	14,753.6	10,859.6	100.8	106.5	90.25	-1,543.2	3,401.5	1,026.4	825.9	200.57	5.118		
14,200.0	10,855.7	14,853.6	10,860.2	103.7	109.1	90.25	-1,543.6	3,501.5	1,026.0	819.8	206.25	4.975		
14,300.0	10,856.3	14,953.6	10,860.7	106.5	111.8	90.25	-1,544.0	3,601.5	1,025.7	813.7	211.95	4.839		
14,400.0	10,856.8	15,053.6	10,861.2	109.3	114.5	90.25	-1,544.4	3,701.5	1,025.3	807.6	217.65	4.711		
14,500.0	10,857.3	15,153.6	10,861.7	112.2	117.3	90.25	-1,544.8	3,801.5	1,024.9	801.5	223.36	4.588		
14,600.0	10,857.8	15,253.6	10,862.3	115.0	120.0	90.25	-1,545.2	3,901.5	1,024.5	795.4	229.08	4.472		
14,700.0	10,858.4	15,353.6	10,862.8	117.9	122.7	90.25	-1,545.7	4,001.5	1,024.1	789.3	234.81	4.361		
14,800.0	10,858.9	15,453.6	10,863.3	120.7	125.5	90.25	-1,546.1	4,101.5	1,023.7	783.1	240.54	4.256		
14,900.0	10,859.4	15,553.6	10,863.8	123.6	128.2	90.25	-1,546.5	4,201.5	1,023.3	777.0	246.28	4.155		
15,000.0	10,859.9	15,653.6	10,864.4	126.4	131.0	90.25	-1,546.9	4,301.5	1,022.9	770.8	252.03	4.059		
15,100.0	10,860.5	15,753.6	10,864.9	129.3	133.7	90.25	-1,547.3	4,401.5	1,022.5	764.7	257.78	3.967		
15,200.0	10,861.0	15,853.6	10,865.4	132.1	136.5	90.25	-1,547.7	4,501.5	1,022.1	758.5	263.53	3.878		
15,300.0	10,861.5	15,953.6	10,865.9	135.0	139.3	90.25	-1,548.1	4,601.5	1,021.7	752.4	269.29	3.794		
15,400.0	10,862.0	16,053.6	10,866.5	137.9	142.1	90.25	-1,548.5	4,701.5	1,021.3	746.2	275.06	3.713		
15,500.0	10,862.6	16,153.6	10,867.0	140.7	144.9	90.25	-1,548.9	4,801.5	1,020.9	740.1	280.83	3.635		
15,600.0	10,863.1	16,253.6	10,867.5	143.6	147.7	90.25	-1,549.3	4,901.5	1,020.5	733.9	286.60	3.561		
15,700.0	10,863.6	16,353.6	10,868.0	146.5	150.5	90.25	-1,549.7	5,001.5	1,020.1	727.7	292.38	3.489		
15,800.0	10,864.1	16,453.6	10,868.5	149.4	153.3	90.25	-1,550.1	5,101.5	1,019.7	721.5	298.15	3.420		
15,900.0	10,864.7	16,553.6	10,869.1	152.3	156.1	90.25	-1,550.5	5,201.5	1,019.3	715.4	303.94	3.354		
16,000.0	10,865.2	16,653.6	10,869.6	155.1	158.9	90.25	-1,551.0	5,301.5	1,018.9	709.2	309.72	3.290		
16,100.0	10,865.7	16,753.6	10,870.1	158.0	161.7	90.25	-1,551.4	5,401.5	1,018.5	703.0	315.51	3.228		
16,200.0	10,866.2	16,853.6	10,870.6	160.9	164.6	90.25	-1,551.8	5,501.5	1,018.1	696.8	321.30	3.169		
16,300.0	10,866.7	16,953.6	10,871.2	163.8	167.4	90.25	-1,552.2	5,601.5	1,017.7	690.6	327.10	3.111		
16,400.0	10,867.3	17,053.6	10,871.7	166.7	170.2	90.25	-1,552.6	5,701.5	1,017.3	684.4	332.89	3.056		
16,500.0	10,867.8	17,153.6	10,872.2	169.6	173.1	90.25	-1,553.0	5,801.5	1,016.9	678.2	338.69	3.003		
16,600.0	10,868.3	17,253.6	10,872.7	172.5	175.9	90.25	-1,553.4	5,901.5	1,016.5	672.0	344.49	2.951		
16,700.0	10,868.8	17,353.6	10,873.3	175.4	178.8	90.25	-1,553.8	6,001.5	1,016.1	665.8	350.29	2.901		
16,800.0	10,869.4	17,453.6	10,873.8	178.3	181.6	90.25	-1,554.2	6,101.5	1,015.7	659.6	356.10	2.852		
16,900.0	10,869.9	17,553.6	10,874.3	181.2	184.4	90.25	-1,554.6	6,201.5	1,015.3	653.4	361.91	2.806		
17,000.0	10,870.4	17,653.6	10,874.8	184.1	187.3	90.25	-1,555.0	6,301.5	1,014.9	647.2	367.71	2.760		
17,100.0	10,870.9	17,753.6	10,875.4	187.0	190.1	90.25	-1,555.4	6,401.4	1,014.6	641.0	373.52	2.716		
17,200.0	10,871.5	17,853.6	10,875.9	189.9	193.0	90.25	-1,555.9	6,501.4	1,014.2	634.8	379.34	2.674		
17,300.0	10,872.0	17,953.6	10,876.4	192.8	195.9	90.25	-1,556.3	6,601.4	1,013.8	628.6	385.15	2.632		
17,400.0	10,872.5	18,053.6	10,876.9	195.7	198.7	90.25	-1,556.7	6,701.4	1,013.4	622.4	390.96	2.592		
17,500.0	10,873.0	18,153.6	10,877.4	198.6	201.6	90.25	-1,557.1	6,801.4	1,013.0	616.2	396.78	2.553		
17,600.0	10,873.6	18,253.6	10,878.0	201.5	204.4	90.25	-1,557.5	6,901.4	1,012.6	610.0	402.60	2.515		
17,700.0	10,874.1	18,353.6	10,878.5	204.4	207.3	90.25	-1,557.9	7,001.4	1,012.2	603.8	408.42	2.478		
17,800.0	10,874.6	18,453.6	10,879.0	207.3	210.2	90.25	-1,558.3	7,101.4	1,011.8	597.5	414.24	2.443		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 0.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)					
17,900.0	10,875.1	18,553.6	10,879.5	210.2	213.0	90.25	-1,558.7	7,201.4	1,011.4	591.3	420.06	2.408	
18,000.0	10,875.6	18,653.6	10,880.1	213.1	215.9	90.25	-1,559.1	7,301.4	1,011.0	585.1	425.88	2.374	
18,100.0	10,876.2	18,753.6	10,880.6	216.0	218.8	90.25	-1,559.5	7,401.4	1,010.6	578.9	431.70	2.341	
18,200.0	10,876.7	18,853.6	10,881.1	218.9	221.7	90.25	-1,559.9	7,501.4	1,010.2	572.7	437.53	2.309	
18,300.0	10,877.2	18,953.6	10,881.6	221.8	224.5	90.25	-1,560.3	7,601.4	1,009.8	566.4	443.35	2.278	
18,400.0	10,877.7	19,053.6	10,882.2	224.7	227.4	90.25	-1,560.7	7,701.4	1,009.4	560.2	449.18	2.247	
18,500.0	10,878.3	19,153.6	10,882.7	227.6	230.3	90.25	-1,561.2	7,801.4	1,009.0	554.0	455.01	2.218	
18,600.0	10,878.8	19,253.6	10,883.2	230.5	233.2	90.25	-1,561.6	7,901.4	1,008.6	547.8	460.84	2.189	
18,700.0	10,879.3	19,353.6	10,883.7	233.4	236.1	90.25	-1,562.0	8,001.4	1,008.2	541.5	466.67	2.160	
18,800.0	10,879.8	19,453.6	10,884.3	236.4	238.9	90.25	-1,562.4	8,101.4	1,007.8	535.3	472.50	2.133	
18,900.0	10,880.4	19,553.6	10,884.8	239.3	241.8	90.25	-1,562.8	8,201.4	1,007.4	529.1	478.33	2.106	
19,000.0	10,880.9	19,653.6	10,885.3	242.2	244.7	90.25	-1,563.2	8,301.4	1,007.0	522.9	484.16	2.080	
19,100.0	10,881.4	19,753.6	10,885.8	245.1	247.6	90.25	-1,563.6	8,401.4	1,006.6	516.6	489.99	2.054	
19,200.0	10,881.9	19,853.6	10,886.3	248.0	250.5	90.25	-1,564.0	8,501.4	1,006.2	510.4	495.82	2.029	
19,300.0	10,882.5	19,953.6	10,886.9	250.9	253.4	90.25	-1,564.4	8,601.4	1,005.8	504.2	501.66	2.005	
19,400.0	10,883.0	20,053.6	10,887.4	253.8	256.3	90.25	-1,564.8	8,701.4	1,005.4	497.9	507.49	1.981	
19,500.0	10,883.5	20,153.6	10,887.9	256.7	259.1	90.25	-1,565.2	8,801.4	1,005.0	491.7	513.33	1.958	
19,600.0	10,884.0	20,253.6	10,888.4	259.7	262.0	90.25	-1,565.6	8,901.4	1,004.6	485.5	519.16	1.935	
19,700.0	10,884.5	20,353.6	10,889.0	262.6	264.9	90.25	-1,566.0	9,001.4	1,004.2	479.2	525.00	1.913	
19,800.0	10,885.1	20,453.6	10,889.5	265.5	267.8	90.25	-1,566.5	9,101.4	1,003.8	473.0	530.84	1.891	
19,900.0	10,885.6	20,553.6	10,890.0	268.4	270.7	90.25	-1,566.9	9,201.4	1,003.5	466.8	536.68	1.870	
20,000.0	10,886.1	20,653.6	10,890.5	271.3	273.6	90.25	-1,567.3	9,301.4	1,003.1	460.5	542.51	1.849	
20,100.0	10,886.6	20,753.6	10,891.1	274.2	276.5	90.25	-1,567.7	9,401.4	1,002.7	454.3	548.35	1.828	
20,200.0	10,887.2	20,853.6	10,891.6	277.2	279.4	90.25	-1,568.1	9,501.4	1,002.3	448.1	554.19	1.809	
20,300.0	10,887.7	20,953.6	10,892.1	280.1	282.3	90.25	-1,568.5	9,601.4	1,001.9	441.8	560.03	1.789	
20,400.0	10,888.2	21,053.6	10,892.6	283.0	285.2	90.25	-1,568.9	9,701.3	1,001.5	435.6	565.87	1.770	
20,500.0	10,888.7	21,153.6	10,893.2	285.9	288.1	90.25	-1,569.3	9,801.3	1,001.1	429.4	571.71	1.751	
20,600.0	10,889.3	21,253.6	10,893.7	288.8	291.0	90.25	-1,569.7	9,901.3	1,000.7	423.1	577.55	1.733	
20,700.0	10,889.8	21,353.6	10,894.2	291.7	293.9	90.25	-1,570.1	10,001.3	1,000.3	416.9	583.39	1.715	
20,717.6	10,889.9	21,371.1	10,894.3	292.3	294.4	90.25	-1,570.2	10,018.9	1,000.2	415.8	584.42	1.711 SF	

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
0.0	0.0	0.0	0.0	0.0	0.0	-155.30	-119.6	-55.0	131.6					
100.0	100.0	100.0	100.0	0.1	0.1	-155.30	-119.6	-55.0	131.6	131.4	0.17	780.729		
200.0	200.0	200.0	200.0	0.3	0.3	-155.30	-119.6	-55.0	131.6	131.0	0.62	212.926		
300.0	300.0	300.0	300.0	0.5	0.5	-155.30	-119.6	-55.0	131.6	130.5	1.07	123.273		
400.0	400.0	400.0	400.0	0.8	0.8	-155.30	-119.6	-55.0	131.6	130.1	1.52	86.748		
500.0	500.0	500.0	500.0	1.0	1.0	-155.30	-119.6	-55.0	131.6	129.6	1.97	66.920		
600.0	600.0	600.0	600.0	1.2	1.2	-155.30	-119.6	-55.0	131.6	129.2	2.42	54.470		
700.0	700.0	700.0	700.0	1.4	1.4	-155.30	-119.6	-55.0	131.6	128.7	2.87	45.925		
800.0	800.0	800.0	800.0	1.7	1.7	-155.30	-119.6	-55.0	131.6	128.3	3.32	39.698		
900.0	900.0	900.0	900.0	1.9	1.9	-155.30	-119.6	-55.0	131.6	127.8	3.76	34.958		
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.1	-155.30	-119.6	-55.0	131.6	127.4	4.21	31.229		
1,100.0	1,100.0	1,100.0	1,100.0	2.3	2.3	-155.30	-119.6	-55.0	131.6	126.9	4.66	28.219		
1,200.0	1,200.0	1,200.0	1,200.0	2.6	2.6	-155.30	-119.6	-55.0	131.6	126.5	5.11	25.738		
1,300.0	1,300.0	1,300.0	1,300.0	2.8	2.8	-155.30	-119.6	-55.0	131.6	126.0	5.56	23.658		
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.0	-155.30	-119.6	-55.0	131.6	125.6	6.01	21.890		
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.2	-155.30	-119.6	-55.0	131.6	125.1	6.46	20.367		
1,600.0	1,600.0	1,600.0	1,600.0	3.5	3.5	-155.30	-119.6	-55.0	131.6	124.7	6.91	19.042		
1,700.0	1,700.0	1,700.0	1,700.0	3.7	3.7	-155.30	-119.6	-55.0	131.6	124.3	7.36	17.879		
1,800.0	1,800.0	1,800.0	1,800.0	3.9	3.9	-155.30	-119.6	-55.0	131.6	123.8	7.81	16.850		
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.1	-155.30	-119.6	-55.0	131.6	123.4	8.26	15.933		
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.4	-155.30	-119.6	-55.0	131.6	122.9	8.71	15.111		
2,100.0	2,100.0	2,100.0	2,100.0	4.6	4.6	-155.30	-119.6	-55.0	131.6	122.5	9.16	14.369		
2,200.0	2,200.0	2,200.0	2,200.0	4.8	4.8	-155.30	-119.6	-55.0	131.6	122.0	9.61	13.697		
2,300.0	2,300.0	2,300.0	2,300.0	5.0	5.0	-155.30	-119.6	-55.0	131.6	121.6	10.06	13.085		
2,400.0	2,400.0	2,400.0	2,400.0	5.3	5.3	-155.30	-119.6	-55.0	131.6	121.1	10.51	12.525		
2,500.0	2,500.0	2,500.0	2,500.0	5.5	5.5	-155.30	-119.6	-55.0	131.6	120.7	10.96	12.011 CC, ES		
2,600.0	2,600.0	2,596.8	2,596.8	5.7	5.7	114.83	-121.1	-54.4	133.5	122.2	11.36	11.757		
2,650.0	2,649.9	2,645.1	2,645.1	5.8	5.8	114.99	-123.0	-53.7	136.0	124.4	11.54	11.779		
2,700.0	2,699.9	2,694.9	2,694.7	5.9	5.8	115.20	-125.5	-52.9	138.9	127.2	11.73	11.842		
2,800.0	2,799.7	2,794.7	2,794.4	6.1	6.0	115.61	-130.4	-51.1	144.8	132.7	12.11	11.958		
2,900.0	2,899.6	2,894.5	2,894.1	6.3	6.2	115.98	-135.3	-49.3	150.8	138.3	12.50	12.060		
3,000.0	2,999.5	2,994.3	2,993.8	6.5	6.4	116.32	-140.2	-47.5	156.7	143.8	12.89	12.151		
3,100.0	3,099.3	3,094.2	3,093.5	6.7	6.6	116.64	-145.1	-45.7	162.6	149.3	13.30	12.231		
3,200.0	3,199.2	3,194.0	3,193.2	7.0	6.8	116.93	-150.0	-43.9	168.6	154.9	13.70	12.301		
3,300.0	3,299.0	3,293.8	3,292.8	7.2	7.0	117.21	-154.9	-42.1	174.5	160.4	14.11	12.364		
3,400.0	3,398.9	3,393.6	3,392.5	7.4	7.2	117.46	-159.8	-40.3	180.4	165.9	14.53	12.419		
3,501.2	3,500.0	3,494.7	3,493.4	7.6	7.4	117.71	-164.8	-38.5	186.5	171.5	14.96	12.468		
3,600.0	3,598.7	3,597.6	3,596.3	7.8	7.6	117.90	-168.3	-37.2	190.5	175.1	15.38	12.389		
3,651.2	3,649.9	3,651.2	3,649.9	7.9	7.7	-152.07	-168.8	-37.1	191.1	175.5	15.51	12.315		
3,700.0	3,698.7	3,700.0	3,698.7	8.0	7.8	-152.07	-168.8	-37.1	191.1	175.3	15.72	12.153		
3,800.0	3,798.7	3,800.0	3,798.7	8.2	8.0	-152.07	-168.8	-37.1	191.1	174.9	16.16	11.821		
3,900.0	3,898.7	3,900.0	3,898.7	8.5	8.2	-152.07	-168.8	-37.1	191.1	174.5	16.60	11.507		
4,000.0	3,998.7	4,000.0	3,998.7	8.7	8.5	-152.07	-168.8	-37.1	191.1	174.0	17.04	11.209		
4,100.0	4,098.7	4,100.0	4,098.7	8.9	8.7	-152.07	-168.8	-37.1	191.1	173.6	17.49	10.926		
4,200.0	4,198.7	4,200.0	4,198.7	9.1	8.9	-152.07	-168.8	-37.1	191.1	173.1	17.93	10.657		
4,300.0	4,298.7	4,300.0	4,298.7	9.4	9.1	-152.07	-168.8	-37.1	191.1	172.7	18.37	10.400		
4,400.0	4,398.7	4,400.0	4,398.7	9.6	9.3	-152.07	-168.8	-37.1	191.1	172.2	18.81	10.155		
4,500.0	4,498.7	4,500.0	4,498.7	9.8	9.6	-152.07	-168.8	-37.1	191.1	171.8	19.26	9.922		
4,600.0	4,598.7	4,600.0	4,598.7	10.0	9.8	-152.07	-168.8	-37.1	191.1	171.4	19.70	9.698		
4,700.0	4,698.7	4,700.0	4,698.7	10.2	10.0	-152.07	-168.8	-37.1	191.1	170.9	20.14	9.485		
4,800.0	4,798.7	4,800.0	4,798.7	10.5	10.2	-152.07	-168.8	-37.1	191.1	170.5	20.59	9.280		
4,900.0	4,898.7	4,900.0	4,898.7	10.7	10.4	-152.07	-168.8	-37.1	191.1	170.0	21.03	9.084		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
5,000.0	4,998.7	5,000.0	4,998.7	10.9	10.7	-152.07	-168.8	-37.1	191.1	169.6	21.48	8.896		
5,100.0	5,098.7	5,100.0	5,098.7	11.1	10.9	-152.07	-168.8	-37.1	191.1	169.1	21.92	8.716		
5,200.0	5,198.7	5,200.0	5,198.7	11.4	11.1	-152.07	-168.8	-37.1	191.1	168.7	22.37	8.542		
5,300.0	5,298.7	5,300.0	5,298.7	11.6	11.3	-152.07	-168.8	-37.1	191.1	168.2	22.81	8.376		
5,400.0	5,398.7	5,400.0	5,398.7	11.8	11.6	-152.07	-168.8	-37.1	191.1	167.8	23.26	8.215		
5,500.0	5,498.7	5,500.0	5,498.7	12.0	11.8	-152.07	-168.8	-37.1	191.1	167.4	23.70	8.061		
5,600.0	5,598.7	5,600.0	5,598.7	12.2	12.0	-152.07	-168.8	-37.1	191.1	166.9	24.15	7.912		
5,700.0	5,698.7	5,700.0	5,698.7	12.5	12.2	-152.07	-168.8	-37.1	191.1	166.5	24.59	7.769		
5,800.0	5,798.7	5,800.0	5,798.7	12.7	12.4	-152.07	-168.8	-37.1	191.1	166.0	25.04	7.631		
5,900.0	5,898.7	5,900.0	5,898.7	12.9	12.7	-152.07	-168.8	-37.1	191.1	165.6	25.48	7.497		
6,000.0	5,998.7	6,000.0	5,998.7	13.1	12.9	-152.07	-168.8	-37.1	191.1	165.1	25.93	7.368		
6,100.0	6,098.7	6,100.0	6,098.7	13.4	13.1	-152.07	-168.8	-37.1	191.1	164.7	26.38	7.243		
6,200.0	6,198.7	6,200.0	6,198.7	13.6	13.3	-152.07	-168.8	-37.1	191.1	164.2	26.82	7.123		
6,300.0	6,298.7	6,300.0	6,298.7	13.8	13.6	-152.07	-168.8	-37.1	191.1	163.8	27.27	7.006		
6,400.0	6,398.7	6,400.0	6,398.7	14.0	13.8	-152.07	-168.8	-37.1	191.1	163.3	27.72	6.893		
6,500.0	6,498.7	6,500.0	6,498.7	14.3	14.0	-152.07	-168.8	-37.1	191.1	162.9	28.16	6.784		
6,600.0	6,598.7	6,600.0	6,598.7	14.5	14.2	-152.07	-168.8	-37.1	191.1	162.4	28.61	6.678		
6,700.0	6,698.7	6,700.0	6,698.7	14.7	14.4	-152.07	-168.8	-37.1	191.1	162.0	29.06	6.576		
6,800.0	6,798.7	6,800.0	6,798.7	14.9	14.7	-152.07	-168.8	-37.1	191.1	161.6	29.50	6.476		
6,900.0	6,898.7	6,900.0	6,898.7	15.2	14.9	-152.07	-168.8	-37.1	191.1	161.1	29.95	6.379		
7,000.0	6,998.7	7,000.0	6,998.7	15.4	15.1	-152.07	-168.8	-37.1	191.1	160.7	30.40	6.286		
7,100.0	7,098.7	7,100.0	7,098.7	15.6	15.3	-152.07	-168.8	-37.1	191.1	160.2	30.84	6.194		
7,200.0	7,198.7	7,200.0	7,198.7	15.8	15.6	-152.07	-168.8	-37.1	191.1	159.8	31.29	6.106		
7,300.0	7,298.7	7,300.0	7,298.7	16.0	15.8	-152.07	-168.8	-37.1	191.1	159.3	31.74	6.020		
7,400.0	7,398.7	7,400.0	7,398.7	16.3	16.0	-152.07	-168.8	-37.1	191.1	158.9	32.19	5.936		
7,500.0	7,498.7	7,500.0	7,498.7	16.5	16.2	-152.07	-168.8	-37.1	191.1	158.4	32.63	5.855		
7,600.0	7,598.7	7,600.0	7,598.7	16.7	16.5	-152.07	-168.8	-37.1	191.1	158.0	33.08	5.776		
7,700.0	7,698.7	7,700.0	7,698.7	16.9	16.7	-152.07	-168.8	-37.1	191.1	157.5	33.53	5.699		
7,800.0	7,798.7	7,800.0	7,798.7	17.2	16.9	-152.07	-168.8	-37.1	191.1	157.1	33.98	5.623		
7,900.0	7,898.7	7,900.0	7,898.7	17.4	17.1	-152.07	-168.8	-37.1	191.1	156.6	34.42	5.550		
8,000.0	7,998.7	8,000.0	7,998.7	17.6	17.3	-152.07	-168.8	-37.1	191.1	156.2	34.87	5.479		
8,100.0	8,098.7	8,100.0	8,098.7	17.8	17.6	-152.07	-168.8	-37.1	191.1	155.7	35.32	5.410		
8,200.0	8,198.7	8,200.0	8,198.7	18.1	17.8	-152.07	-168.8	-37.1	191.1	155.3	35.77	5.342		
8,300.0	8,298.7	8,300.0	8,298.7	18.3	18.0	-152.07	-168.8	-37.1	191.1	154.8	36.21	5.276		
8,400.0	8,398.7	8,400.0	8,398.7	18.5	18.2	-152.07	-168.8	-37.1	191.1	154.4	36.66	5.211		
8,500.0	8,498.7	8,500.0	8,498.7	18.7	18.5	-152.07	-168.8	-37.1	191.1	153.9	37.11	5.149		
8,600.0	8,598.7	8,600.0	8,598.7	19.0	18.7	-152.07	-168.8	-37.1	191.1	153.5	37.56	5.087		
8,700.0	8,698.7	8,700.0	8,698.7	19.2	18.9	-152.07	-168.8	-37.1	191.1	153.1	38.01	5.027		
8,800.0	8,798.7	8,800.0	8,798.7	19.4	19.1	-152.07	-168.8	-37.1	191.1	152.6	38.45	4.969		
8,900.0	8,898.7	8,900.0	8,898.7	19.6	19.4	-152.07	-168.8	-37.1	191.1	152.2	38.90	4.911		
9,000.0	8,998.7	9,000.0	8,998.7	19.9	19.6	-152.07	-168.8	-37.1	191.1	151.7	39.35	4.855		
9,100.0	9,098.7	9,100.0	9,098.7	20.1	19.8	-152.07	-168.8	-37.1	191.1	151.3	39.80	4.801		
9,200.0	9,198.7	9,200.0	9,198.7	20.3	20.0	-152.07	-168.8	-37.1	191.1	150.8	40.25	4.747		
9,300.0	9,298.7	9,300.0	9,298.7	20.5	20.3	-152.07	-168.8	-37.1	191.1	150.4	40.69	4.695		
9,400.0	9,398.7	9,400.0	9,398.7	20.8	20.5	-152.07	-168.8	-37.1	191.1	149.9	41.14	4.644		
9,500.0	9,498.7	9,500.0	9,498.7	21.0	20.7	-152.07	-168.8	-37.1	191.1	149.5	41.59	4.594		
9,600.0	9,598.7	9,600.0	9,598.7	21.2	20.9	-152.07	-168.8	-37.1	191.1	149.0	42.04	4.545		
9,700.0	9,698.7	9,700.0	9,698.7	21.4	21.2	-152.07	-168.8	-37.1	191.1	148.6	42.49	4.497		
9,800.0	9,798.7	9,800.0	9,798.7	21.6	21.4	-152.07	-168.8	-37.1	191.1	148.1	42.93	4.450		
9,900.0	9,898.7	9,900.0	9,898.7	21.9	21.6	-152.07	-168.8	-37.1	191.1	147.7	43.38	4.404		
10,000.0	9,998.7	10,000.0	9,998.7	22.1	21.8	-152.07	-168.8	-37.1	191.1	147.2	43.83	4.359		
10,100.0	10,098.7	10,100.0	10,098.7	22.3	22.0	-152.07	-168.8	-37.1	191.1	146.8	44.28	4.315		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
10,200.0	10,198.7	10,200.0	10,198.7	22.5	22.3	-152.07	-168.8	-37.1	191.1	146.3	44.73	4.272		
10,259.7	10,258.4	10,259.7	10,258.4	22.7	22.4	-152.07	-168.8	-37.1	191.1	146.1	45.00	4.246		
10,300.0	10,298.7	10,295.4	10,294.0	22.8	22.5	-152.20	-169.2	-36.8	191.4	146.2	45.16	4.238		
10,363.3	10,362.0	10,347.0	10,345.4	22.9	22.6	-153.48	-173.7	-34.3	194.8	149.4	45.39	4.292		
10,375.0	10,373.7	10,356.4	10,354.7	22.9	22.6	86.05	-175.0	-33.5	195.9	150.4	45.52	4.303		
10,400.0	10,398.7	10,375.0	10,372.9	23.0	22.6	85.13	-178.1	-31.7	198.5	152.9	45.60	4.353		
10,425.0	10,423.5	10,396.3	10,393.6	23.0	22.7	84.19	-182.4	-29.2	201.7	156.0	45.70	4.414		
10,450.0	10,448.2	10,416.0	10,412.6	23.1	22.7	83.36	-187.1	-26.5	205.3	159.5	45.78	4.485		
10,475.0	10,472.7	10,435.6	10,431.2	23.1	22.8	82.59	-192.4	-23.5	209.4	163.6	45.87	4.566		
10,500.0	10,496.8	10,455.1	10,449.4	23.2	22.8	81.88	-198.3	-20.0	214.0	168.0	45.96	4.656		
10,525.0	10,520.6	10,475.0	10,467.8	23.3	22.8	81.23	-205.1	-16.2	218.9	172.9	46.05	4.755		
10,550.0	10,544.0	10,493.5	10,484.5	23.3	22.9	80.64	-211.9	-12.2	224.3	178.2	46.13	4.862		
10,575.0	10,566.8	10,512.4	10,501.3	23.4	22.9	80.10	-219.5	-7.8	230.0	183.8	46.21	4.977		
10,600.0	10,589.1	10,531.2	10,517.6	23.5	23.0	79.61	-227.5	-3.2	236.1	189.8	46.29	5.100		
10,625.0	10,610.8	10,550.0	10,533.5	23.5	23.0	79.17	-236.2	1.8	242.5	196.1	46.38	5.229		
10,650.0	10,631.8	10,568.3	10,548.6	23.6	23.1	78.76	-245.1	7.0	249.3	202.8	46.46	5.365		
10,675.0	10,652.0	10,586.6	10,563.3	23.7	23.1	78.38	-254.5	12.4	256.3	209.8	46.55	5.506		
10,700.0	10,671.5	10,604.7	10,577.5	23.8	23.2	78.03	-264.3	18.1	263.7	217.0	46.64	5.653		
10,725.0	10,690.1	10,625.0	10,592.8	23.9	23.3	77.78	-275.8	24.7	271.3	224.6	46.75	5.803		
10,750.0	10,707.8	10,640.5	10,604.1	24.0	23.3	77.38	-285.0	30.0	279.2	232.3	46.85	5.960		
10,775.0	10,724.5	10,658.2	10,616.6	24.1	23.4	77.09	-295.8	36.3	287.3	240.4	46.96	6.119		
10,800.0	10,740.3	10,675.0	10,628.1	24.3	23.5	76.77	-306.5	42.4	295.7	248.7	47.08	6.281		
10,825.0	10,755.0	10,693.1	10,639.9	24.4	23.5	76.52	-318.4	49.3	304.4	257.1	47.23	6.445		
10,850.0	10,768.7	10,710.4	10,650.7	24.6	23.6	76.24	-330.1	56.0	313.2	265.8	47.38	6.611		
10,875.0	10,781.2	10,725.0	10,659.4	24.8	23.7	75.83	-340.2	61.9	322.3	274.8	47.53	6.781		
10,900.0	10,792.6	10,744.6	10,670.6	25.0	23.8	75.69	-354.2	70.0	331.5	283.8	47.74	6.945		
10,925.0	10,802.8	10,761.6	10,679.7	25.2	23.9	75.42	-366.6	77.1	341.0	293.0	47.94	7.112		
10,950.0	10,811.8	10,778.5	10,688.2	25.4	24.0	75.14	-379.3	84.4	350.5	302.4	48.16	7.278		
10,975.0	10,819.6	10,795.4	10,696.2	25.7	24.1	74.86	-392.1	91.8	360.3	311.9	48.41	7.443		
11,000.0	10,826.1	10,812.2	10,703.7	25.9	24.2	74.58	-405.1	99.4	370.2	321.5	48.67	7.606		
11,025.0	10,831.3	10,828.9	10,710.6	26.2	24.3	74.30	-418.3	107.0	380.2	331.2	48.95	7.767		
11,050.0	10,835.3	10,845.7	10,717.0	26.5	24.4	74.02	-431.8	114.7	390.3	341.1	49.26	7.924		
11,075.0	10,837.9	10,862.4	10,722.8	26.8	24.6	73.74	-445.3	122.6	400.5	351.0	49.58	8.078		
11,100.0	10,839.3	10,879.2	10,728.1	27.1	24.7	73.46	-459.1	130.5	410.8	360.9	49.93	8.228		
11,110.8	10,839.5	10,886.4	10,730.2	27.2	24.8	73.34	-465.1	134.0	415.3	365.2	50.09	8.292		
11,125.8	10,839.5	10,896.6	10,733.0	27.4	24.8	73.94	-473.6	138.9	421.6	371.2	50.45	8.357		
11,200.0	10,839.9	10,947.6	10,743.7	28.5	25.3	76.13	-516.8	163.8	455.8	403.6	52.15	8.740		
11,300.0	10,840.5	11,016.9	10,749.7	30.1	26.0	77.72	-576.5	198.3	508.9	454.5	54.36	9.362		
11,400.0	10,841.0	11,121.3	10,750.3	31.9	27.2	78.93	-665.9	252.1	566.5	509.6	56.92	9.952		
11,500.0	10,841.5	11,239.1	10,750.9	33.9	28.8	80.04	-763.2	318.5	623.8	564.0	59.76	10.438		
11,600.0	10,842.1	11,363.2	10,751.6	36.0	30.7	80.98	-861.1	394.8	680.3	617.5	62.79	10.835		
11,700.0	10,842.6	11,494.4	10,752.4	38.1	32.9	81.78	-958.9	482.3	735.7	669.7	65.99	11.149		
11,800.0	10,843.2	11,633.6	10,753.2	40.4	35.6	82.47	-1,055.8	582.1	789.7	720.3	69.39	11.381		
11,900.0	10,843.7	11,781.7	10,754.1	42.7	38.6	83.07	-1,150.7	695.7	841.8	768.8	72.95	11.539		
12,000.0	10,844.2	11,939.6	10,755.0	45.0	42.0	83.60	-1,241.8	824.6	891.5	814.8	76.71	11.623		
12,100.0	10,844.8	12,108.2	10,756.0	47.4	45.7	84.07	-1,327.1	970.1	938.4	857.8	80.65	11.637		
12,110.4	10,844.8	12,126.5	10,756.2	47.6	46.1	84.11	-1,335.5	986.3	943.1	862.1	81.08	11.632		
12,200.0	10,845.3	12,290.1	10,757.1	49.8	49.9	84.56	-1,404.2	1,134.7	980.0	893.4	86.62	11.314		
12,300.0	10,845.8	12,487.1	10,758.3	52.2	54.5	84.93	-1,469.7	1,320.4	1,012.4	918.8	93.56	10.821		
12,400.0	10,846.3	12,696.3	10,759.5	54.7	59.3	85.17	-1,517.9	1,523.9	1,034.6	933.4	101.20	10.224		
12,500.0	10,846.8	12,913.7	10,760.7	57.3	64.3	85.29	-1,544.1	1,739.5	1,045.9	936.5	109.34	9.565		
12,600.0	10,847.4	13,077.7	10,761.6	59.9	68.0	85.30	-1,548.2	1,903.5	1,047.0	930.7	116.26	9.005		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6													Offset Site Error:	0.0 usft
Survey Program: 0-MWID													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance							Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
12,700.0	10,847.9	13,177.7	10,762.1	62.5	70.2	85.30	-1,548.4	2,003.5	1,046.4	924.8	121.57	8.608		
12,800.0	10,848.4	13,277.7	10,762.6	65.1	72.6	85.29	-1,548.7	2,103.5	1,045.9	918.9	126.93	8.240		
12,900.0	10,848.9	13,377.7	10,763.2	67.8	74.9	85.29	-1,548.9	2,203.5	1,045.3	913.0	132.33	7.900		
13,000.0	10,849.5	13,477.7	10,763.7	70.4	77.3	85.29	-1,549.2	2,303.5	1,044.8	907.0	137.76	7.584		
13,100.0	10,850.0	13,577.7	10,764.2	73.1	79.8	85.29	-1,549.5	2,403.5	1,044.3	901.0	143.23	7.291		
13,200.0	10,850.5	13,677.7	10,764.7	75.9	82.2	85.28	-1,549.7	2,503.5	1,043.7	895.0	148.73	7.018		
13,300.0	10,851.0	13,777.7	10,765.3	78.6	84.7	85.28	-1,550.0	2,603.5	1,043.2	888.9	154.25	6.763		
13,400.0	10,851.6	13,877.7	10,765.8	81.3	87.3	85.28	-1,550.3	2,703.5	1,042.6	882.9	159.79	6.525		
13,500.0	10,852.1	13,977.7	10,766.3	84.1	89.8	85.28	-1,550.5	2,803.5	1,042.1	876.8	165.36	6.302		
13,600.0	10,852.6	14,077.7	10,766.8	86.9	92.4	85.27	-1,550.8	2,903.5	1,041.6	870.6	170.94	6.093		
13,700.0	10,853.1	14,177.7	10,767.3	89.6	95.0	85.27	-1,551.1	3,003.5	1,041.0	864.5	176.54	5.897		
13,800.0	10,853.7	14,277.7	10,767.9	92.4	97.6	85.27	-1,551.3	3,103.5	1,040.5	858.3	182.15	5.712		
13,900.0	10,854.2	14,377.7	10,768.4	95.2	100.2	85.27	-1,551.6	3,203.5	1,040.0	852.2	187.78	5.538		
14,000.0	10,854.7	14,477.7	10,768.9	98.0	102.9	85.26	-1,551.9	3,303.5	1,039.4	846.0	193.42	5.374		
14,100.0	10,855.2	14,577.7	10,769.4	100.8	105.5	85.26	-1,552.1	3,403.5	1,038.9	839.8	199.08	5.218		
14,200.0	10,855.7	14,677.7	10,770.0	103.7	108.2	85.26	-1,552.4	3,503.5	1,038.3	833.6	204.74	5.071		
14,300.0	10,856.3	14,777.7	10,770.5	106.5	110.9	85.26	-1,552.6	3,603.5	1,037.8	827.4	210.41	4.932		
14,400.0	10,856.8	14,877.7	10,771.0	109.3	113.6	85.25	-1,552.9	3,703.5	1,037.3	821.2	216.10	4.800		
14,500.0	10,857.3	14,977.7	10,771.5	112.2	116.3	85.25	-1,553.2	3,803.4	1,036.7	814.9	221.79	4.674		
14,600.0	10,857.8	15,077.7	10,772.1	115.0	119.0	85.25	-1,553.4	3,903.4	1,036.2	808.7	227.49	4.555		
14,700.0	10,858.4	15,177.7	10,772.6	117.9	121.8	85.25	-1,553.7	4,003.4	1,035.7	802.5	233.19	4.441		
14,800.0	10,858.9	15,277.7	10,773.1	120.7	124.5	85.24	-1,554.0	4,103.4	1,035.1	796.2	238.90	4.333		
14,900.0	10,859.4	15,377.7	10,773.6	123.6	127.3	85.24	-1,554.2	4,203.4	1,034.6	790.0	244.62	4.229		
15,000.0	10,859.9	15,477.7	10,774.2	126.4	130.0	85.24	-1,554.5	4,303.4	1,034.0	783.7	250.34	4.130		
15,100.0	10,860.5	15,577.7	10,774.7	129.3	132.8	85.24	-1,554.8	4,403.4	1,033.5	777.4	256.07	4.036		
15,200.0	10,861.0	15,677.7	10,775.2	132.1	135.6	85.23	-1,555.0	4,503.4	1,033.0	771.2	261.81	3.946		
15,300.0	10,861.5	15,777.7	10,775.7	135.0	138.3	85.23	-1,555.3	4,603.4	1,032.4	764.9	267.55	3.859		
15,400.0	10,862.0	15,877.7	10,776.3	137.9	141.1	85.23	-1,555.5	4,703.4	1,031.9	758.6	273.29	3.776		
15,500.0	10,862.6	15,977.7	10,776.8	140.7	143.9	85.23	-1,555.8	4,803.4	1,031.3	752.3	279.04	3.696		
15,600.0	10,863.1	16,077.7	10,777.3	143.6	146.7	85.22	-1,556.1	4,903.4	1,030.8	746.0	284.79	3.620		
15,700.0	10,863.6	16,177.7	10,777.8	146.5	149.5	85.22	-1,556.3	5,003.4	1,030.3	739.7	290.54	3.546		
15,800.0	10,864.1	16,277.7	10,778.3	149.4	152.3	85.22	-1,556.6	5,103.4	1,029.7	733.4	296.30	3.475		
15,900.0	10,864.7	16,377.7	10,778.9	152.3	155.1	85.22	-1,556.9	5,203.4	1,029.2	727.1	302.06	3.407		
16,000.0	10,865.2	16,477.7	10,779.4	155.1	158.0	85.21	-1,557.1	5,303.4	1,028.7	720.8	307.82	3.342		
16,100.0	10,865.7	16,577.7	10,779.9	158.0	160.8	85.21	-1,557.4	5,403.4	1,028.1	714.5	313.59	3.279		
16,200.0	10,866.2	16,677.7	10,780.4	160.9	163.6	85.21	-1,557.7	5,503.4	1,027.6	708.2	319.36	3.218		
16,300.0	10,866.7	16,777.7	10,781.0	163.8	166.4	85.21	-1,557.9	5,603.4	1,027.0	701.9	325.13	3.159		
16,400.0	10,867.3	16,877.7	10,781.5	166.7	169.3	85.20	-1,558.2	5,703.4	1,026.5	695.6	330.90	3.102		
16,500.0	10,867.8	16,977.7	10,782.0	169.6	172.1	85.20	-1,558.5	5,803.4	1,026.0	689.3	336.68	3.047		
16,600.0	10,868.3	17,077.7	10,782.5	172.5	174.9	85.20	-1,558.7	5,903.4	1,025.4	683.0	342.46	2.994		
16,700.0	10,868.8	17,177.7	10,783.1	175.4	177.8	85.20	-1,559.0	6,003.4	1,024.9	676.6	348.24	2.943		
16,800.0	10,869.4	17,277.7	10,783.6	178.3	180.6	85.19	-1,559.2	6,103.4	1,024.3	670.3	354.02	2.893		
16,900.0	10,869.9	17,377.7	10,784.1	181.2	183.5	85.19	-1,559.5	6,203.4	1,023.8	664.0	359.81	2.845		
17,000.0	10,870.4	17,477.7	10,784.6	184.1	186.3	85.19	-1,559.8	6,303.4	1,023.3	657.7	365.59	2.799		
17,100.0	10,870.9	17,577.7	10,785.2	187.0	189.2	85.19	-1,560.0	6,403.4	1,022.7	651.4	371.38	2.754		
17,200.0	10,871.5	17,677.7	10,785.7	189.9	192.0	85.18	-1,560.3	6,503.4	1,022.2	645.0	377.17	2.710		
17,300.0	10,872.0	17,777.7	10,786.2	192.8	194.9	85.18	-1,560.6	6,603.4	1,021.7	638.7	382.96	2.668		
17,400.0	10,872.5	17,877.7	10,786.7	195.7	197.8	85.18	-1,560.8	6,703.4	1,021.1	632.4	388.75	2.627		
17,500.0	10,873.0	17,977.7	10,787.2	198.6	200.6	85.18	-1,561.1	6,803.4	1,020.6	626.0	394.55	2.587		
17,600.0	10,873.6	18,077.7	10,787.8	201.5	203.5	85.17	-1,561.4	6,903.3	1,020.0	619.7	400.34	2.548		
17,700.0	10,874.1	18,177.7	10,788.3	204.4	206.3	85.17	-1,561.6	7,003.3	1,019.5	613.4	406.14	2.510		
17,800.0	10,874.6	18,277.7	10,788.8	207.3	209.2	85.17	-1,561.9	7,103.3	1,019.0	607.0	411.94	2.474		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

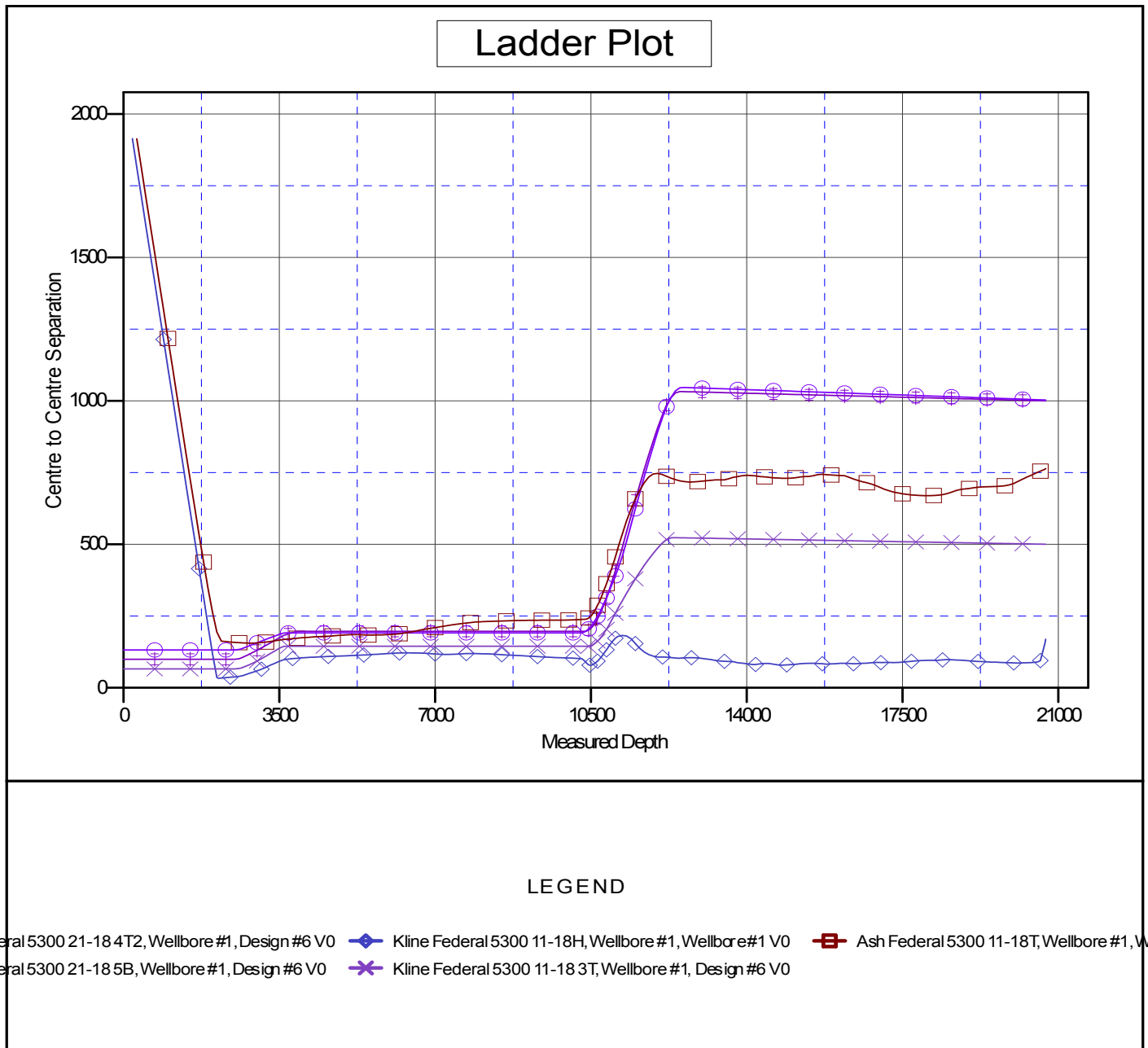
Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
17,900.0	10,875.1	18,377.7	10,789.3	210.2	212.1	85.17	-1,562.2	7,203.3	1,018.4	600.7	417.74	2.438	
18,000.0	10,875.6	18,477.7	10,789.9	213.1	215.0	85.16	-1,562.4	7,303.3	1,017.9	594.4	423.54	2.403	
18,100.0	10,876.2	18,577.7	10,790.4	216.0	217.8	85.16	-1,562.7	7,403.3	1,017.4	588.0	429.34	2.370	
18,200.0	10,876.7	18,677.7	10,790.9	218.9	220.7	85.16	-1,562.9	7,503.3	1,016.8	581.7	435.14	2.337	
18,300.0	10,877.2	18,777.7	10,791.4	221.8	223.6	85.16	-1,563.2	7,603.3	1,016.3	575.3	440.94	2.305	
18,400.0	10,877.7	18,877.7	10,792.0	224.7	226.5	85.15	-1,563.5	7,703.3	1,015.7	569.0	446.75	2.274	
18,500.0	10,878.3	18,977.7	10,792.5	227.6	229.3	85.15	-1,563.7	7,803.3	1,015.2	562.7	452.55	2.243	
18,600.0	10,878.8	19,077.7	10,793.0	230.5	232.2	85.15	-1,564.0	7,903.3	1,014.7	556.3	458.36	2.214	
18,700.0	10,879.3	19,177.7	10,793.5	233.4	235.1	85.15	-1,564.3	8,003.3	1,014.1	550.0	464.16	2.185	
18,800.0	10,879.8	19,277.7	10,794.1	236.4	238.0	85.14	-1,564.5	8,103.3	1,013.6	543.6	469.97	2.157	
18,900.0	10,880.4	19,377.7	10,794.6	239.3	240.9	85.14	-1,564.8	8,203.3	1,013.0	537.3	475.78	2.129	
19,000.0	10,880.9	19,477.7	10,795.1	242.2	243.7	85.14	-1,565.1	8,303.3	1,012.5	530.9	481.59	2.102	
19,100.0	10,881.4	19,577.7	10,795.6	245.1	246.6	85.14	-1,565.3	8,403.3	1,012.0	524.6	487.40	2.076	
19,200.0	10,881.9	19,677.7	10,796.1	248.0	249.5	85.13	-1,565.6	8,503.3	1,011.4	518.2	493.21	2.051	
19,300.0	10,882.5	19,777.7	10,796.7	250.9	252.4	85.13	-1,565.8	8,603.3	1,010.9	511.9	499.02	2.026	
19,400.0	10,883.0	19,877.6	10,797.2	253.8	255.3	85.13	-1,566.1	8,703.3	1,010.4	505.5	504.83	2.001	
19,500.0	10,883.5	19,977.6	10,797.7	256.7	258.2	85.13	-1,566.4	8,803.3	1,009.8	499.2	510.64	1.978	
19,600.0	10,884.0	20,077.6	10,798.2	259.7	261.1	85.12	-1,566.6	8,903.3	1,009.3	492.8	516.45	1.954	
19,700.0	10,884.5	20,177.6	10,798.8	262.6	264.0	85.12	-1,566.9	9,003.3	1,008.7	486.5	522.27	1.931	
19,800.0	10,885.1	20,277.6	10,799.3	265.5	266.9	85.12	-1,567.2	9,103.3	1,008.2	480.1	528.08	1.909	
19,900.0	10,885.6	20,377.6	10,799.8	268.4	269.7	85.11	-1,567.4	9,203.3	1,007.7	473.8	533.89	1.887	
20,000.0	10,886.1	20,477.6	10,800.3	271.3	272.6	85.11	-1,567.7	9,303.3	1,007.1	467.4	539.71	1.866	
20,100.0	10,886.6	20,577.6	10,800.9	274.2	275.5	85.11	-1,568.0	9,403.3	1,006.6	461.1	545.52	1.845	
20,200.0	10,887.2	20,677.6	10,801.4	277.2	278.4	85.11	-1,568.2	9,503.3	1,006.1	454.7	551.34	1.825	
20,300.0	10,887.7	20,777.6	10,801.9	280.1	281.3	85.10	-1,568.5	9,603.3	1,005.5	448.4	557.15	1.805	
20,400.0	10,888.2	20,877.6	10,802.4	283.0	284.2	85.10	-1,568.8	9,703.3	1,005.0	442.0	562.97	1.785	
20,500.0	10,888.7	20,977.6	10,803.0	285.9	287.1	85.10	-1,569.0	9,803.3	1,004.4	435.7	568.79	1.766	
20,600.0	10,889.3	21,077.6	10,803.5	288.8	290.0	85.10	-1,569.3	9,903.3	1,003.9	429.3	574.61	1.747	
20,700.0	10,889.8	21,177.6	10,804.0	291.7	292.9	85.09	-1,569.5	10,003.3	1,003.4	422.9	580.42	1.729	
20,717.6	10,889.9	21,195.2	10,804.1	292.3	293.4	85.09	-1,569.6	10,020.8	1,003.3	421.8	581.44	1.725 SF	

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Reference Depths are relative to WELL @ 2078.0usft (Original Well Ele
Offset Depths are relative to Offset Datum
Central Meridian is 100° 30' 0.000 W

Coordinates are relative to: Kline Federal 5300 11-18 2T2
Coordinate System is US State Plane 1983, North Dakota Northern Zone
Grid Convergence at Surface is: -2.31°



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 2T2
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 2T2	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Reference Depths are relative to WELL @ 2078.0usft (Original Well Ele
Offset Depths are relative to Offset Datum
Central Meridian is 100° 30' 0.000 W

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